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1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ. SCHEM, MLB, X1036 LAST_MODIFICATION=Thu Jul 12 17:54:35 2018												REV	ECN	DESCRIPTION OF REVISION	CK APPD / DATE		
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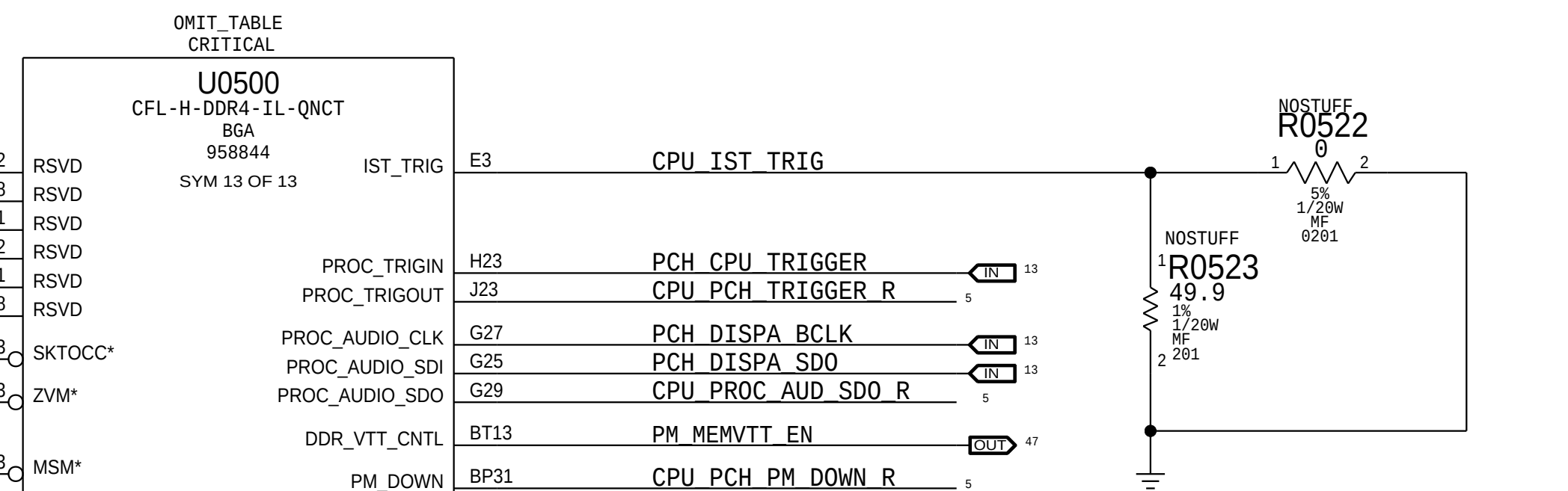
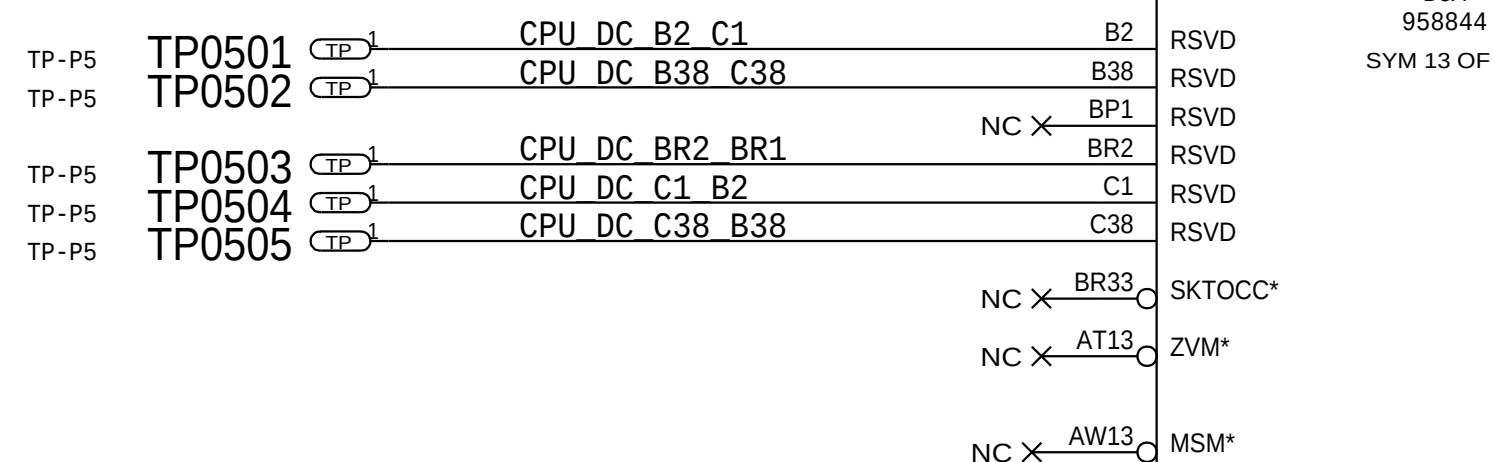
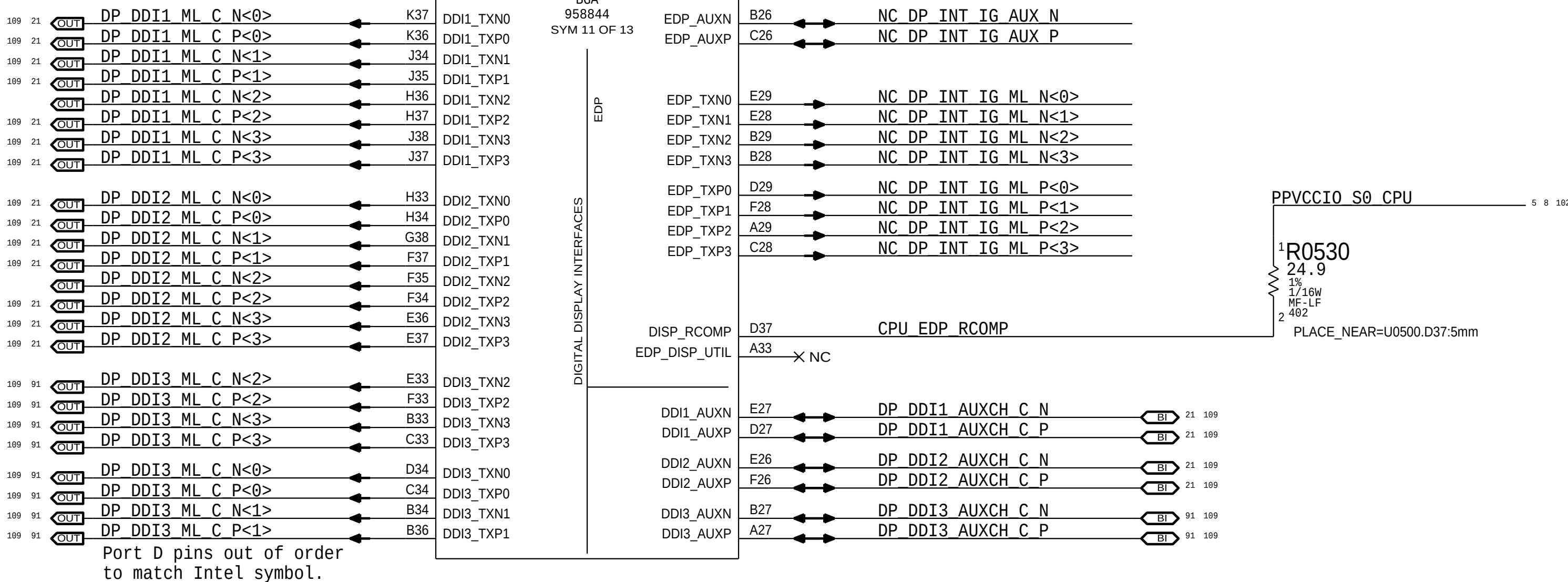
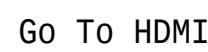
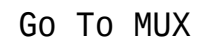
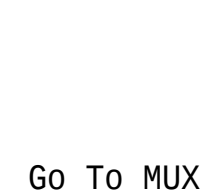
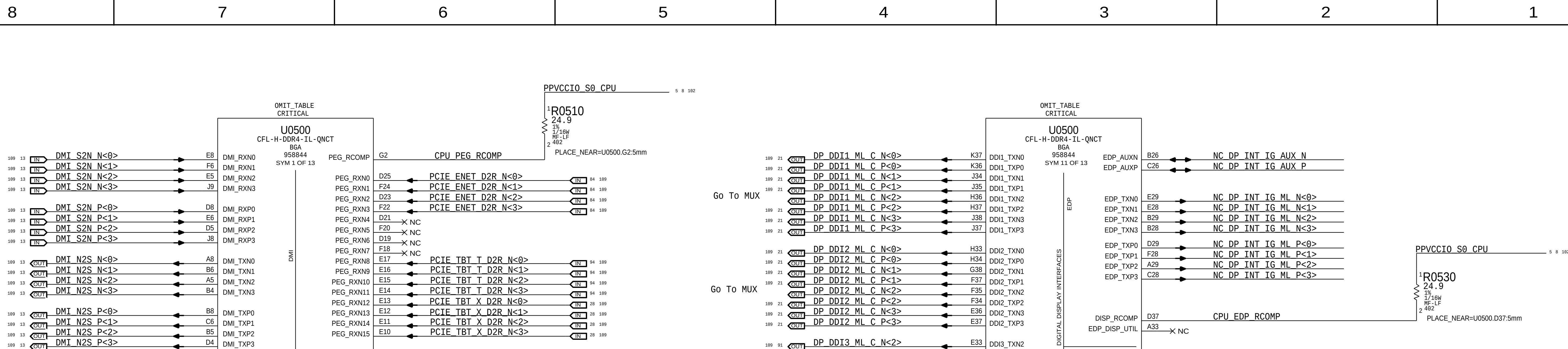
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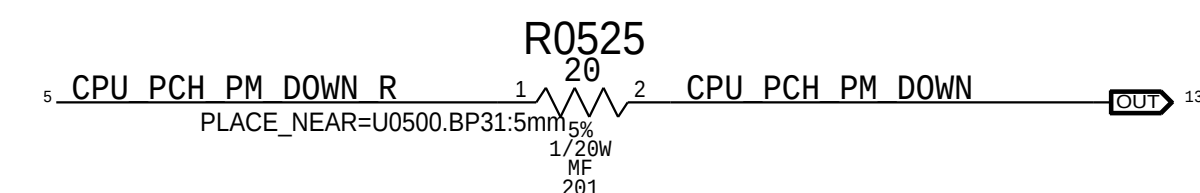
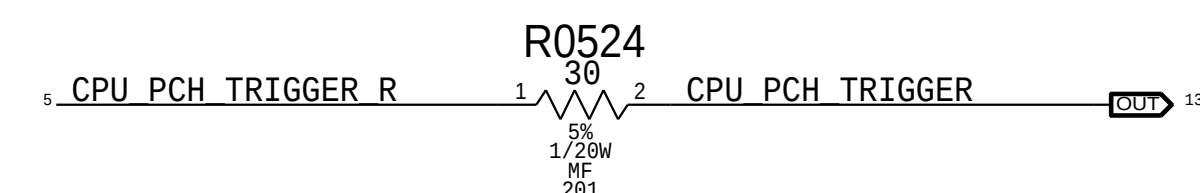
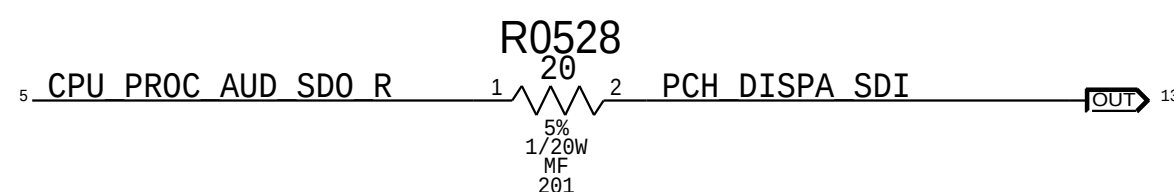
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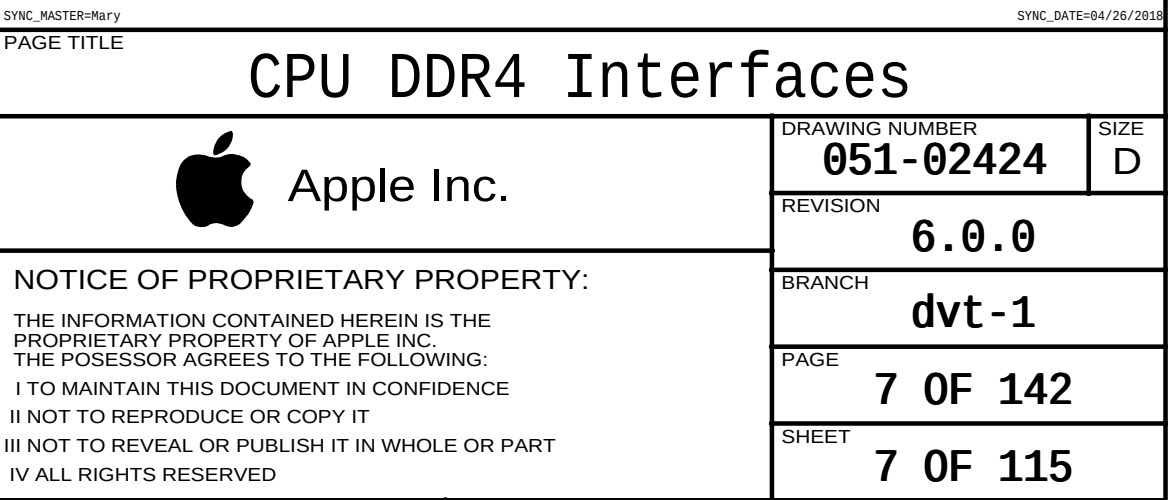
CPU Daisy-Chain Strategy:

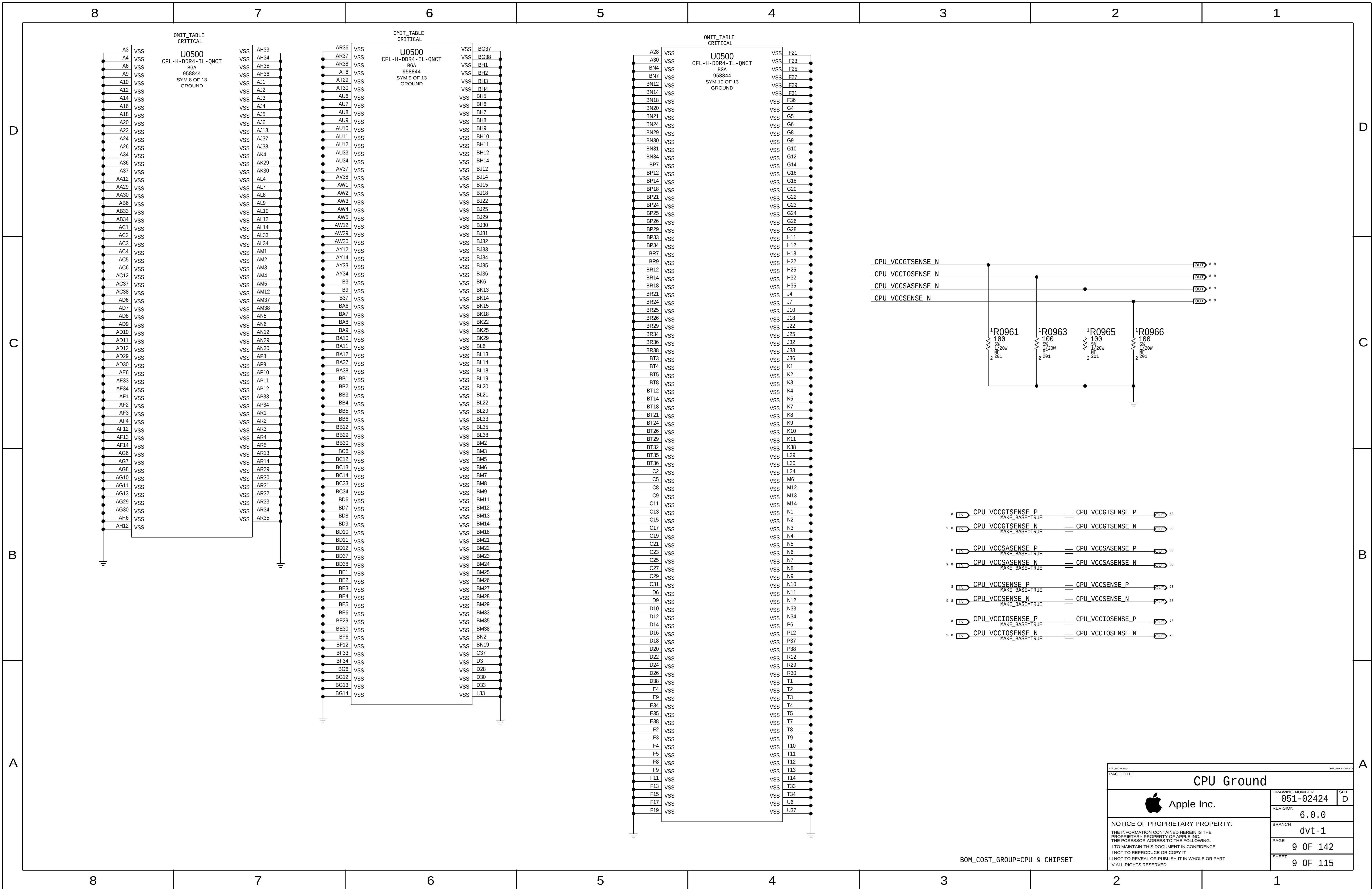
Each corner of CPU has two testpoints. Other corner test signals connected in daisy-chain fashion. Continuity should exist between both TP's on each corner.



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BOM_COST_GROUP=CPU & CHIPSET



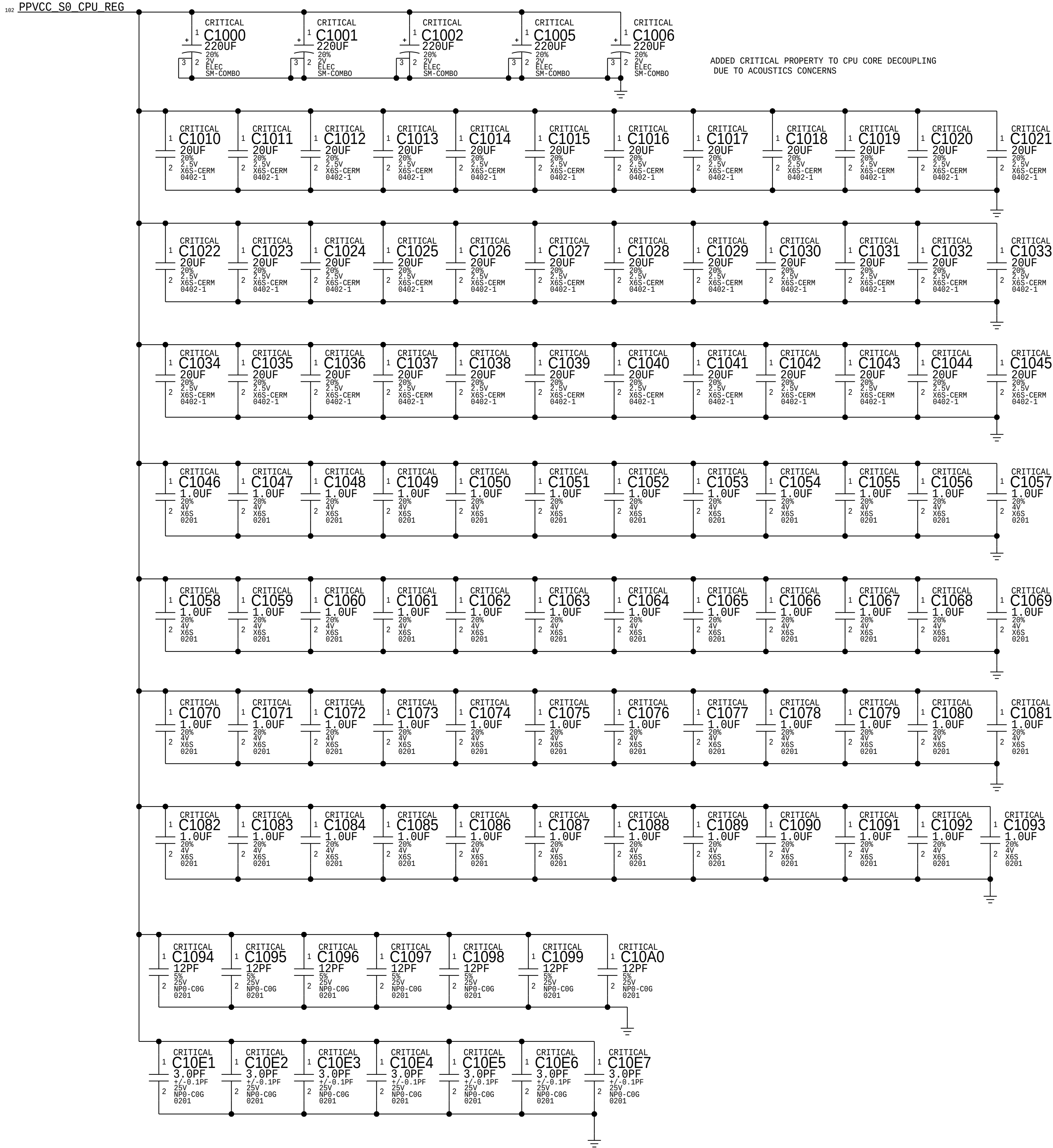


CPU VCORE DECOUPLING

Intel Recommendation:3x 330uF near CPU
5x 47uF 0805 @ board edge
12x 22uF 0603,21x 10uF 0402,24x 1uF 0201,24x 0201(placeholder) @ back side

Apple Implementation:5x 220uF
36x 20uF
48x 1uF
7x 12pF and 3pF

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

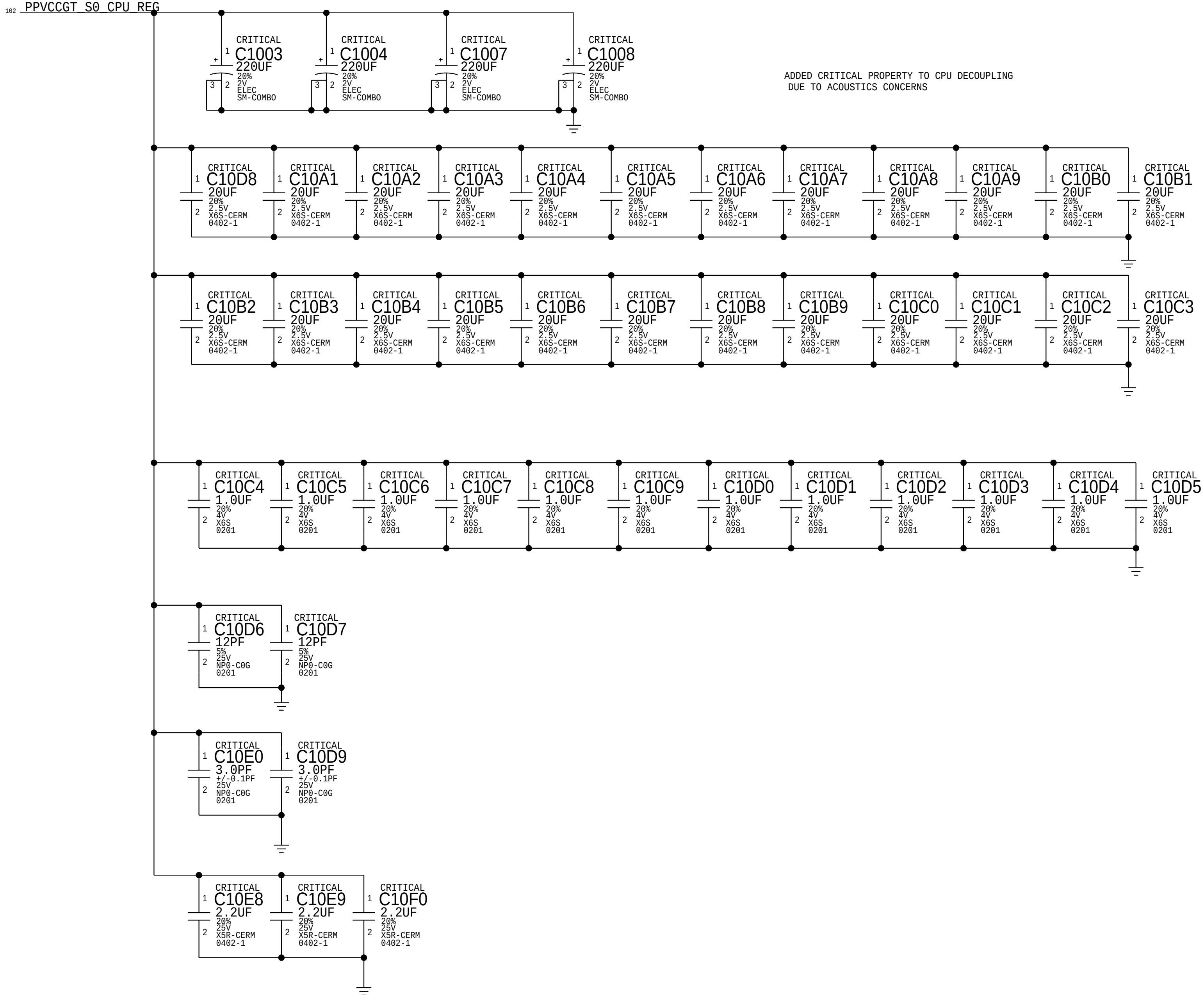


CPU GT DECOUPLING

INTEL RECOMMENDATION:2X 220uF Near CPU
3X 47uF 0805,7x 22uF 0603 @ board edge
10x 10uF 0402,12x 1uF 0201 @ back side

Apple Implementation: 4x 220uF
24x 20uF
12x 1uF
2x 12pF and 3pF

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



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BOM_COST_GROUP=CPU & CHIPSET

Memory (CPU VCCDDR) DECOUPLING

INTEL RECOMMENDATION:4X 22uF 0603,11x 10uF 0402 (back side)

Apple Implementation:10x 20uF

2x 12pF and 3pF (NO STUFF)

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

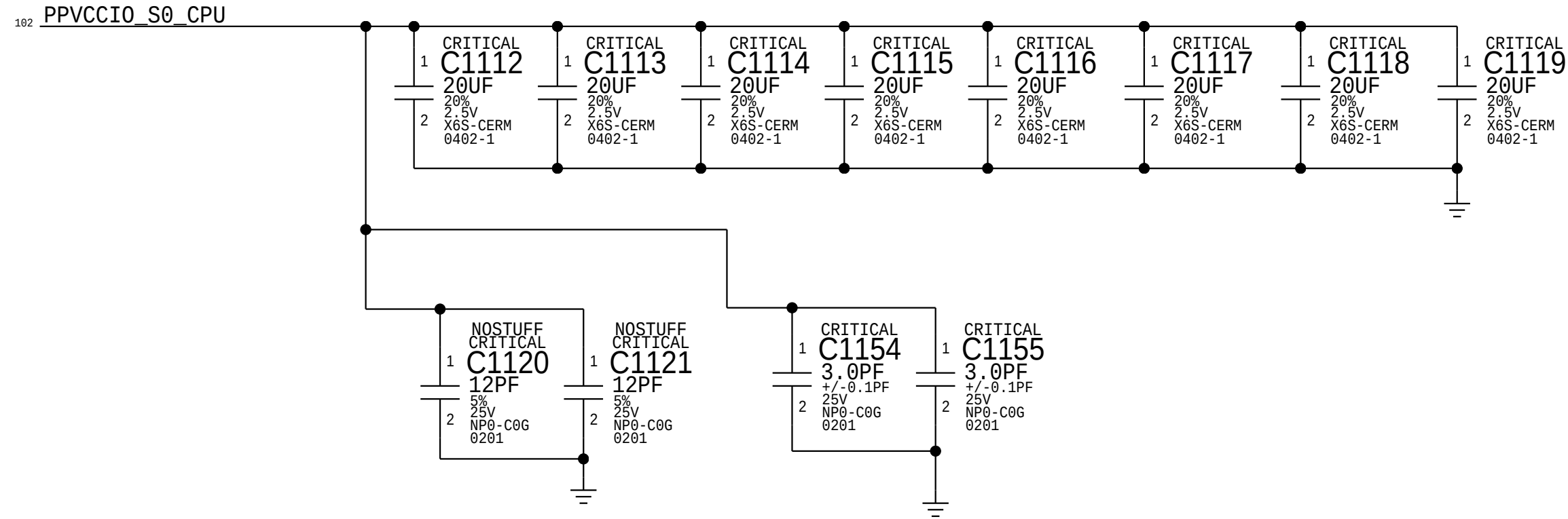
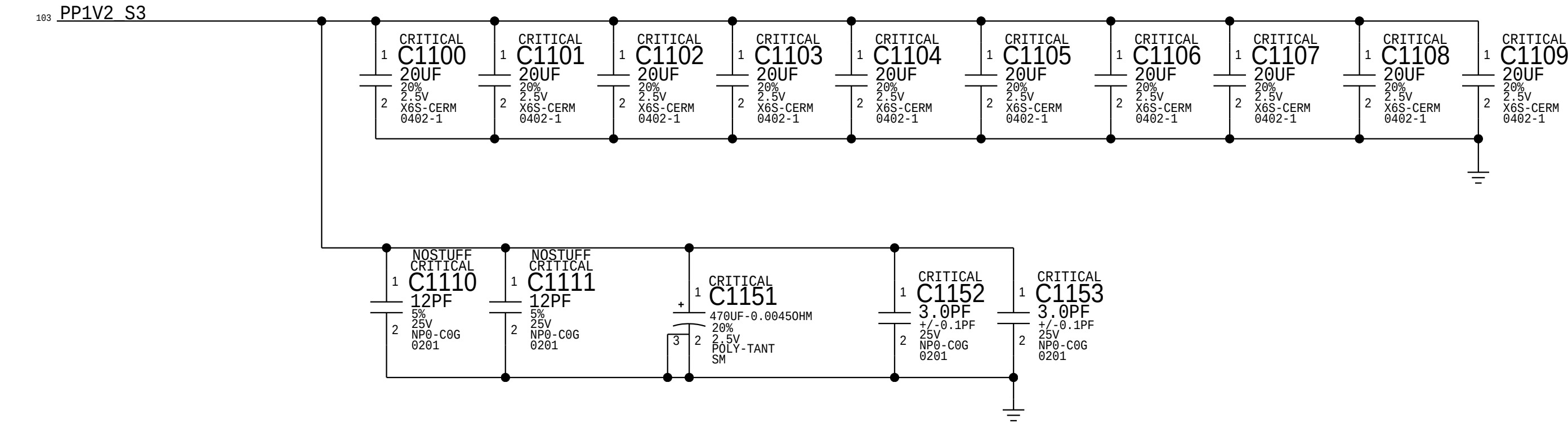
CPU VCCIO DECOUPLING

Intel Recommendation:2x 47uF 0805 near VR ouput
3x 10uF 0402,3x 0402 placeholder @ back side

Apple Implementation:8x 20uF

2x 12pF and 3pF (NO STUFF)

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



CPU VCCST / VCCSTG / VCCPLL / VCCPLL_OC DECOUPLING

Intel Recommendation: VCCST: 1x 1uF 0201
VCCSTG: 1x 1uF 0201
VCCPLL: 1x 1uF 0201,1x 22uF/47uF 0805 placeholder
VCCPLL_OC: 2x 1uF 0201

Apple Implementation: VCCST: 2x 1uF
VCCSTG: 2x 1uF
VCCPLL: 2x 20uF,2x 1uF
VCCPLL_OC: 2x 1uF

Layout Note: All these Must be Ground referenced.
VCCSTG board routing resistance from BGA to gate should less than 10mOhm.
Do not route VCCSTG closet adjacent layer over any power net other than GND.

CPU VCCSA DECOUPLING

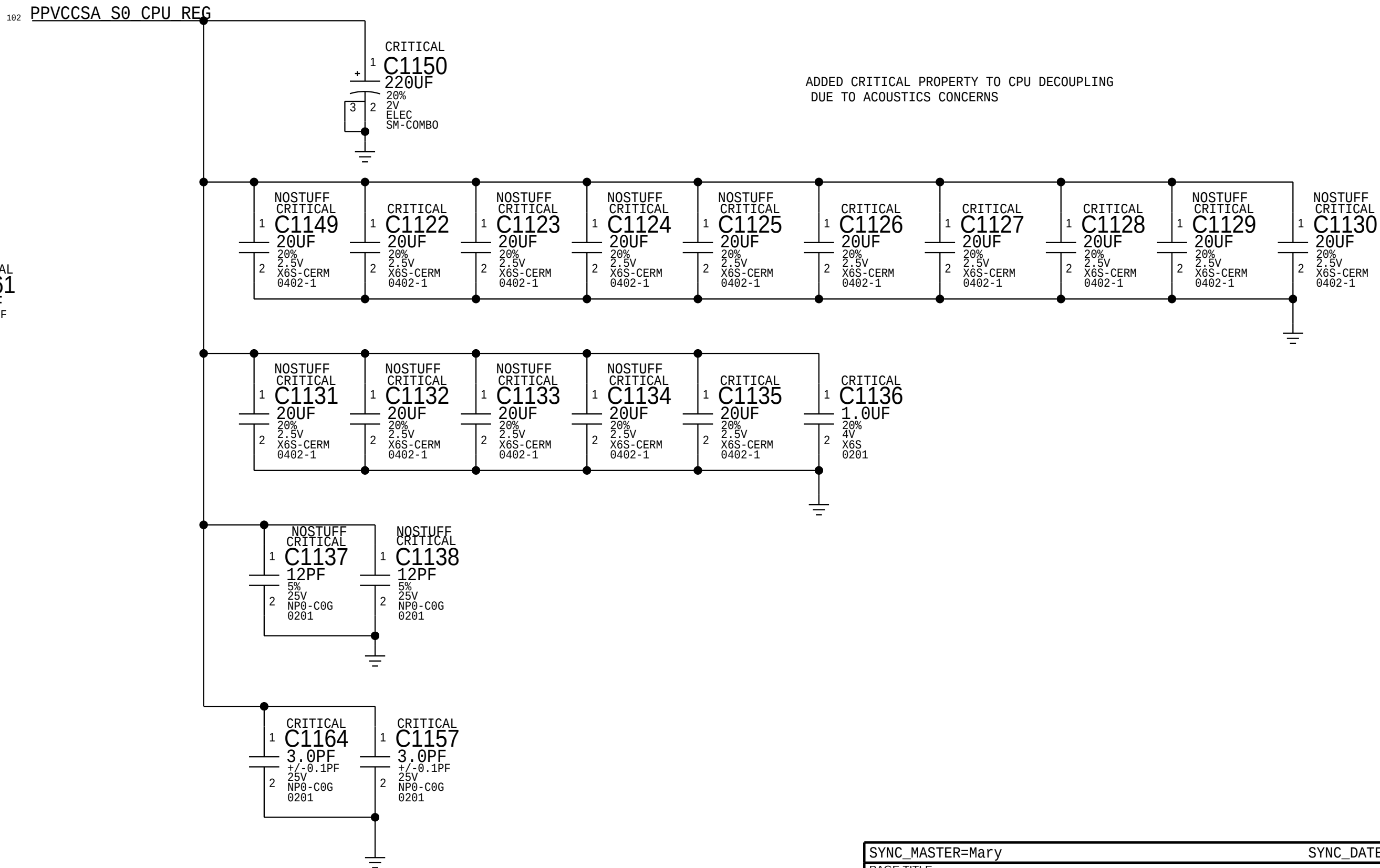
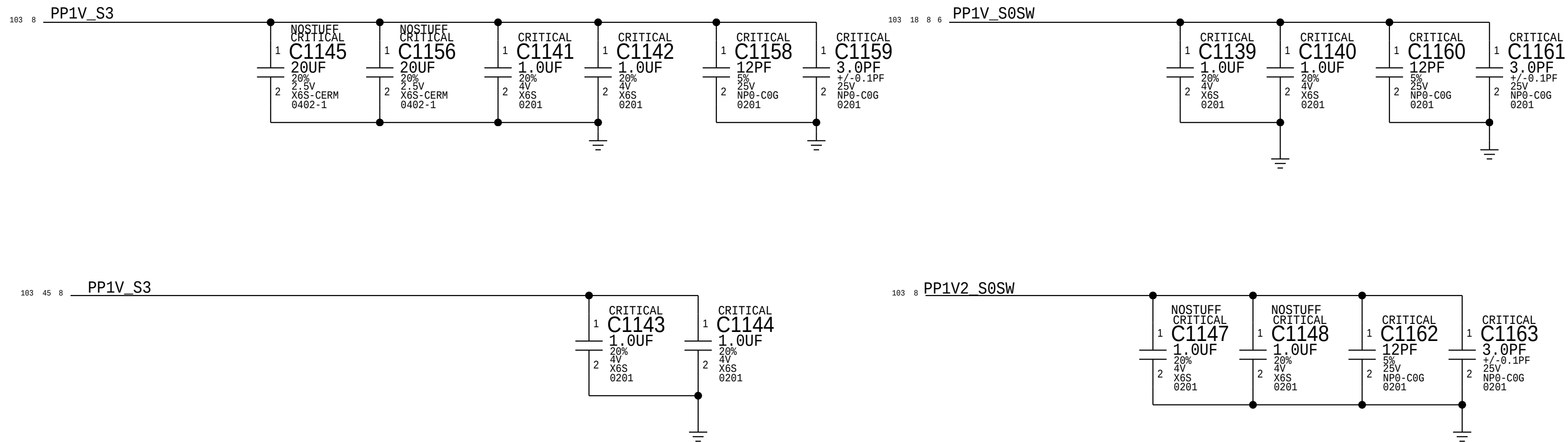
Intel Recommendation:2x 47uF 0805 near CPU
2x 47uF 0805,2x 22uF 0603 @ board edge
7x 10uF 0402,1x 1uF 0201 @ backside

Apple Implementation:1x 220uF

5x 20uF

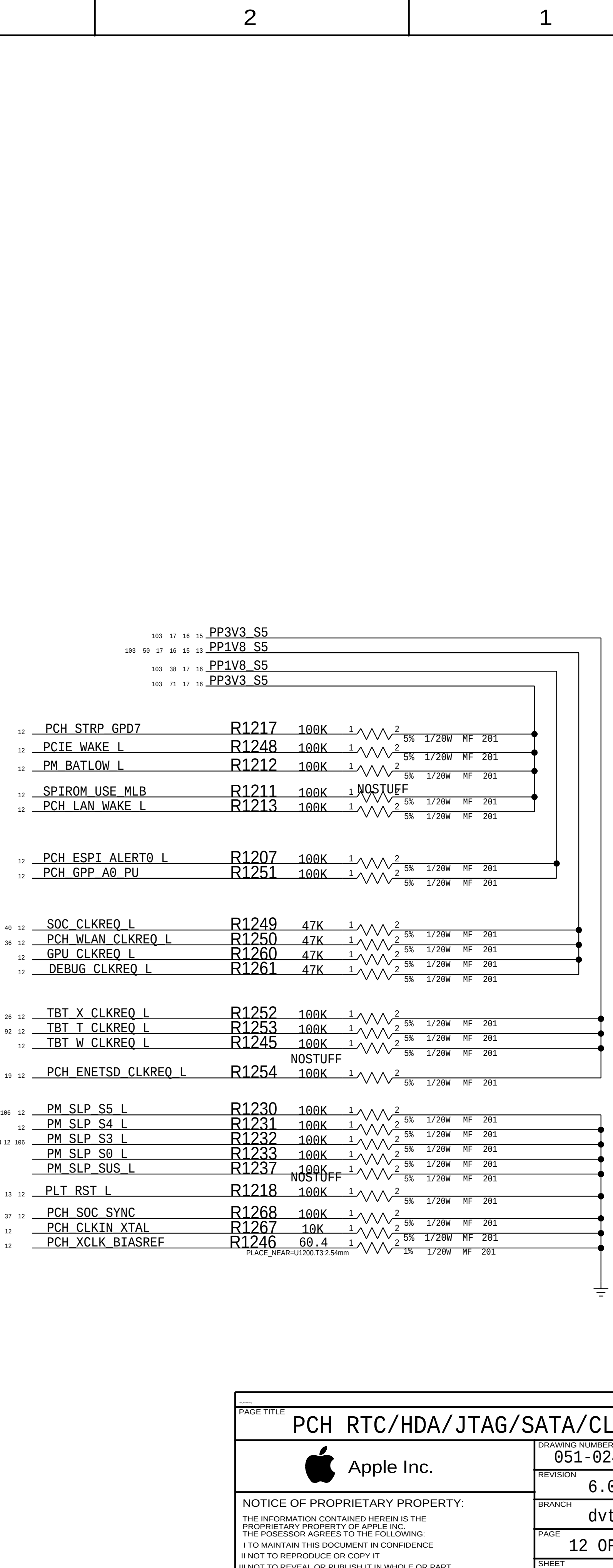
1x 1uF

2x 12pF and 3pF (NO STUFF)

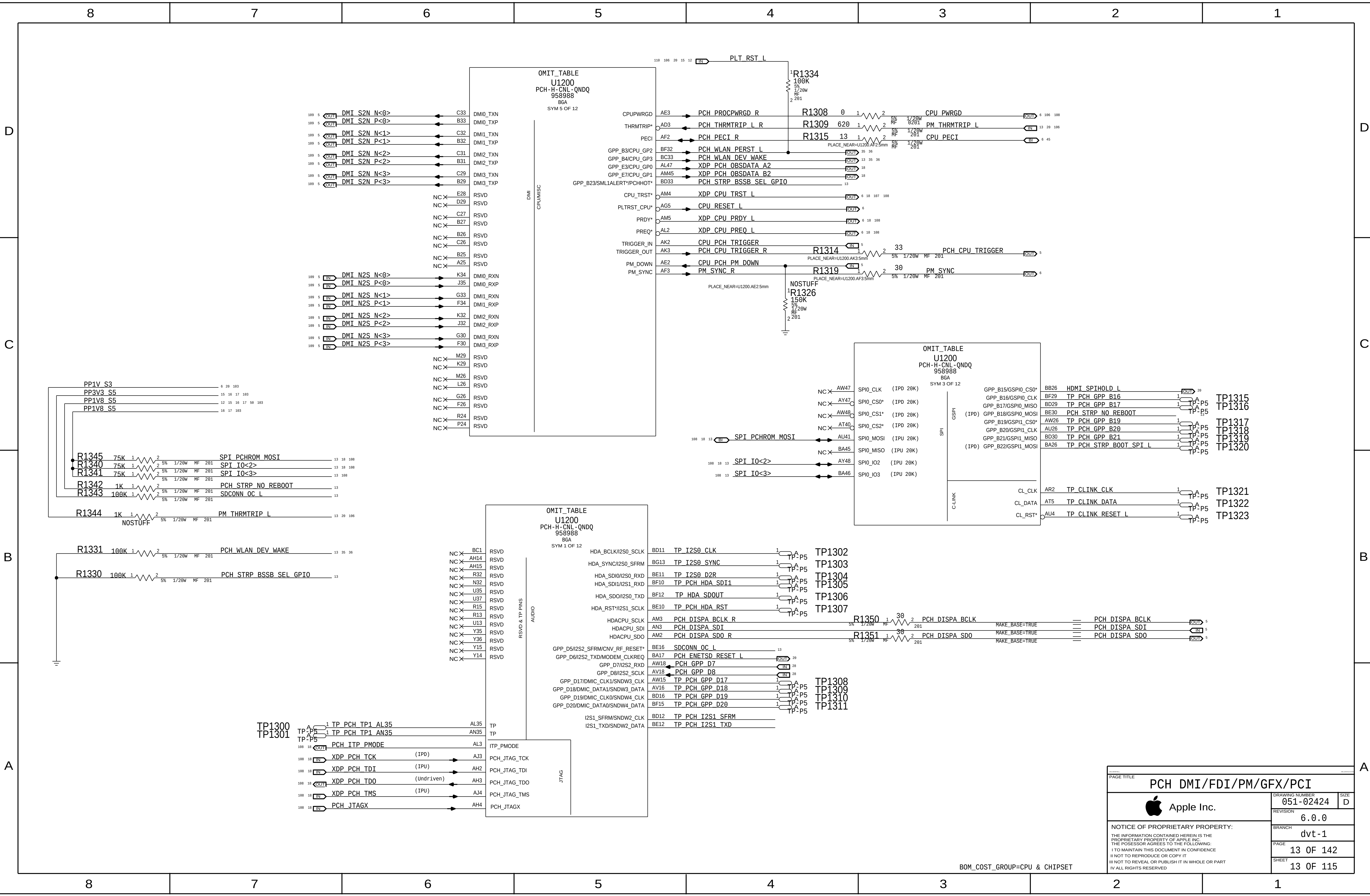


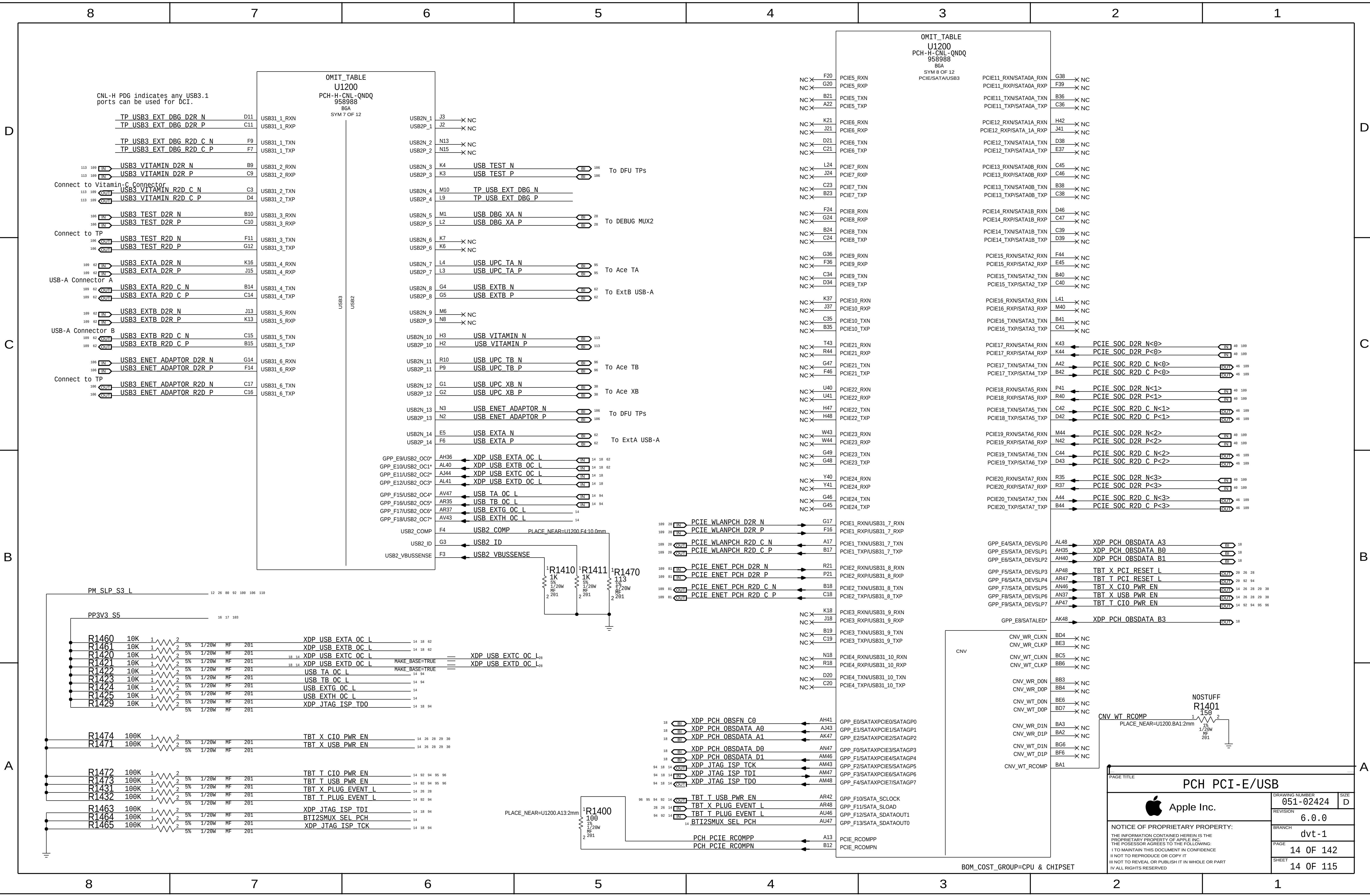
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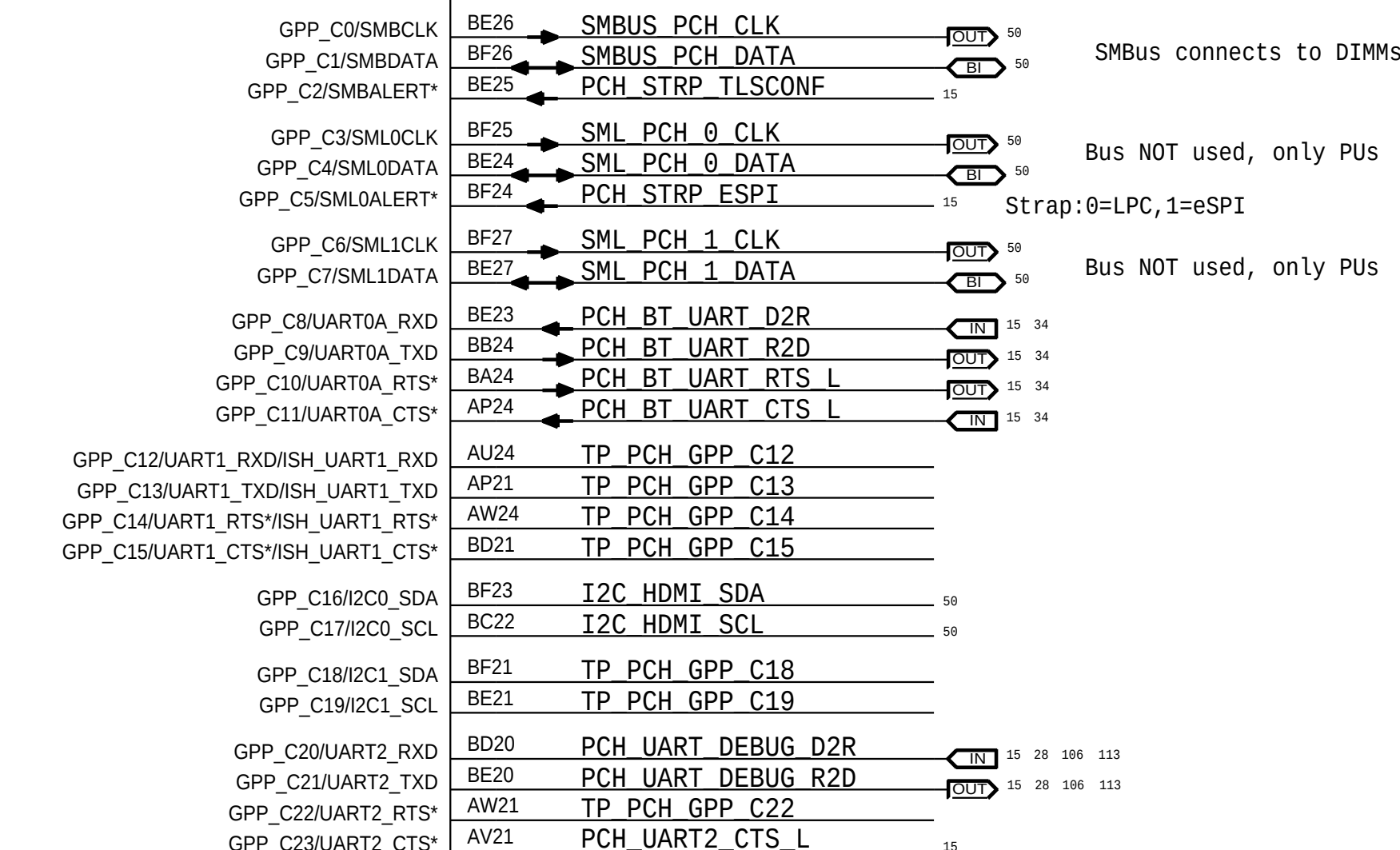
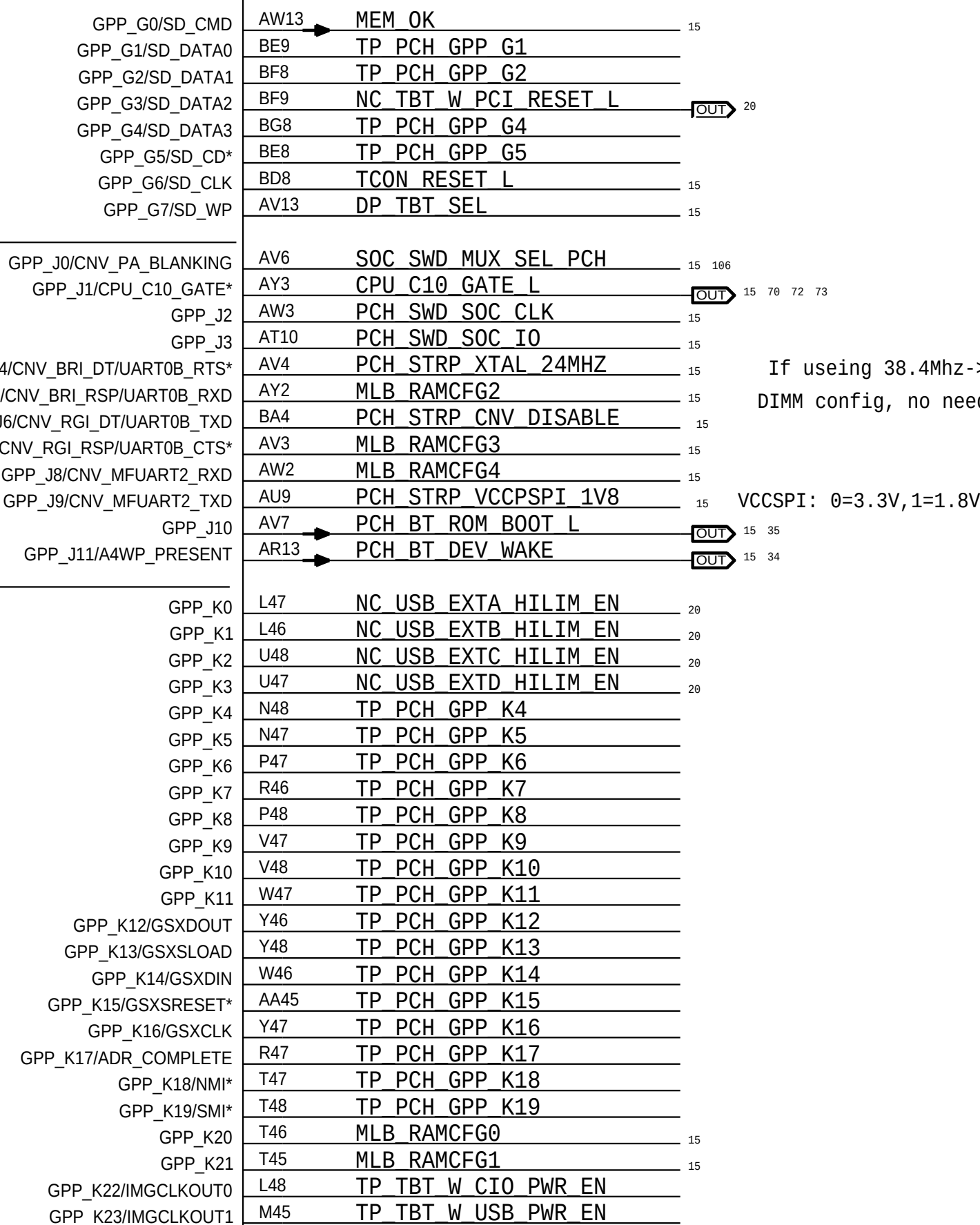
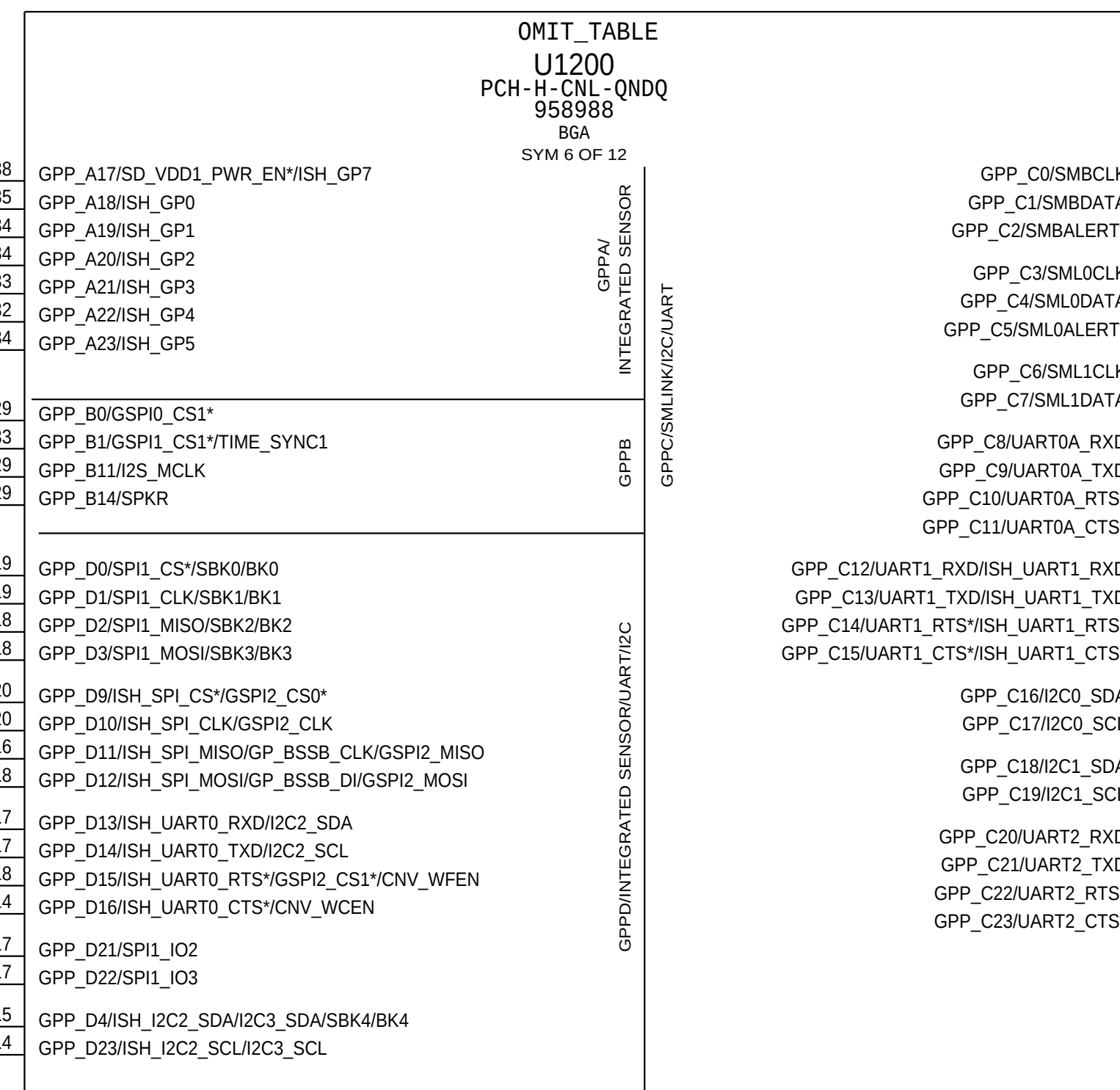
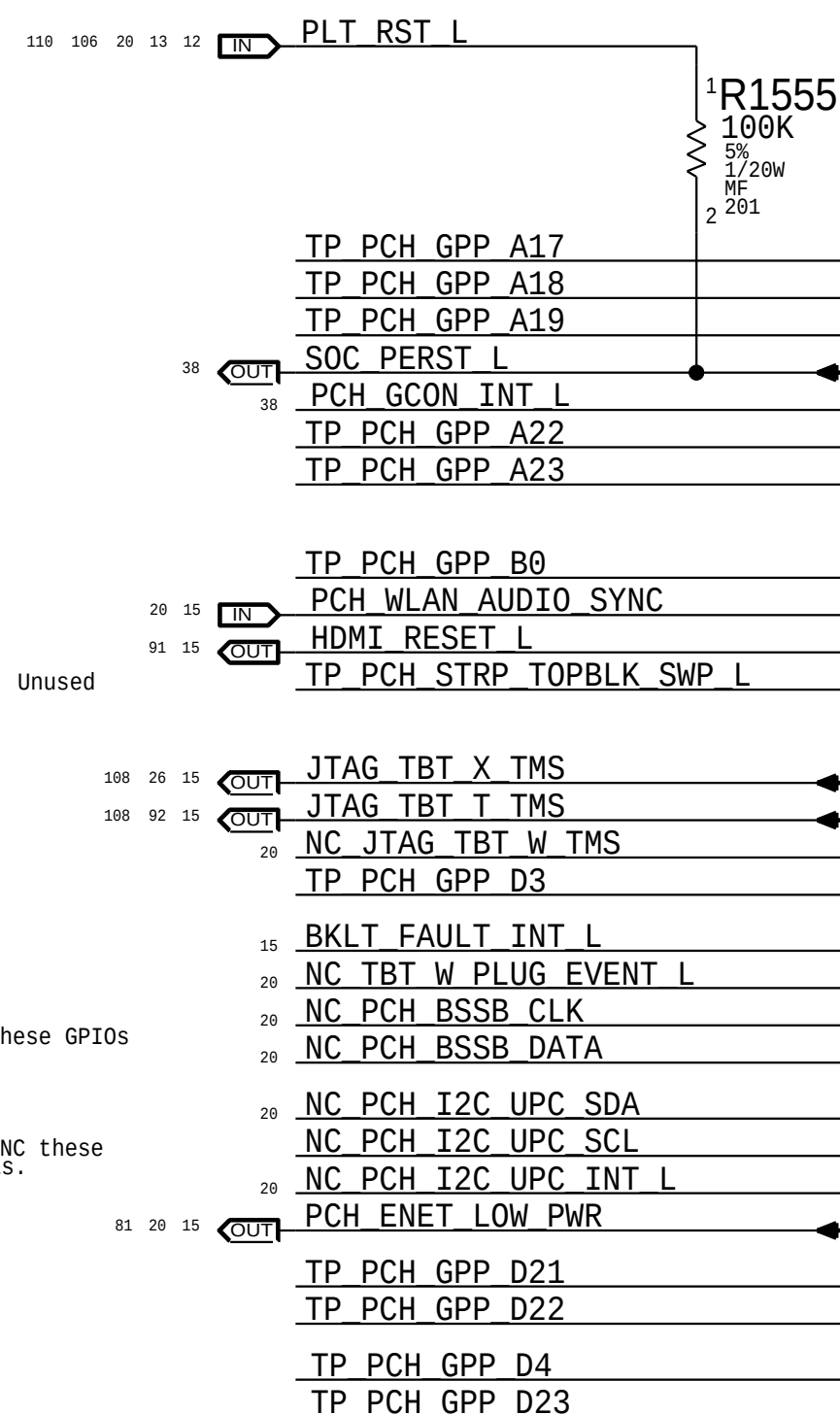
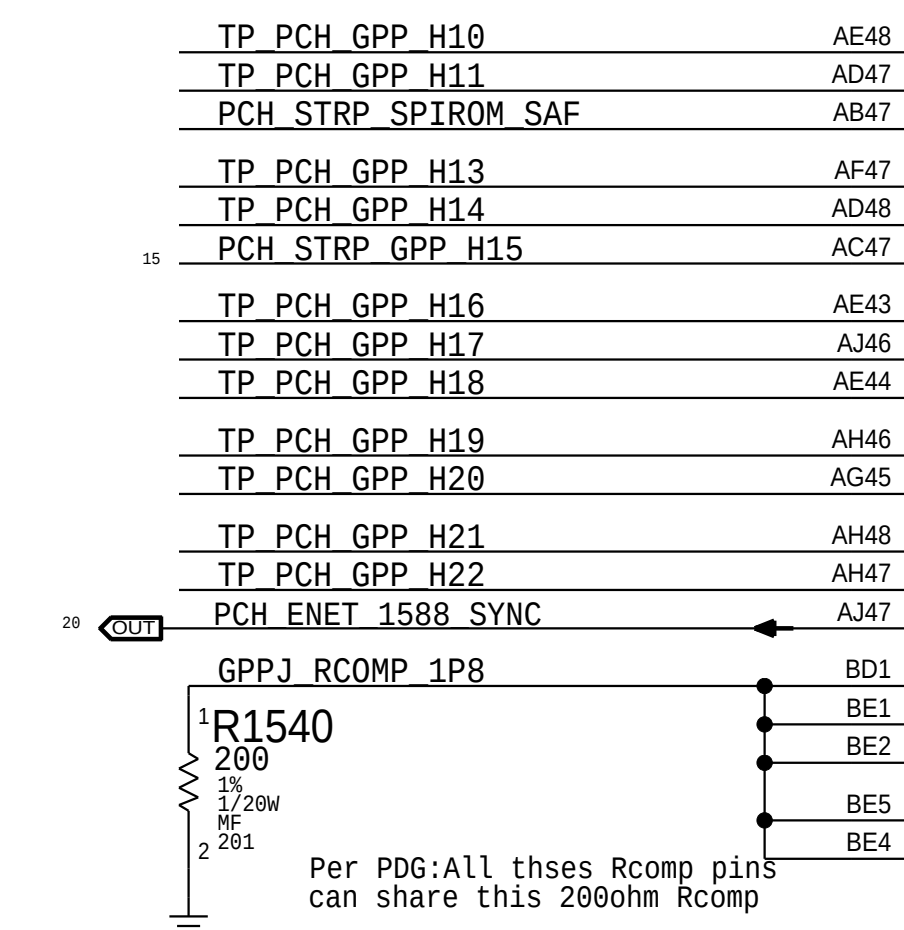
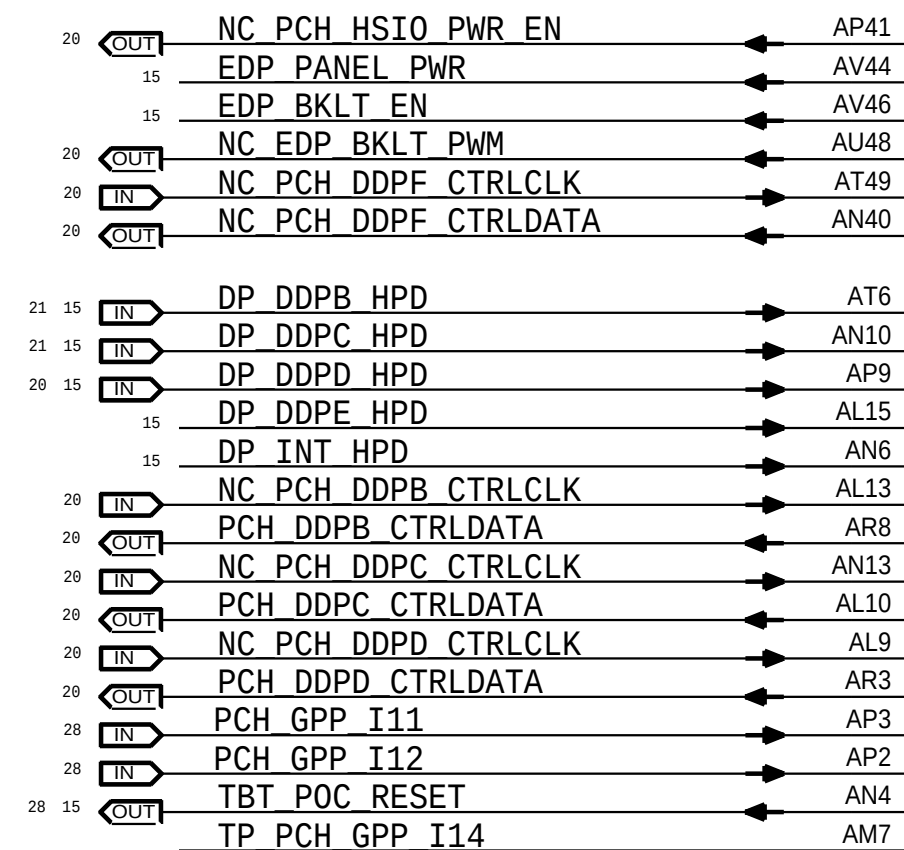
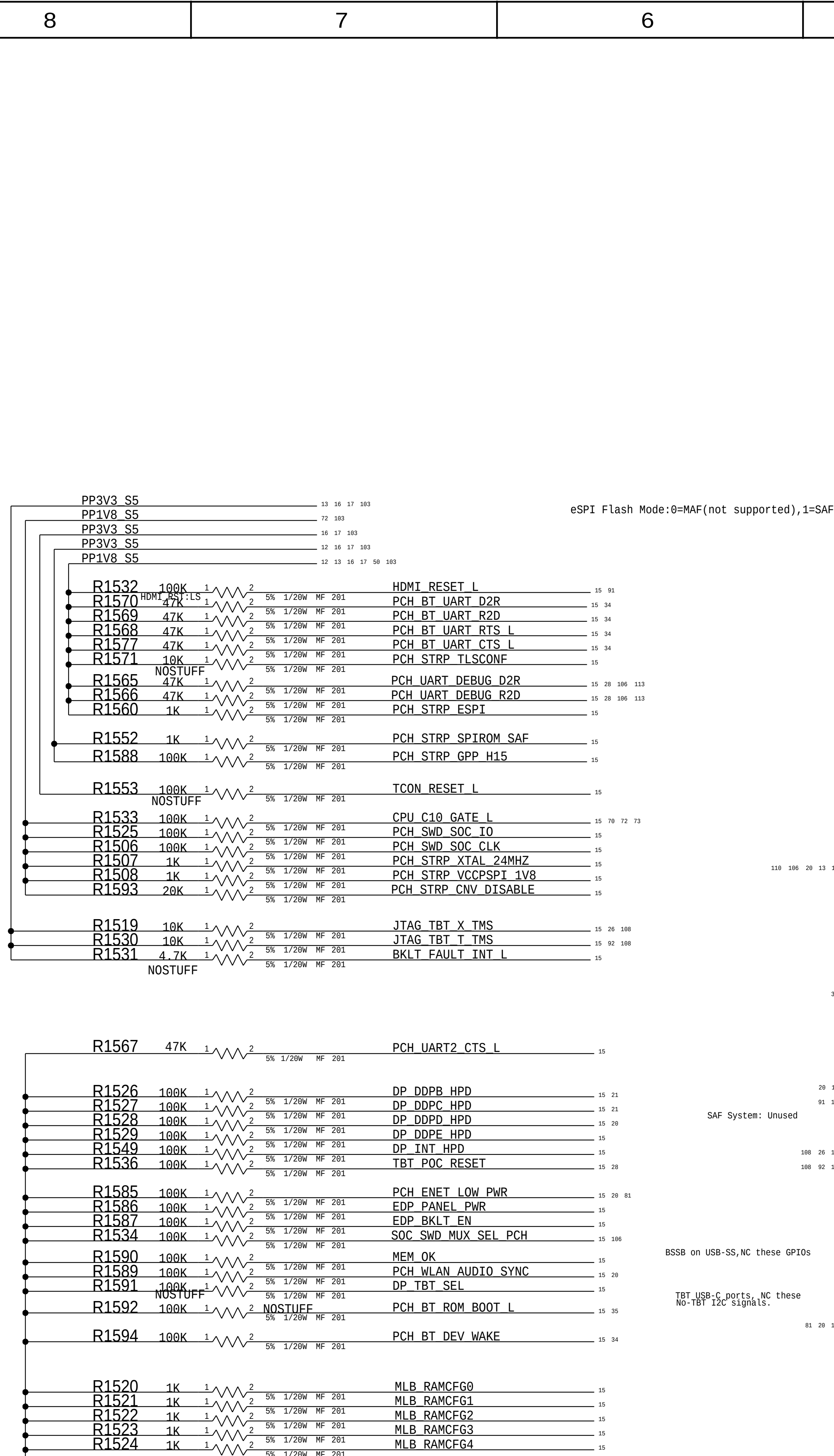
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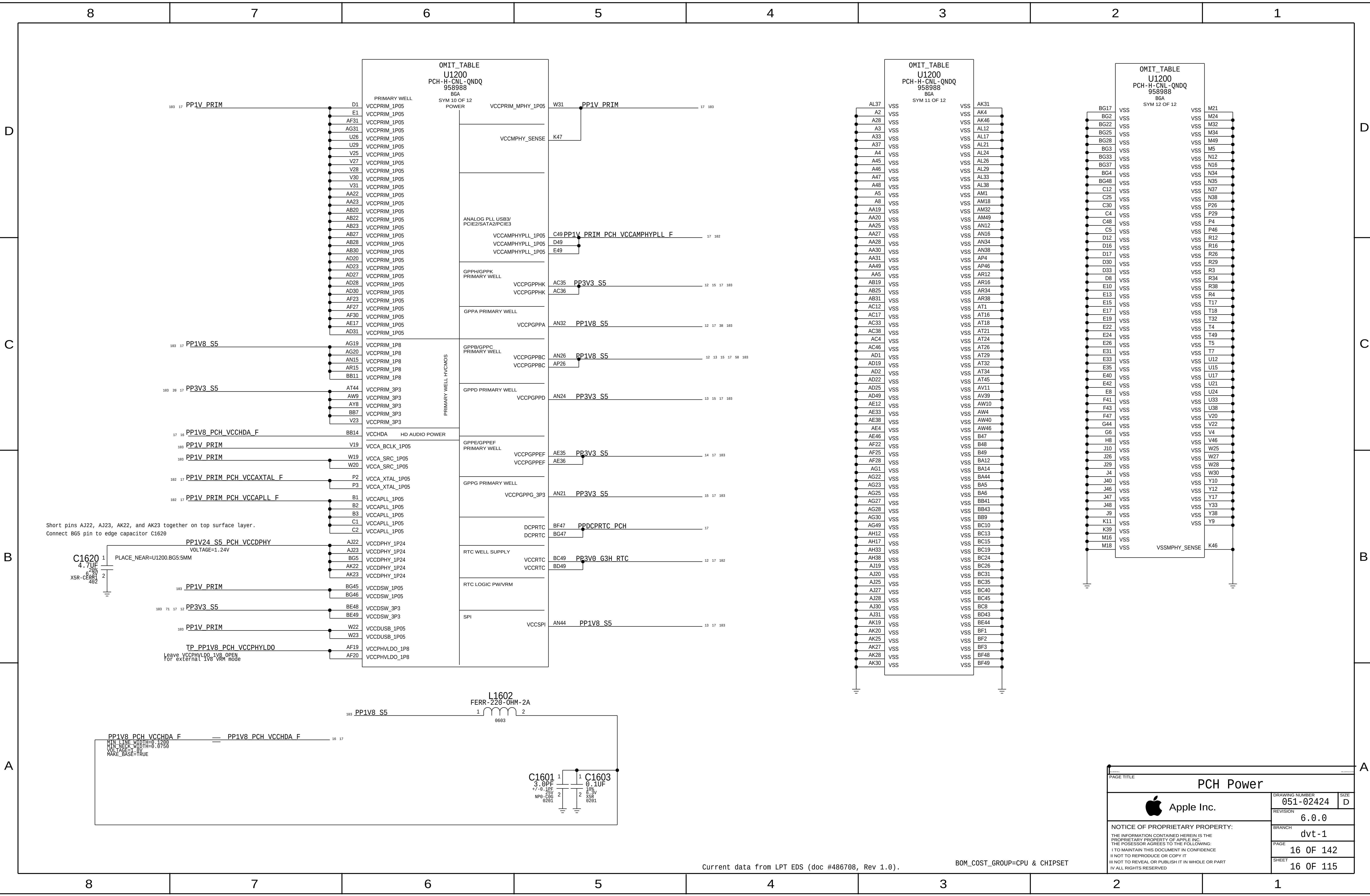
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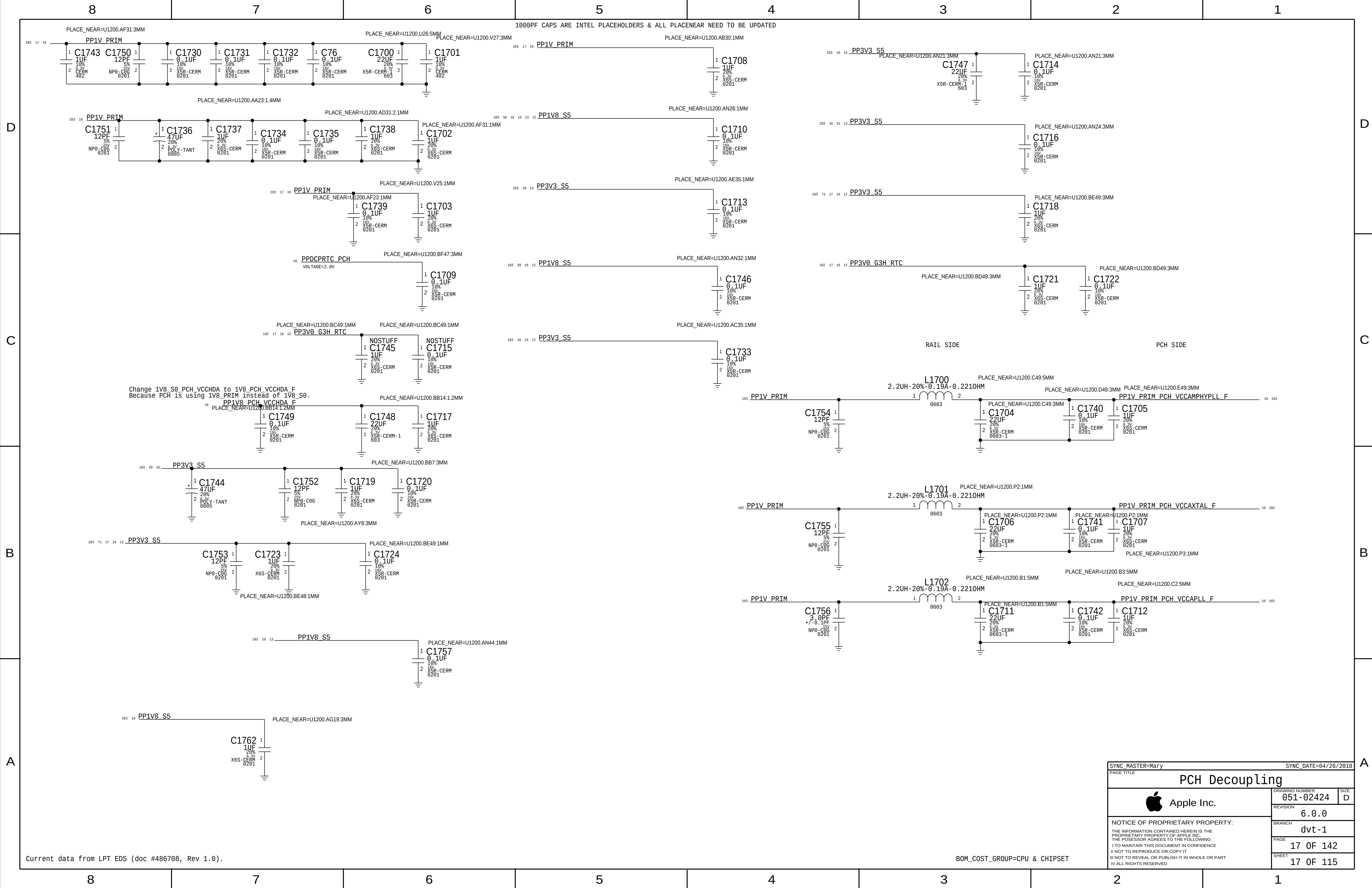
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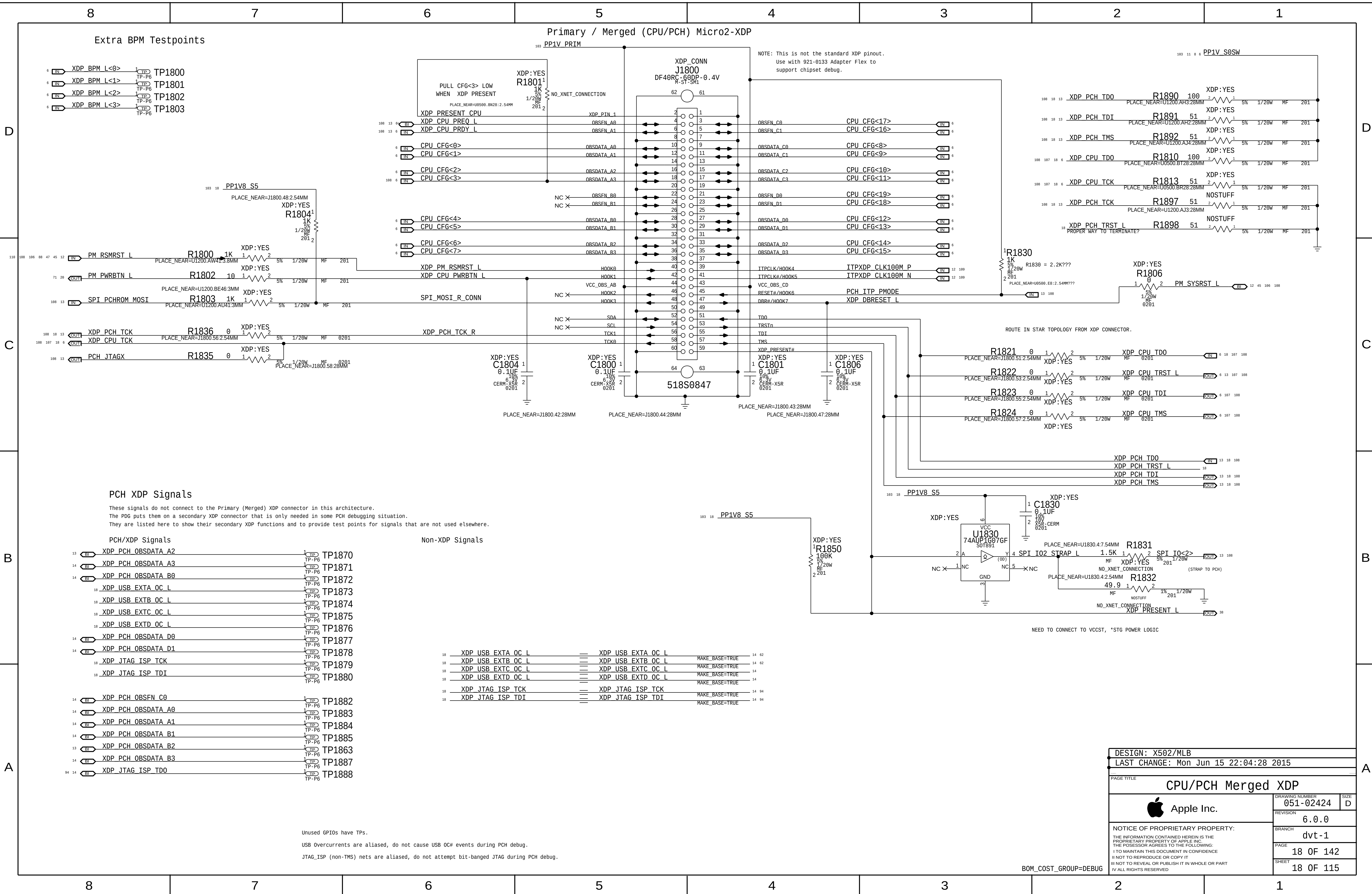


PAGE TITLE		
PCH Power		
	DRAWING NUMBER	051-02424
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Current data from LPT EDS (doc #486708, Rev 1.0).

BOM_COST_GROUP=CPU & CHIPSET





PDG recommends 18pF, check in charz.

Coin-Cell Holder

PPVBATT G3 RTC

BT1900
BB10201-C34A3-7H

Per radar://32938827,
coin connector is back to 511s0074

R1902
1K

PP3V3 G3H

PPVBATT G3 RTC R

C1975
0.1uF

SP1900
1.97X2.02MM-NSP

SP1901
1.97X2.02MM-NSP

NOSTUFF
D1900
BAT54DW-X-G

Desktop should use diode for RTC power

PP3V3 G3 RTC R

NOSTUFF
C1976
0.1uF

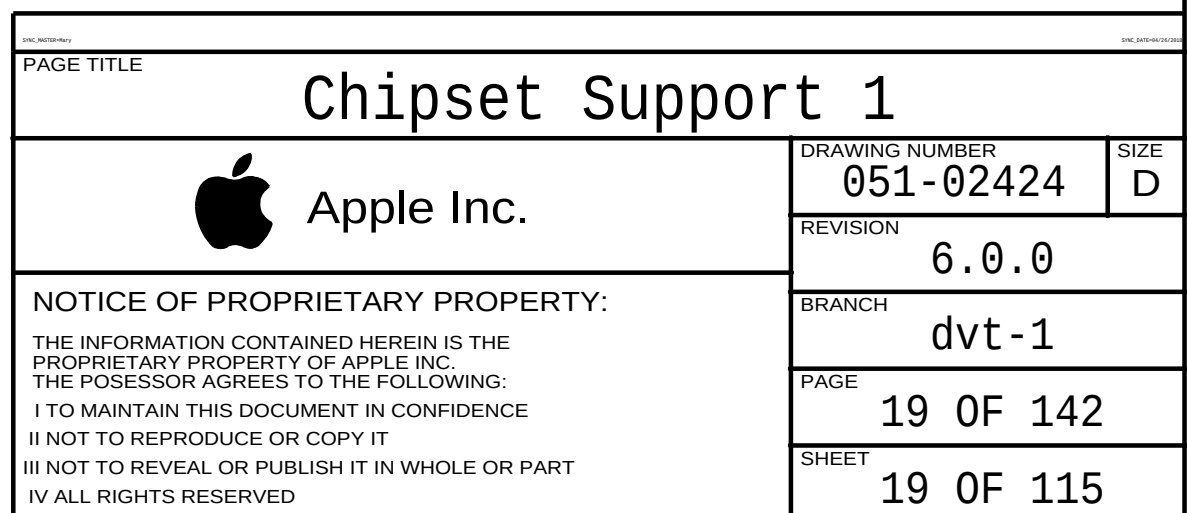
PPVRTC G3 PCH REG R

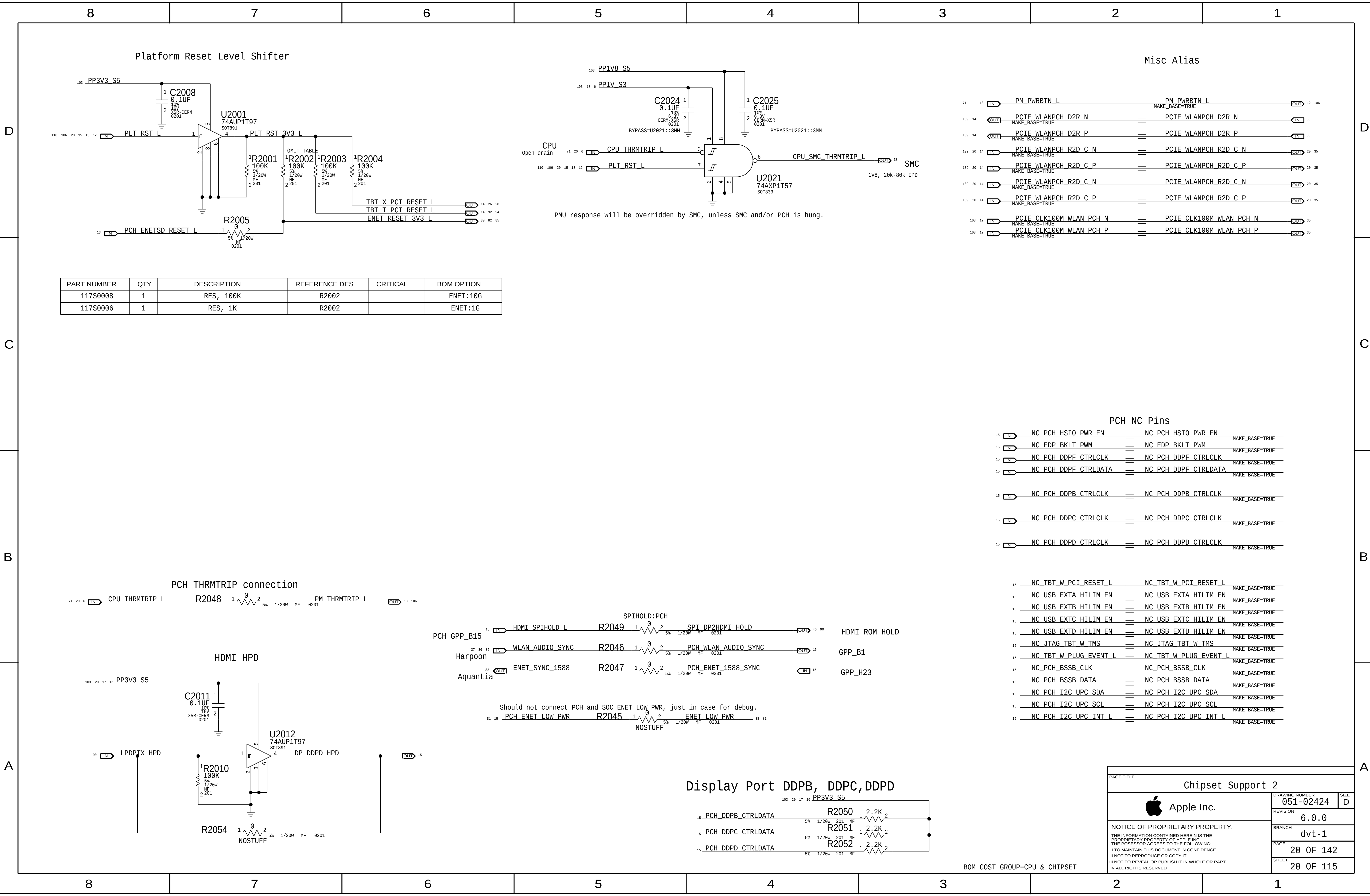
NOSTUFF
R1932

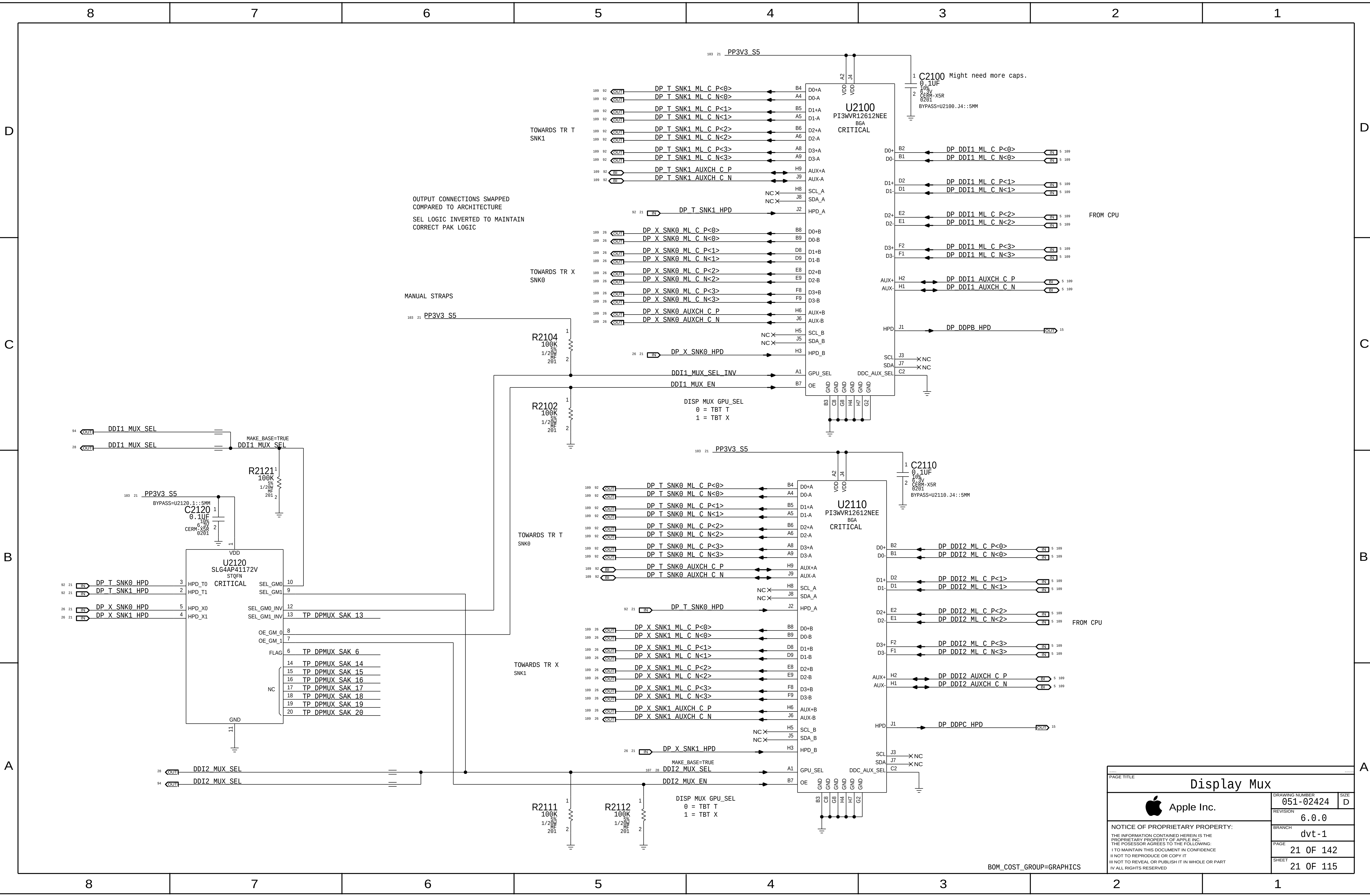
R1931

PP3V0 G3H RTC

To PCH VCCRTC







PAGE TITLE		
Display Mux		
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Page Notes

Power aliases required by this page

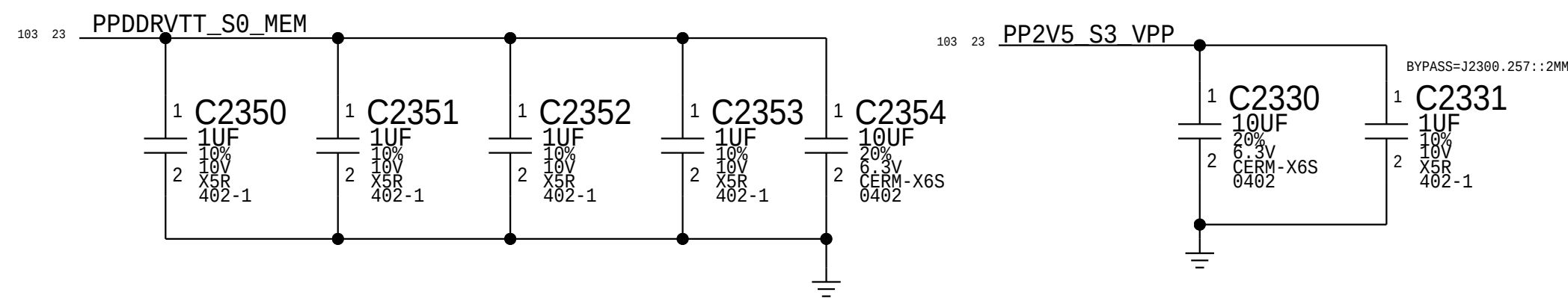
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- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A

Signal aliases required by this page:

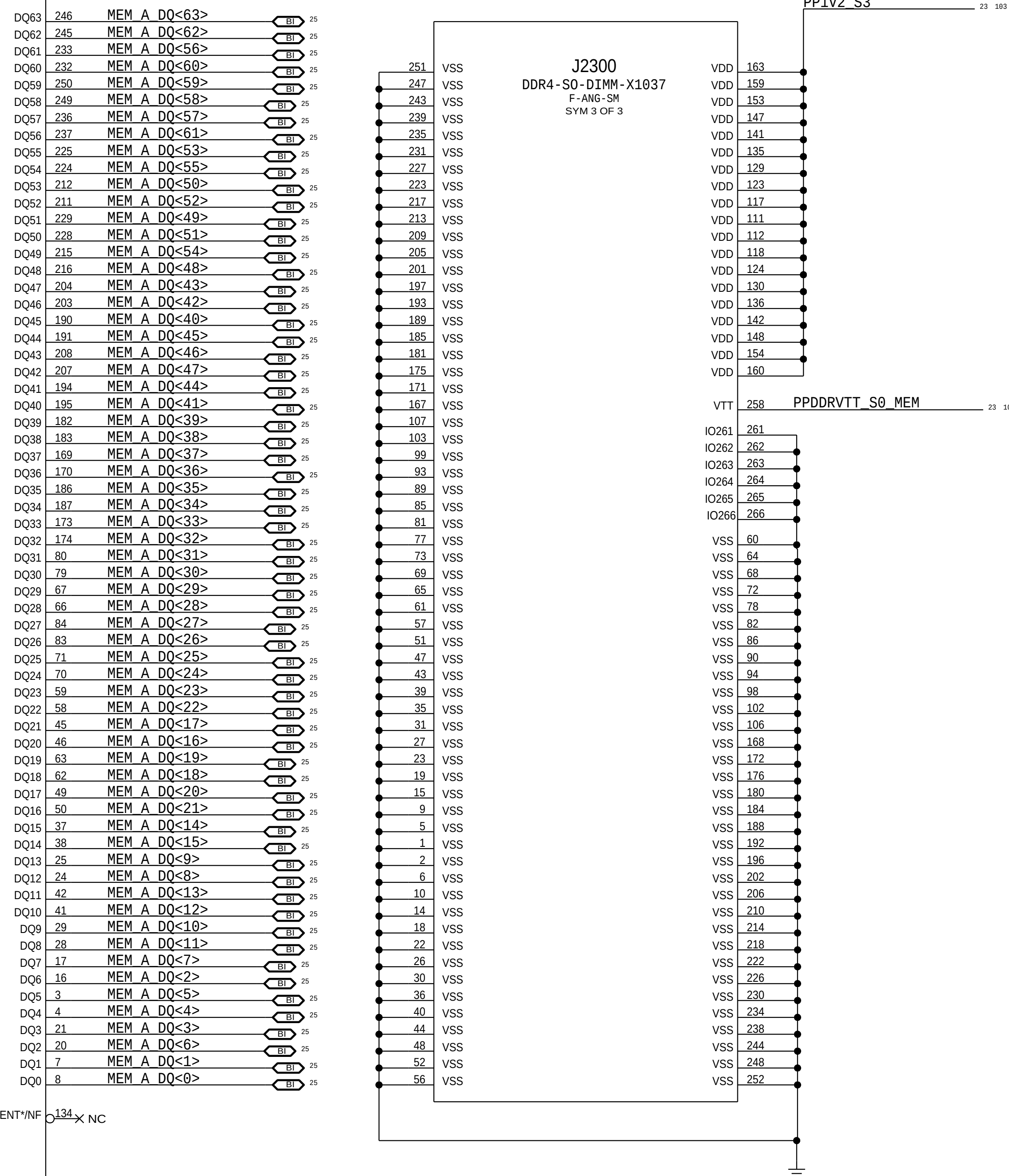
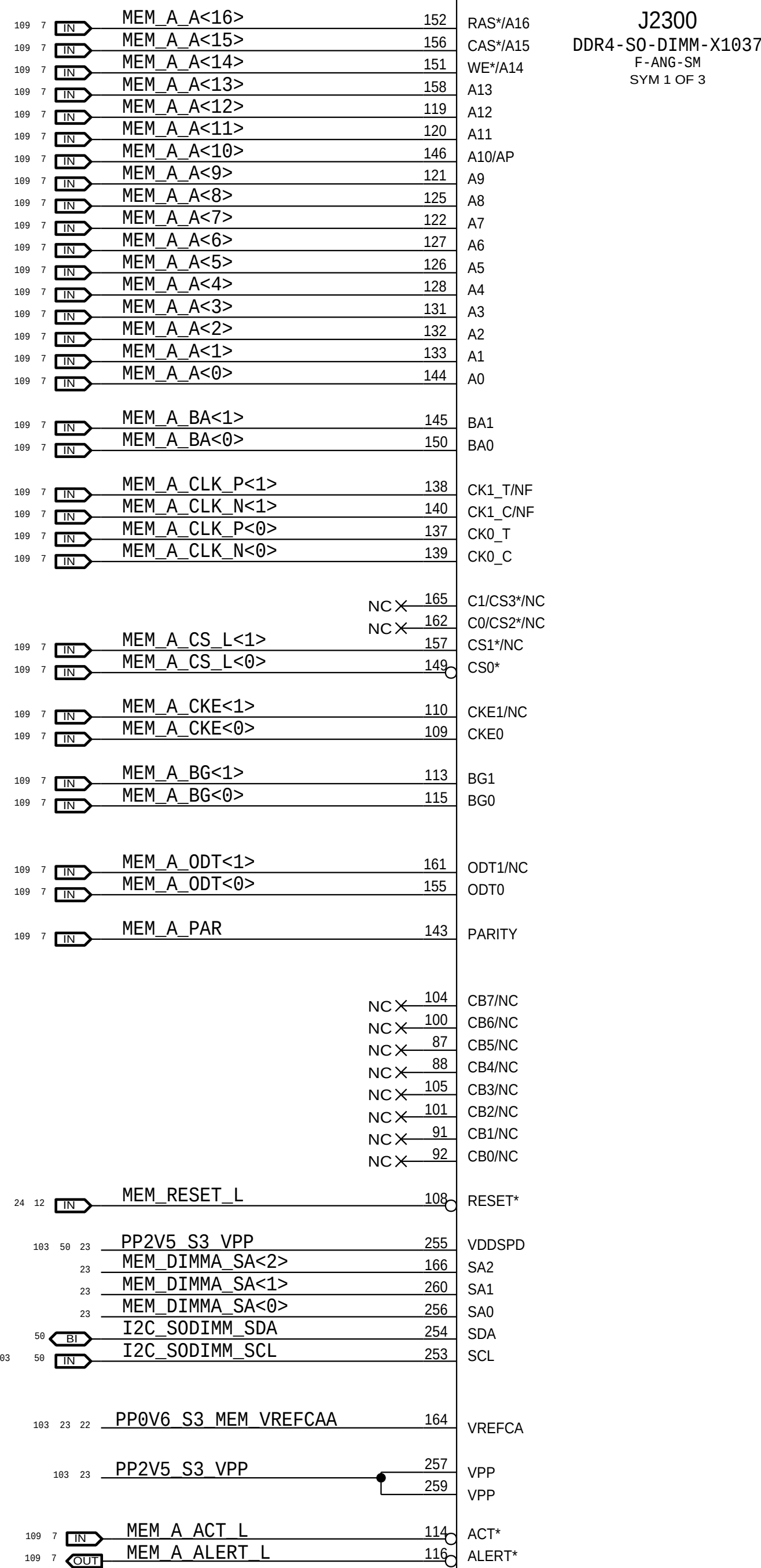
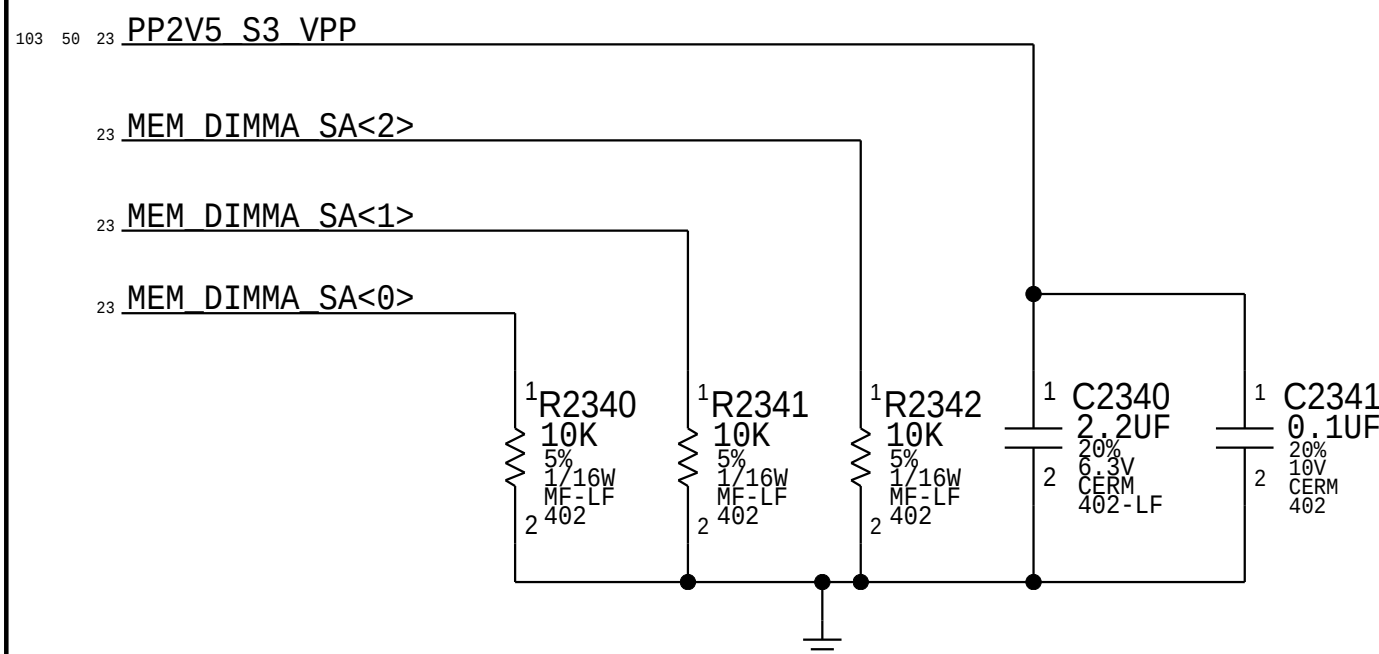
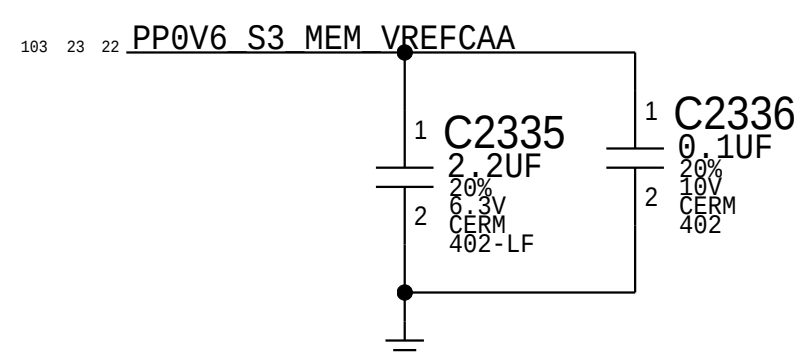
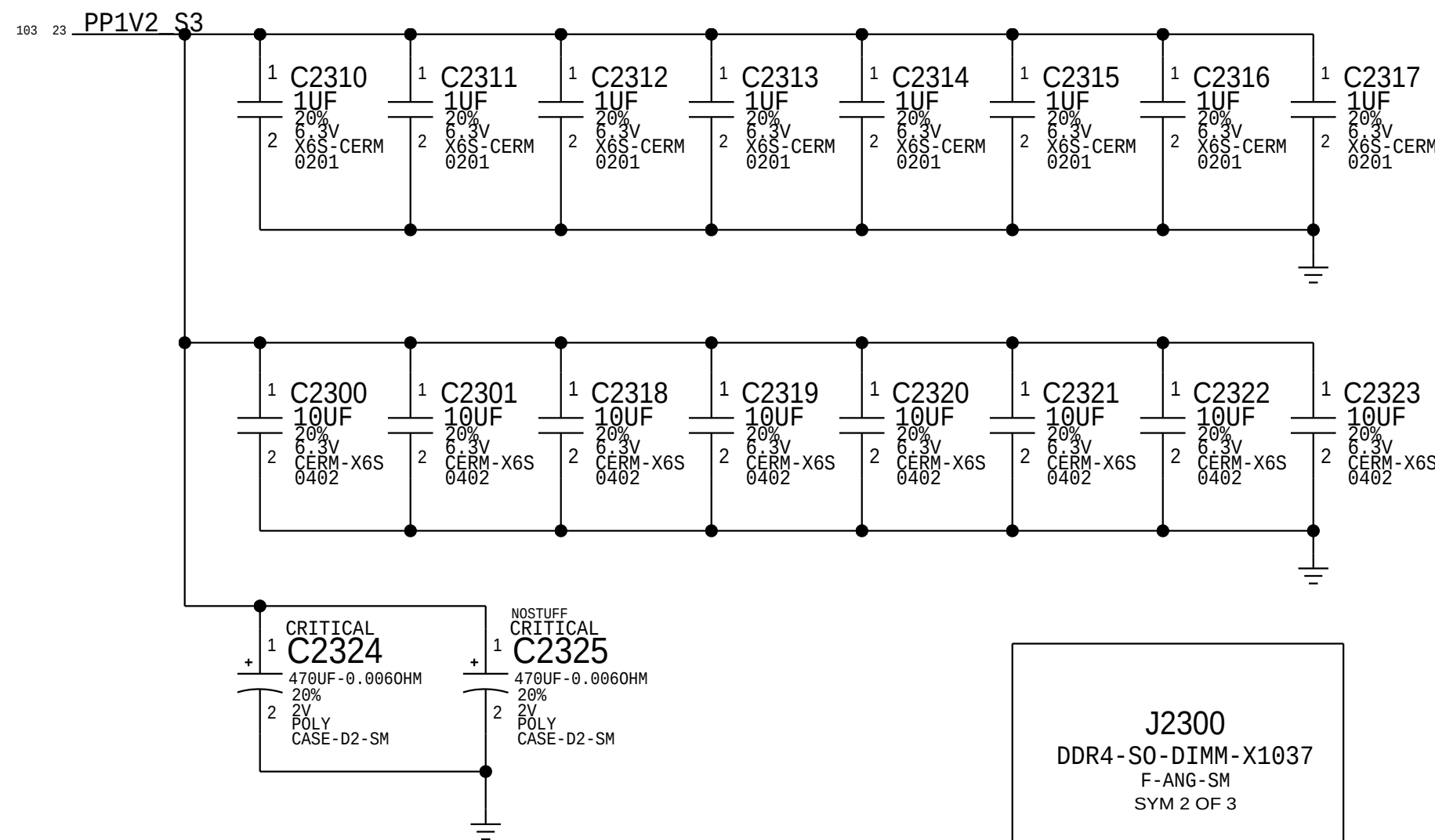
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- =I2C_SODIMMA_SDA

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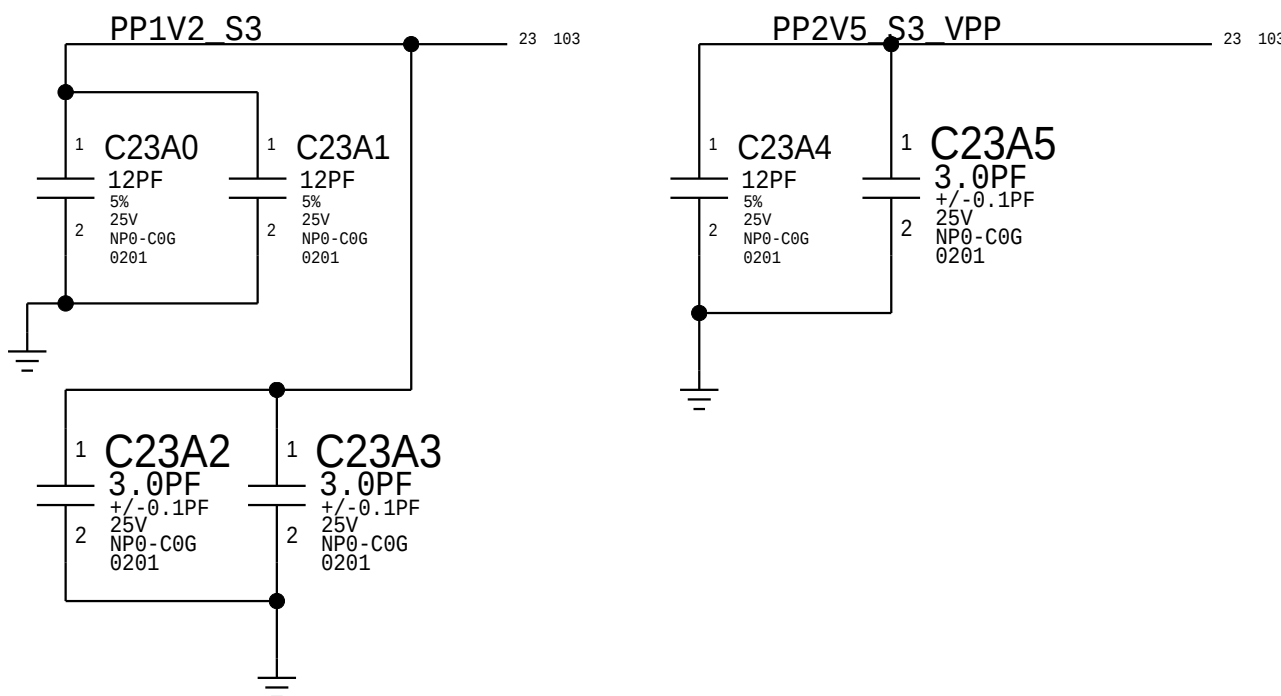
(NONE)



DDR4 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)



Memory desense caps



SYNC_MASTER=Mary		SYNC_DATE=04/26/2018	
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		051-02424	
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8		7		6		5		4		3		2		1	
Memory Bit/Byte Swizzle															
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	109	7	MEM_A_DQS_P<0>	MAKE_BASE=TRUE	MEM_A_DQS_P<0>	23	109	7	MEM_B_DQS_P<0>	MAKE_BASE=TRUE	MEM_B_DQS_P<0>	24			
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SYNC_MASTER=Mary

SYNC_DATE=04/26/2018

DRAM: ALIASES AND BITSWAPS

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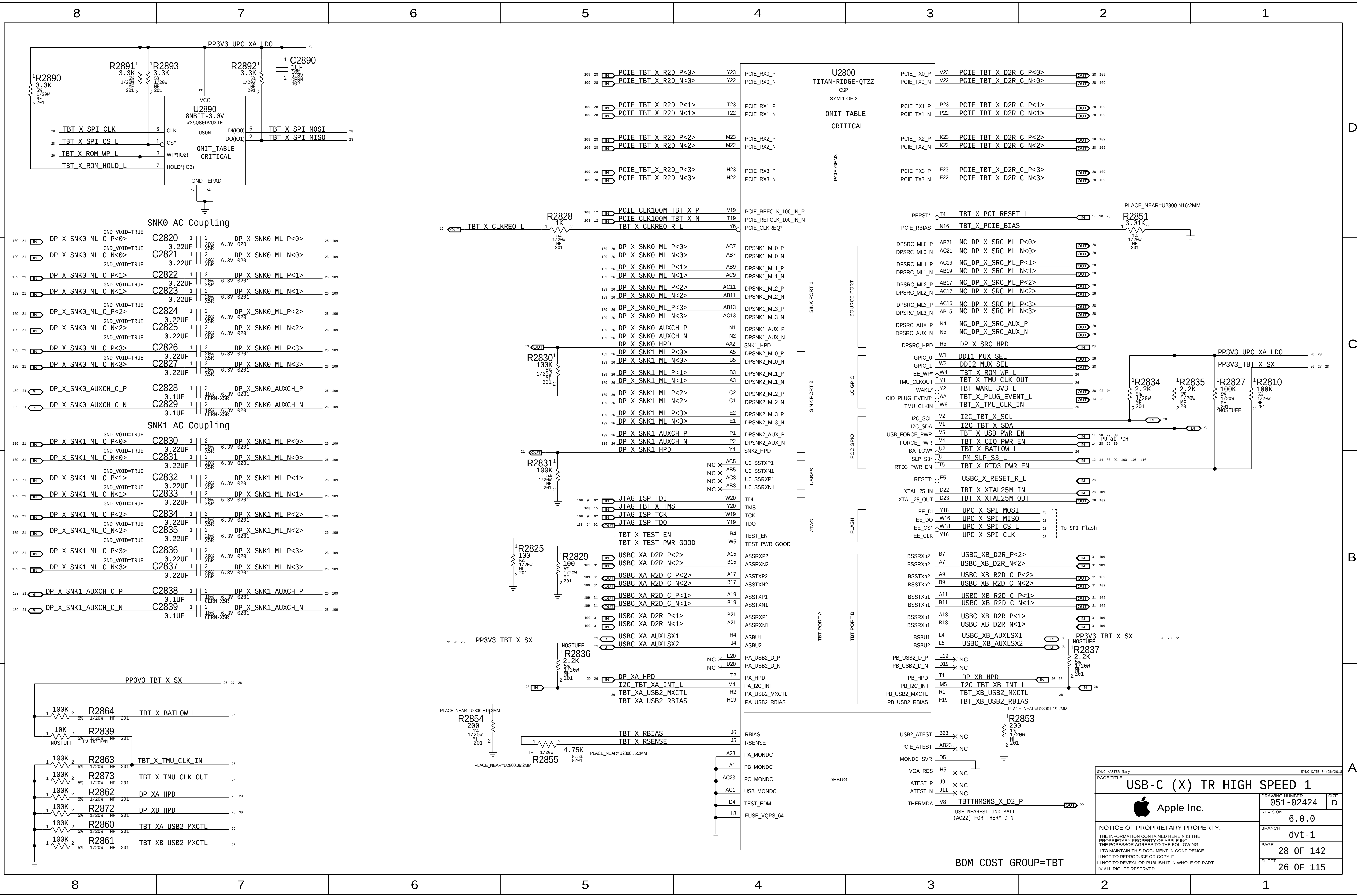
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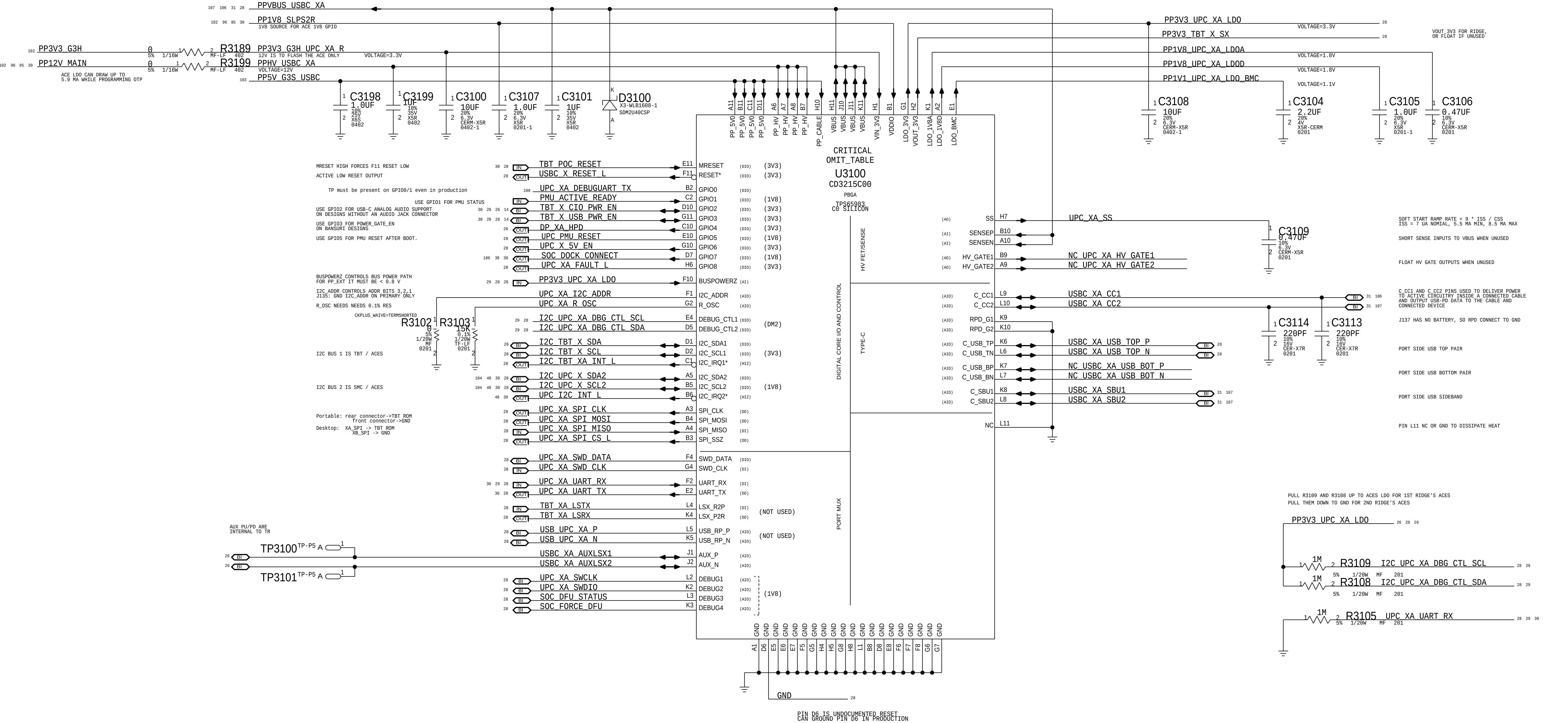
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X PRIMARY ACE USB-C PORT CONTROLLER (UPC)

J137 USB-C SUPPORTS 5V @ 3A
PP12V IS FOR PROGRAMMING ACE ONLY

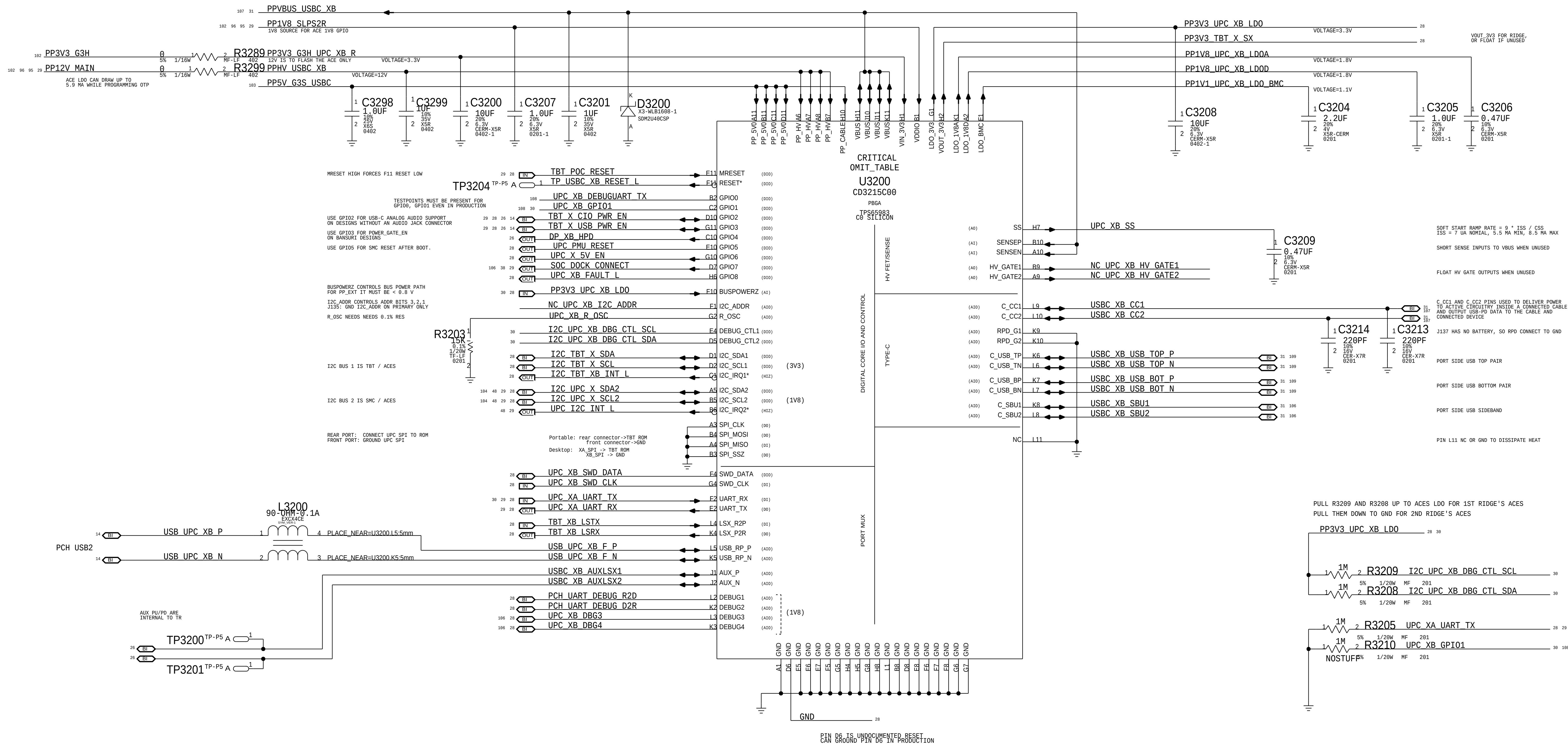


NOTE:
DEBUG_MUX IS CONNECTED TO
PRIMARY ACE OF 1ST RIDGE (XA ACE)
R3752-R3755, R3790-R3792 ARE OPTIONAL

BOM_COST_GROUP=USB-C

PAGE TITLE		NOT SYNCED TO OTHER PLATFORMS		SYNC_DATE=04/26/2018	
PAGE_TITLE=USB-C (X) PORT CONTROLLER A		Mary			
 Apple Inc.	DRAWING NUMBER	051-02424	SIZE		
	REVISION	6.0.0	D		
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X SECONDARY ACE1 USB-C PORT CONTROLLER (UPC)

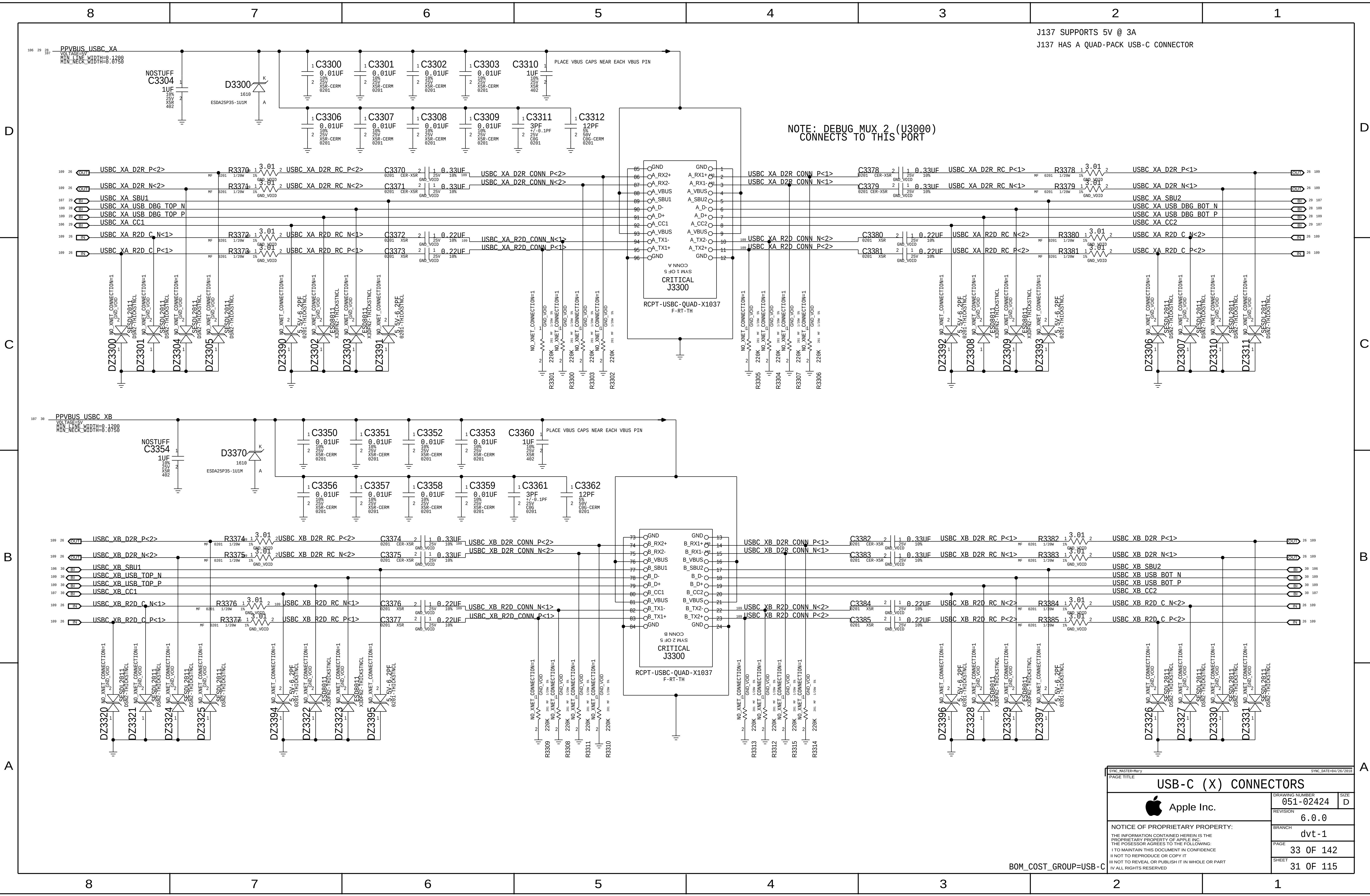


PIN D6 IS UNDOCUMENTED RESET
CAN GROUND PIN D6 IN PRODUCTION

~~NOT SYNCED TO OTHER PLATFORMS~~
SYNC DATE=04/26/2018

PAGE TITLE		DRAWING NUMBER		SIZE	
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BOM_COST_GROUP=USB-C



J137 SUPPORTS 5V @ 3A
J137 HAS A QUAD-PACK USB-C CONNECTOR

NOTE: DEBUG_MUX_2 (U3000)
CONNECTS TO THIS PORT

PAGE TITLE		PAGE NUMBER	
USB-C (X) CONNECTORS		051-02424	
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BOM_COST_GROUP=USB-C

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D																D	
C		PLACE HOLDER														C	
B																B	
A																A	
8		7		6		5		4		3		2		1			

Empty

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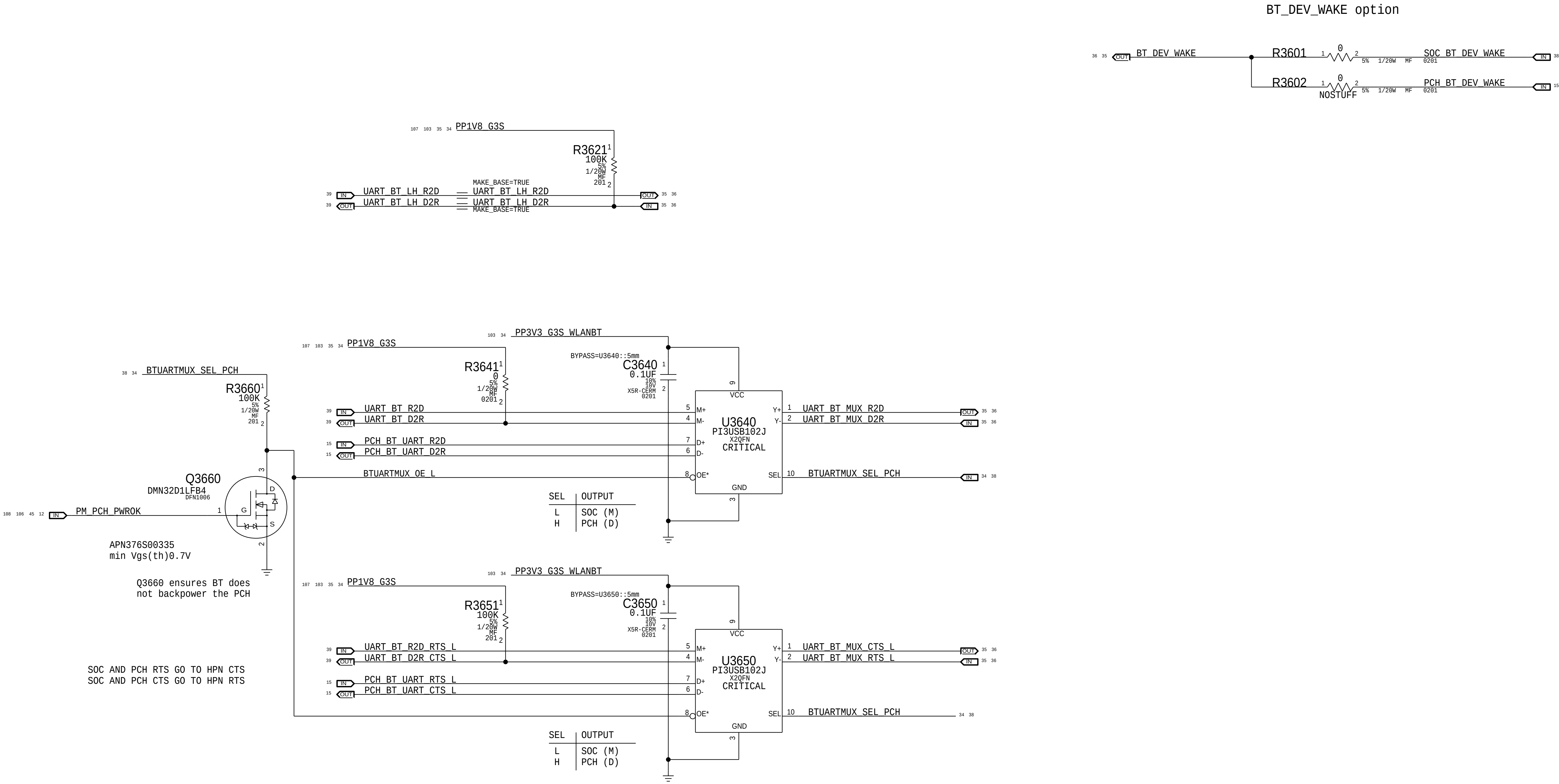
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8		7		6		5		4		3		2		1	
D														D	
C		PLACE HOLDER												C	
B														B	
A														A	
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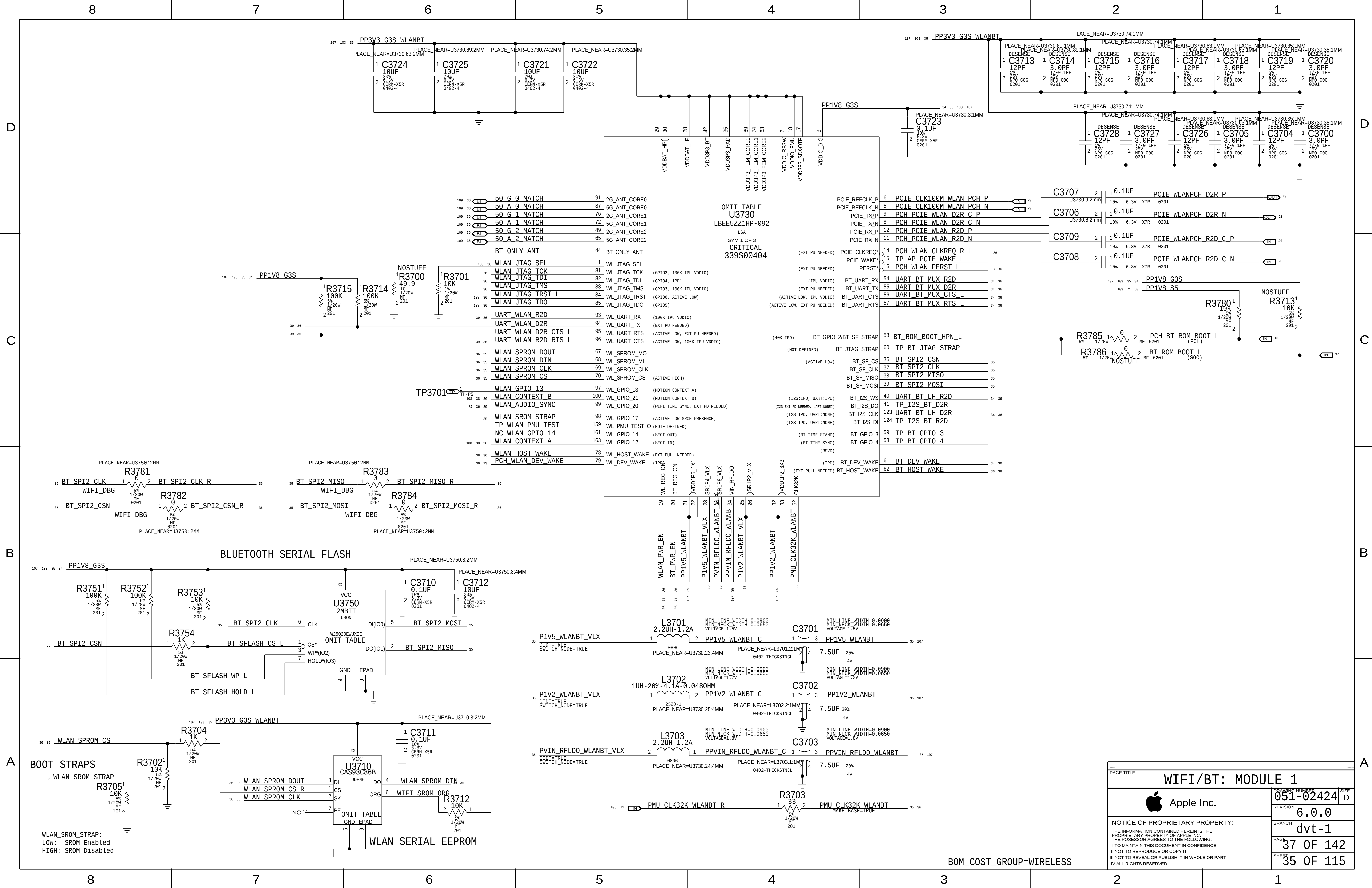
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Gen1 (macOS)	0	1	UART (PCH)
Gen2	X	0	UART (SOC)

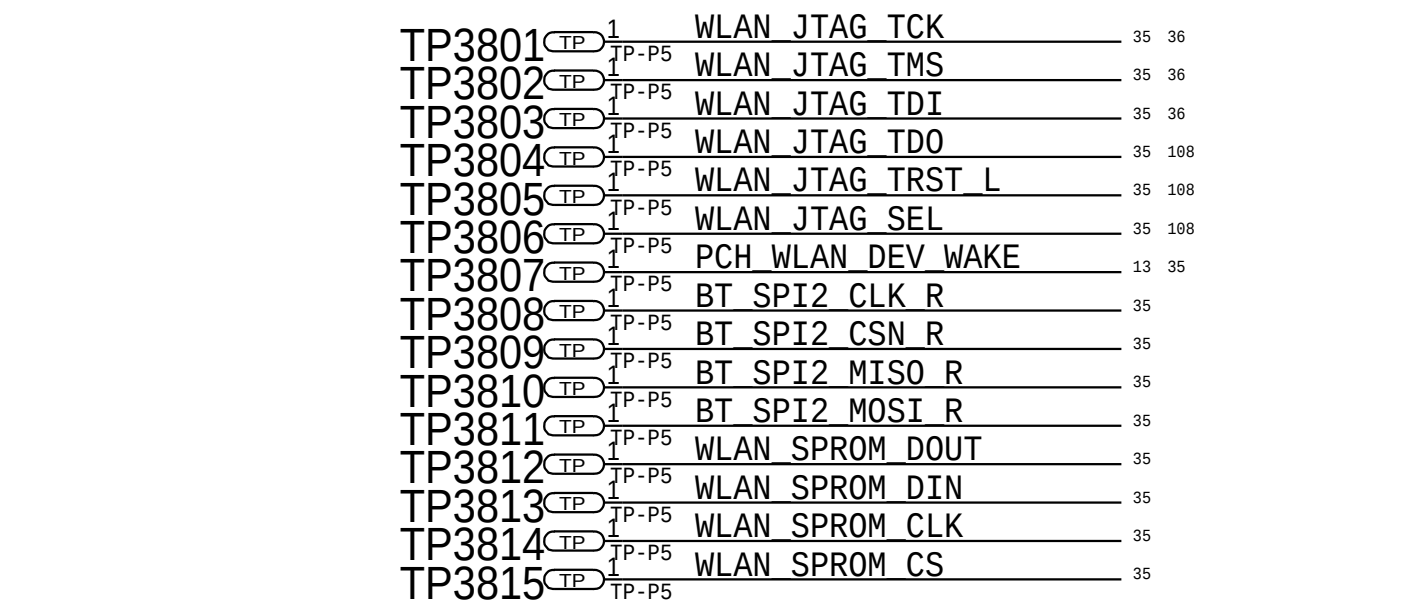
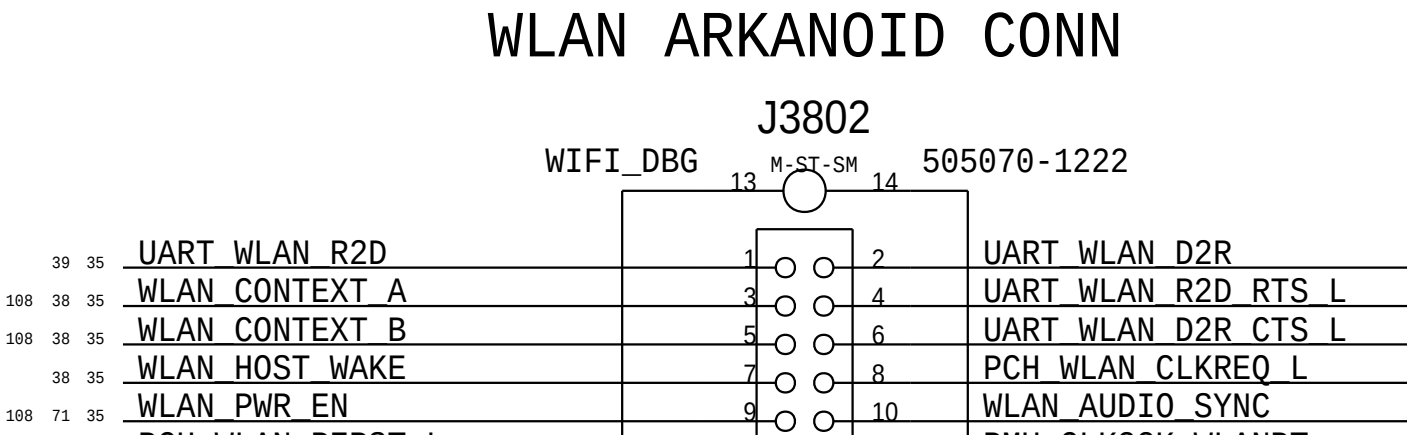
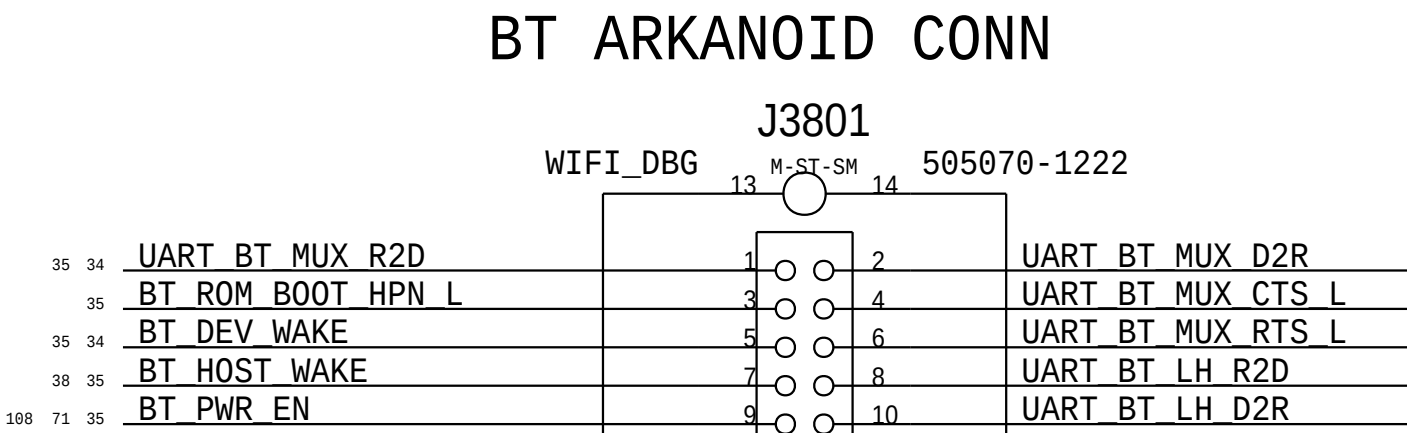
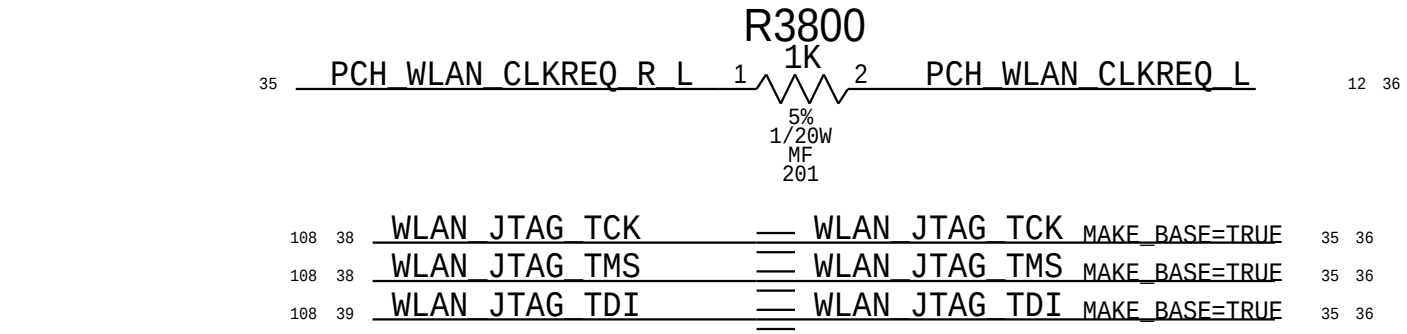
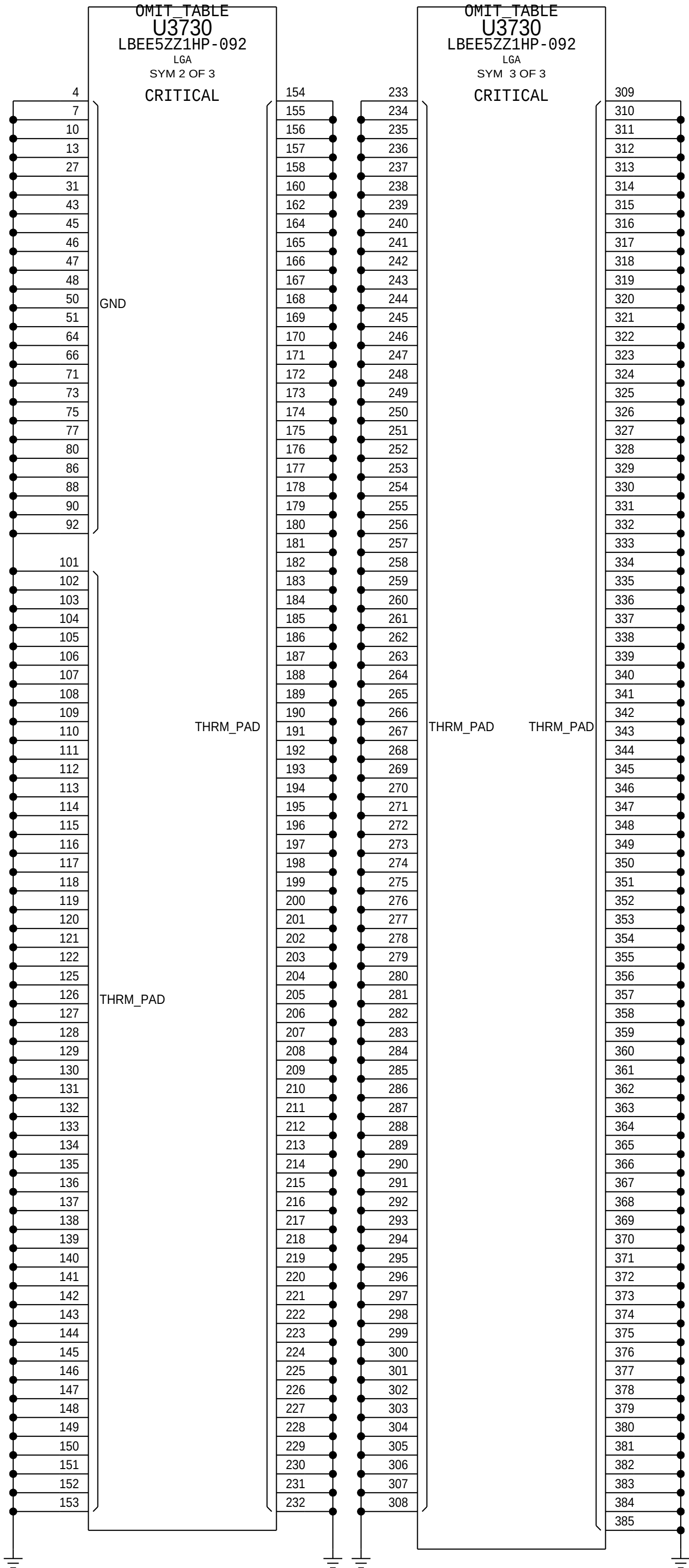


WIFI/BT: Support		
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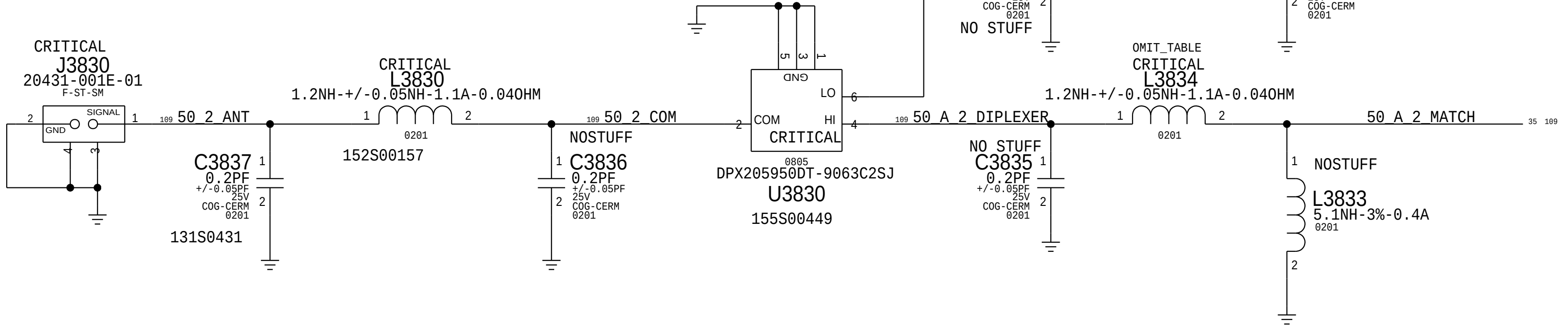
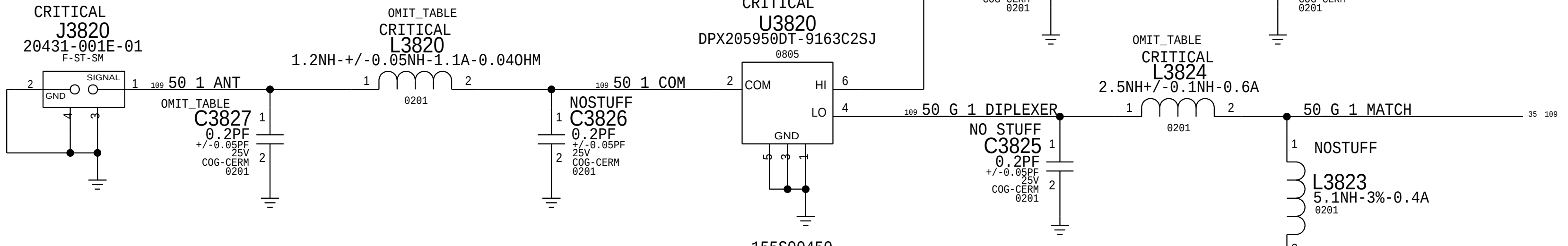
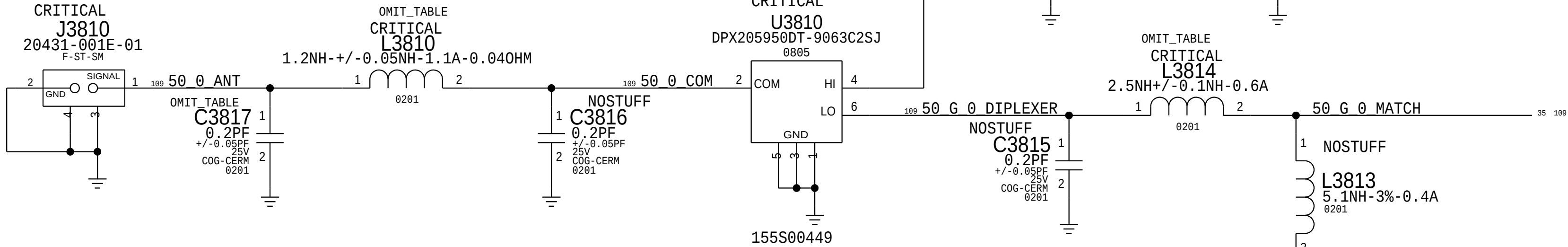
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00152	1	IND, 0.9NH, 1.1A, 0201	L3810	CRITICAL	
131S0631	1	CAP, CER, 0.3pF, 0201	C3817, C3827	CRITICAL	
152S00153	1	IND, 1NH, 1.1A, 0201	L3820	CRITICAL	
152S1222	1	IND, 3NH, 1.1A, 0201	C3812	CRITICAL	
131S0555	1	cap, CER, 1pF, 0201	L3811	CRITICAL	



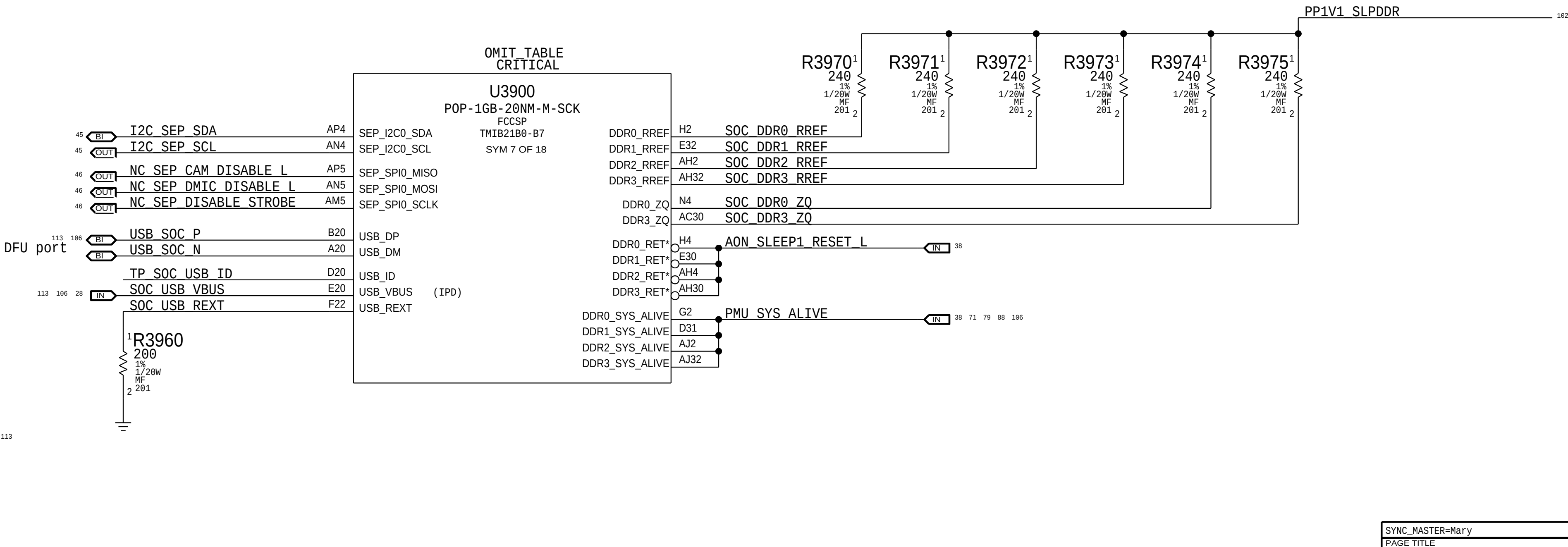
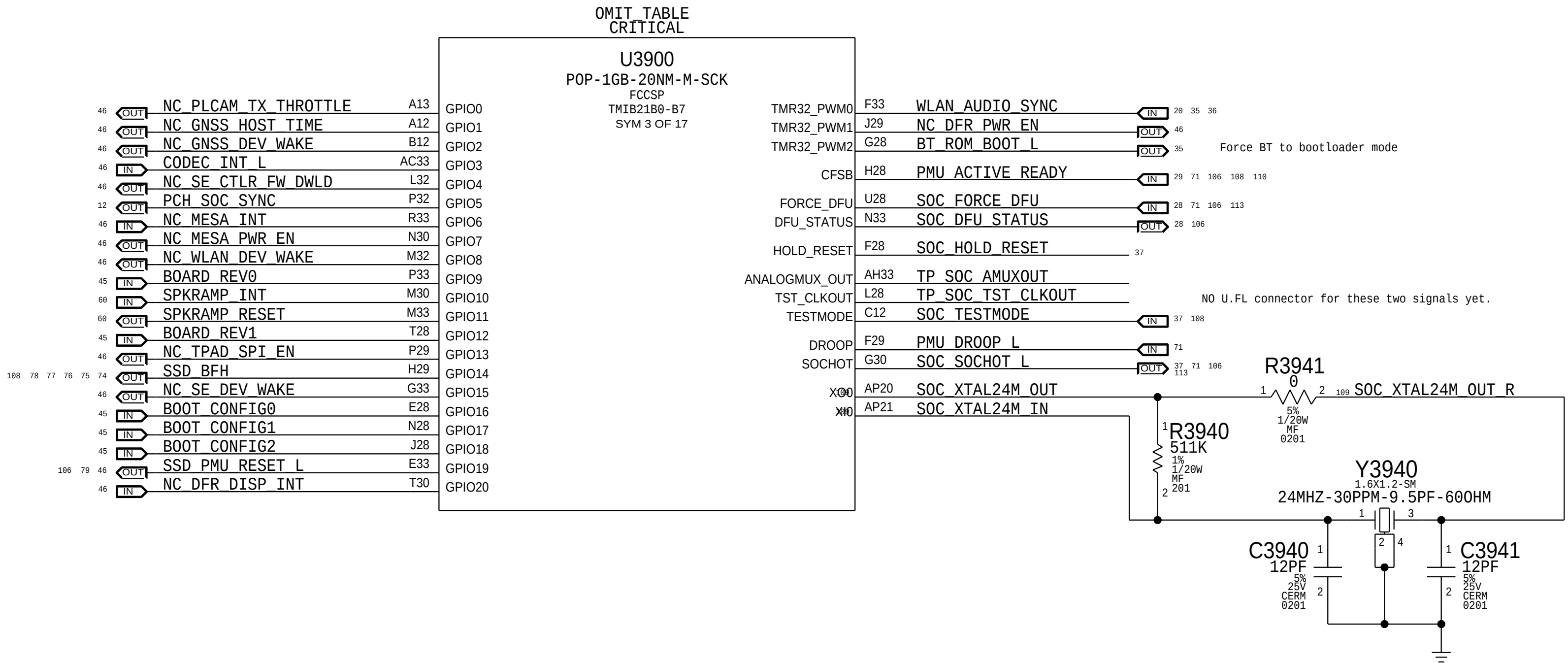
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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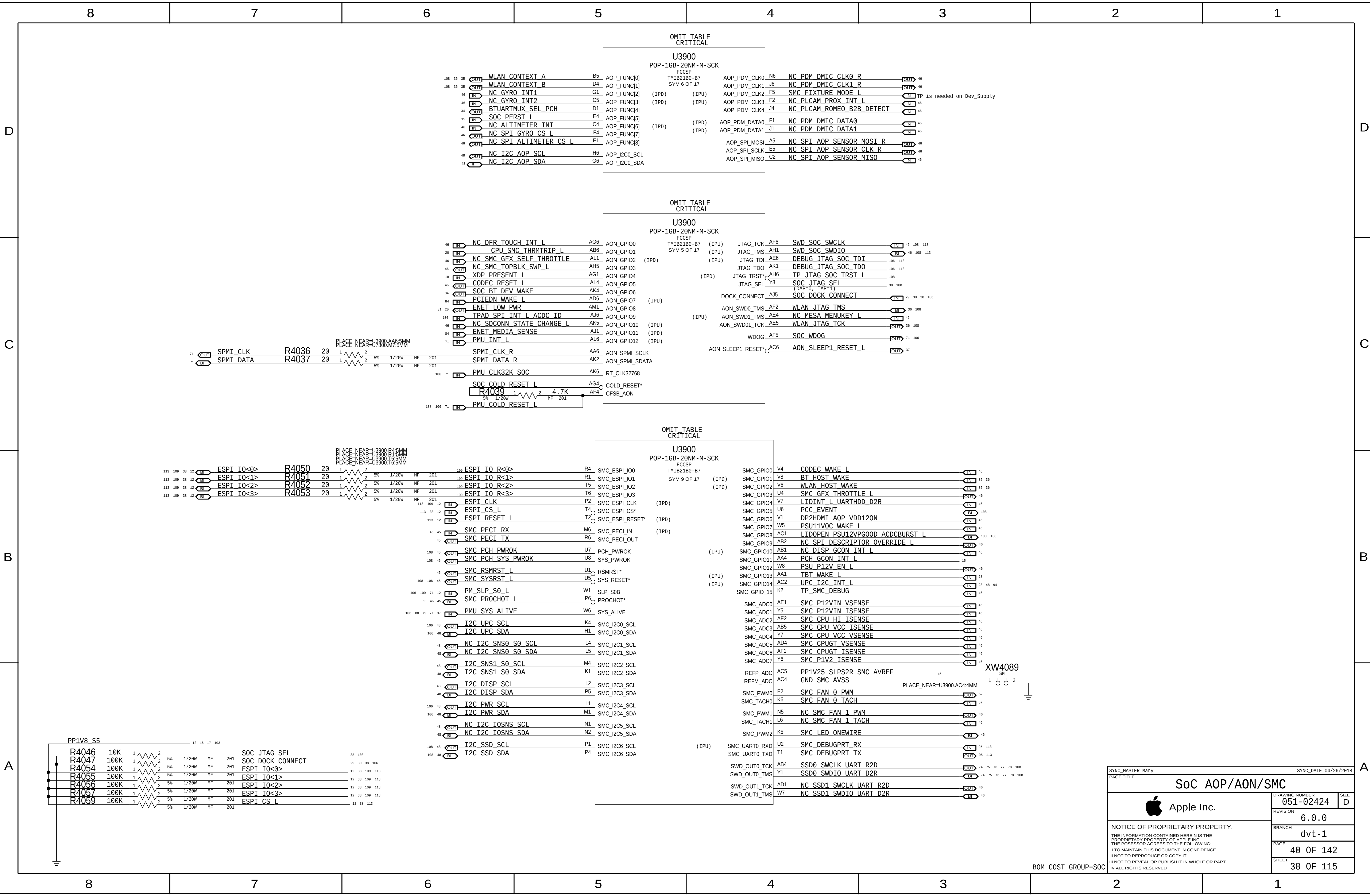
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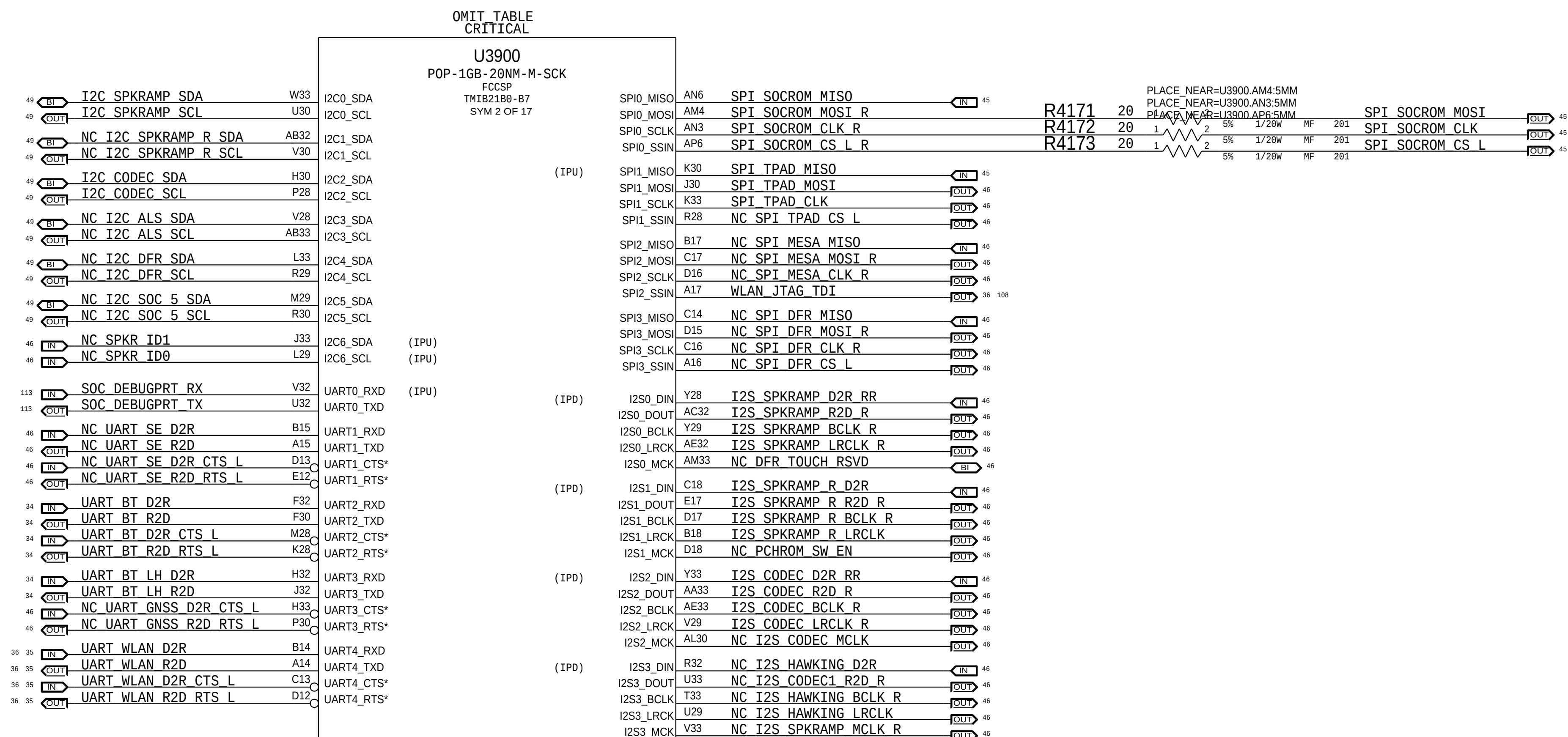
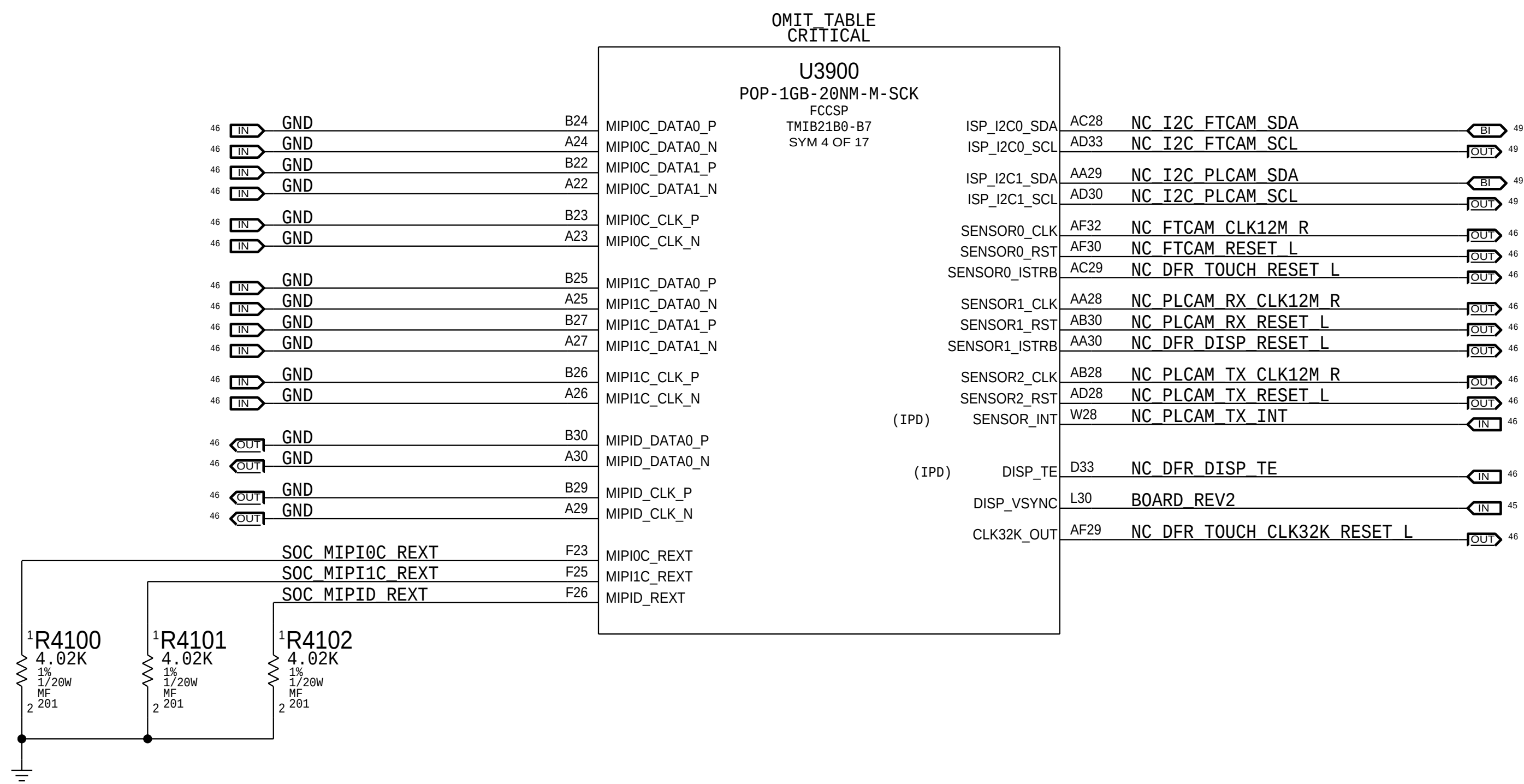
Note 1) IPU represents SW configured state, not HW default



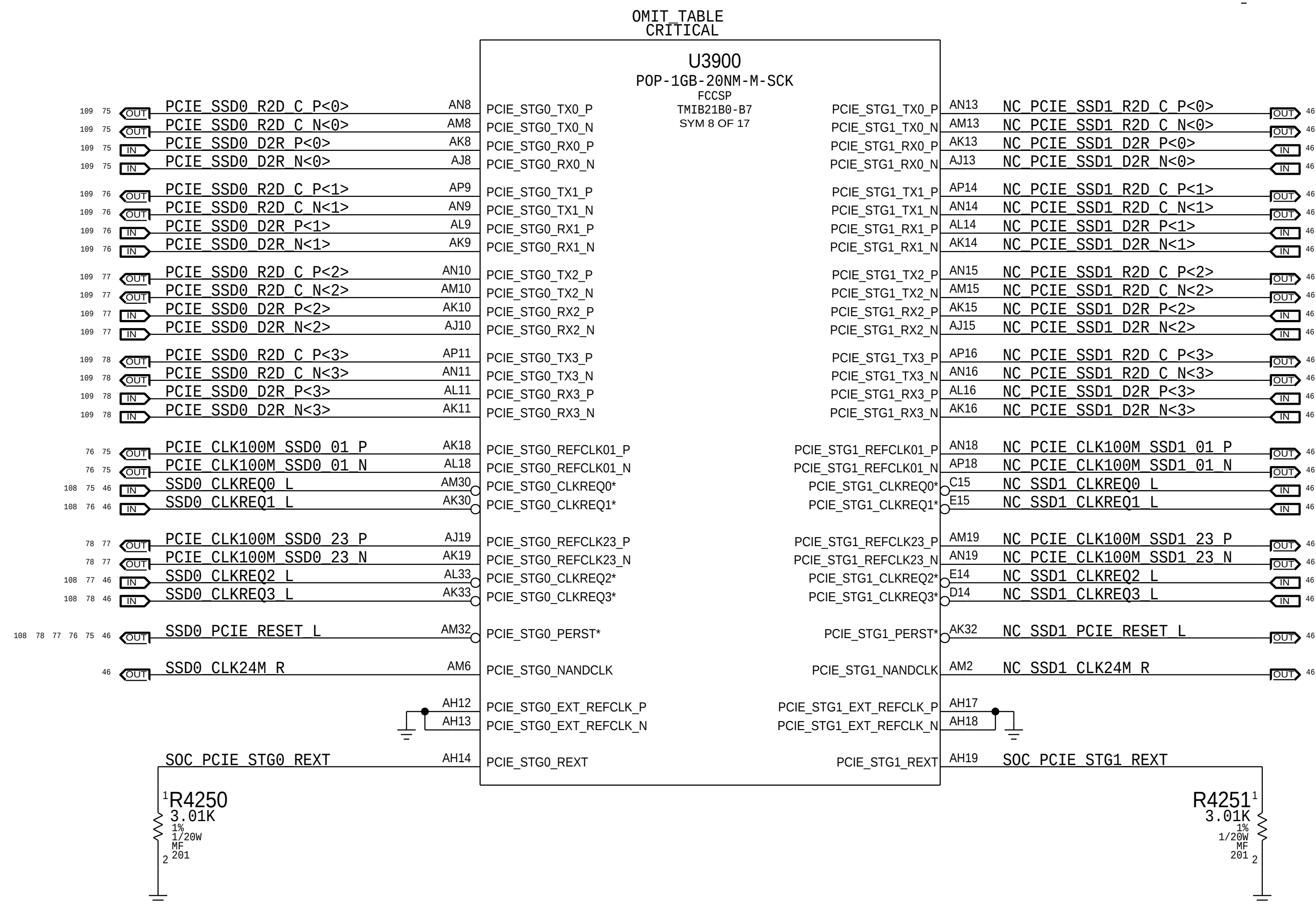
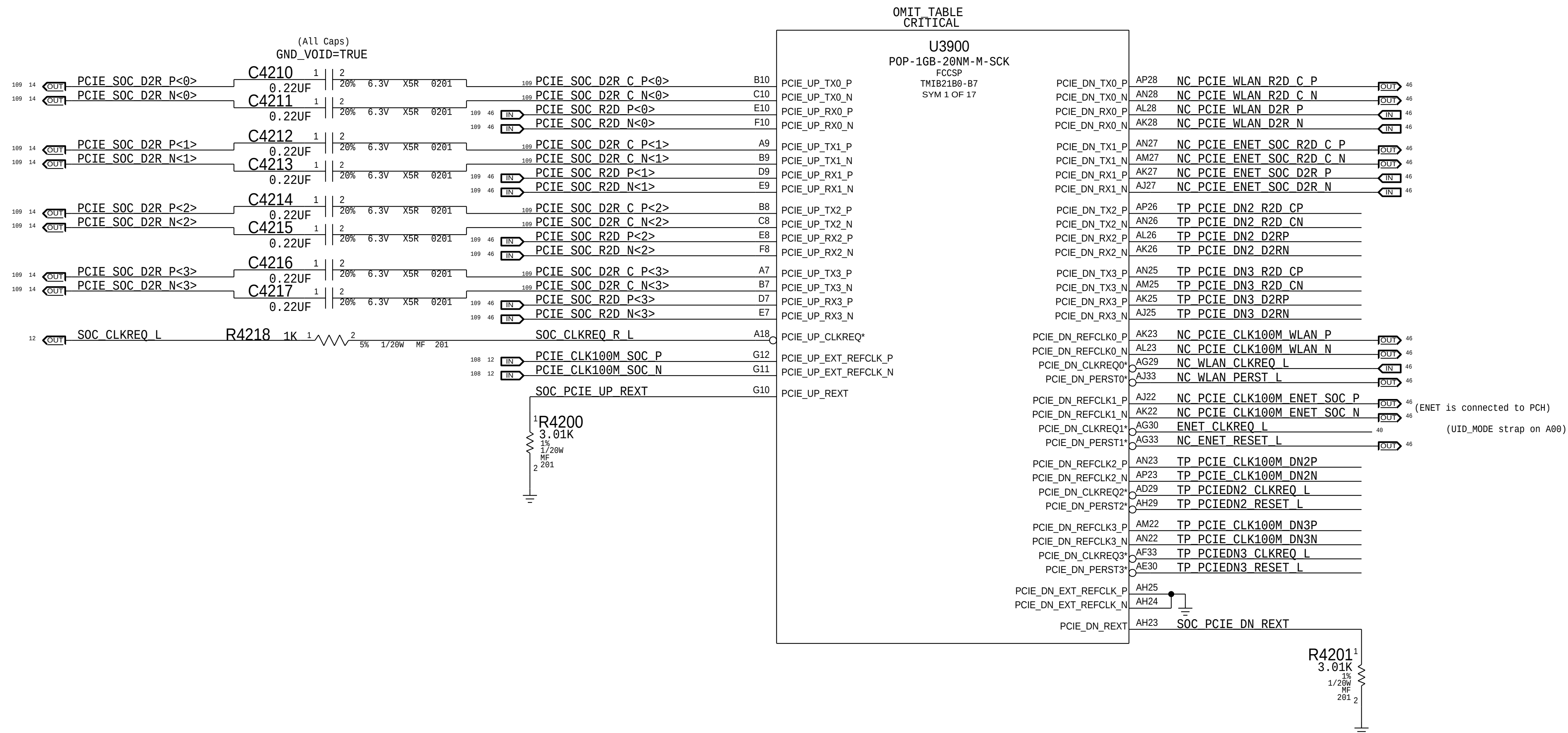
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	BRANCH		
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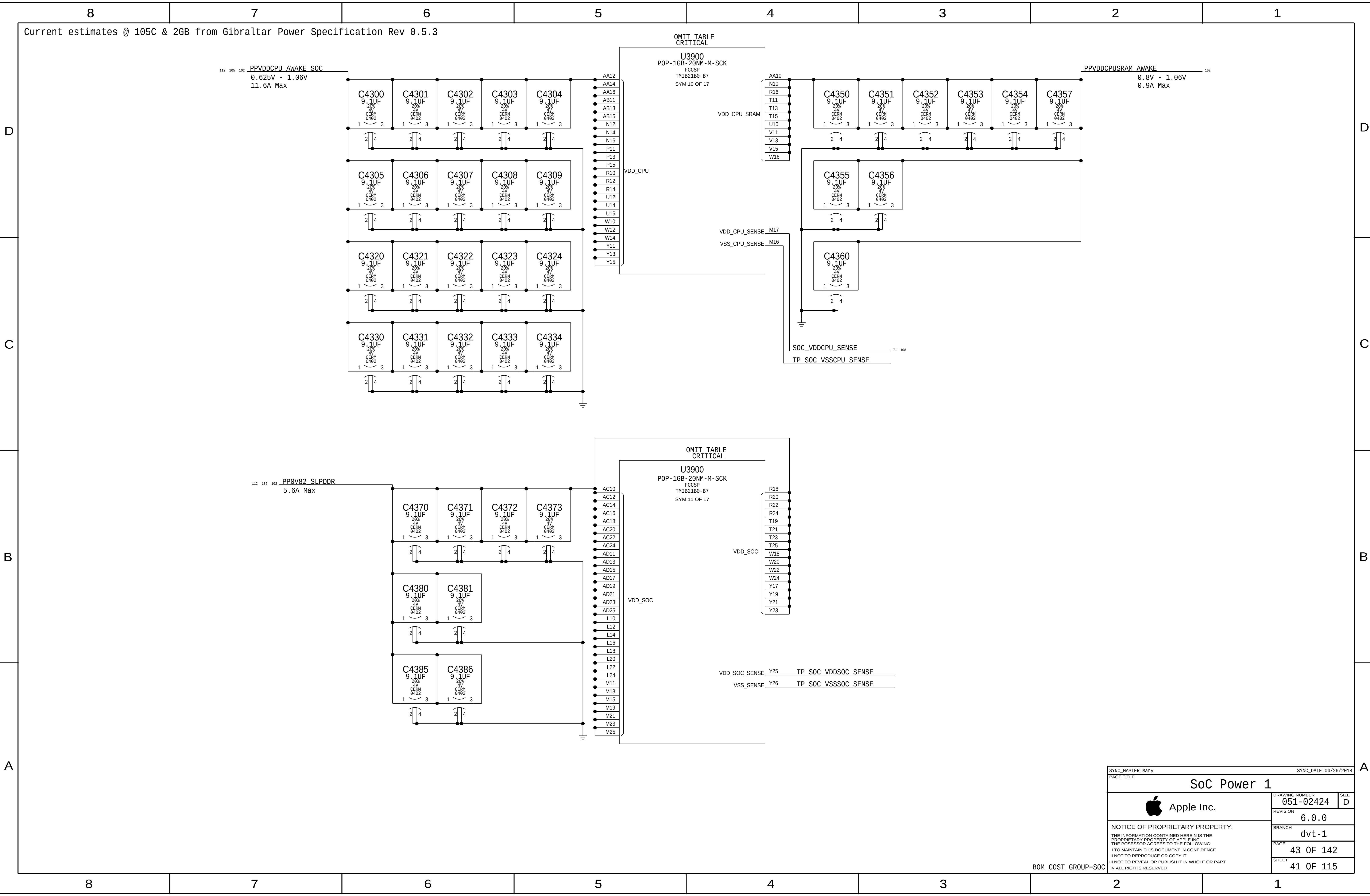
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BOM_COST_GROUP=SOC

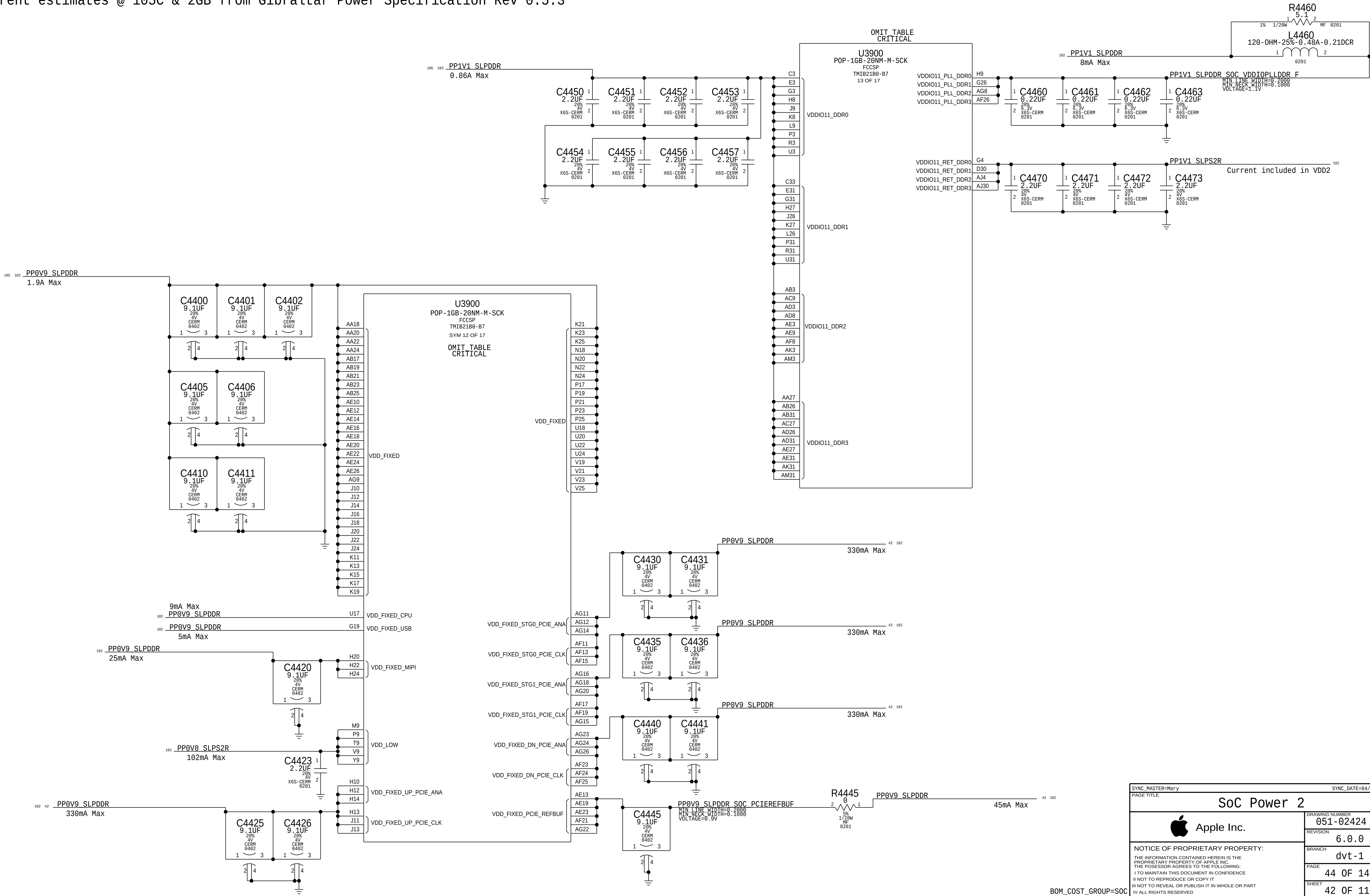




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BOM_COST_GROUP=SOC

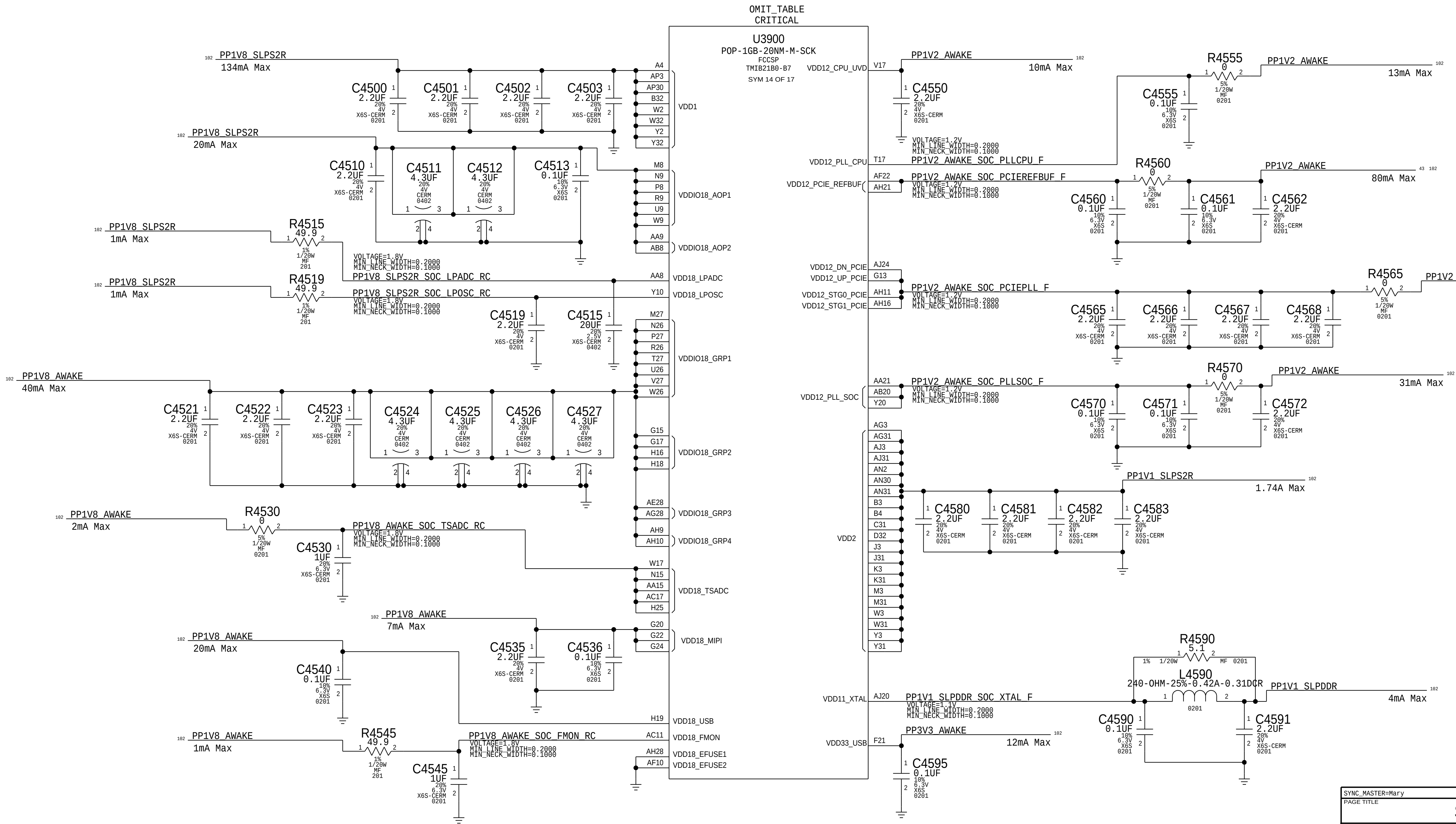
Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3



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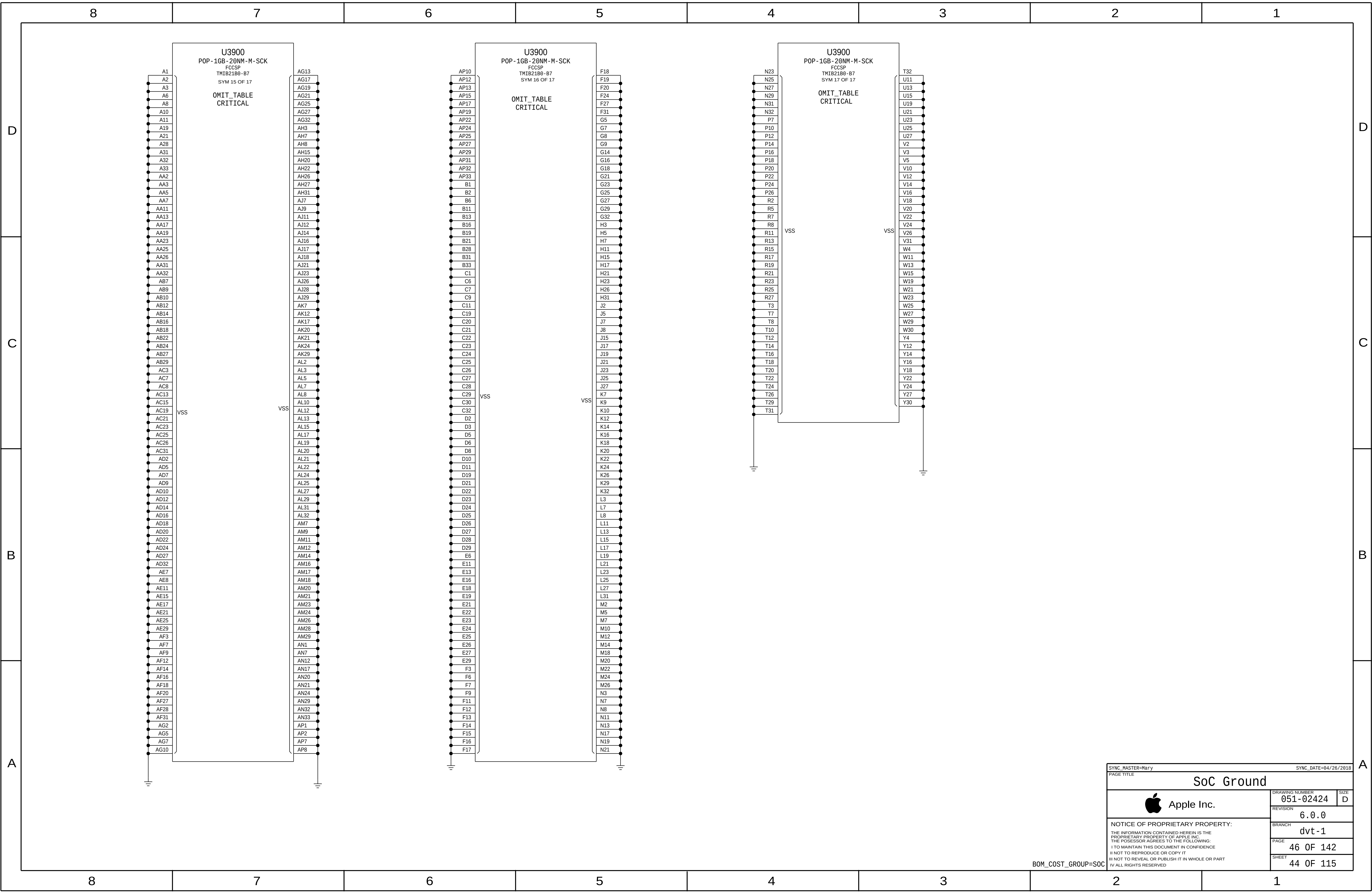
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Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3



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	BRANCH		
	dvt-1		
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BOM_COST_GROUP=SOC



T32

U11

U13

U15

U19

U21

U23

U25

U27

V2

V3

V5

V10

V12

V14

V16

V18

V20

V22

V24

V26

V31

W4

W11

W13

W15

W19

W21

W23

W25

W27

W29

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3

2

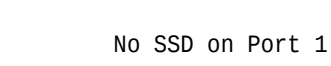
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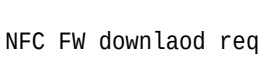
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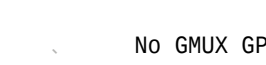
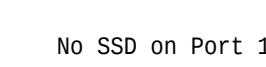
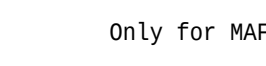
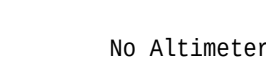
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PMU_VDD_I PMU_P3V3_CAPLE_ISENSE = PMU_P3V3_CAPLE_ISENSE 71
MAKE_BASE=TRUE
PMU_CRUSA_ISENSE PMU_CRUSA_ISENSE



3	2	1
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SYNC_MASTER=Mary		SYNC_DATE=04/26/2018	
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		051-02424	
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BRANCH		dvt-1	
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SHEET		46 OF 115	

12V_G3H uses ACDC

3V3_G3H_PG00D is to PMIC.GPIO

P5V_G3S is enabled by PMIC.GPIO

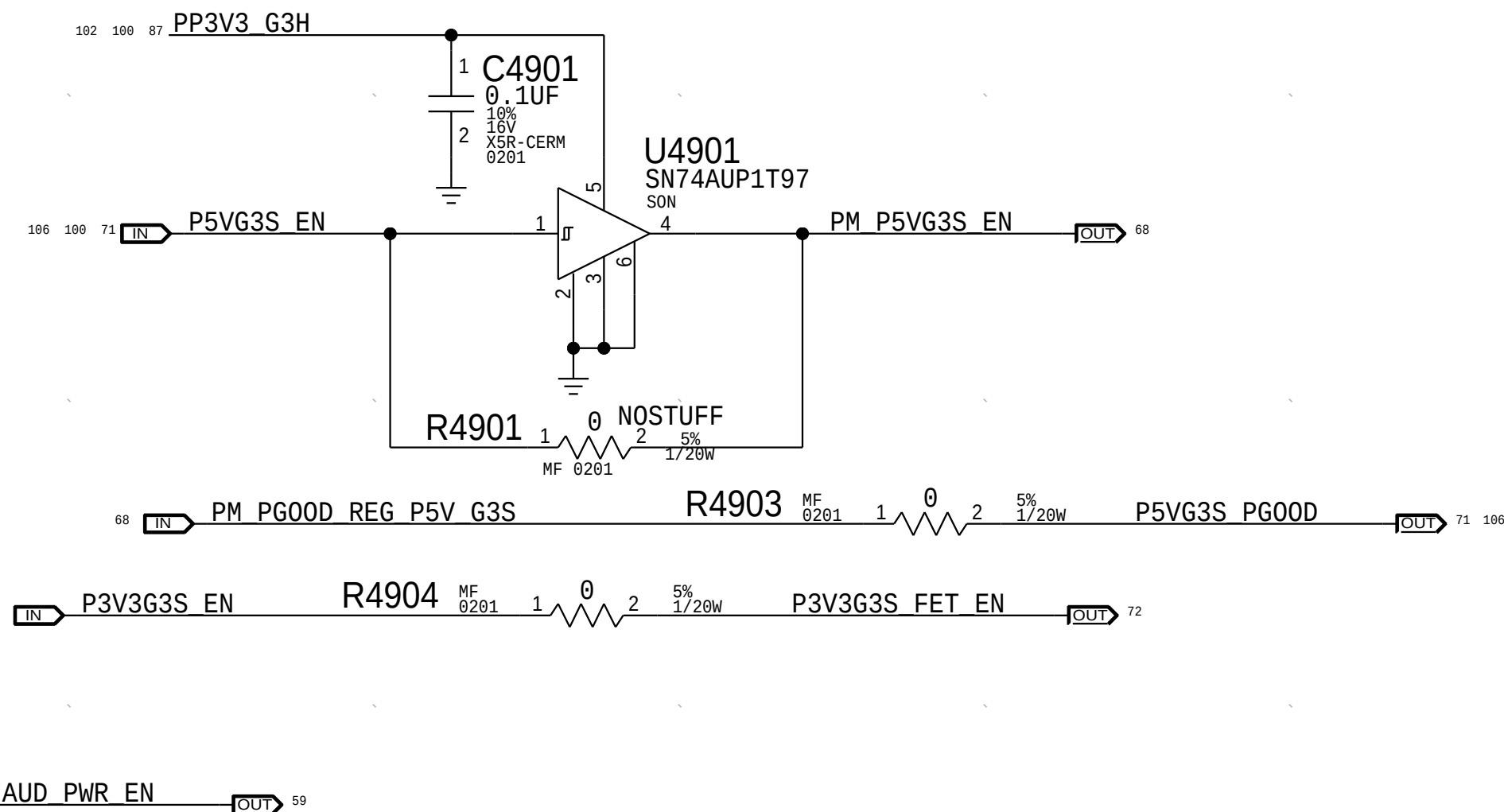
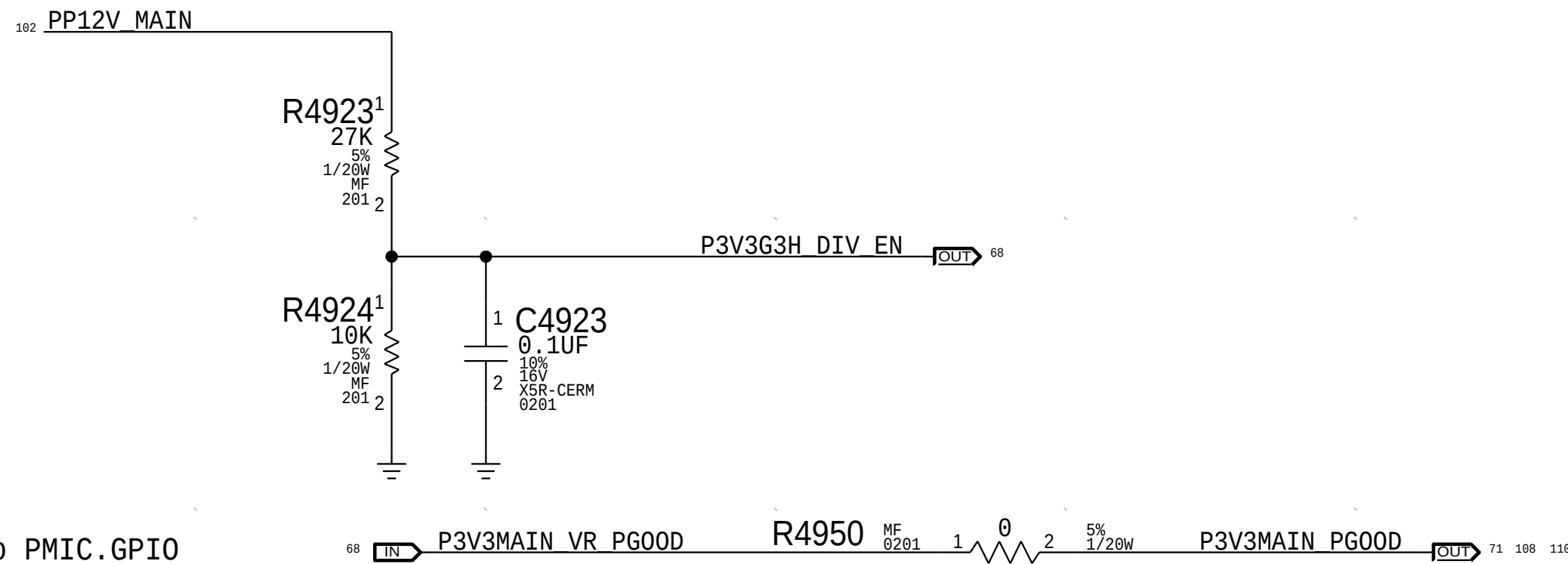
P3V3_G3S is enabled by PMIC.GPIO

P1V8_G3S is through BUCK3_SW4.

3V3_S5 uses PMU Sw

1V8_S5 uses PMU SW

1V05_S5 uses PMU SW

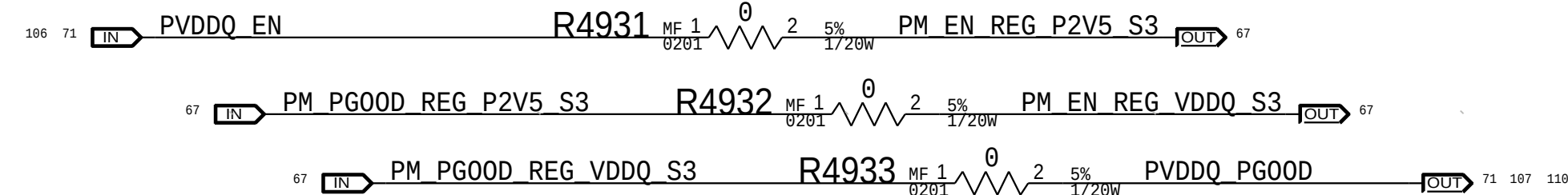


Timing diagram for R4920 and R4921. The diagram shows two signal traces. The top trace, labeled R4920, shows a signal labeled PM RSMRST L (input) and a signal labeled PM EN USB A PWR (output). The bottom trace, labeled R4921, shows a signal labeled PP1V8 S5 (input) and a signal labeled NOSTUFF (output). Both signals have a 5% 1/20W delay specification.

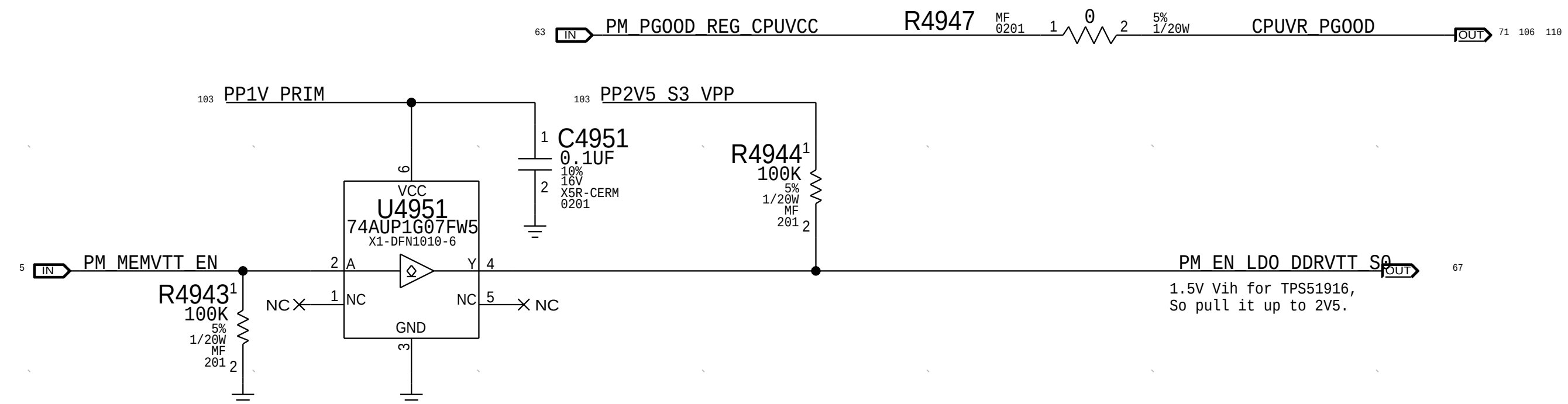
1V2_S3 sequencing

Calpe VDDQ_EN -> 2V5_VTT -> 2V5_PG00D -> Enable 1V2 VDDQ -> VDDQ_PGDDD

DDR4 Requirement:
VPP > VDDQ at all times
=> Ensure VPP discharges more slowly than VDDQ

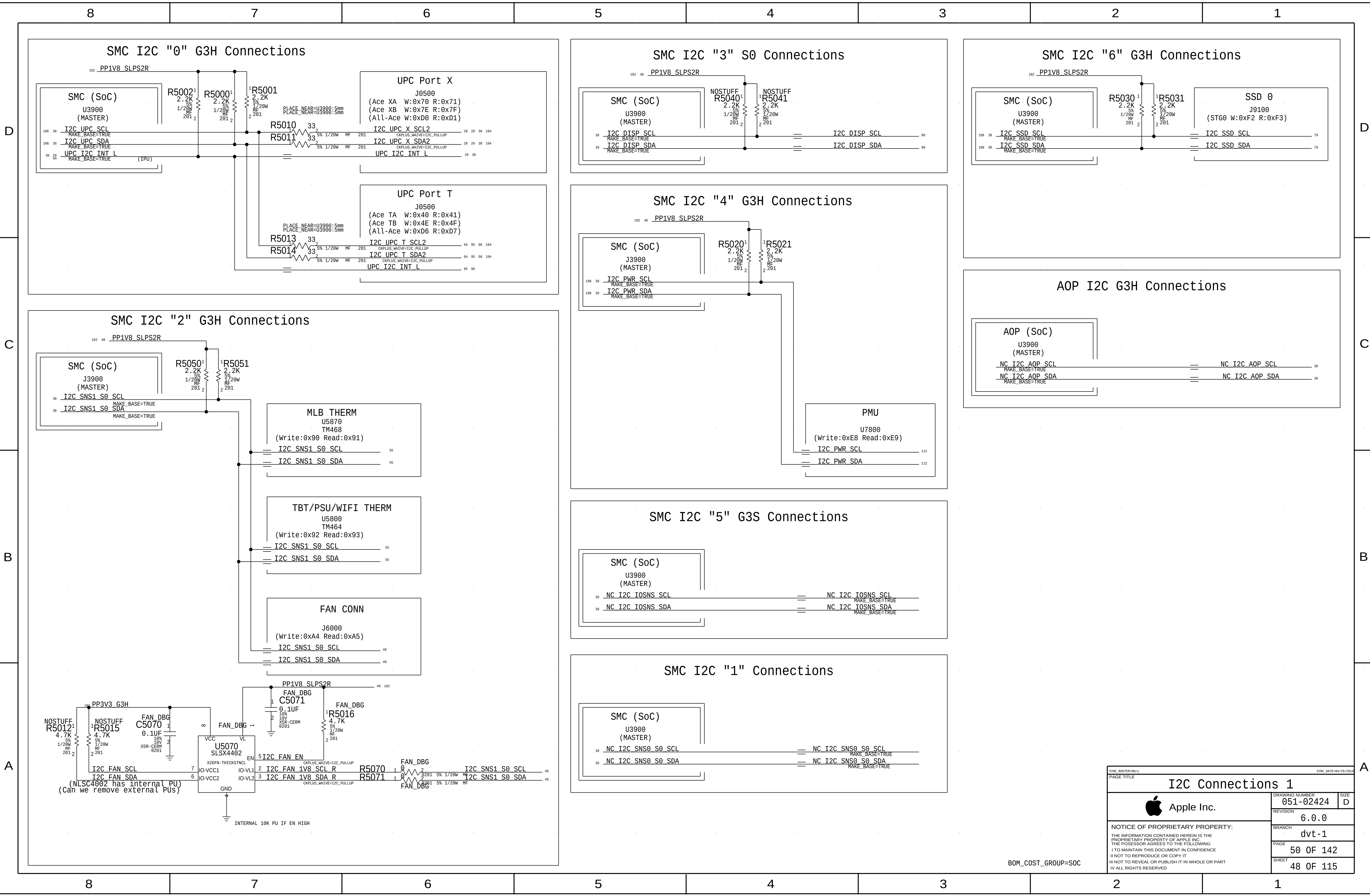


1V05_S3 uses PMU SW

[illegible]

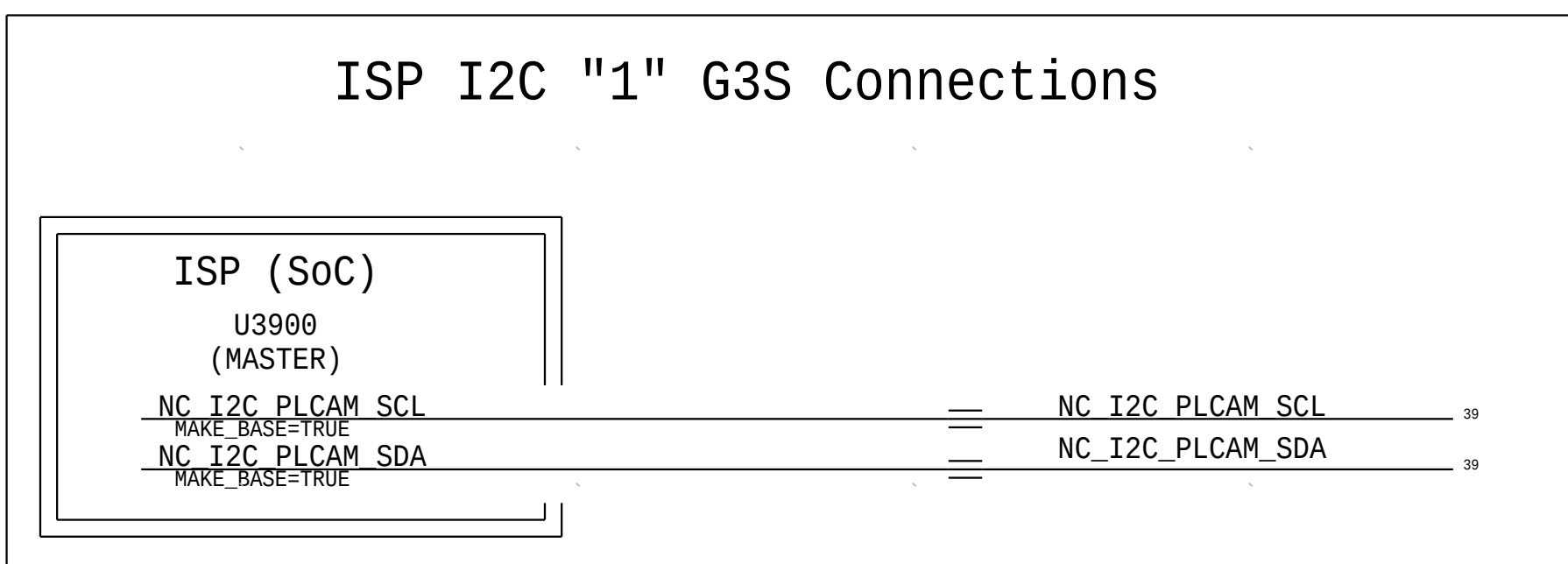
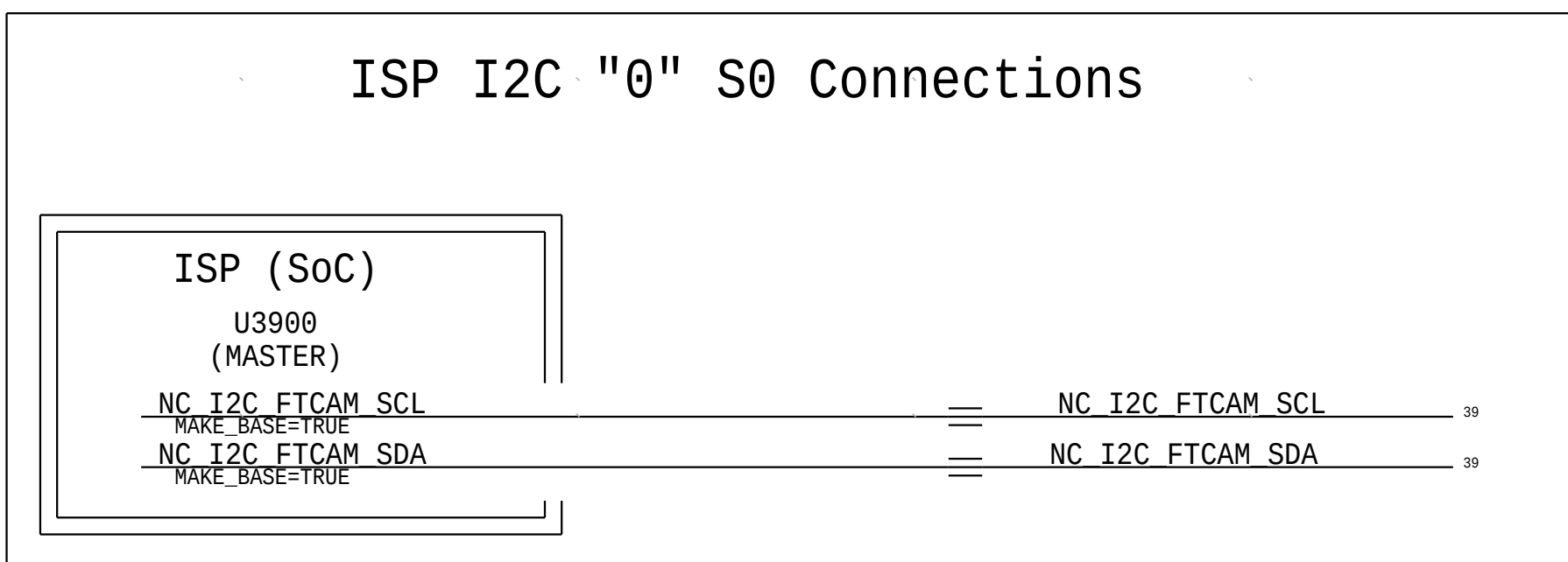
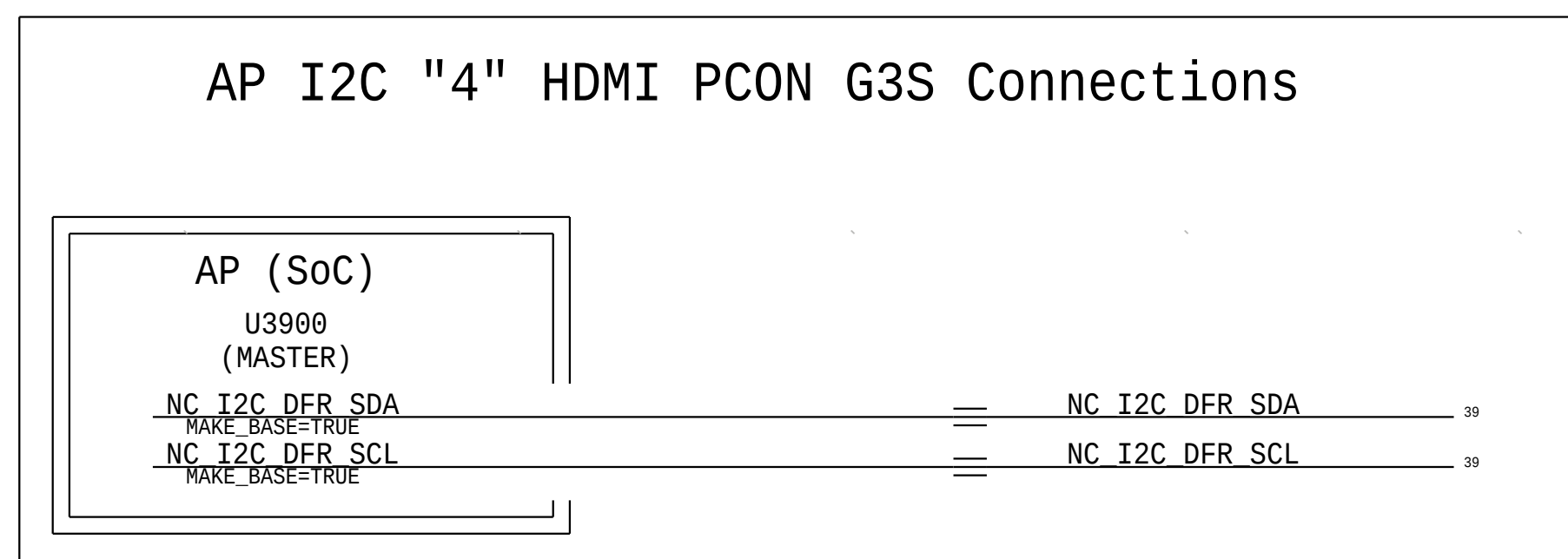
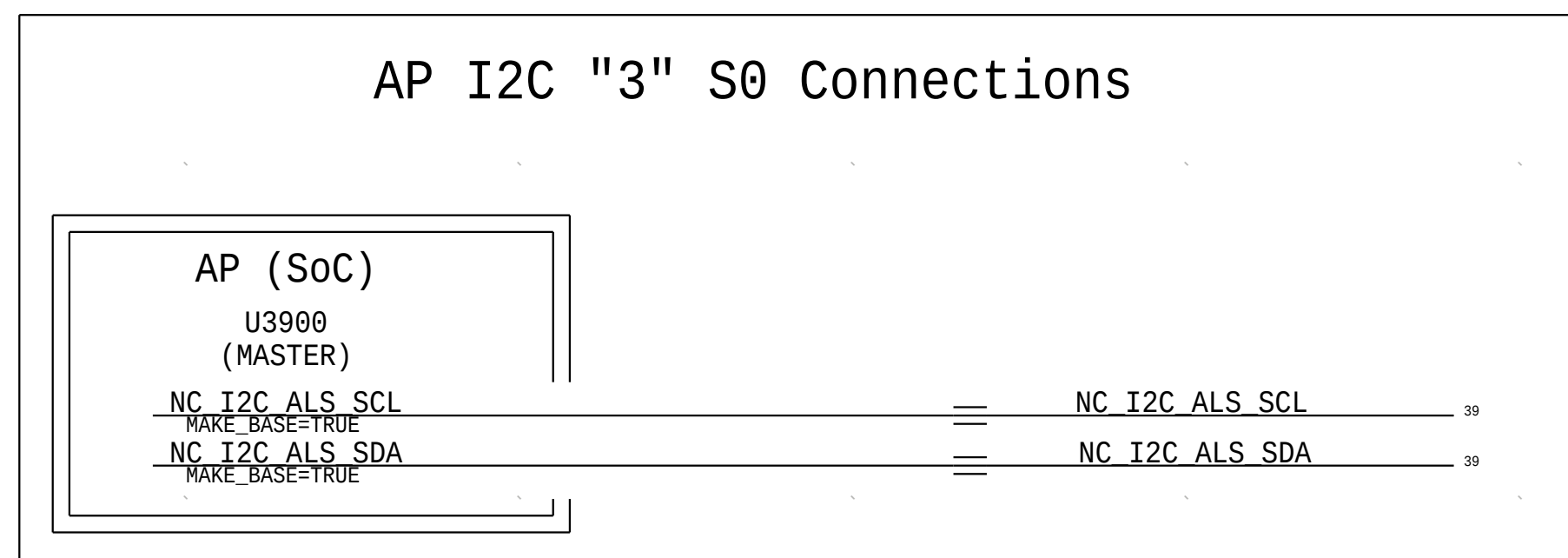
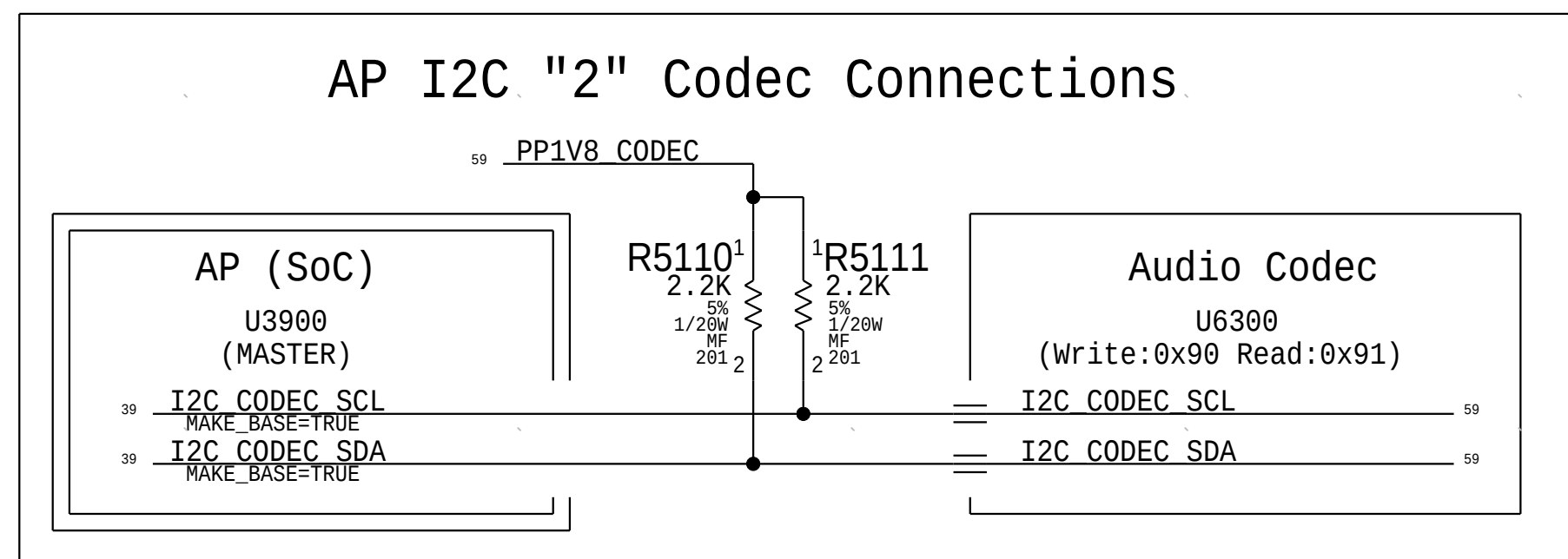
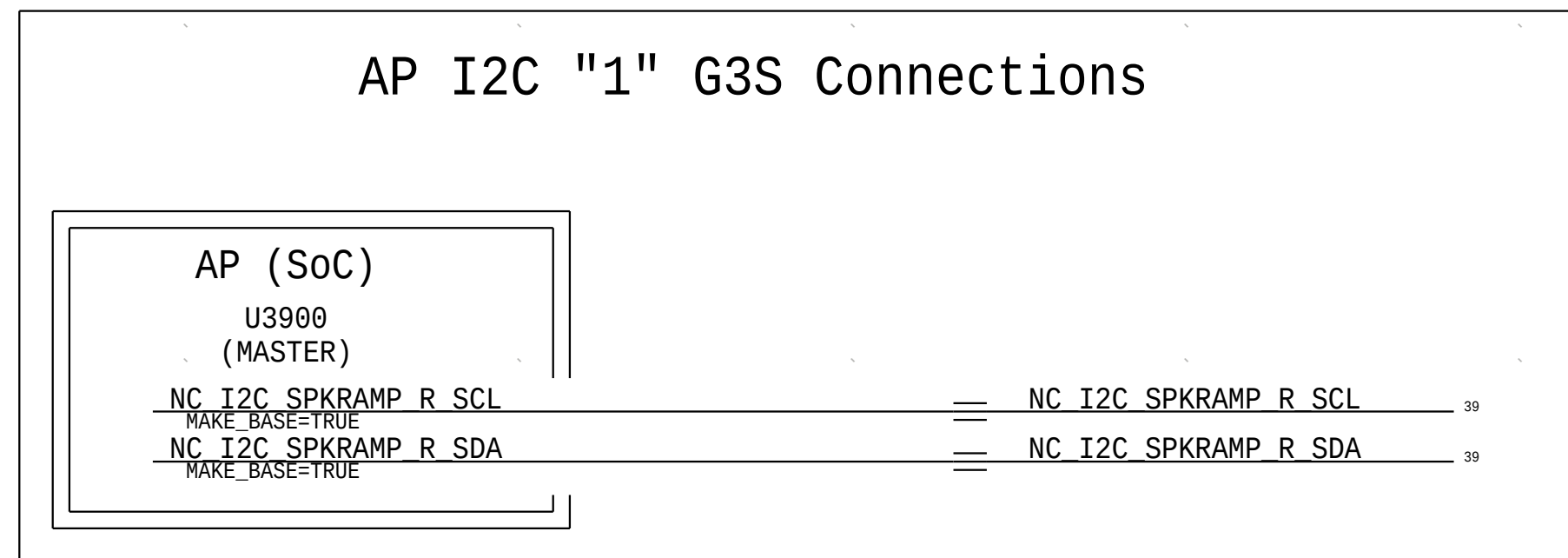
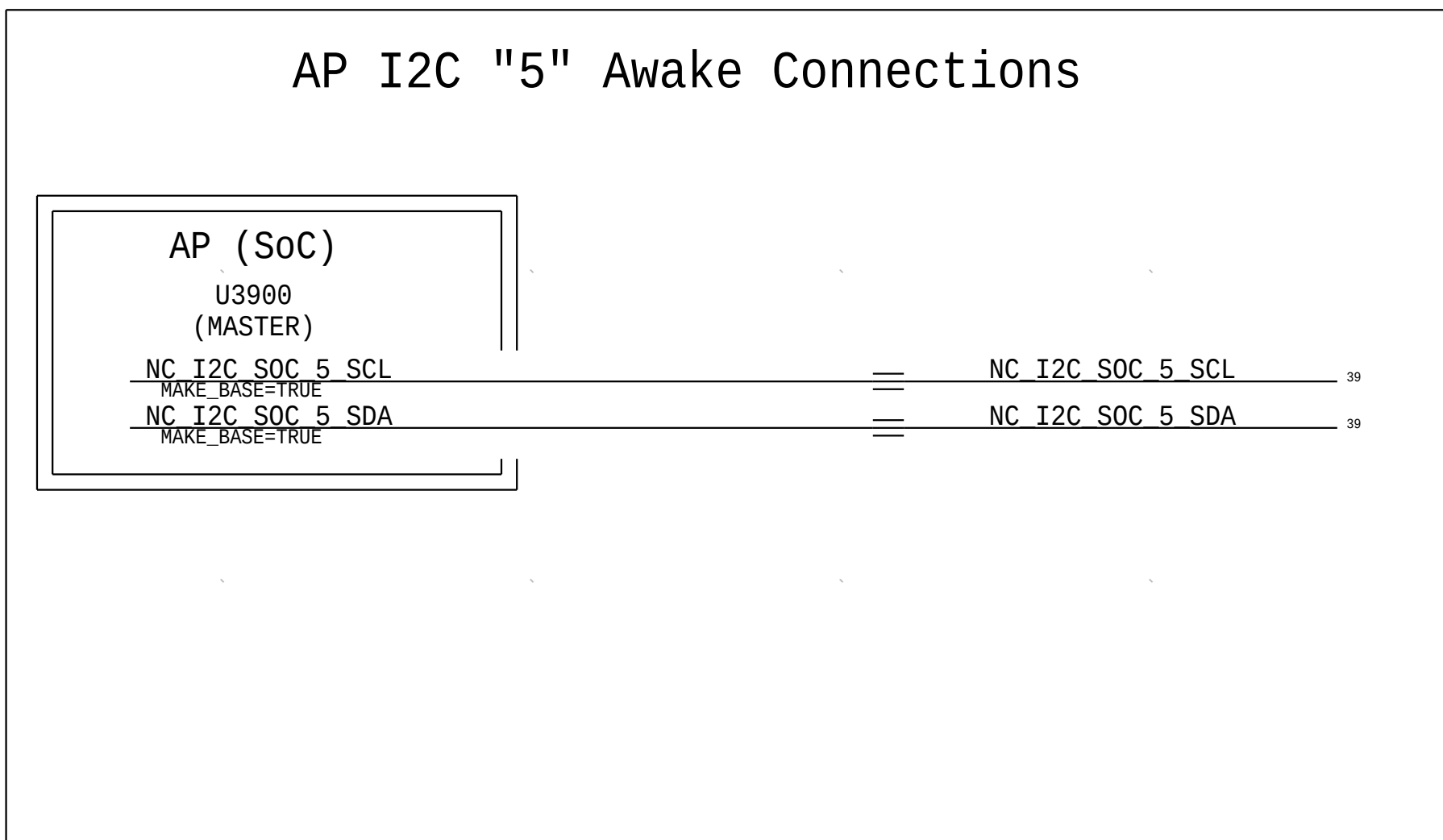
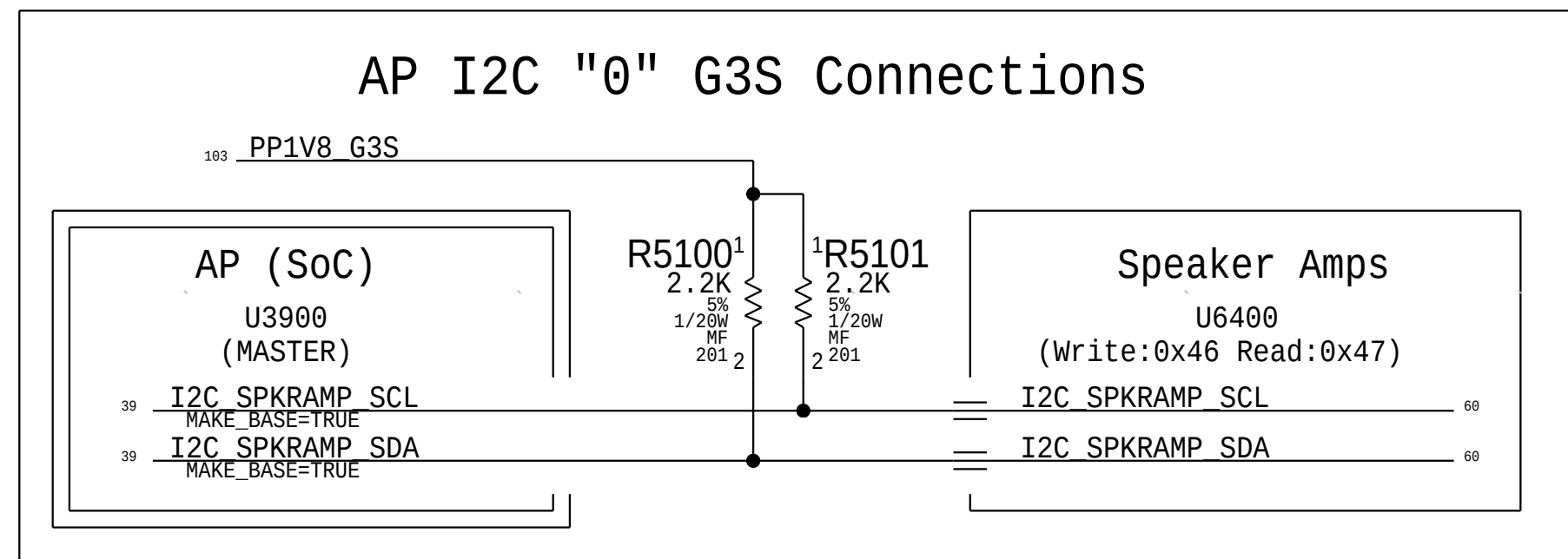
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		SHEET		47 OF 115	

BOM_COST_GROUP=PLATFORM POWER

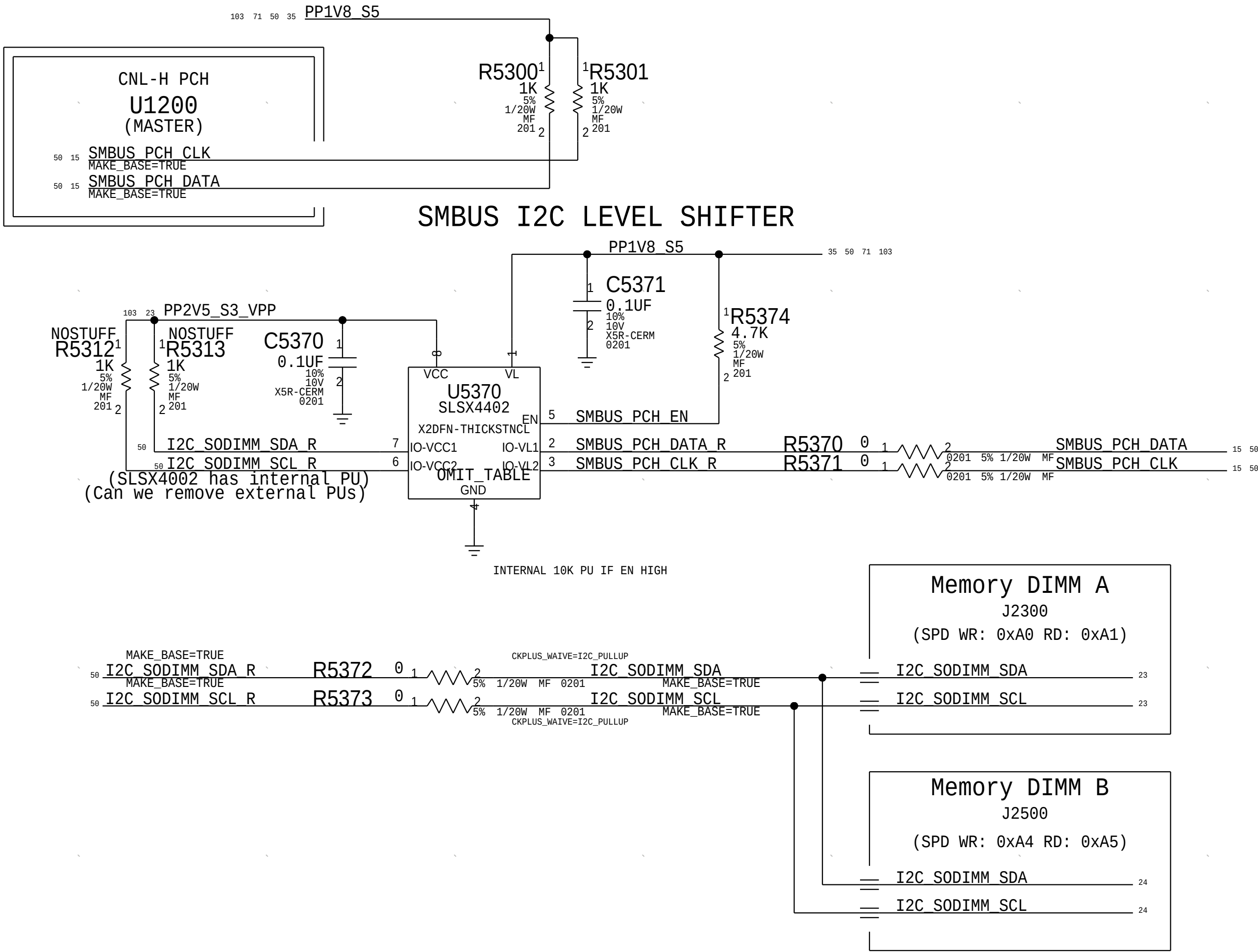


I2C Connections 1		
	DRAWING NUMBER	051-02424
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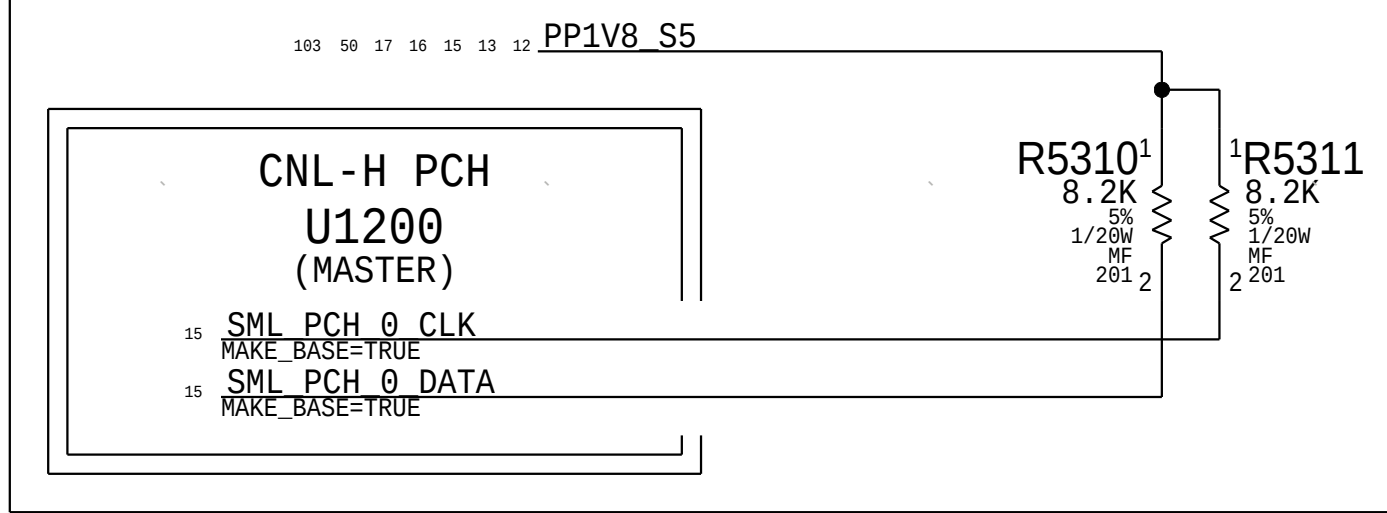
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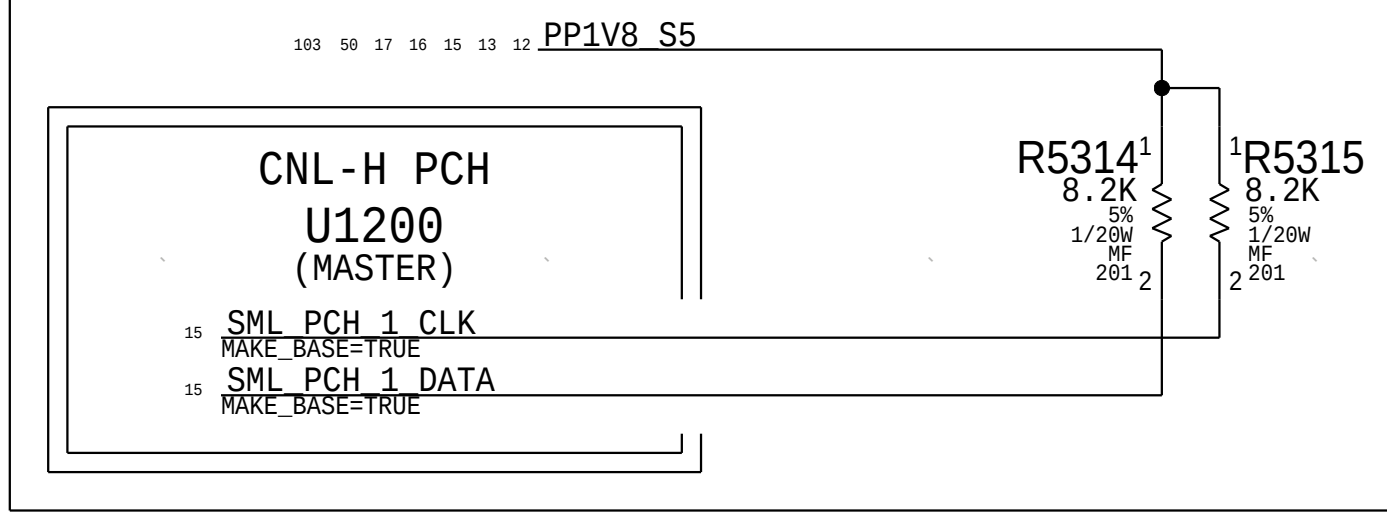
CNL-H PCH "SMBus 0" Connections



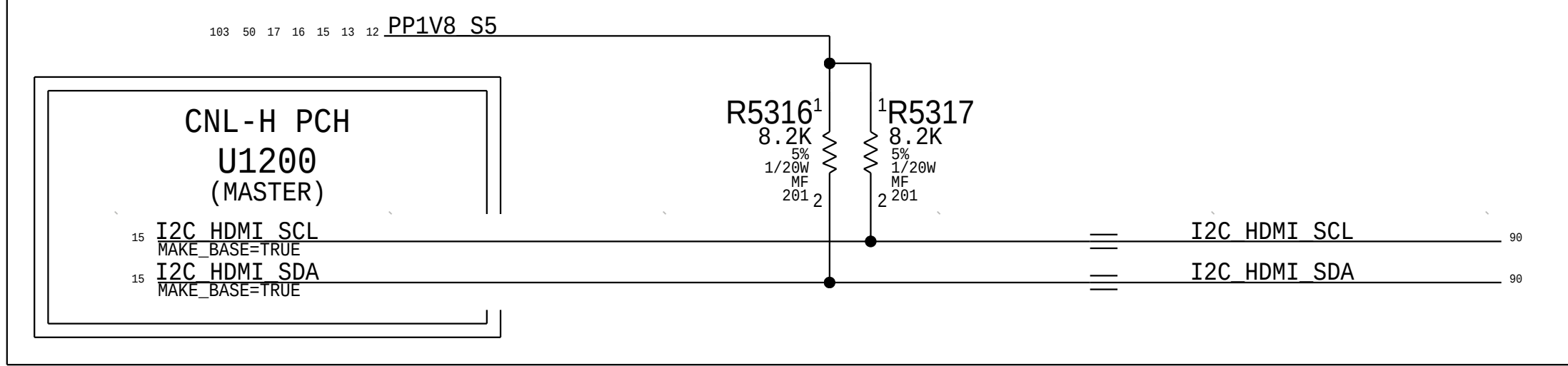
CNL-H PCH "SMLink 0"



CNL-H PCH "SMLink 1"



CNL-H PCH "I2C 0"



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
311S00179	1	IC, SLSX4402, 2BIT 0-LVL XLTR, 1.8X1.2, DFN8	U5379	CRITICAL	

PAGE TITLE		
SMBus Connections		
 Apple Inc.	DRAWING NUMBER	051-02424
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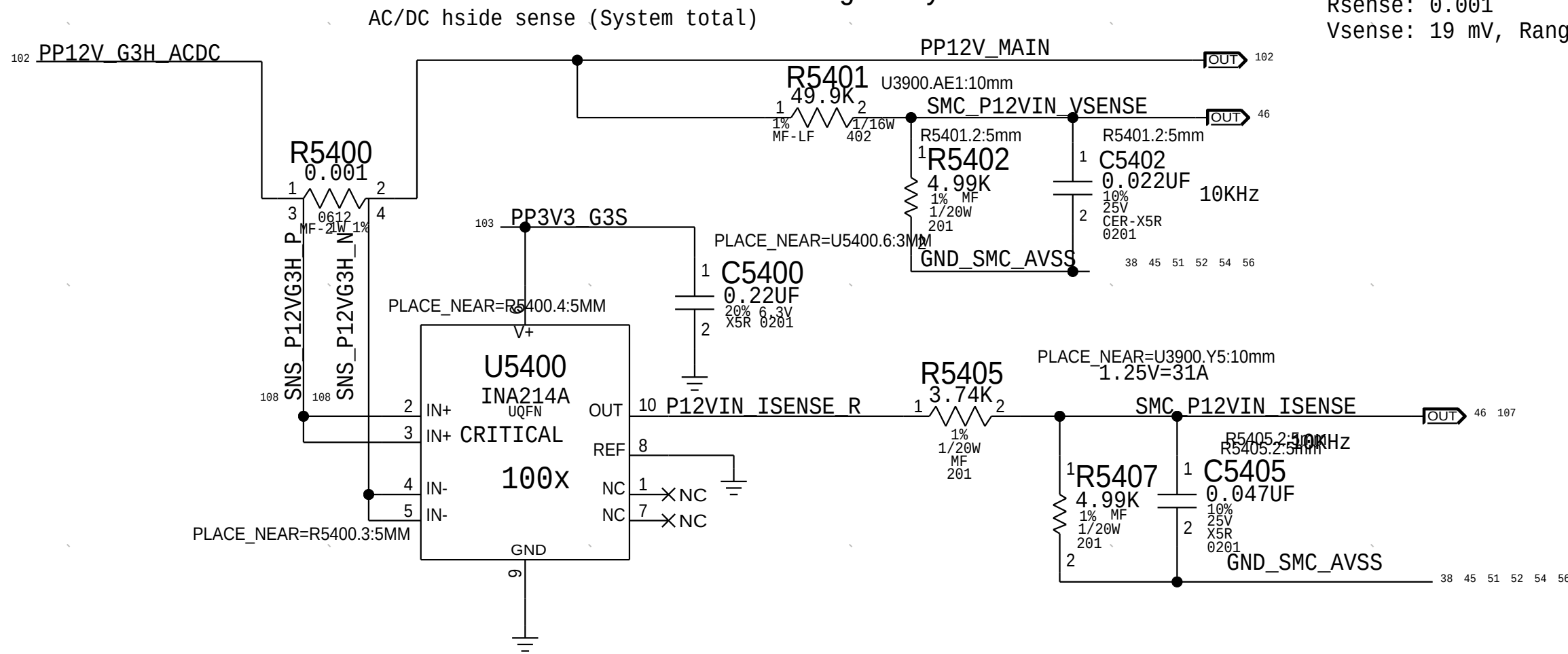
BOM_COST_GROUP=CPU & CHIPSET

J680 SENSOR SETTINGS				
CHIP	Vref(V)	Vmax	SMC sample Freq.	ADR RC filter
H9M	1.25	1.8	10khz	0.1ms
CALPE	1.5	5	10hz	1ms
EADC	2.5	5	1-2hz(10hz)	100ms

PP12V_G3H

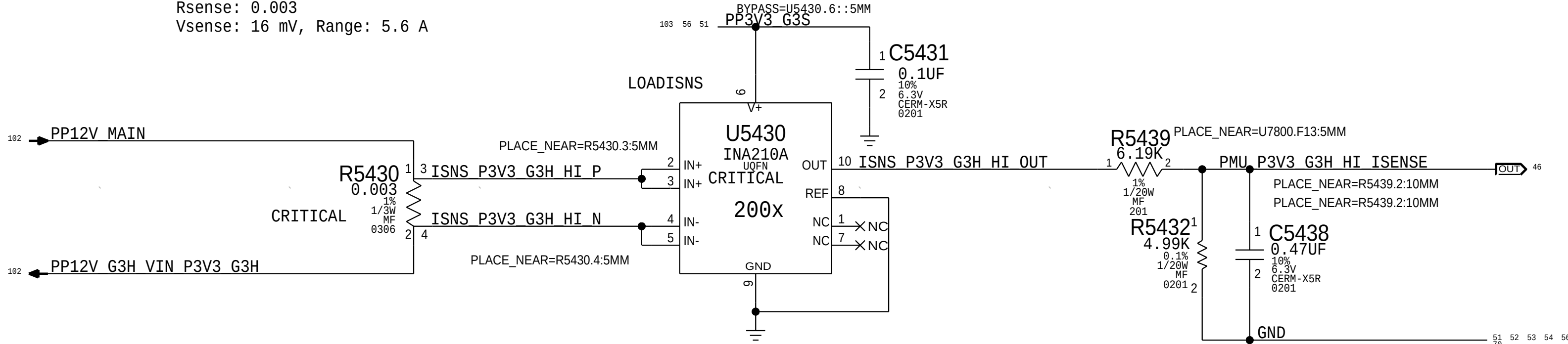
Current key:ID0M
Voltage key:VD0R

Gain: 100x, EDP: 18.75 A
Rsense: 0.001
Vsense: 19 mV, Range: 21.8 A



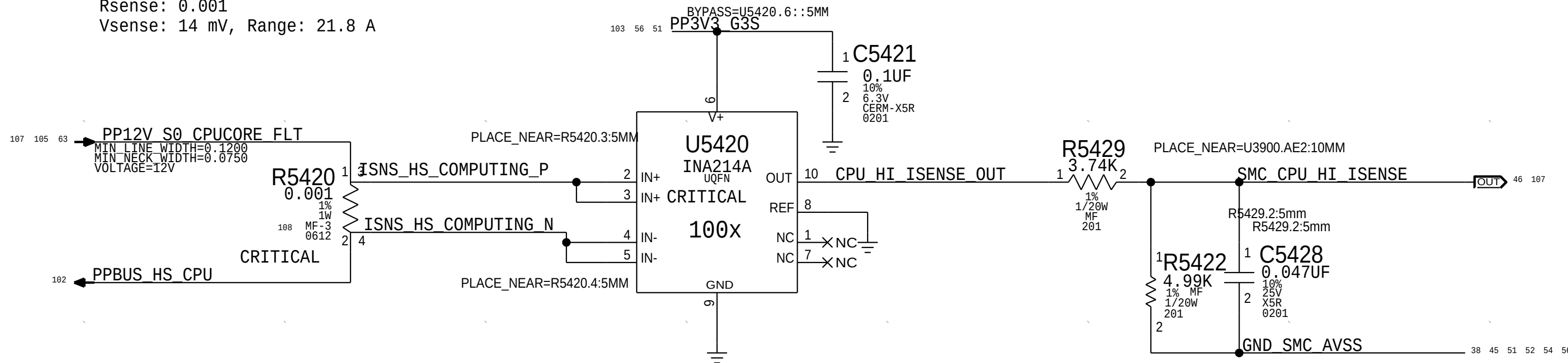
P3V3_G3H High Side Current Sense (I03R)

Gain: 200x, EDP: 5.31 A
Rsense: 0.003
Vsense: 16 mV, Range: 5.6 A



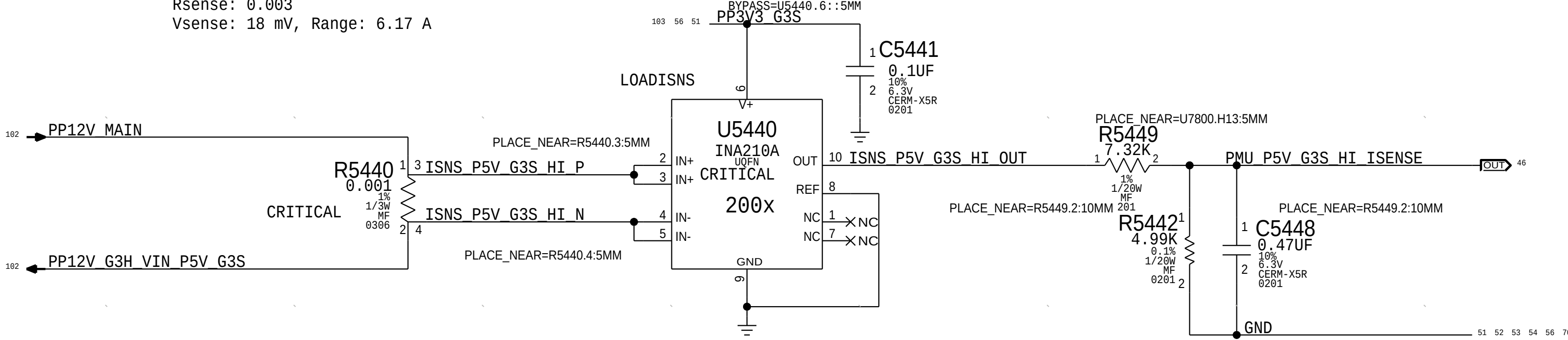
CPU High Side Currrent Sense (IC0R)

Gain: 100x, EDP: 14.41 A
Rsense: 0.001
Vsense: 14 mV, Range: 21.8 A



P5V_G3H High Side Current Sense (I05R)

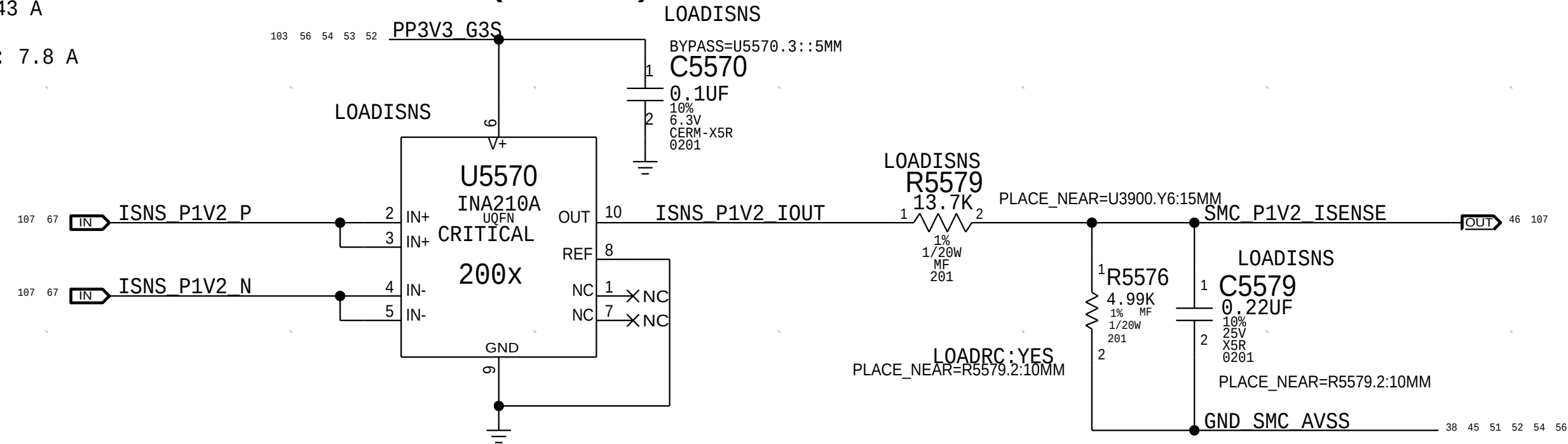
Gain: 200x, EDP: 5.89 A
Rsense: 0.003
Vsense: 18 mV, Range: 6.17 A



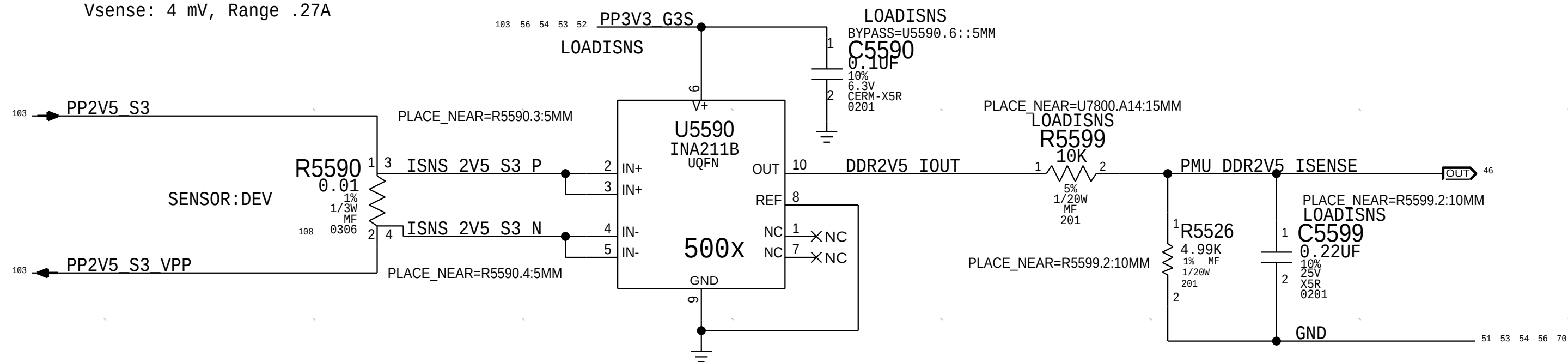
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BOM_COST_GROUP=SENSORS

Gain: 200x, EDP: 7.43 A
Rsense: 0.001
Vsense: 7 mV, Range: 7.8 A

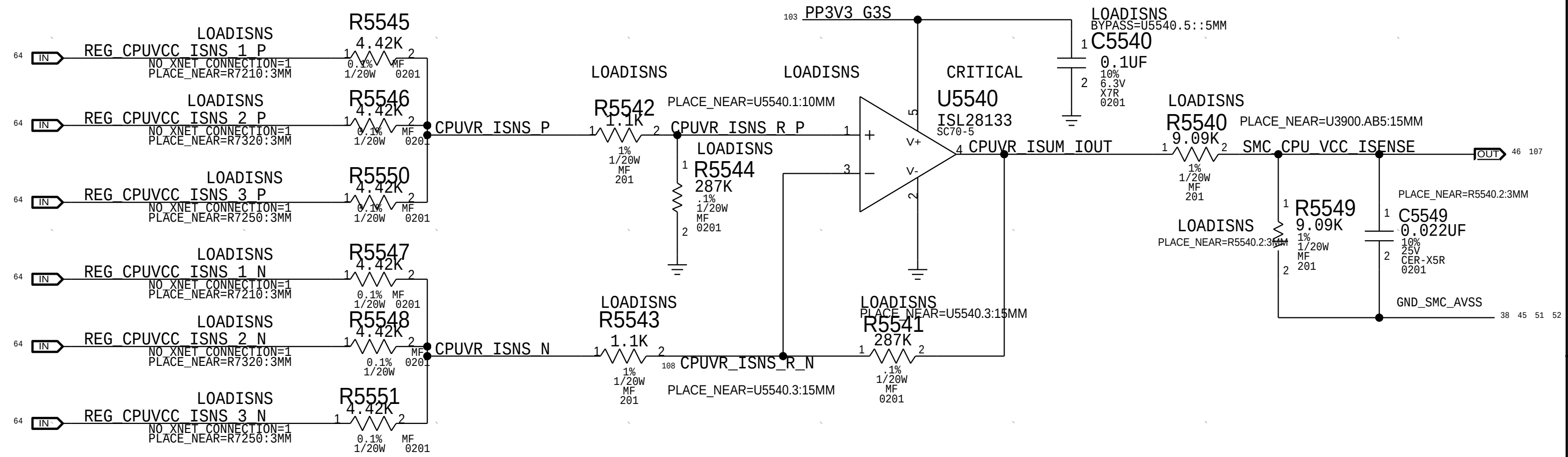


Gain: 500x, EDP: .24 A
Rsense: 0.015
Vsense: 4 mV, Range .27A



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	5	RES, MTL FLTM, 100K, 1/16W, 0201, SMD, LF	R5576, R5519, R5549, C5539, R5529		LOADRC:NO

Gain: 111.6x, EDP: 128 A
Rsense: 3x of 0.0005
Vsense: 22.4 mV, Range: 134.4 A

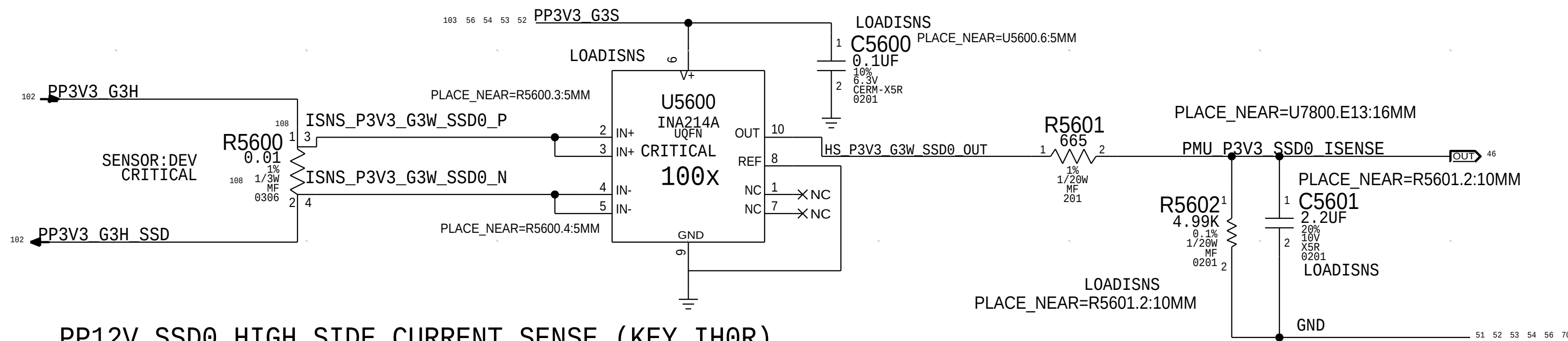


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Power Sensors Load Side			
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BOM_COST_GROUP=SENSORS

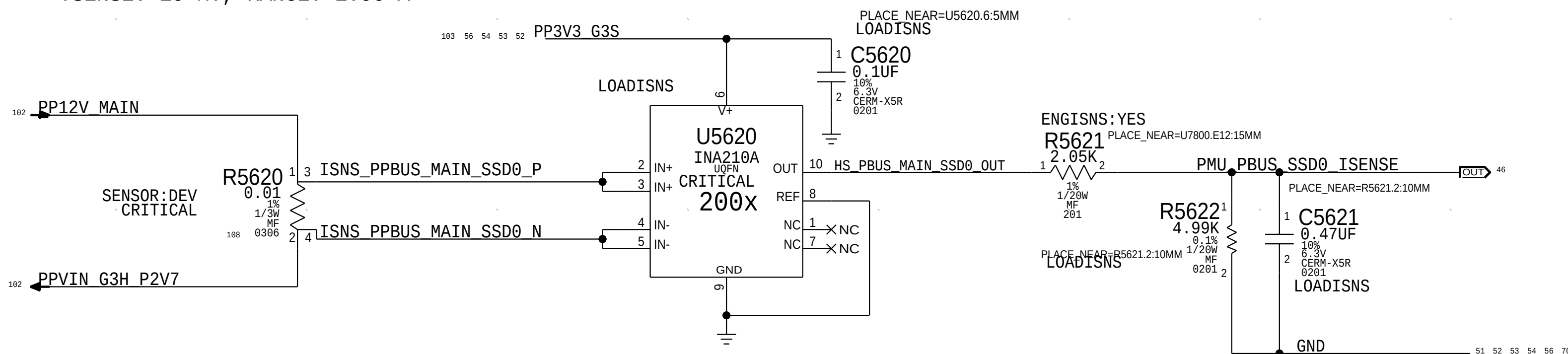
PP3V3_G3W_SSD0 HIGH SIDE CURRENT SENSE (IH0C)

GAIN: 100X, EDP: 1.6 A
RSENSE: 0.01
VSENSE: 16 MV, RANGE: 1.7 A



PP12V SSD0 HIGH SIDE CURRENT SENSE (KEY IH0R)

GAIN: 200X, EDP: 1 A
RSENSE: 0.01
VSENSE: 10 MV, RANGE: 1.06 A

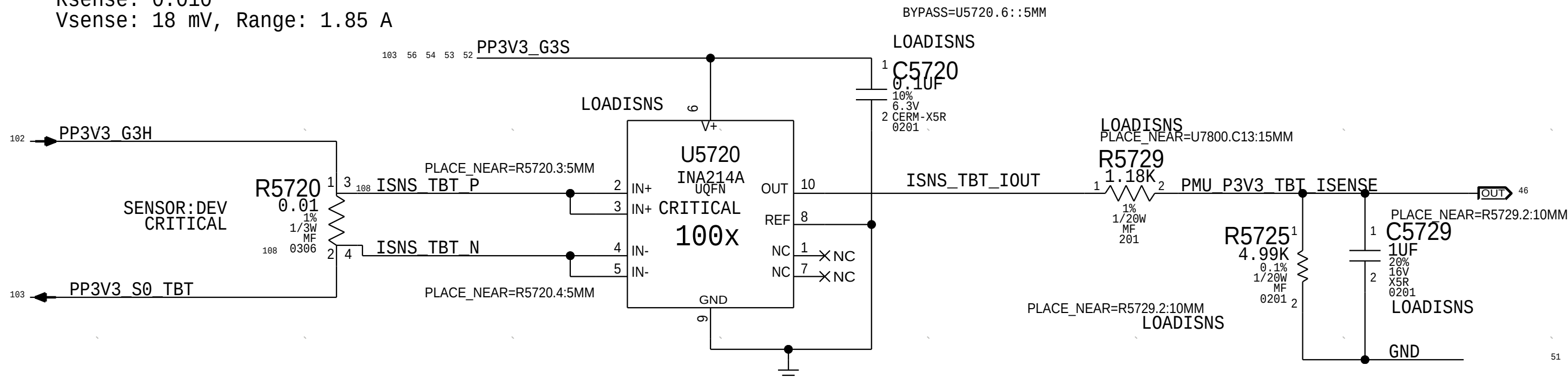


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BOM_COST_GROUP=SENSORS

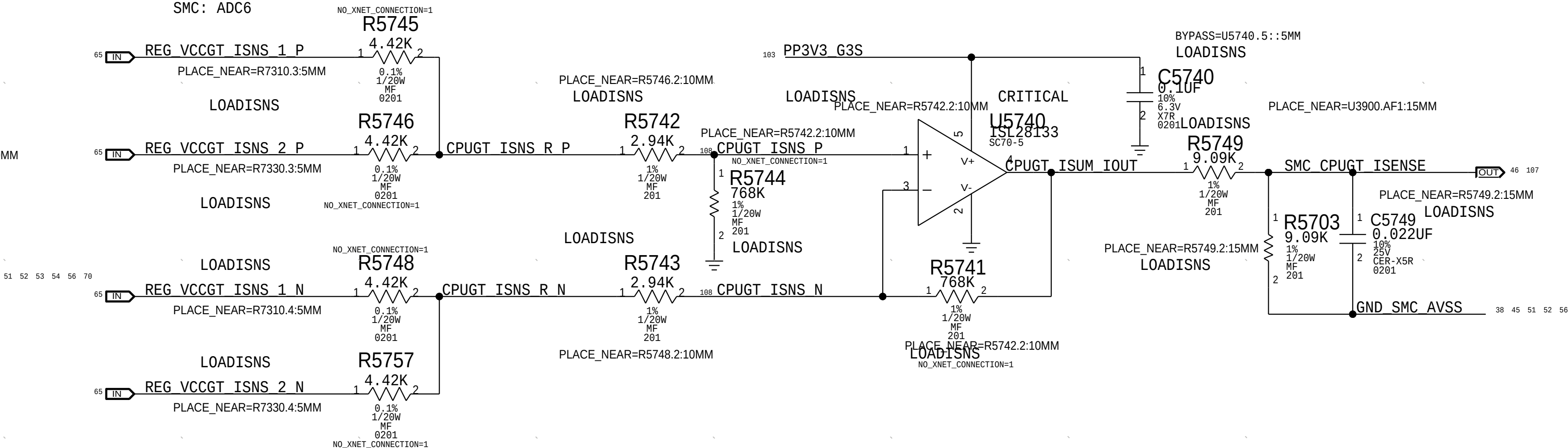
Thunderbolt TBT Current Sense (IU3C)

Gain: 100x, EDP: 1.76 A
Rsense: 0.010
Vsense: 18 mV, Range: 1.85 A



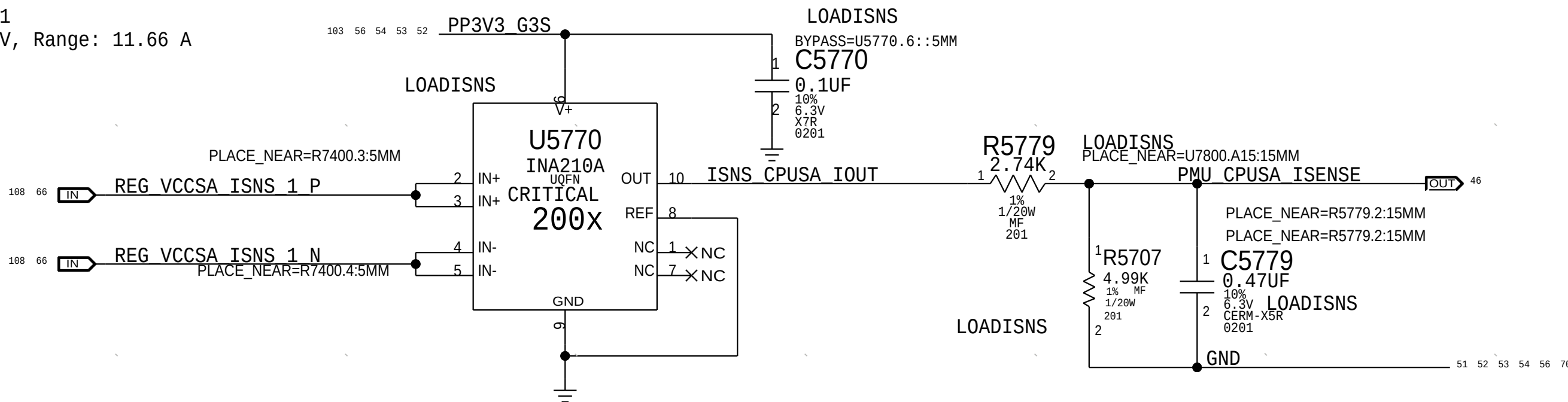
CPU GT Current Sense (ICTC)

Gain: 148.8x, EDP: 32 A
Rsense: 2x of 0.001
Vsense: 16.8 mV, Range: 33.6 A
SMC: ADC6



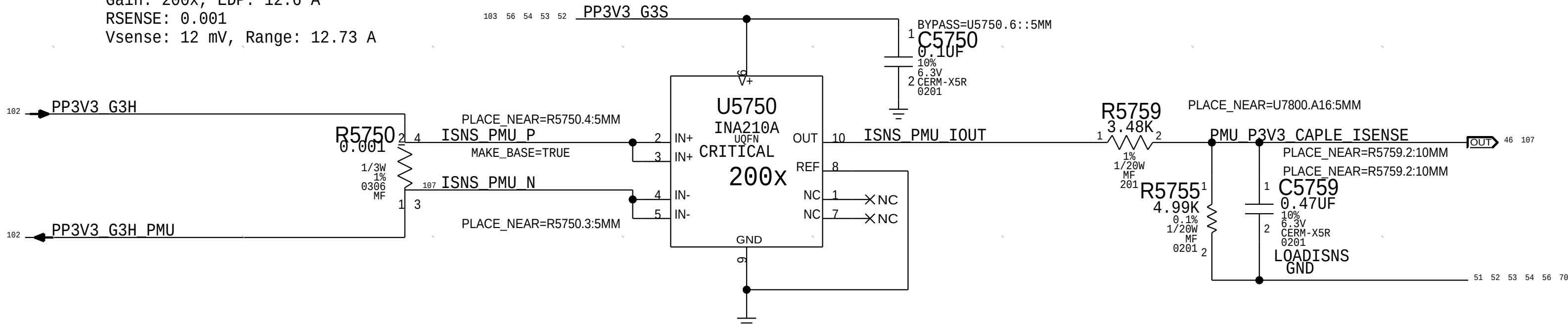
CPU SA Current Sense (ICSC)

Gain: 200x, EDP: 11.1 A
Rsense: 0.001
Vsense: 11 mV, Range: 11.66 A



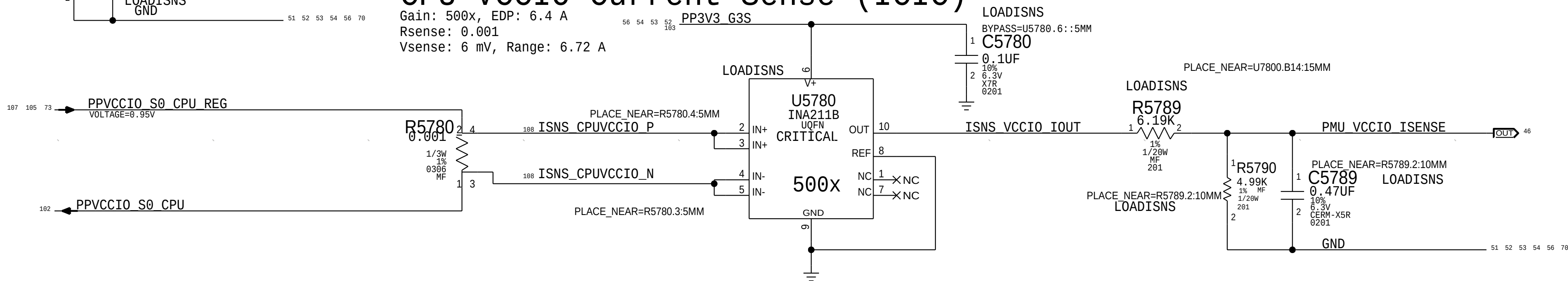
Calpe 3V3 Current Sense (ISLC)

Gain: 200x, EDP: 12.6 A
RSENSE: 0.001
Vsense: 12 mV, Range: 12.73 A



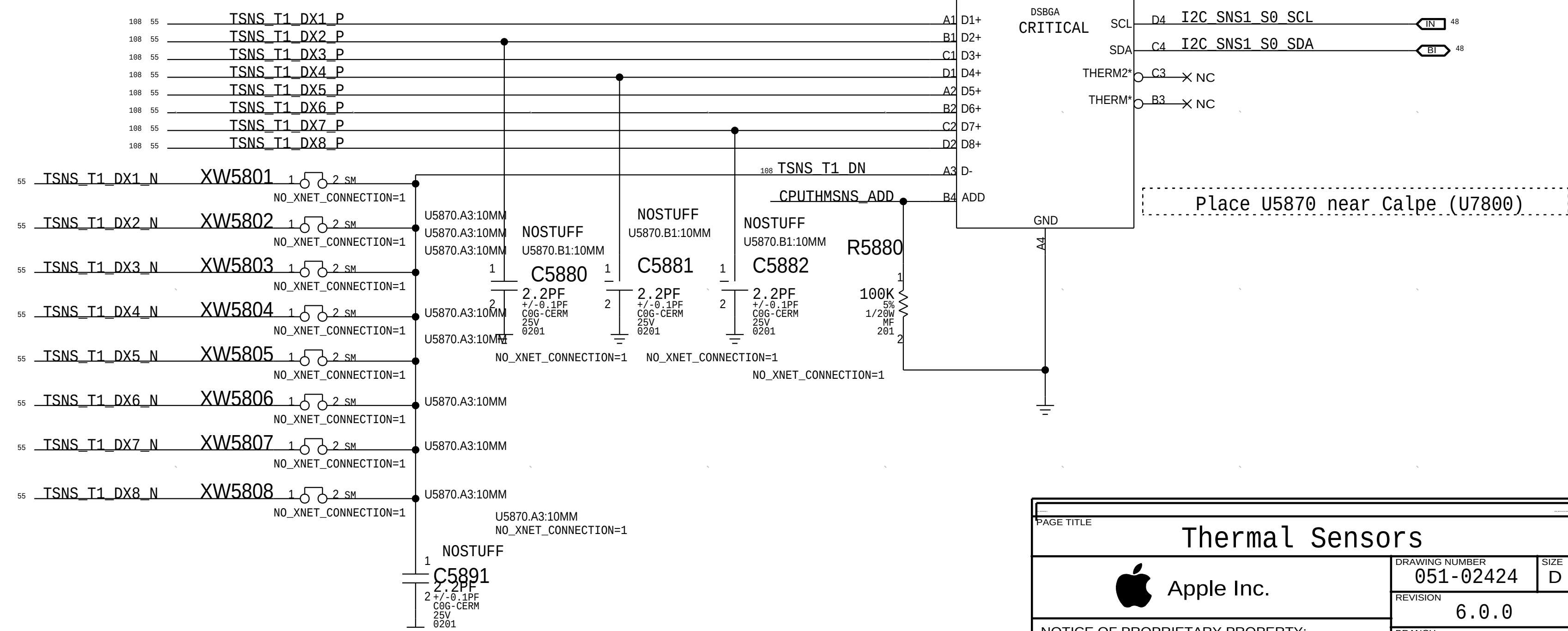
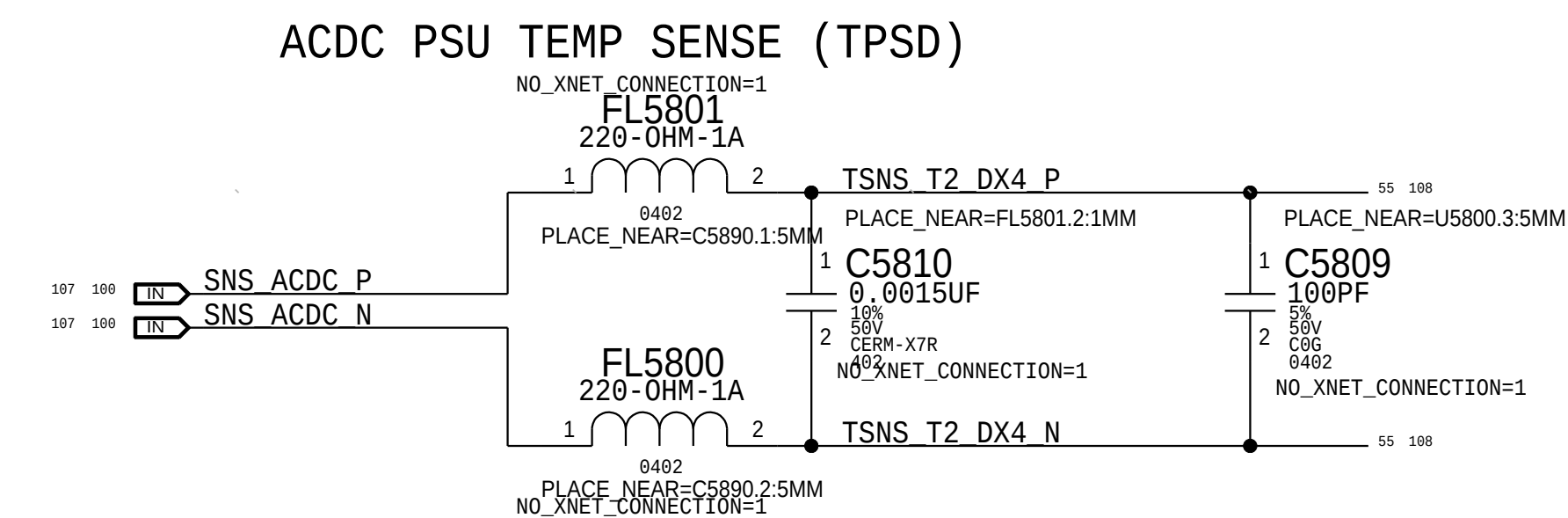
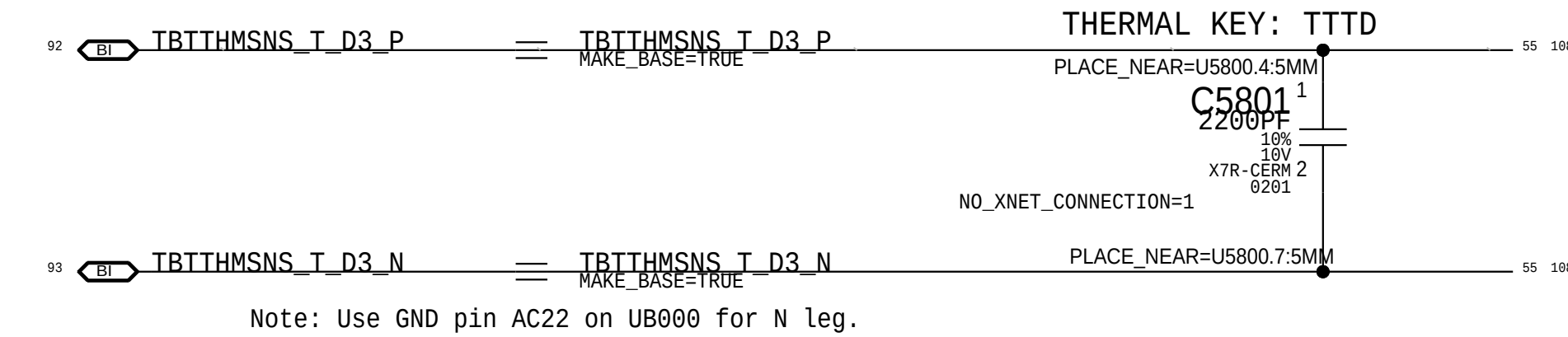
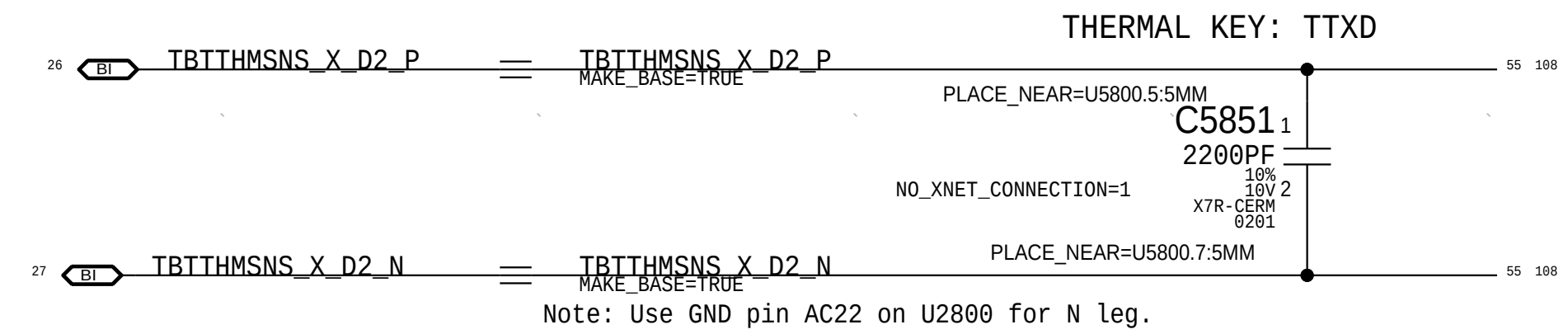
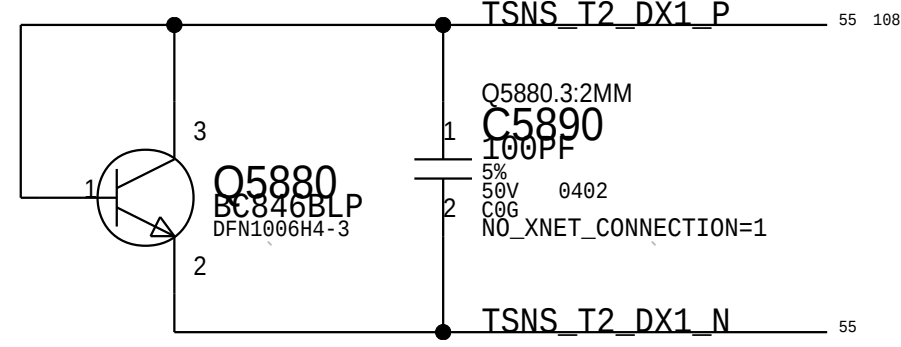
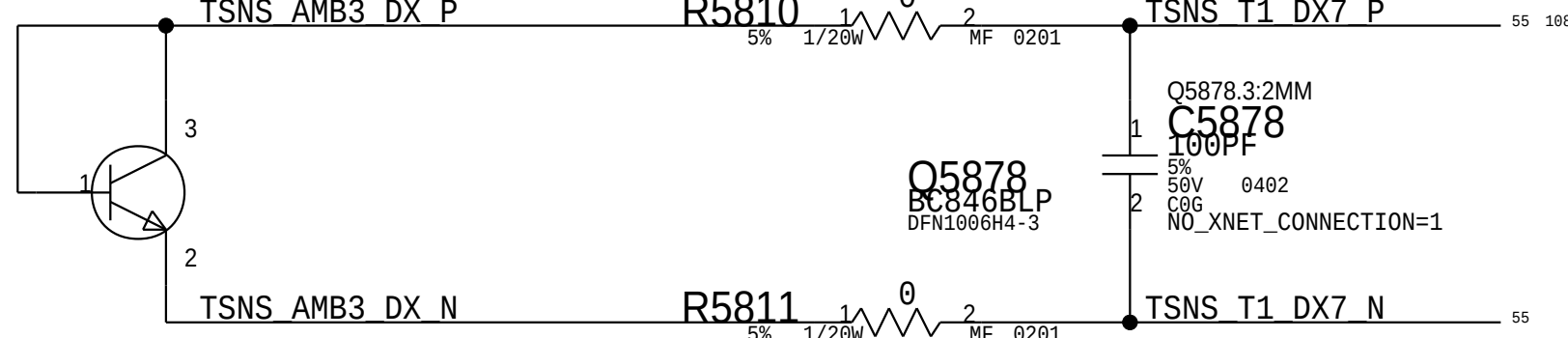
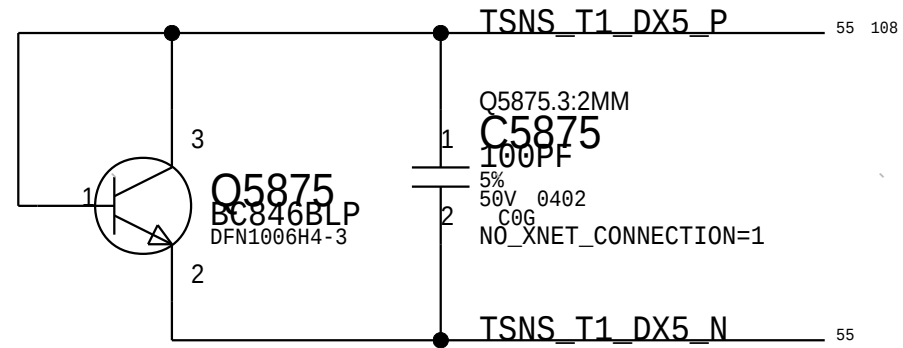
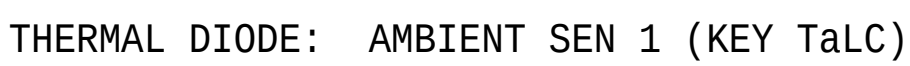
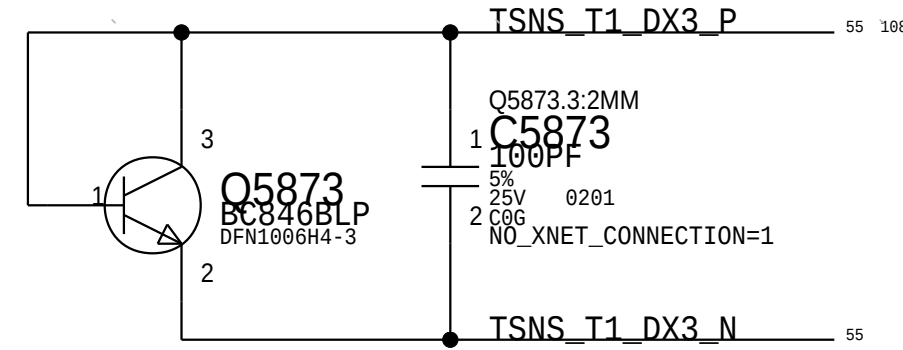
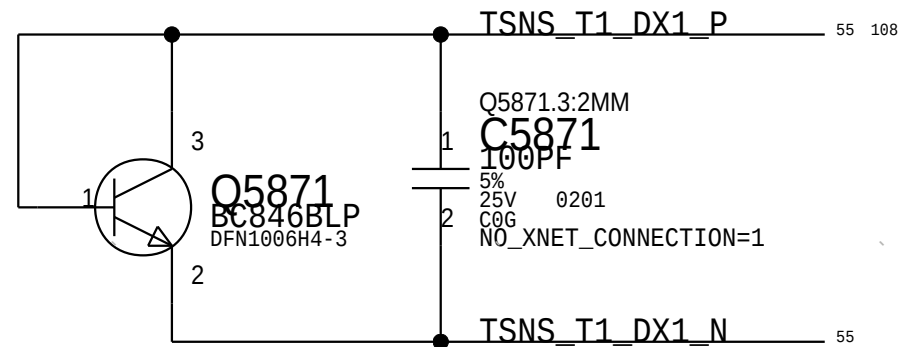
CPU VCCIO Current Sense (ICIC)

Gain: 500x, EDP: 6.4 A
Rsense: 0.001
Vsense: 6 mV, Range: 6.72 A



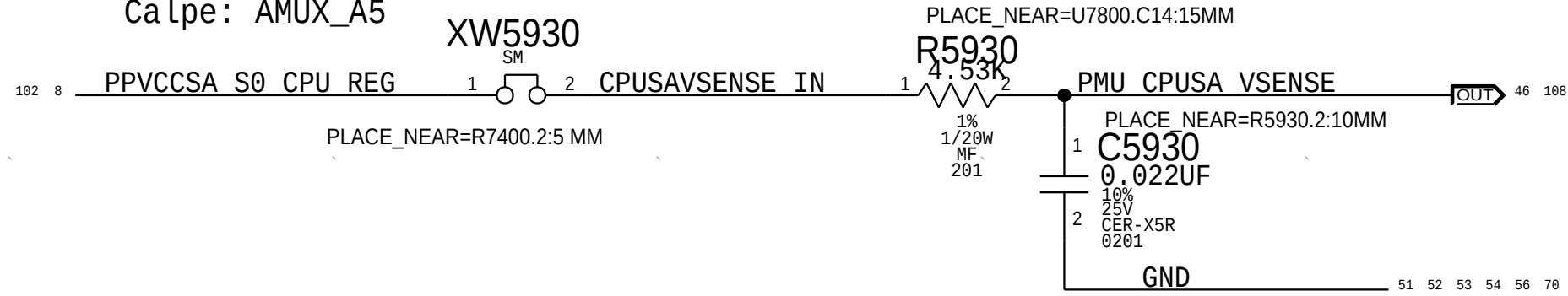
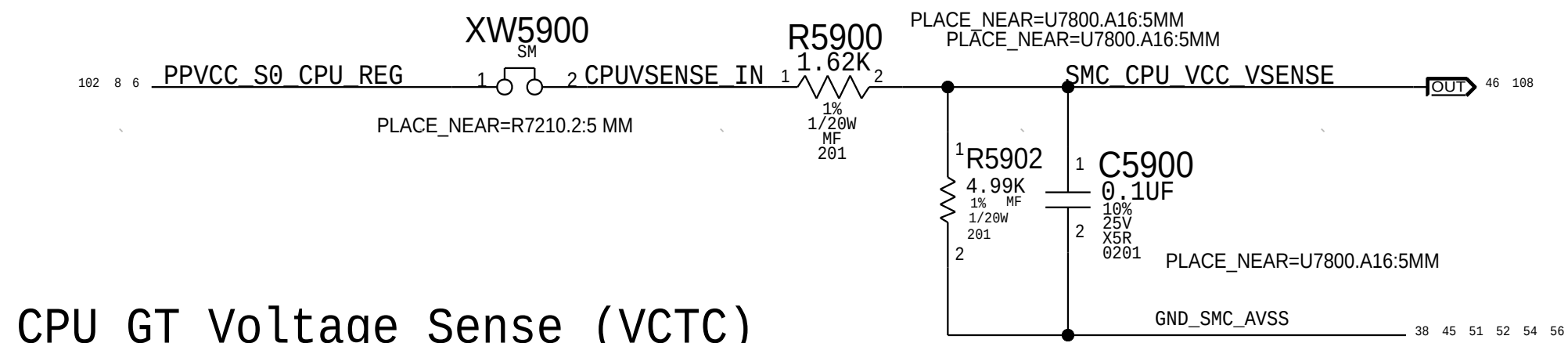
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Power Sensors Extended 2		
 Apple Inc.	DRAWING NUMBER	051-02424
	REVISION	6.0.0
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BOM_COST_GROUP=SENSORS

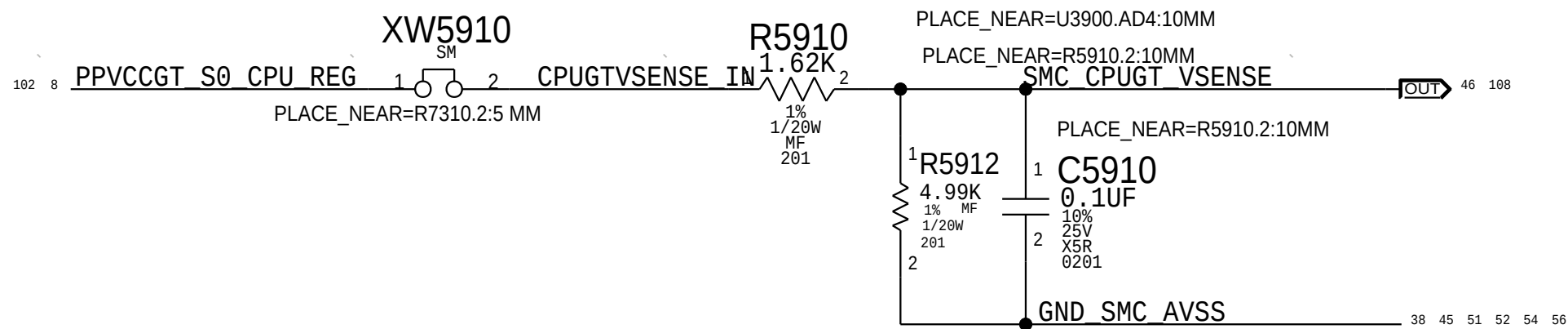


CPU Core Voltage Sense (VCAC)

CPU SA Voltage Sense (VCSC)

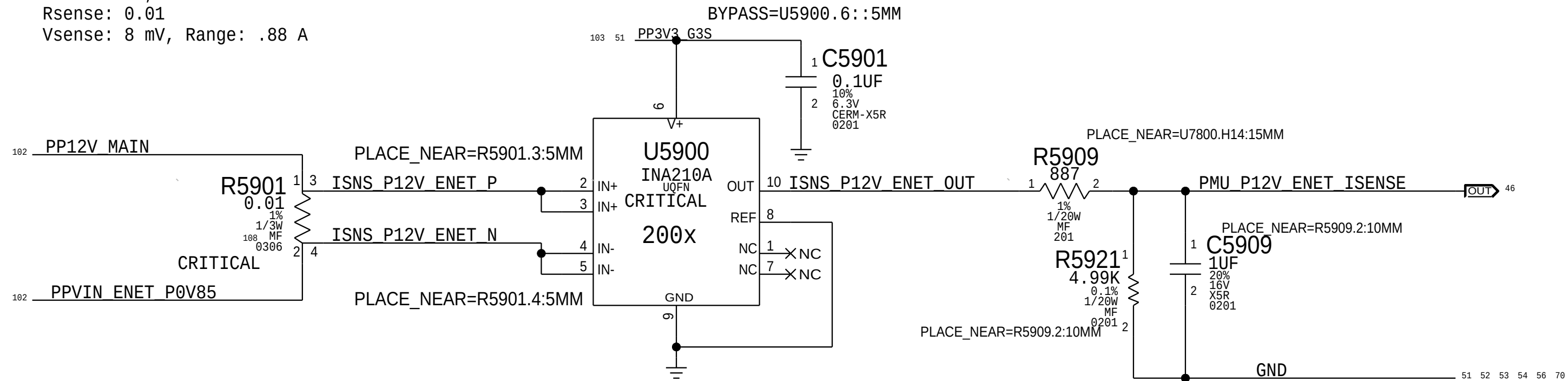


CPU GT Voltage Sense (VCTC)



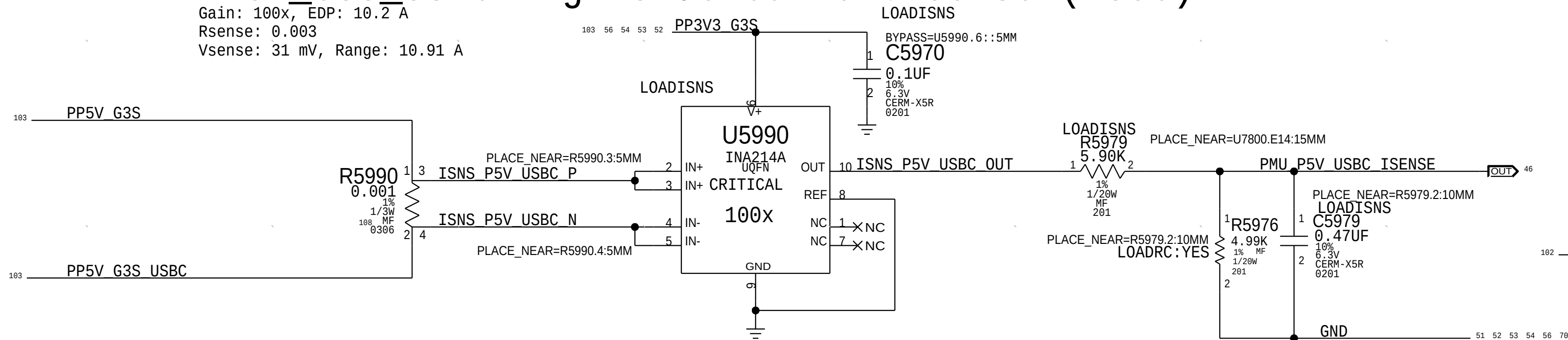
ENET 0V85 VR P12V Input Current Sense (IE2R)

Gain: 200x, EDP: .8 A
Rsense: 0.01
Vsense: 8 mV, Range: .88 A



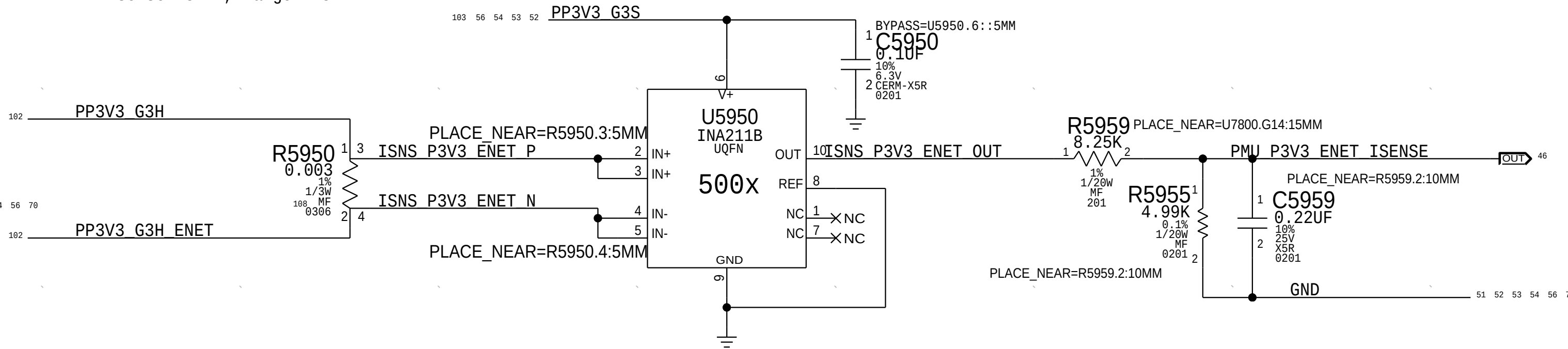
PP5V_G3S_USBC High side Current Sense (IU5C)

Gain: 100x, EDP: 10.2 A
Rsense: 0.003
Vsense: 31 mV, Range: 10.91 A



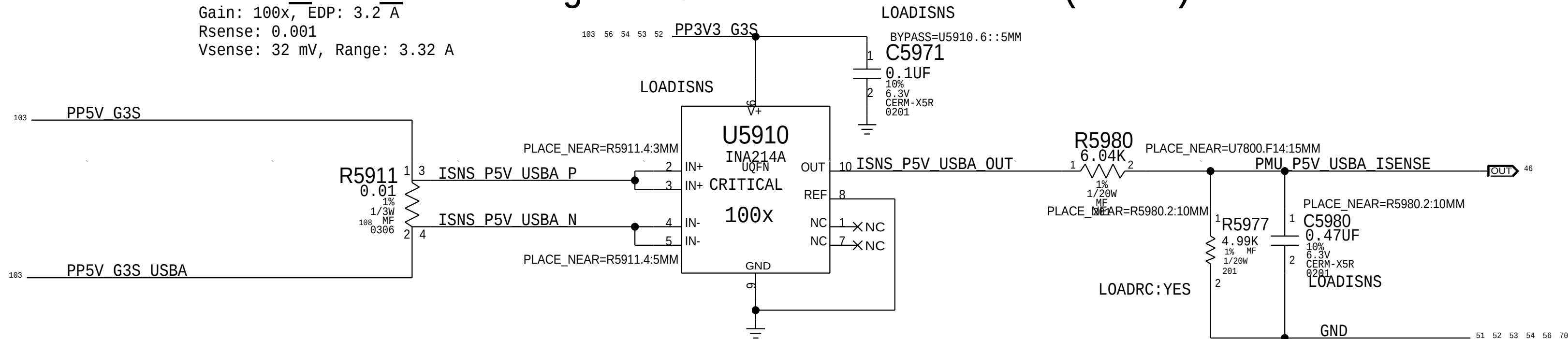
ENET 3V3 Current Sense (IE3R)

Gain: 500x, EDP: .79 A
RSENSE: 0.01
Vsense: 8 mV, Range: .8 A



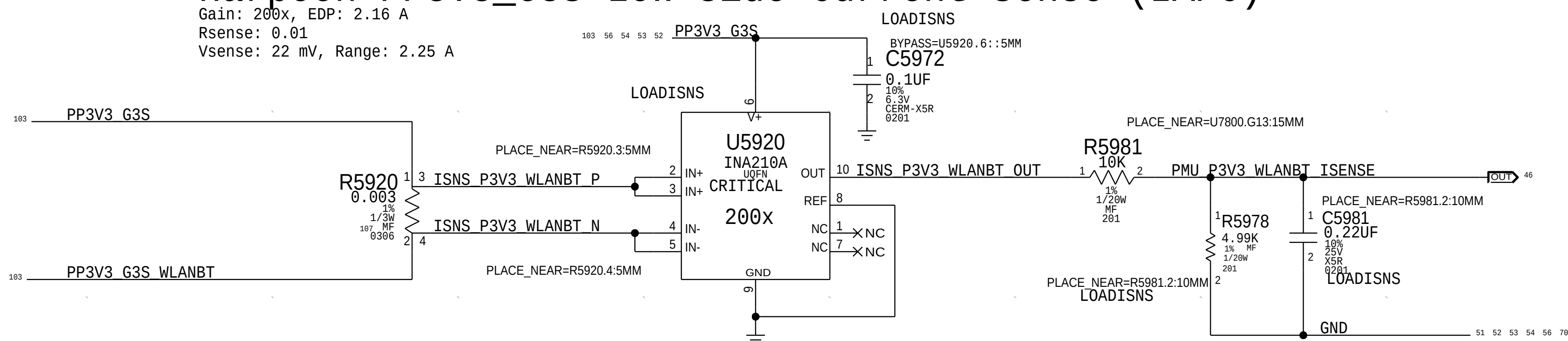
PP5V_G3S_USBA High side Current Sense (IUAC)

Gain: 100x, EDP: 3.2 A
Rsense: 0.001
Vsense: 32 mV, Range: 3.32 A



Harpoon PP3V3_G3S Low side Current Sense (IAPC)

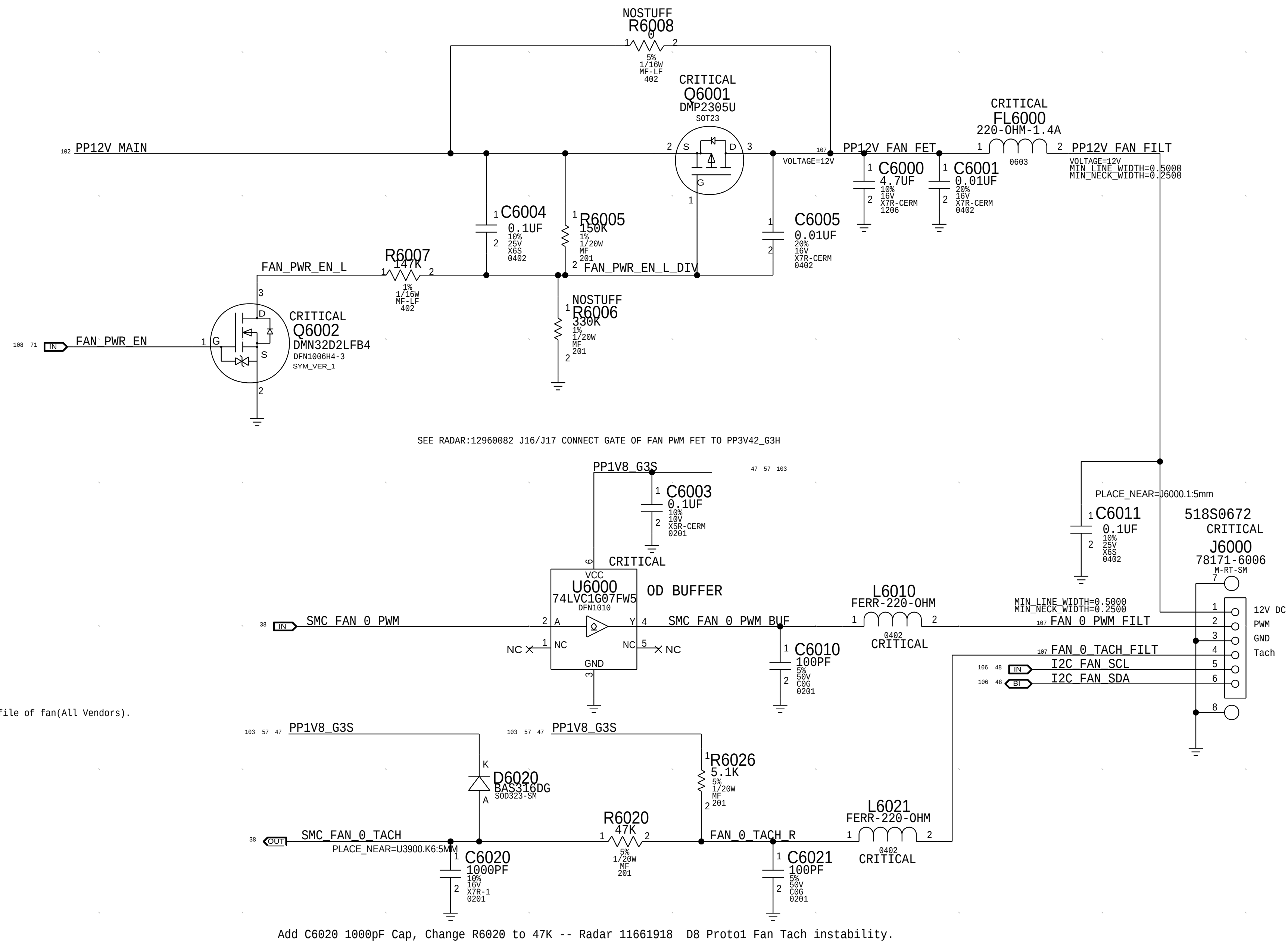
Gain: 200x, EDP: 2.16 A
Rsense: 0.01
Vsense: 22 mV, Range: 2.25 A



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More V/I Sensing			
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		BRANCH	dvt-1
		PAGE	59 OF 142
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BOM_COST_GROUP=SENSORS

SMC Fan 0 (System)



Note:

The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET. See RADAR: 10565825 - D7: Need schematic and PCB file of fan(All Vendors).

SYNC_MASTER=Mary		SYNC_DATE=64/26/2018	
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System Fan Connector			
 Apple Inc.	DRAWING NUMBER		SIZE
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8		7		6		5		4		3		2		1			
D																D	
C																C	
B																B	
A																A	
8		7		6		5		4		3		2		1			

SYNC_MASTER=Mary

SYNC_DATE=84/26/2018

PAGE TITLE

SPI Debug Connector

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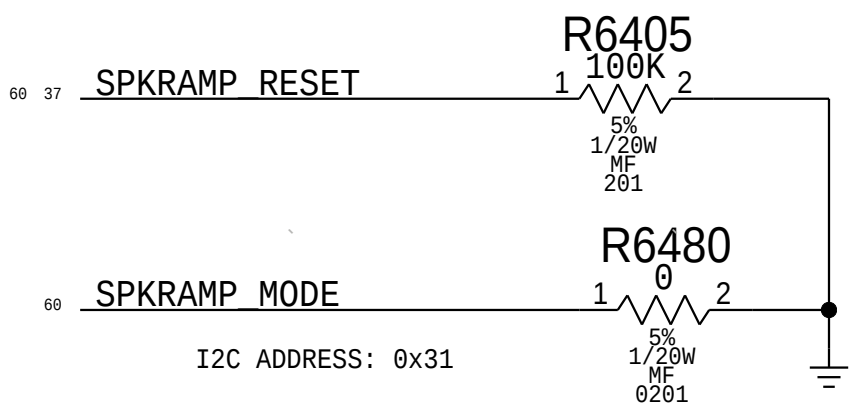
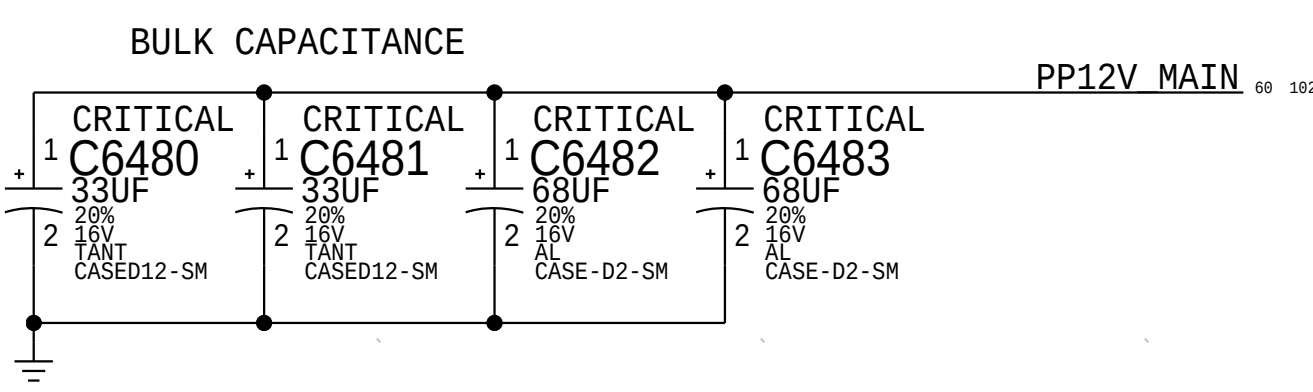
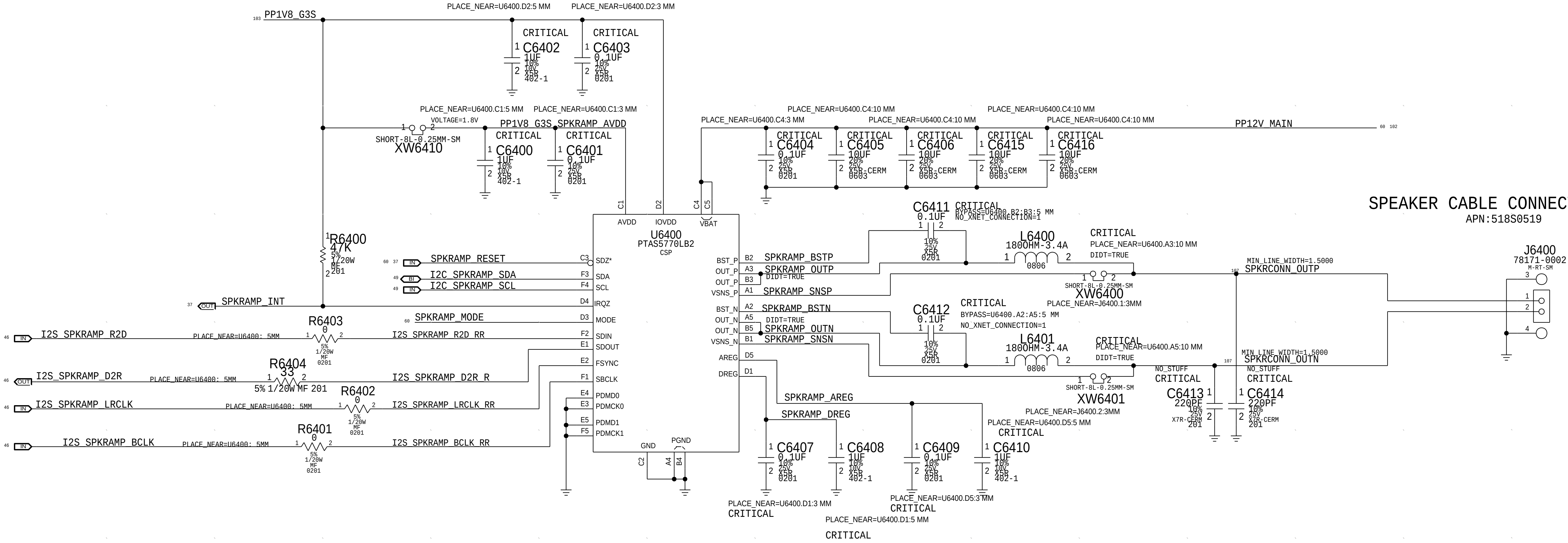
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BOM_COST_GROUP=DEBUG

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		SHEET	
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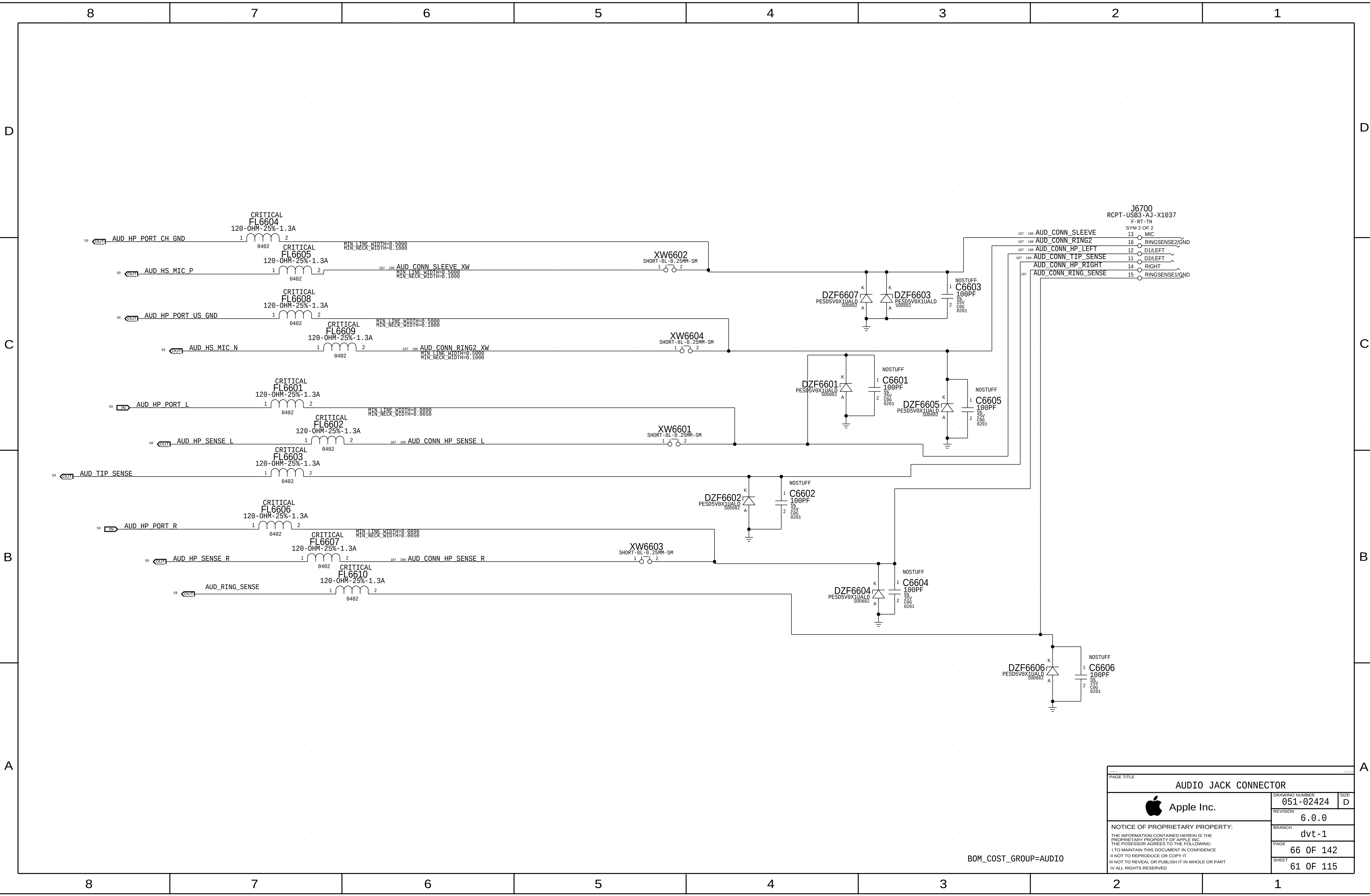
1x MONO SPEAKER AMPLIFIER (TAS5770)
APN: 353S01629



SPEAKER CABLE CONNECTOR
APN:518S0519

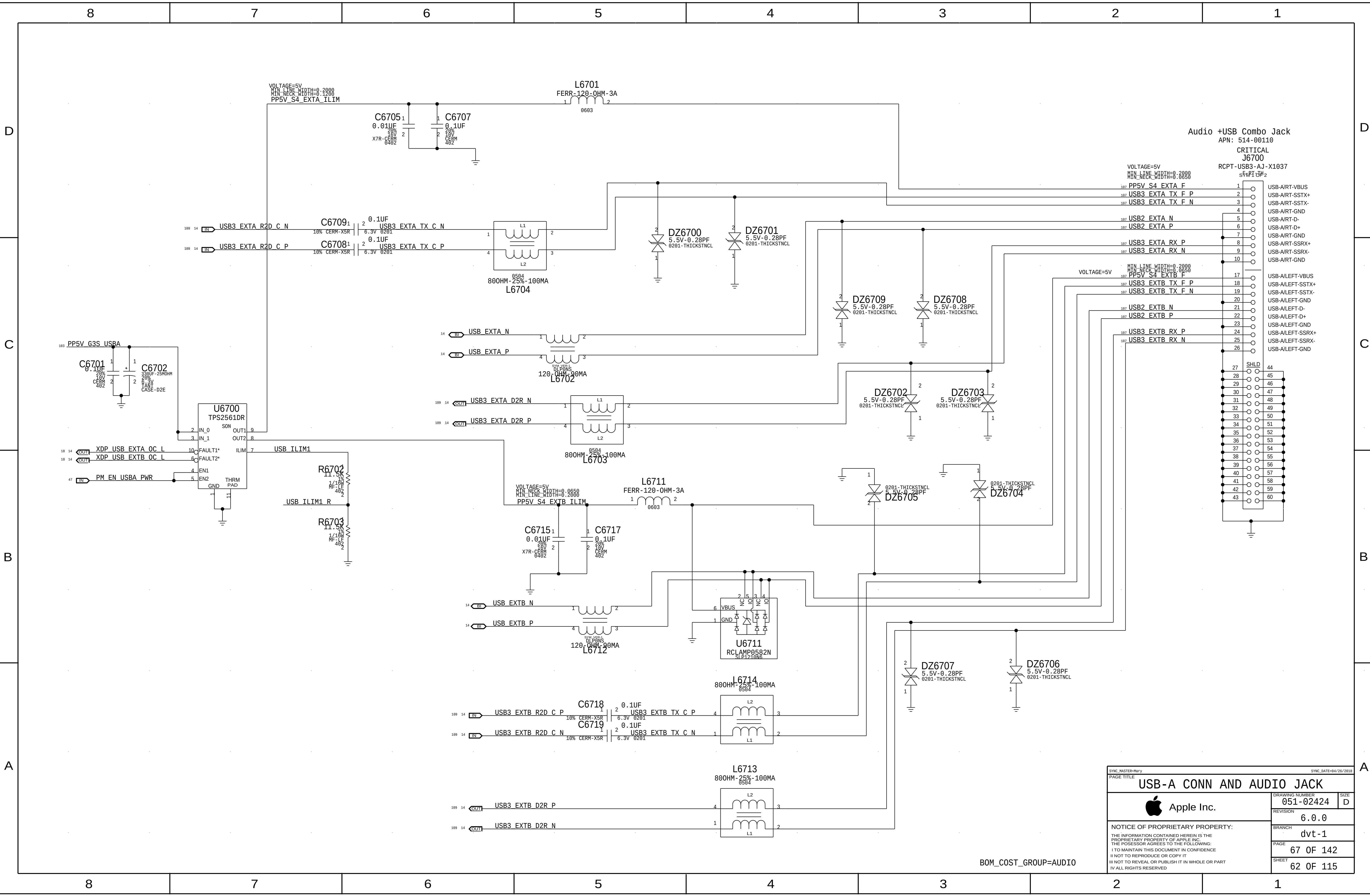
AUDIO SKPR AMP		
	DRAWING NUMBER	051-02424
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BOM_COST_GROUP=AUDIO



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AUDIO JACK CONNECTOR		
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	PAGE	66 OF 142
	SHEET	61 OF 115

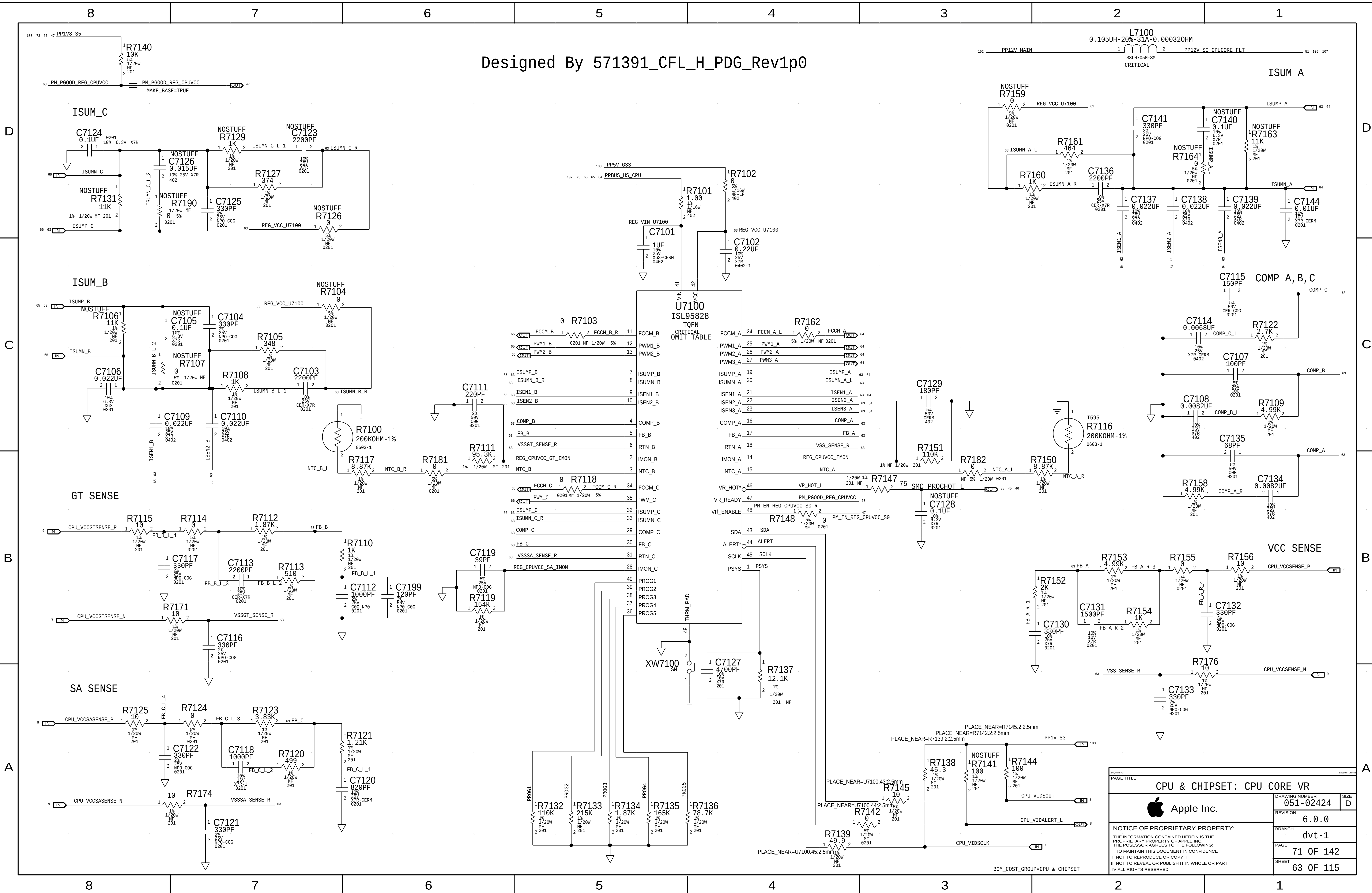
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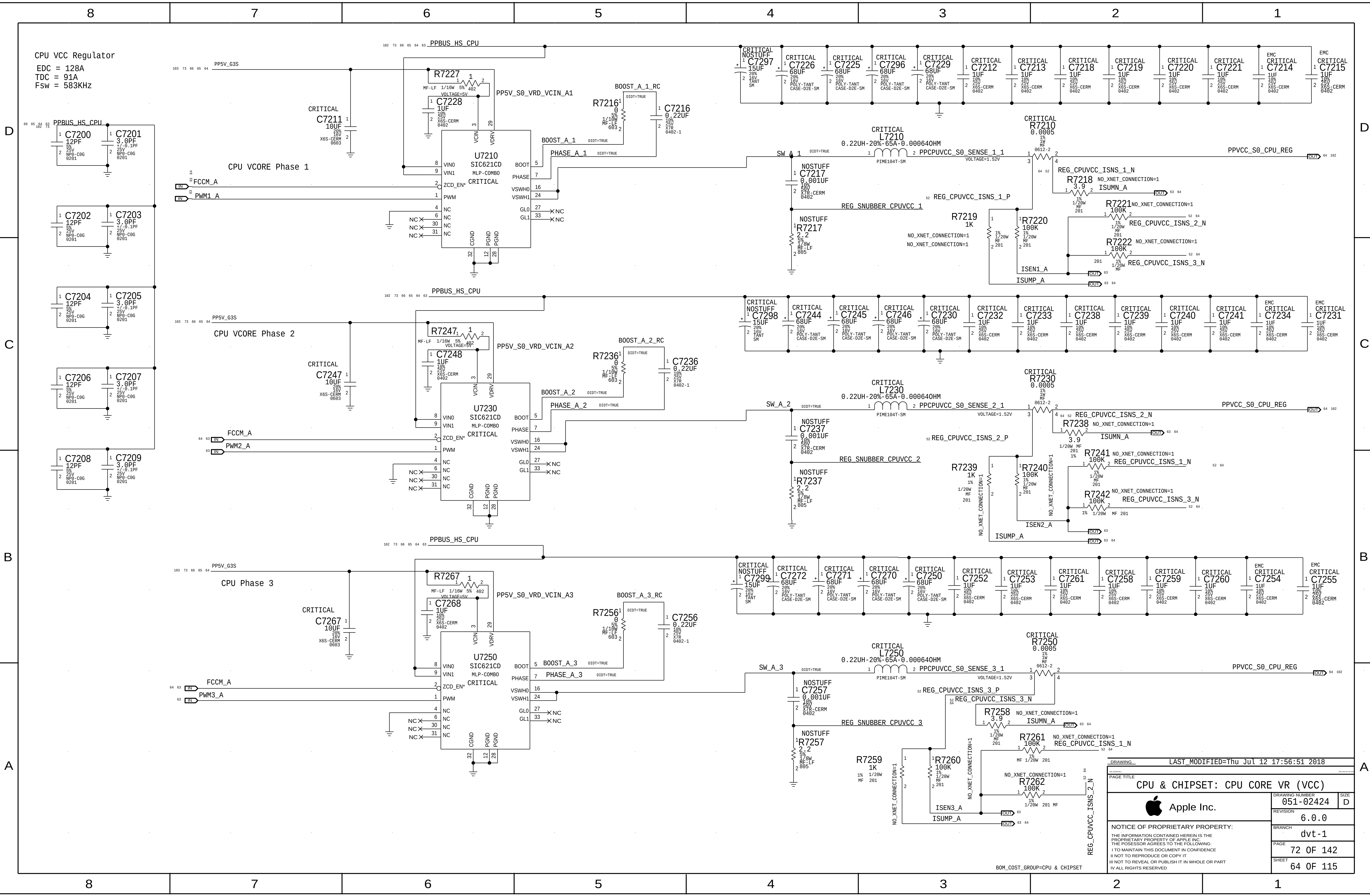


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USB-A CONN AND AUDIO JACK			051-02424		
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I TO NOT REVEAL OR PUBLISH IT IN WHOLE OR PART			62 OF 115		

BOM_COST_GROUP=AUDIO10

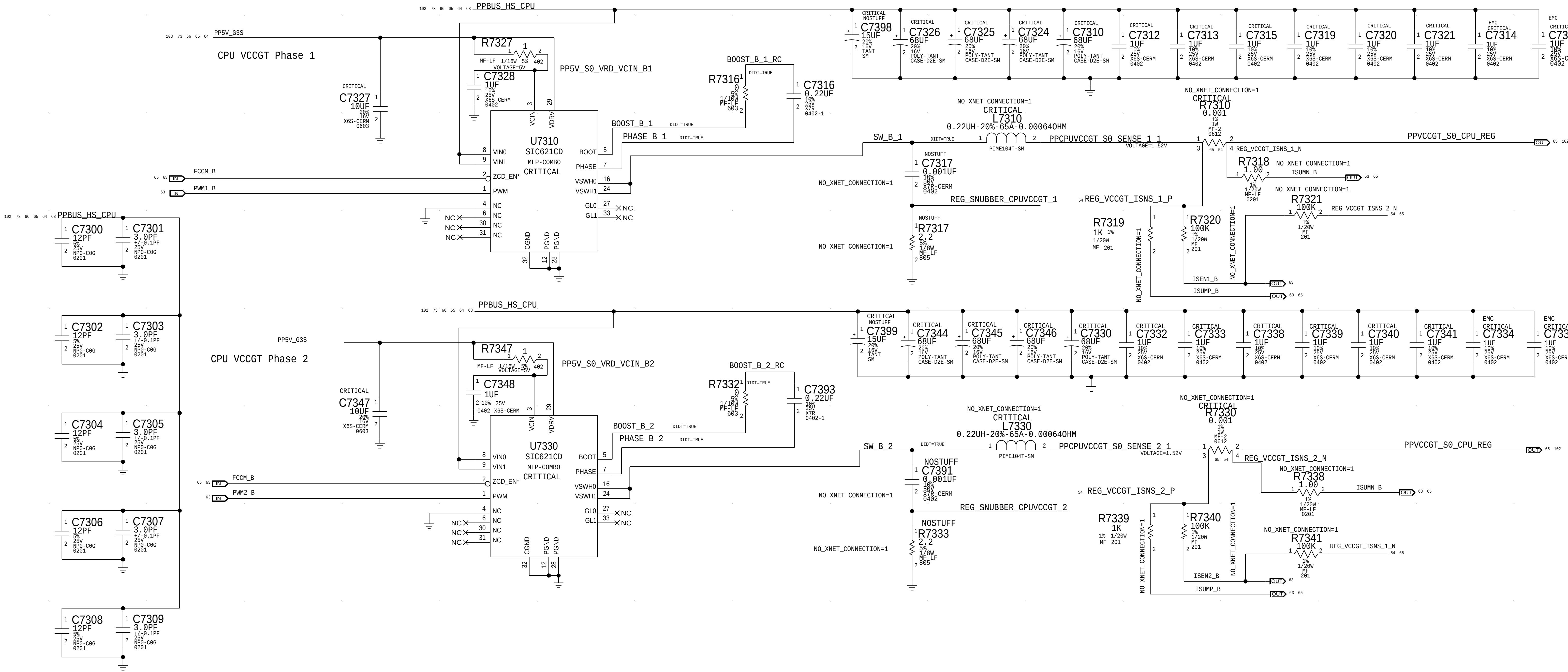
Designed By 571391_CFL_H_PDG_Rev1p0





CPU VCCGT Regulator

EDC = 32A
TDC = 25A
Fsw = 583KHz

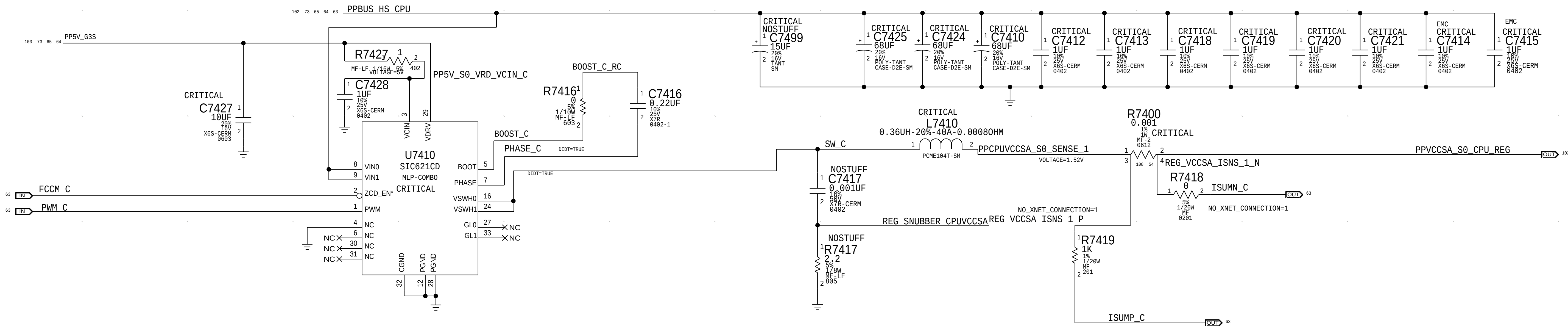


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PAGE TITLE			CPU & CHIPSET: CPU CORE VR (VCCGT)		
			DRAWING NUMBER	051-02424	SIZE
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BOM_COST_GROUP=CPU & CHIPSET

CPU VCCSA Regulator

EDC = 11.1A
TDC = 10A
FSW = 583KHz



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CPU & CHIPSET: CPU CORE VR (VCCSA)		
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BOM_COST_GROUP=CPU & CHIPSET

D

C

B

A



C

B

A

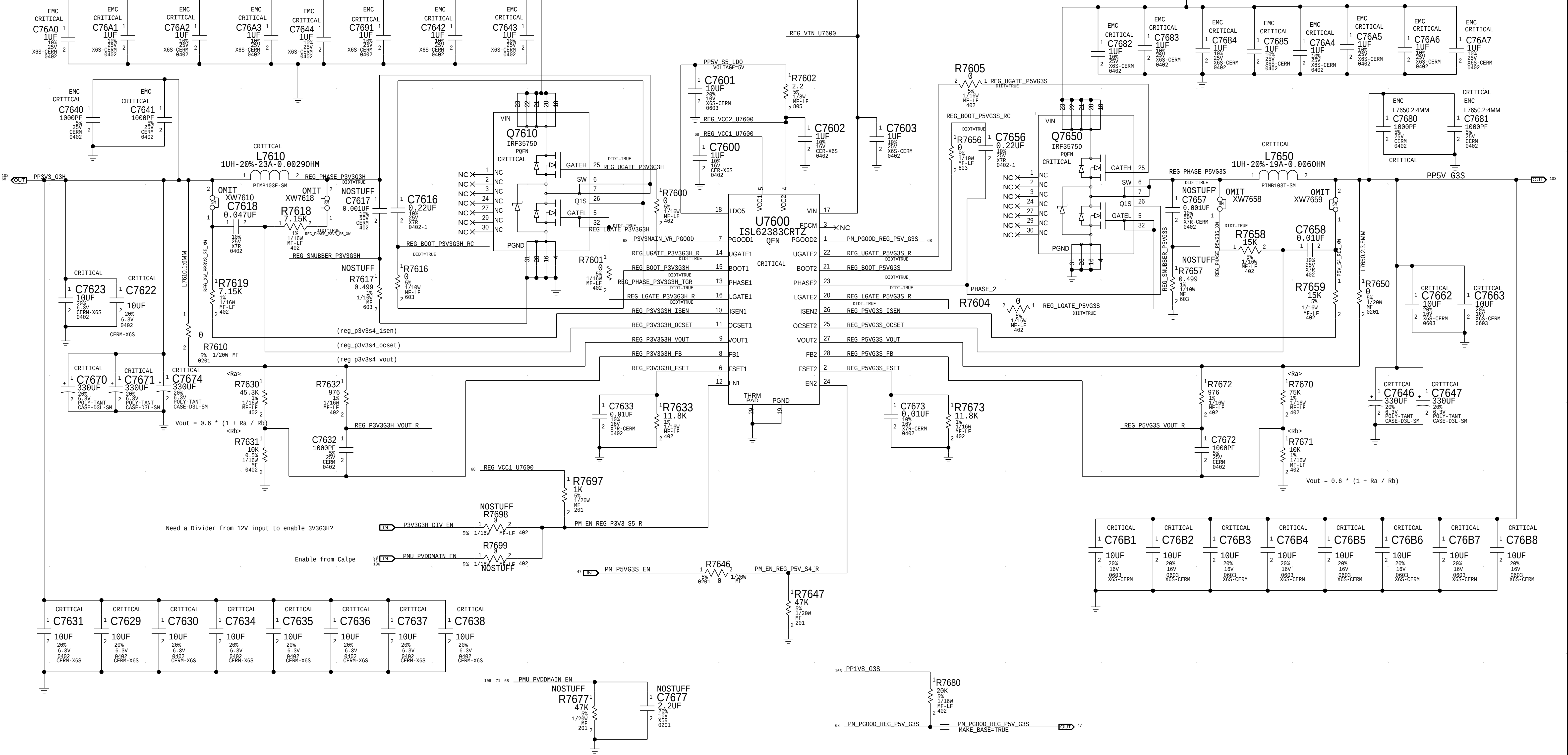
BOM_COST_GROUP=DRAM

3.3V G3H Regulator

EDC = 16A
TDC = 13.6A
Fsw = 500KHZ

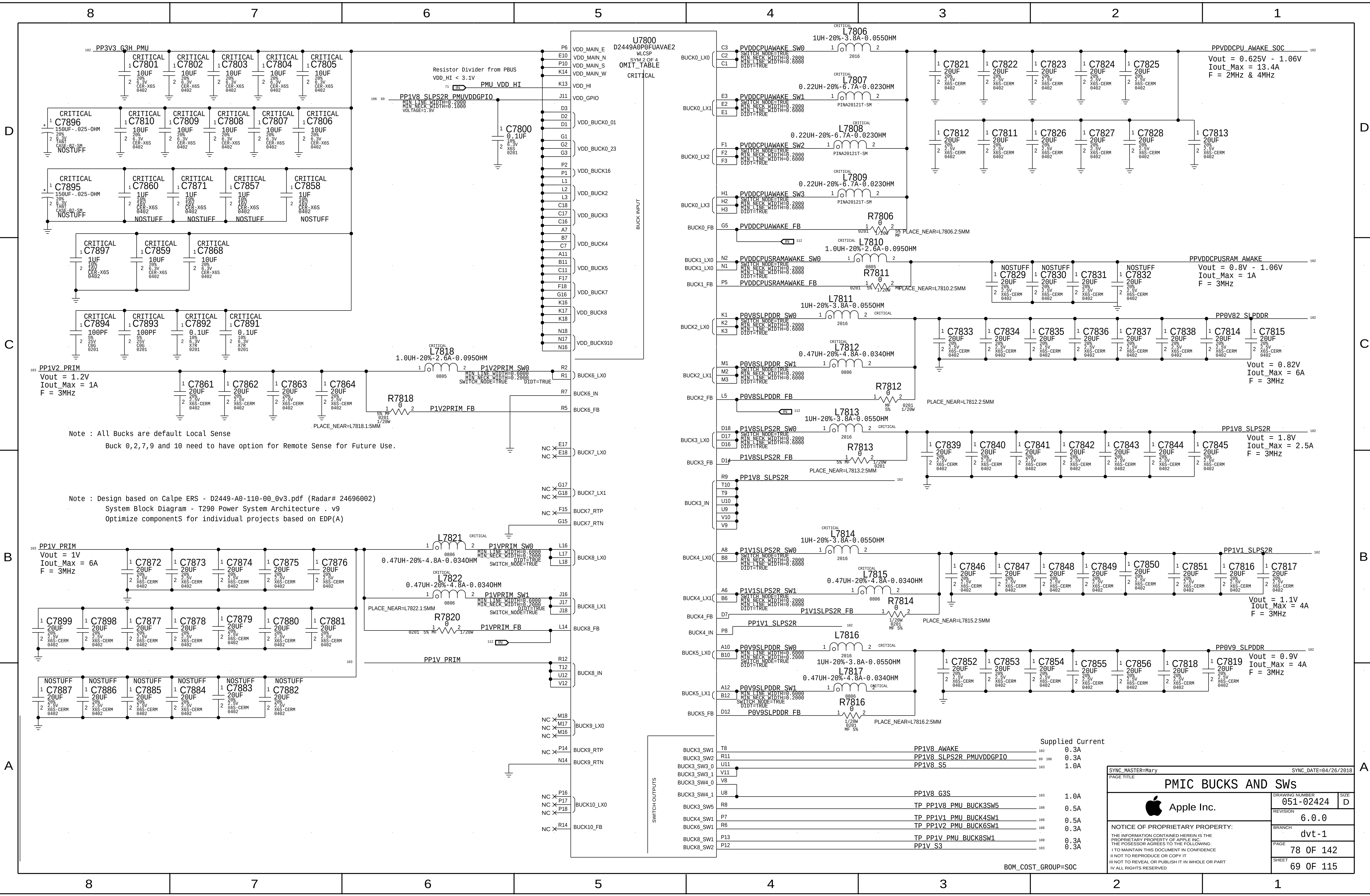
5V G3S Regulator

EDC = 13.5A
TDC = 13.5A
Fsw = 500KHZ



PAGE TITLE		
PLATFORM POWER: 3.3V G3H/5V G3S VR		
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BOM_COST_GROUP=PLATFORM POWER



Note : All Bucks are default Local Sense
Buck 0,2,7,9 and 10 need to have option for Remote Sense for Future Use.

Note : Design based on Calpe ERS - D2449-A0-110-00_0v3.pdf (Radar# 24696002)
System Block Diagram - T290 Power System Architecture . v9
Optimize components for individual projects based on EDP(A)

Supplied Current	
BUCK3_SW1	PP1V8 AWAKE 0.3A
BUCK3_SW2	PP1V8 SLPS2R PMUVDDGPIO 0.3A
BUCK3_SW3_0	PP1V8 S5 1.0A
BUCK3_SW3_1	
BUCK3_SW4_0	
BUCK3_SW4_1	PP1V8 G3S 1.0A
BUCK3_SW5	TP PP1V8 PMU BUCK3SW5 0.5A
BUCK4_SW1	TP PP1V1 PMU BUCK4SW1 0.5A
BUCK6_SW1	TP PP1V2 PMU BUCK6SW1 0.3A
BUCK8_SW1	TP PP1V PMU BUCK8SW1 0.3A
BUCK8_SW2	PP1V S3 0.3A

BOM_COST_GROUP=SOC

SYNC_MASTER=Mary

SYNC_DATE=04/26/2018

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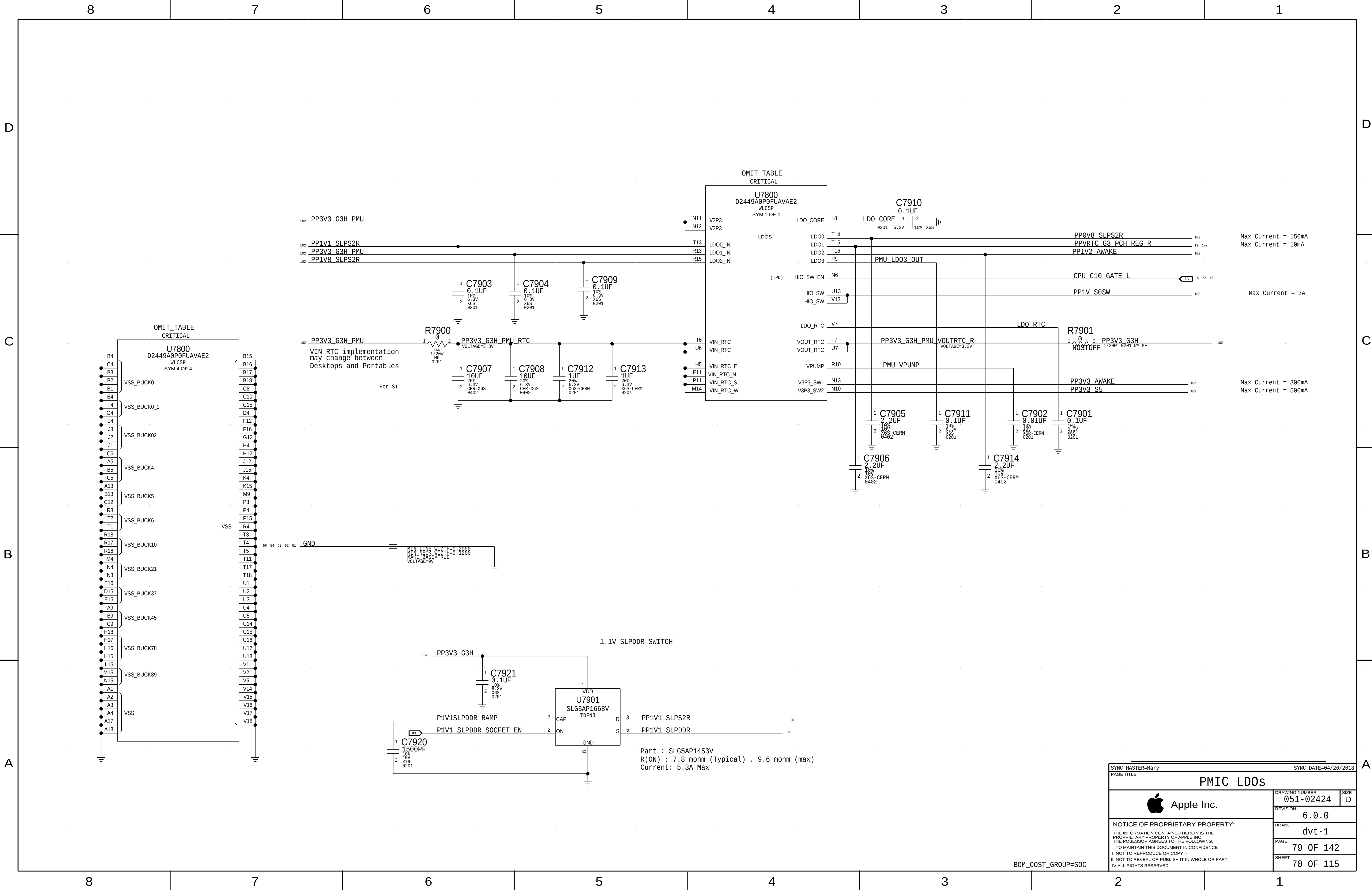
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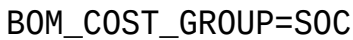
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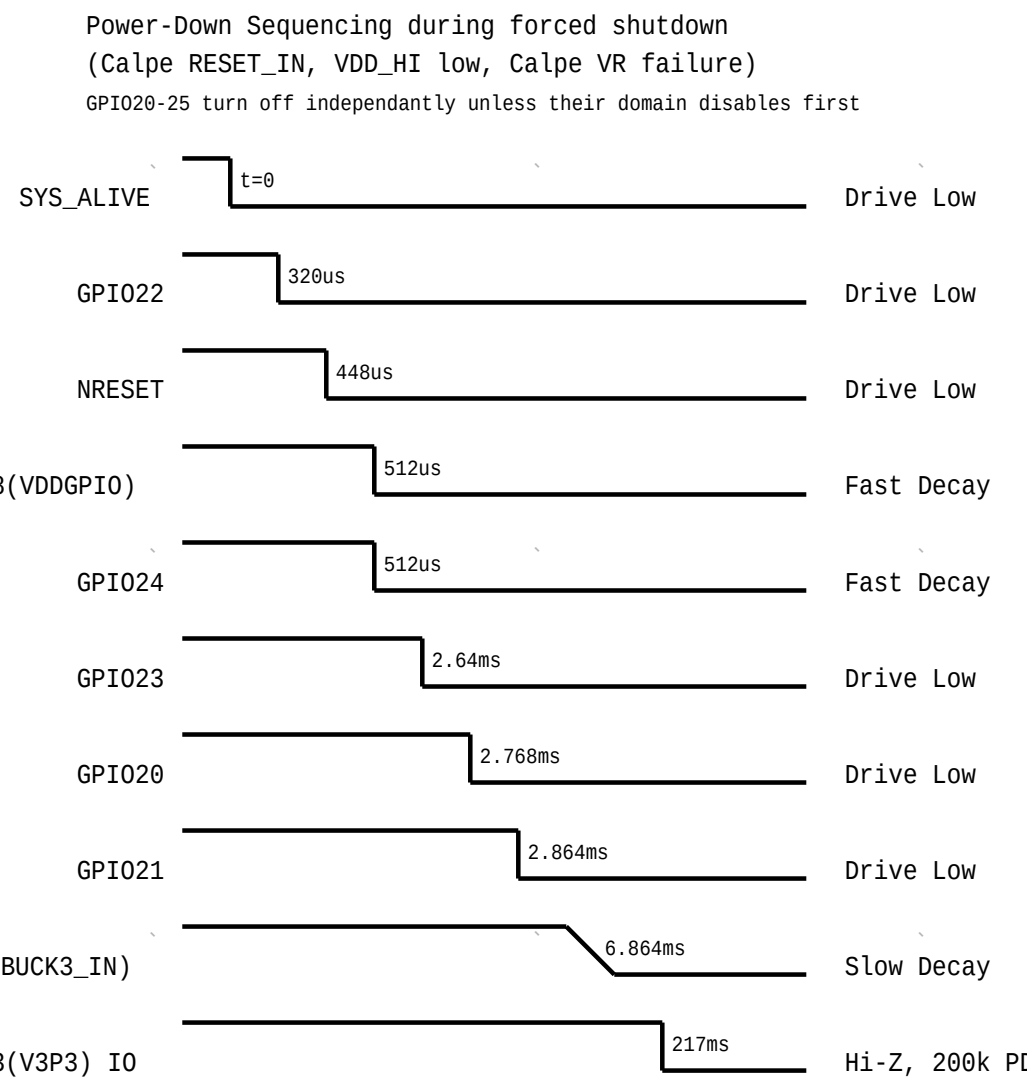


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BOM_COST_GROUP=SOC



PBUS	Rising Vth	Falling Vth
2S	6.059V	5.636V
3S	9.076V	8.443V
12V	9.076V	8.443V



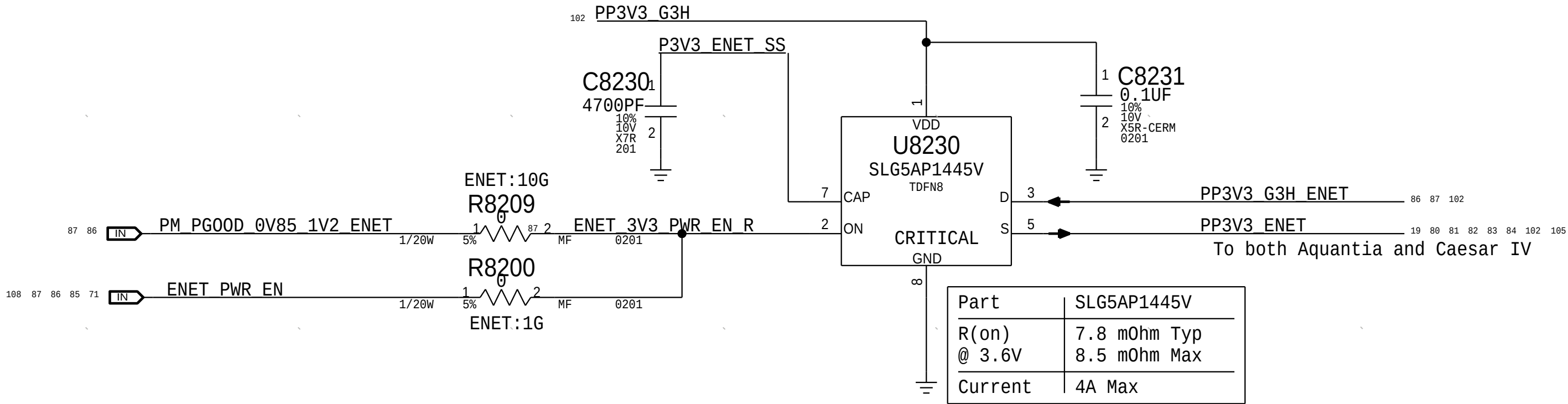
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PMIC GPIOs Controls			
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System Power States

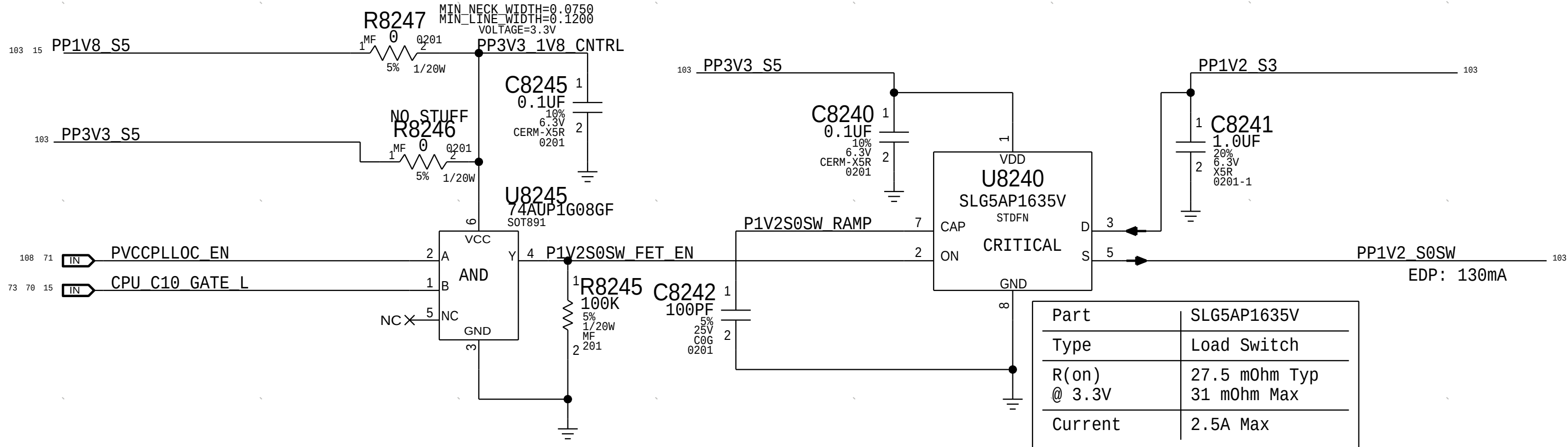
System State:	Shutdown (G3H)		Standby (G3S)		Standby (S4)		Sleep (S0i/S3)		Run (S0)
CPU/PCH State:	Off (RTC Only)		Off (RTC Only)		Standby		Sleep		Run
Rails	SoC State:		S2R	Awake	S2R	Awake	S2R	Awake	Awake
PP*_S2R (0.8,1.1,1.8V)			On	On	On	On	On	On	On
PP*_DDR (0.8,0.9,1.1V)			Off	On	Off	On	Off	On	On
PP*_AWAKE (CPU,SRAM,1.2,1.8,3.3V)			Off	On	Off	On	Off	On	On
PP3V3_G3H (VR1)			On	On	On	On	On	On	On
PP1S_G3H			On	On	On	On	On	On	On
PP3V3_G3W			Off	On	On	On	On	On	On
PP*_G3S (1.8,3.3,5V)			Off	On	On	On	On	On	On
PP*_S5 (1.8,3.3V)			Off	Off	Off	Off	On	On	On
CPU/PCH VRs			Off	Off	Off	Off	Off/On	Off/On	On

- * System: Shutdown Awake is a transition state only.
* SoC: SLP_DDR is a transition state only.
* CPU/PCH: S4 is only used by desktops for USB wakes.
* CPU/PCH: S5 is a transition state. May also be used for RTC wakes.

3.3V ENET Switch

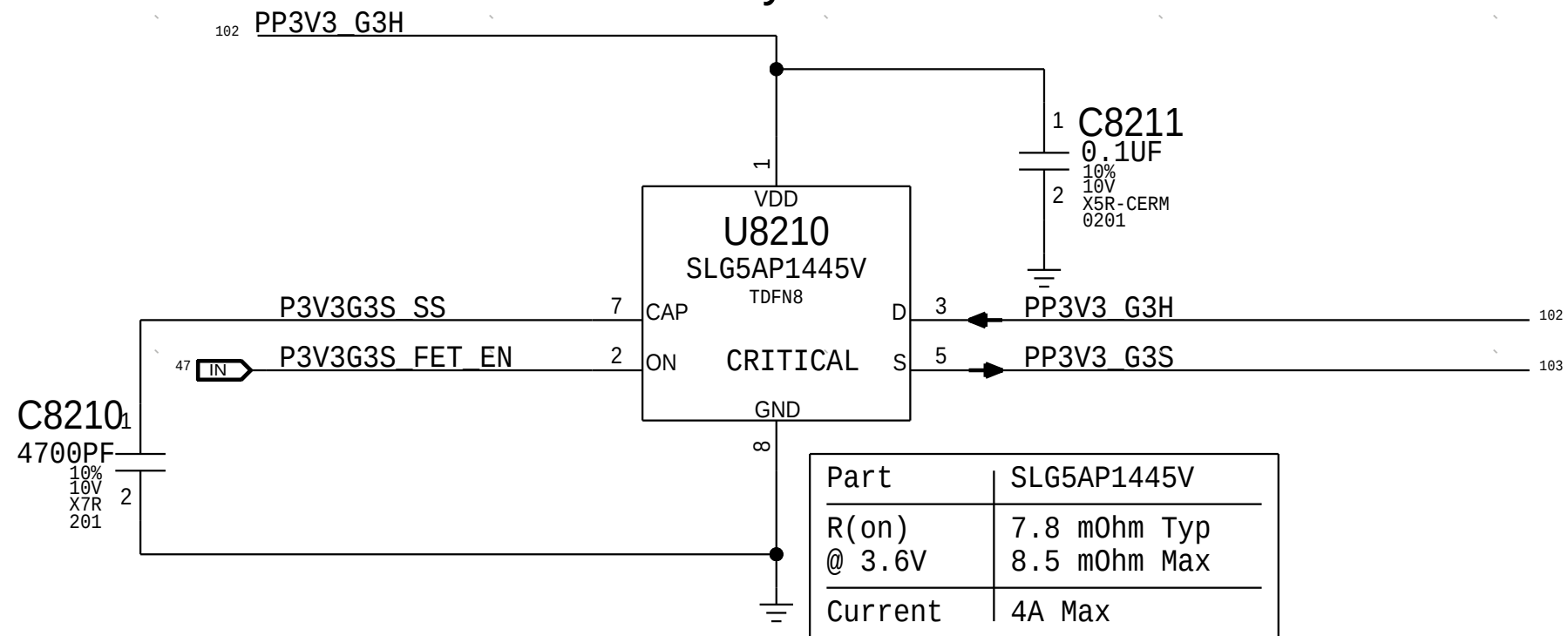


1.2V S0SW VCCPLL_OC Switch



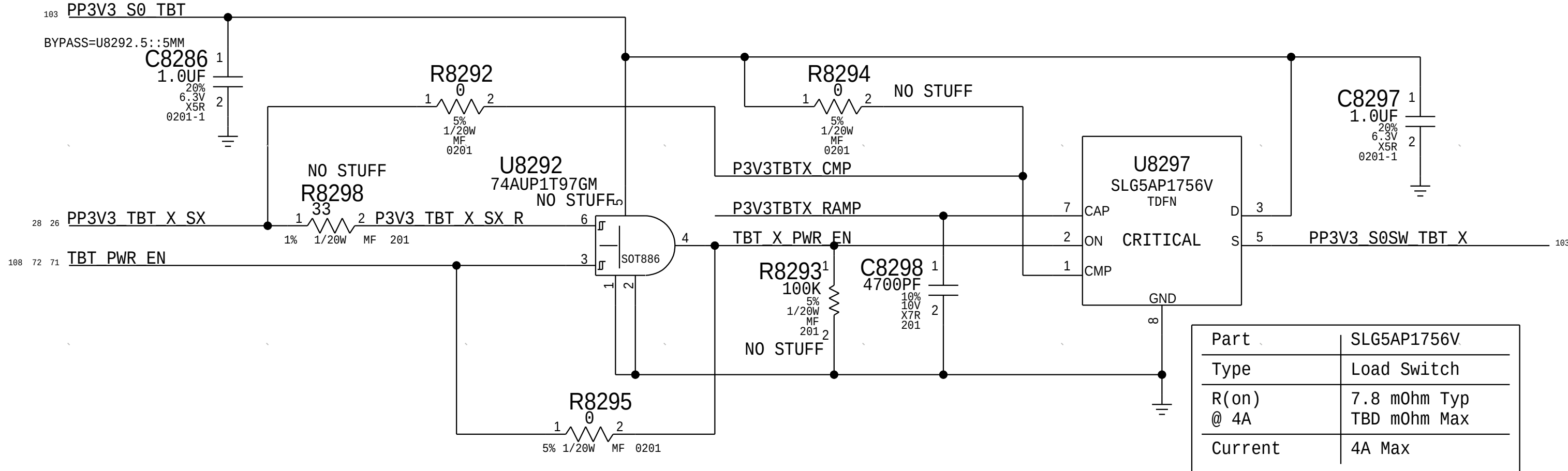
VCCPLL_OC has turn-on requirement of 11uS min and 240uS max from EN to 1.1V

3.3V G3 Standby Switch

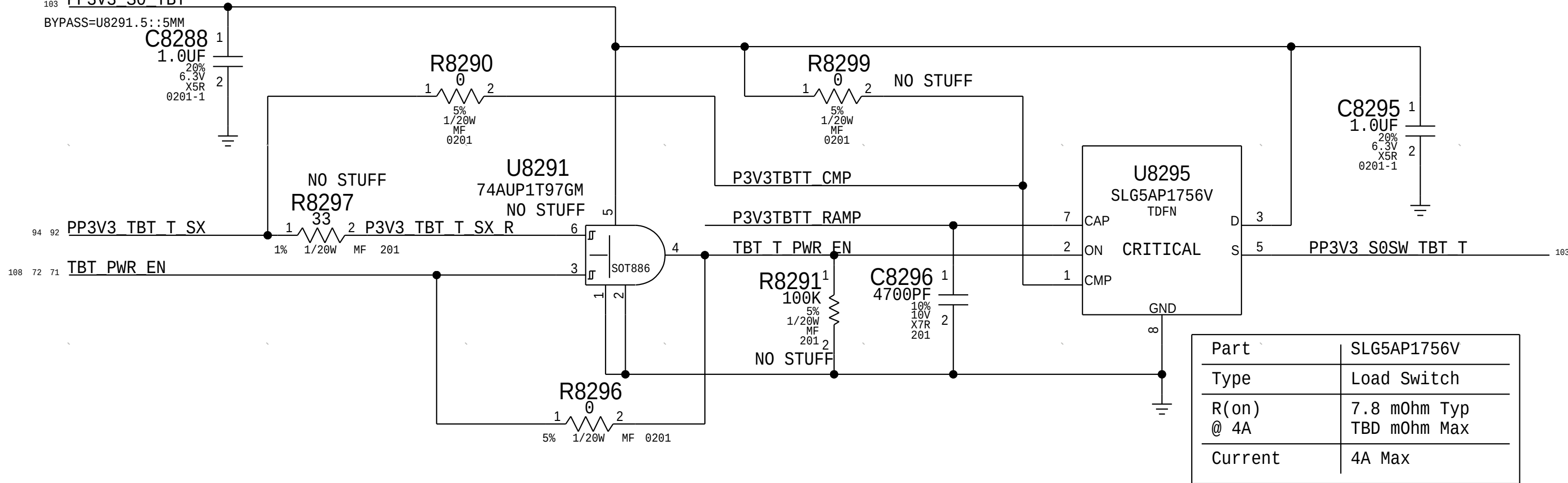


Based off rdar://30356539. 3V3_G3W can be combined into 3V3_G3H,

3.3V S0SW TBT X Switch



3.3V S0SW TBT T Switch



Power FETs

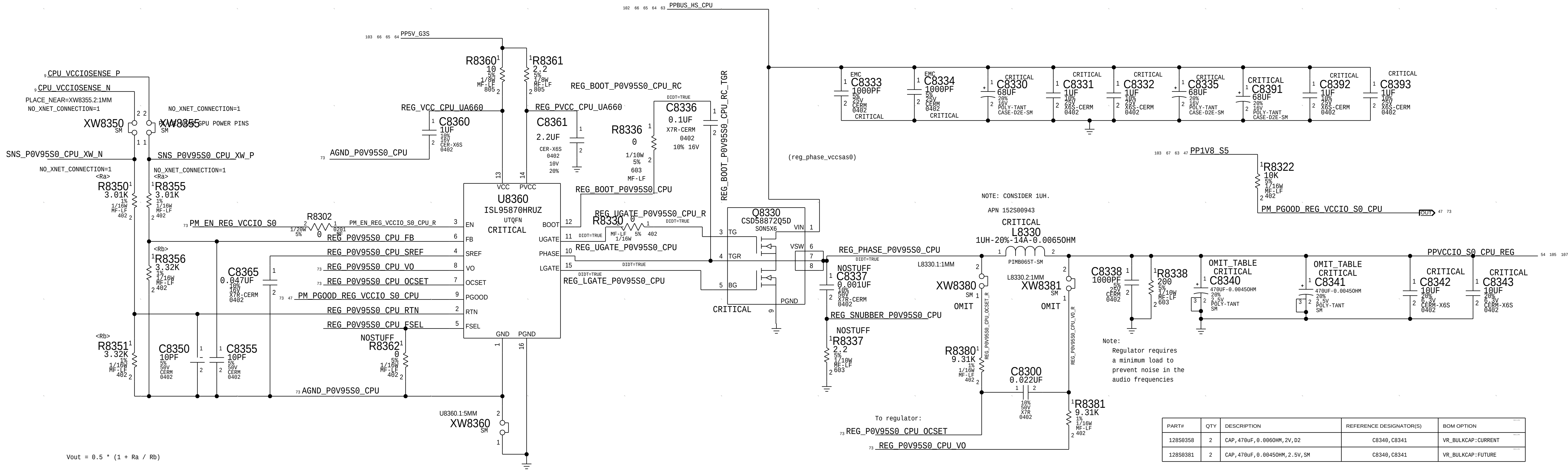


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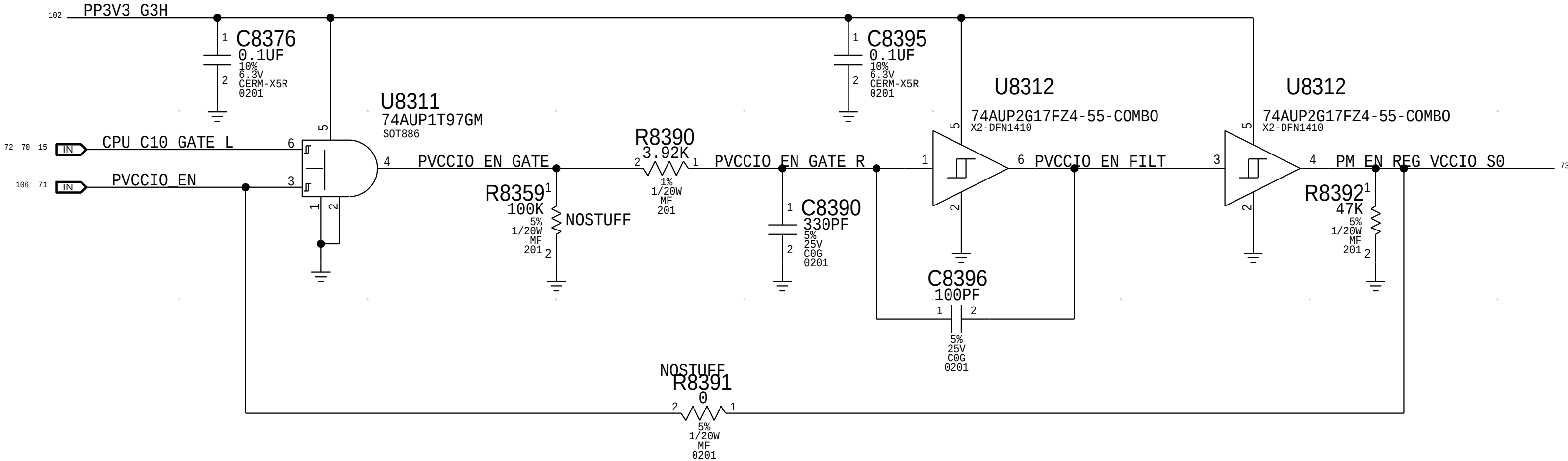
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VCCIO S0 REGULATOR

EDC = 6.4A
TDC = 6.4A
Fsw = 500KHZ



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0358	2	CAP, 470uF, 0.0060HM, 2V, D2	C8340, C8341	VR_BULKCAP:CURRENT
128S0381	2	CAP, 470uF, 0.00450HM, 2.5V, SM	C8340, C8341	VR_BULKCAP:FUTURE



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CPU & CHIPSET: CPU VCCIO VR			
Apple Inc.		DRAWING NUMBER	051-02424
		REVISION	6.0.0
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		PAGE	83 OF 142
		SHEET	73 OF 115

BOM_COST_GROUP=CPU & CHIPSET

S4E0

D

D

C

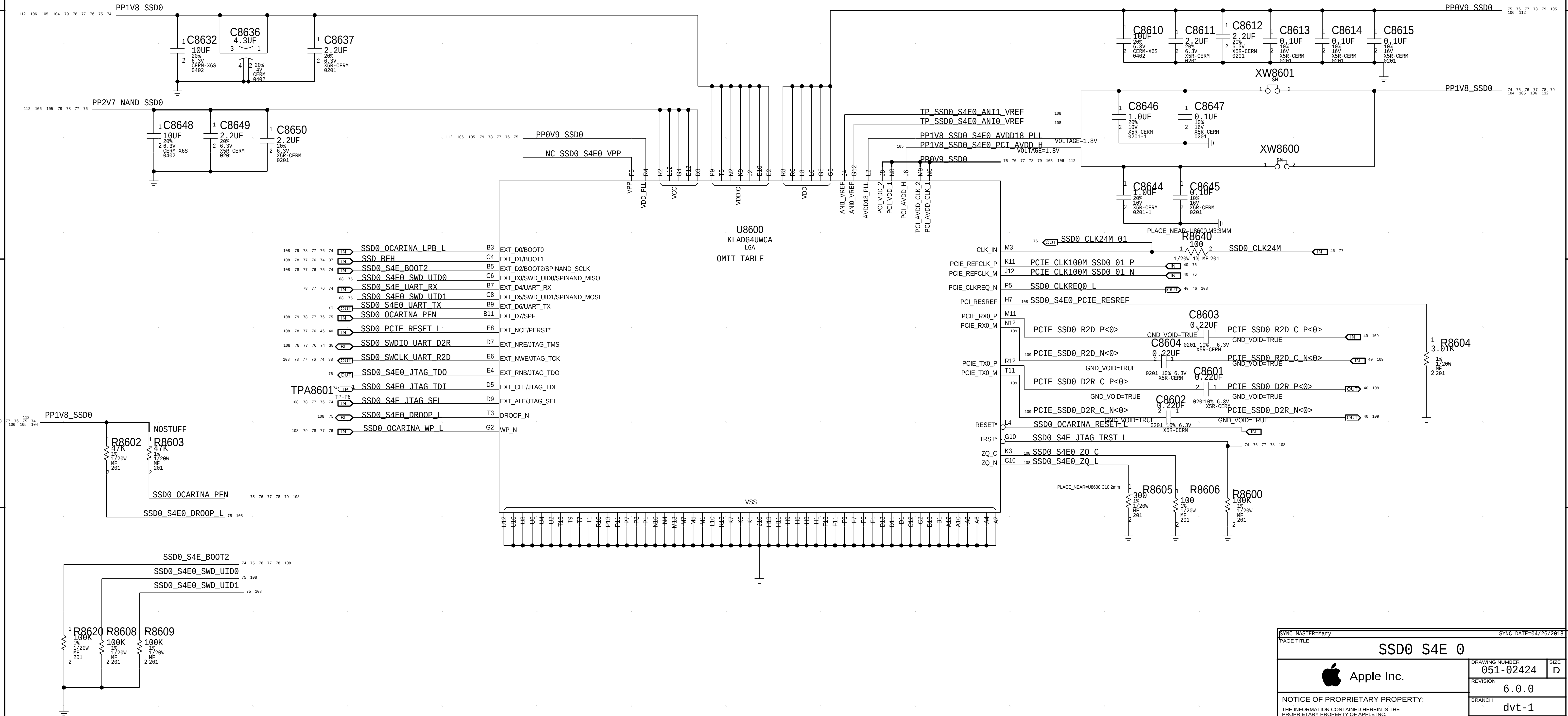
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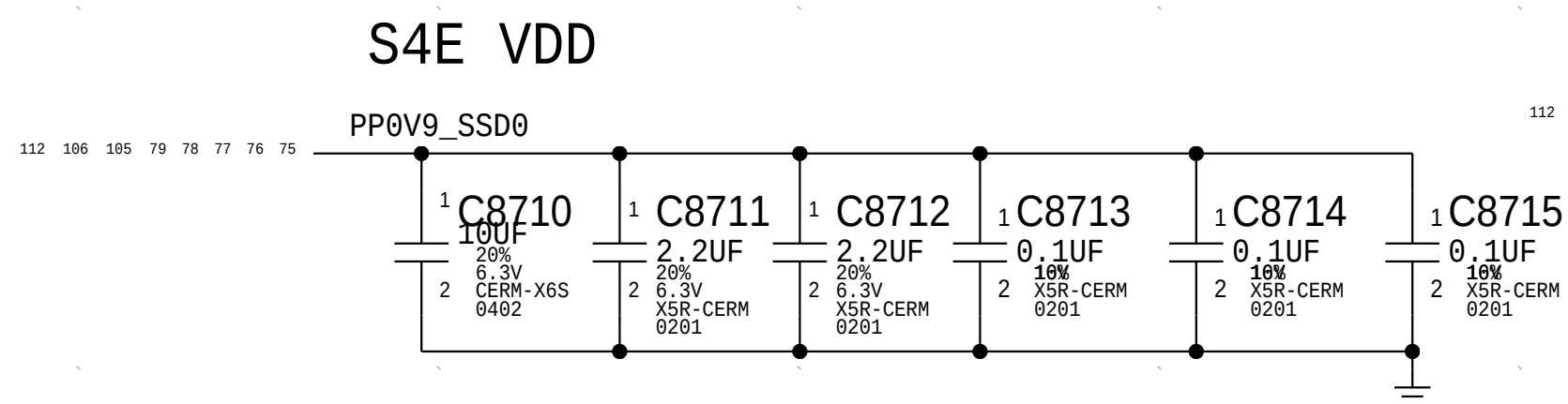
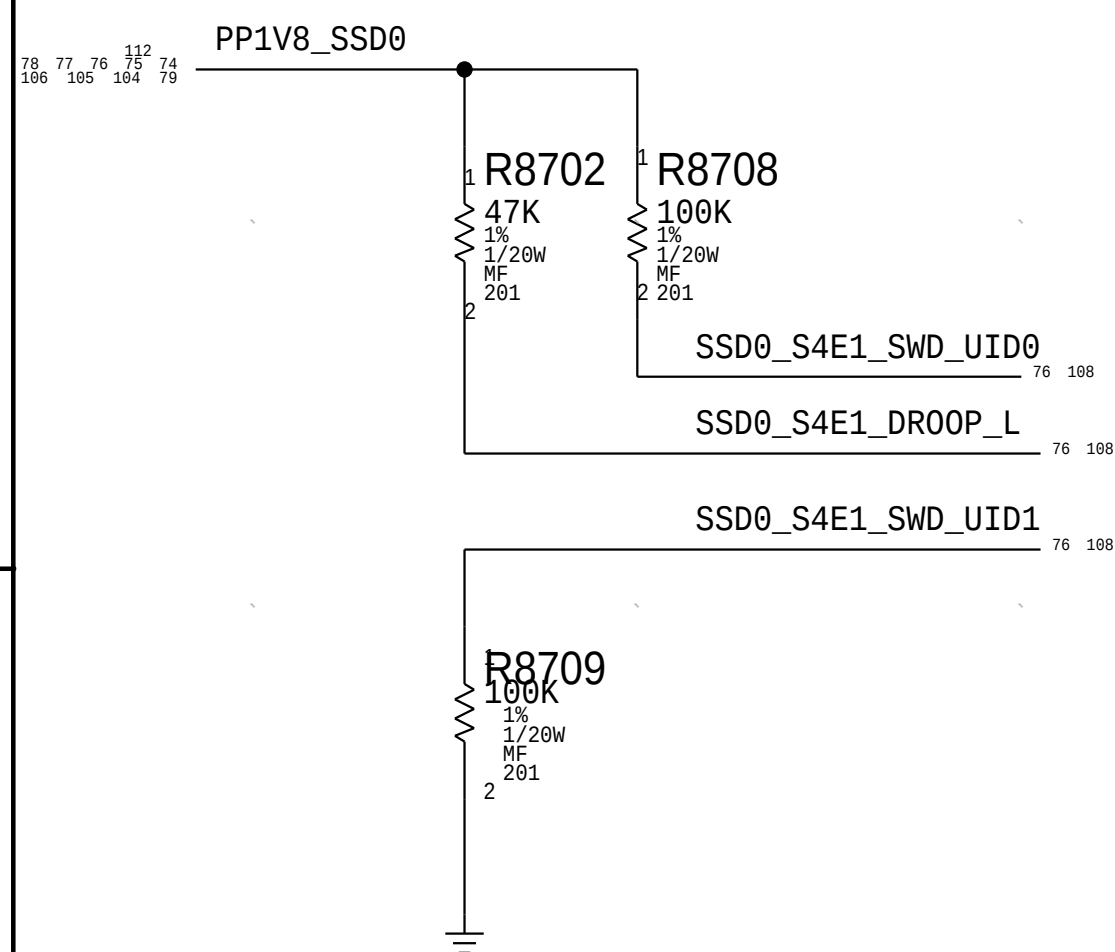
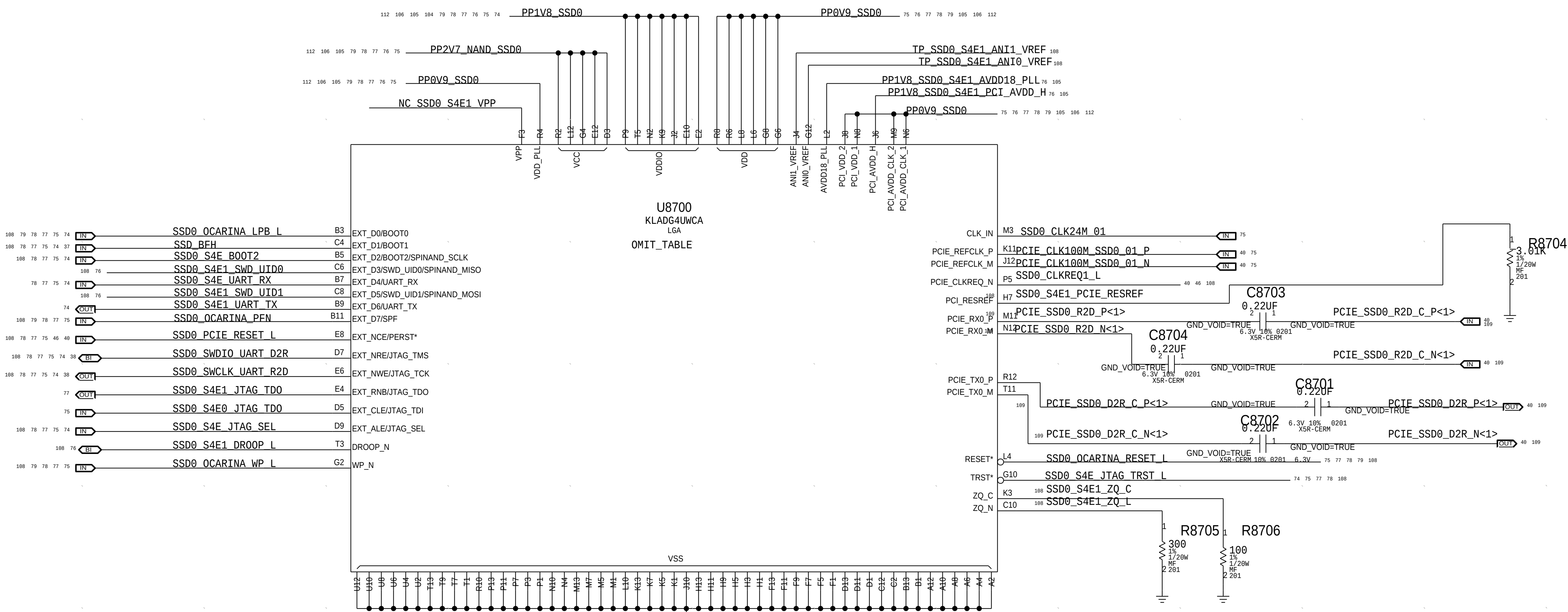
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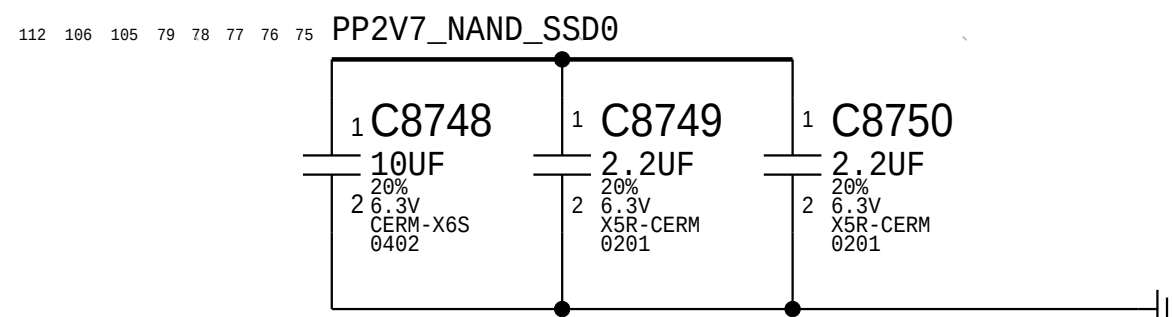
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	PAGE	86 OF 142	
	SHEET	75 OF 115	

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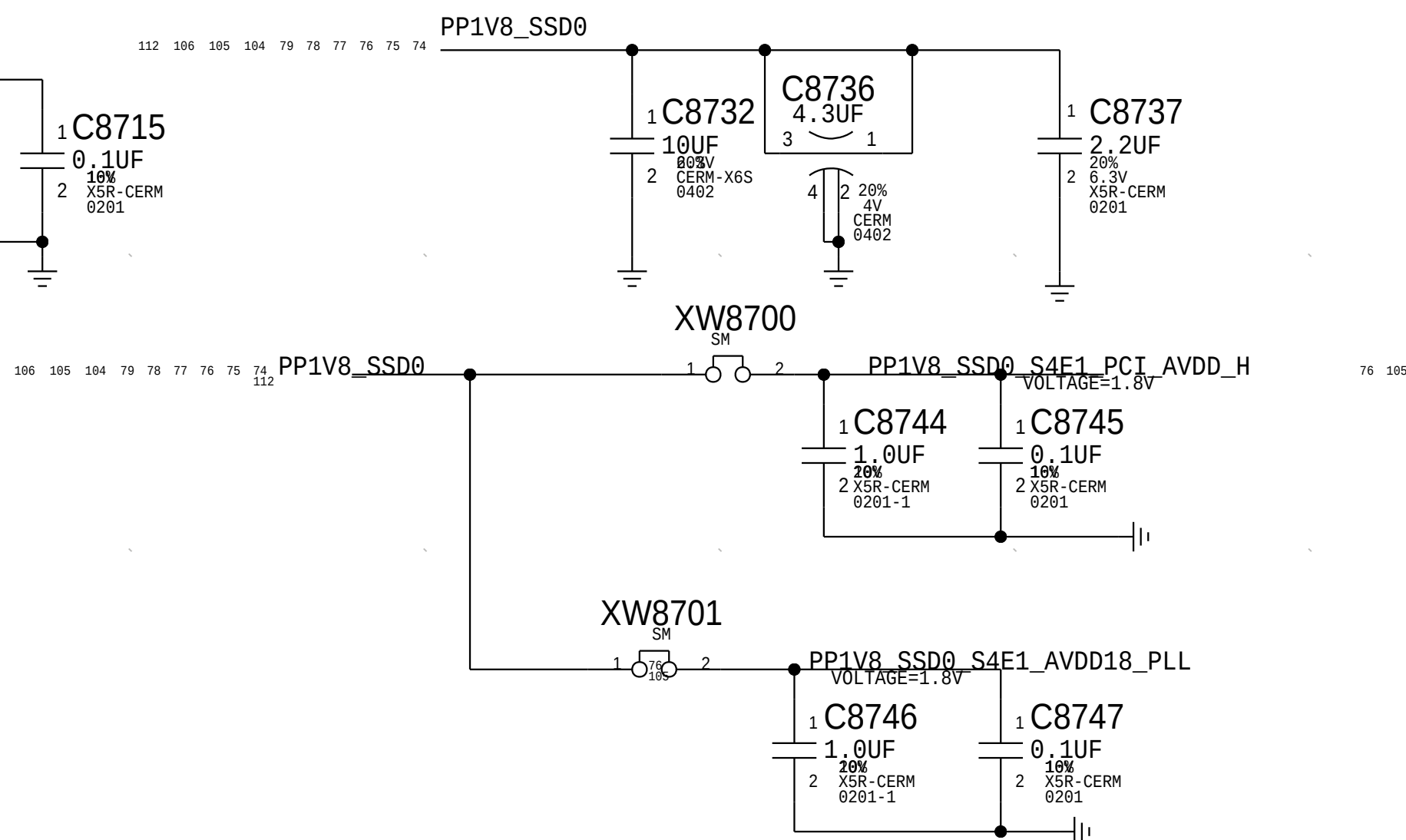
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VCC CAP



S4E VDDIO



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			PAGE	87 OF 142	
			SHEET	76 OF 115	

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S4E2

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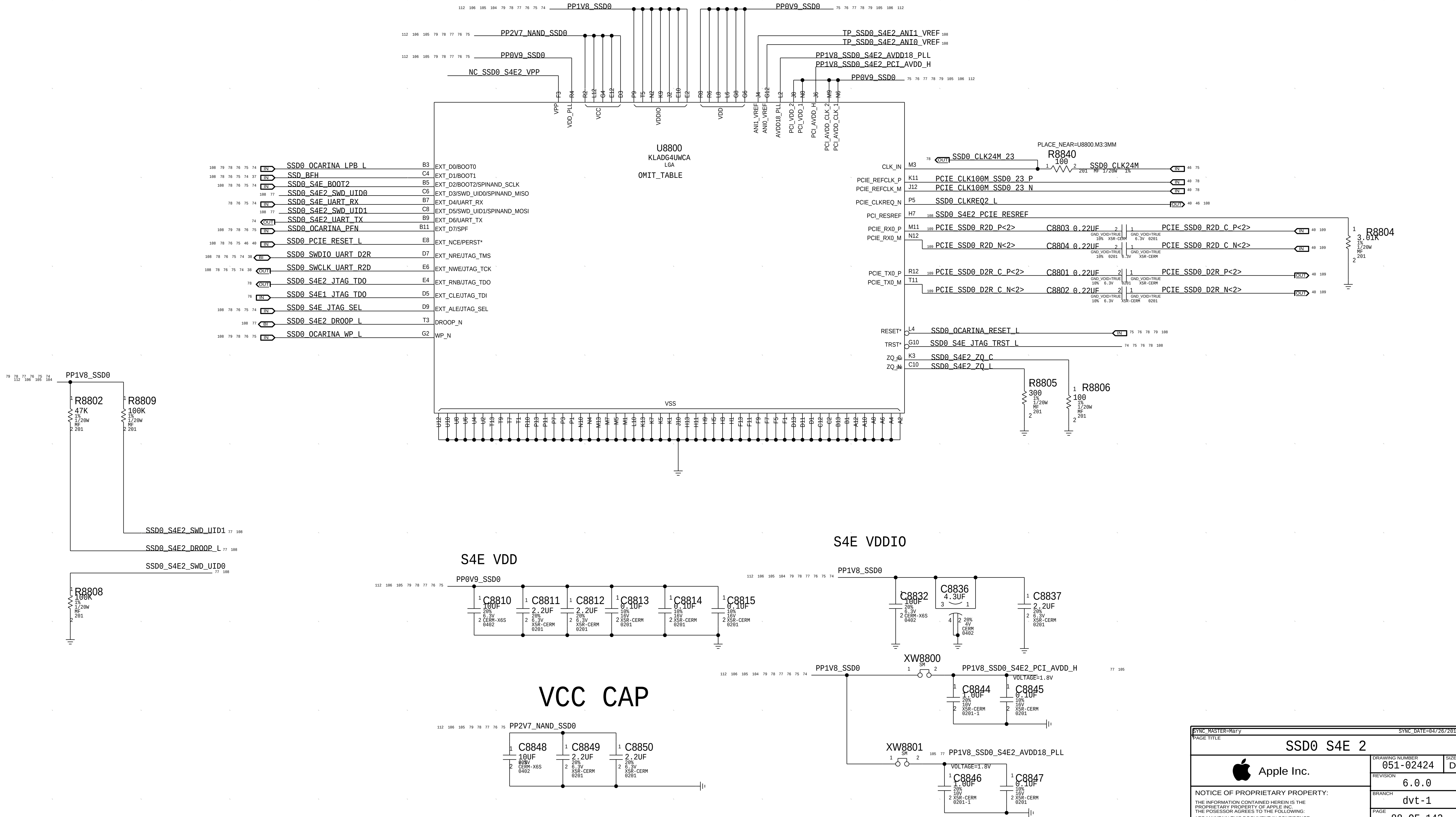
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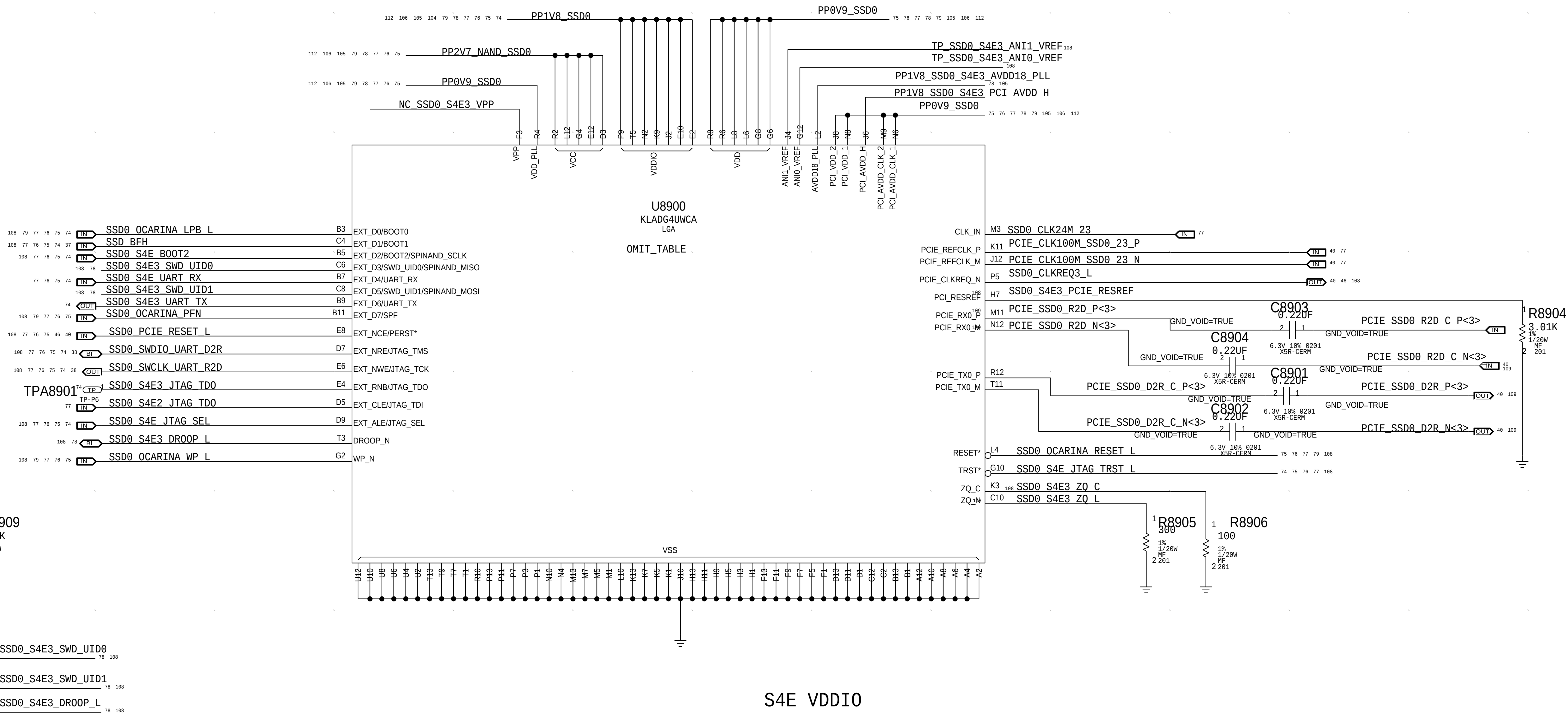
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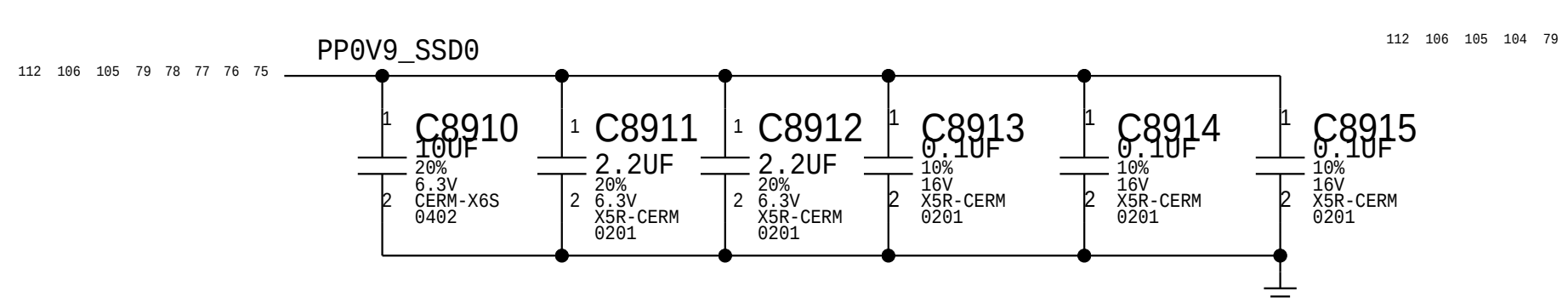
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	051-02424		D
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	BRANCH		
	dvt-1		
	PAGE		
	88 OF 142		
	SHEET		
	77 OF 115		

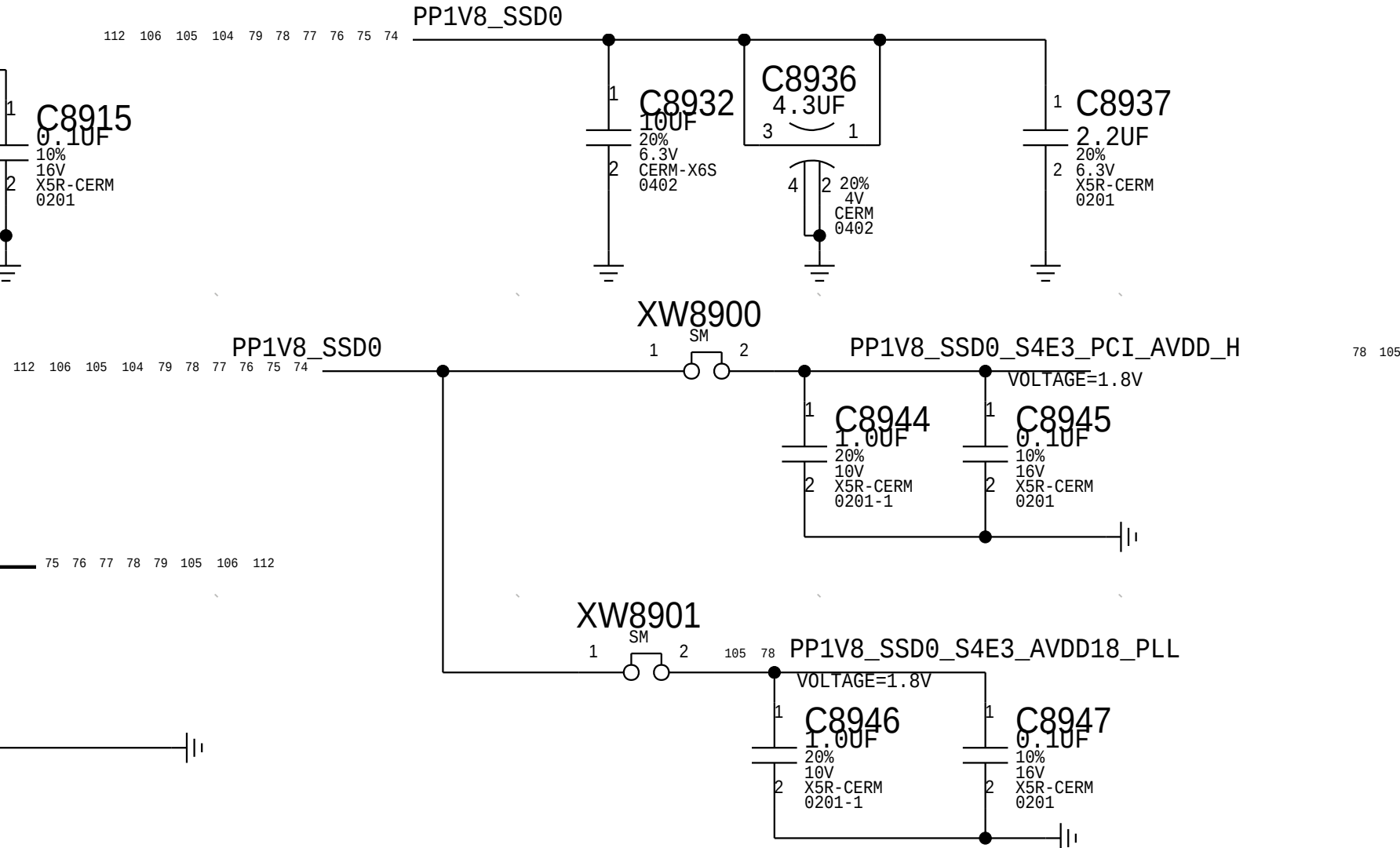
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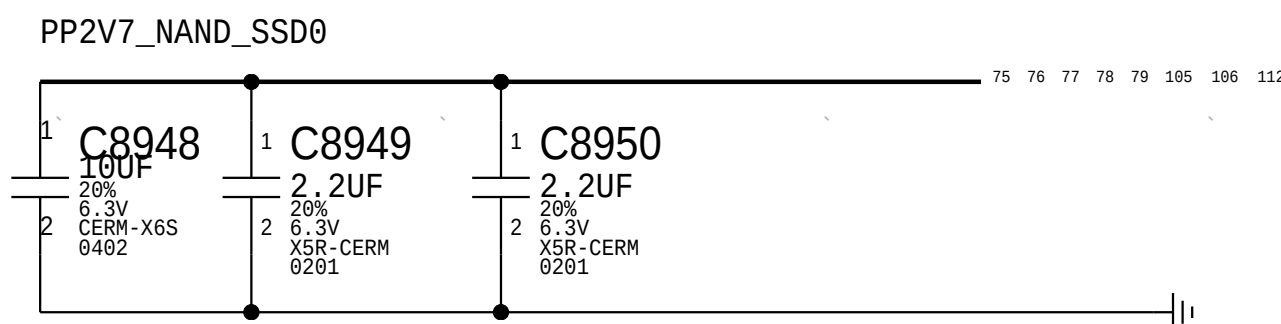
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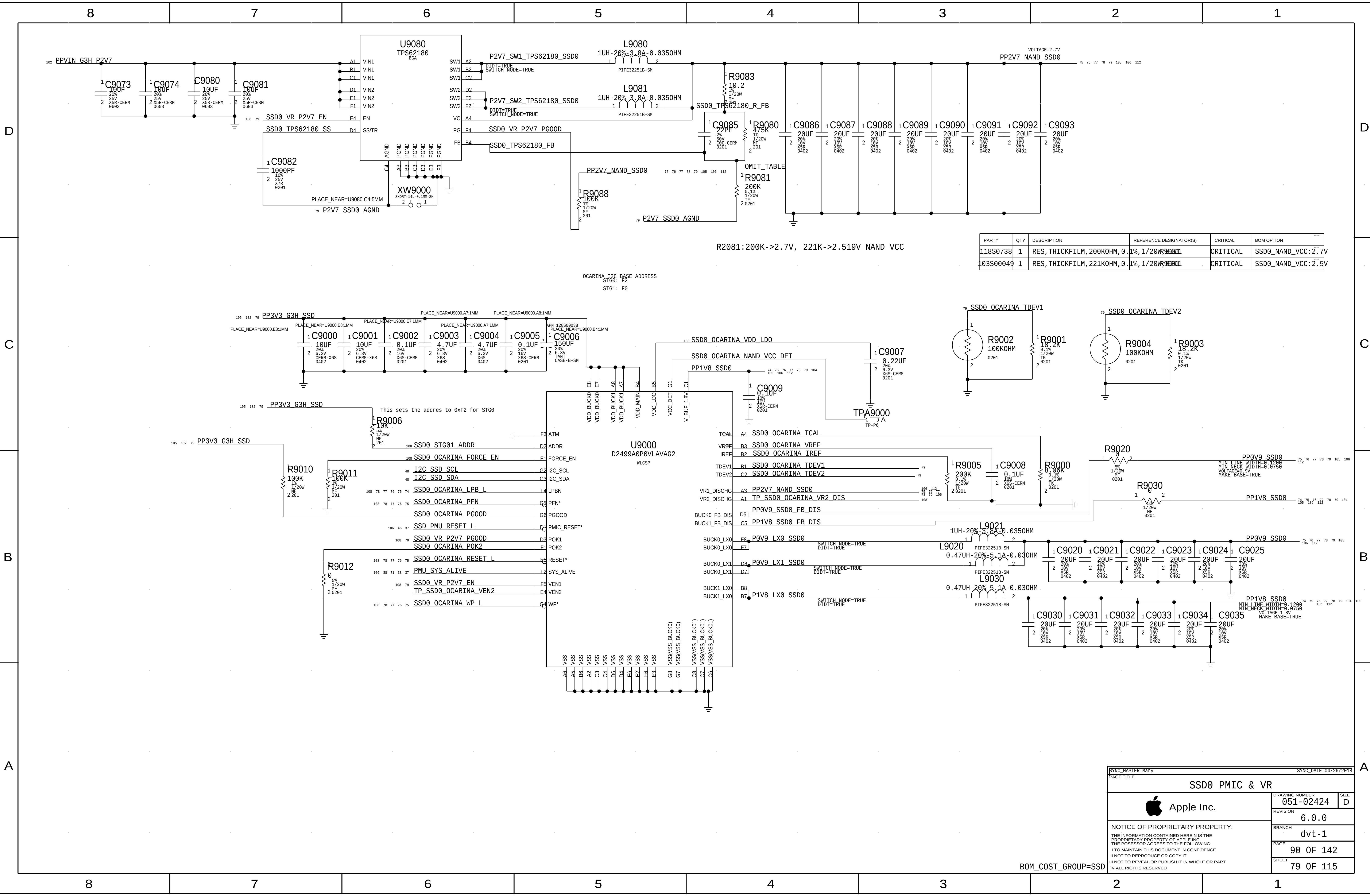


VCC CAP



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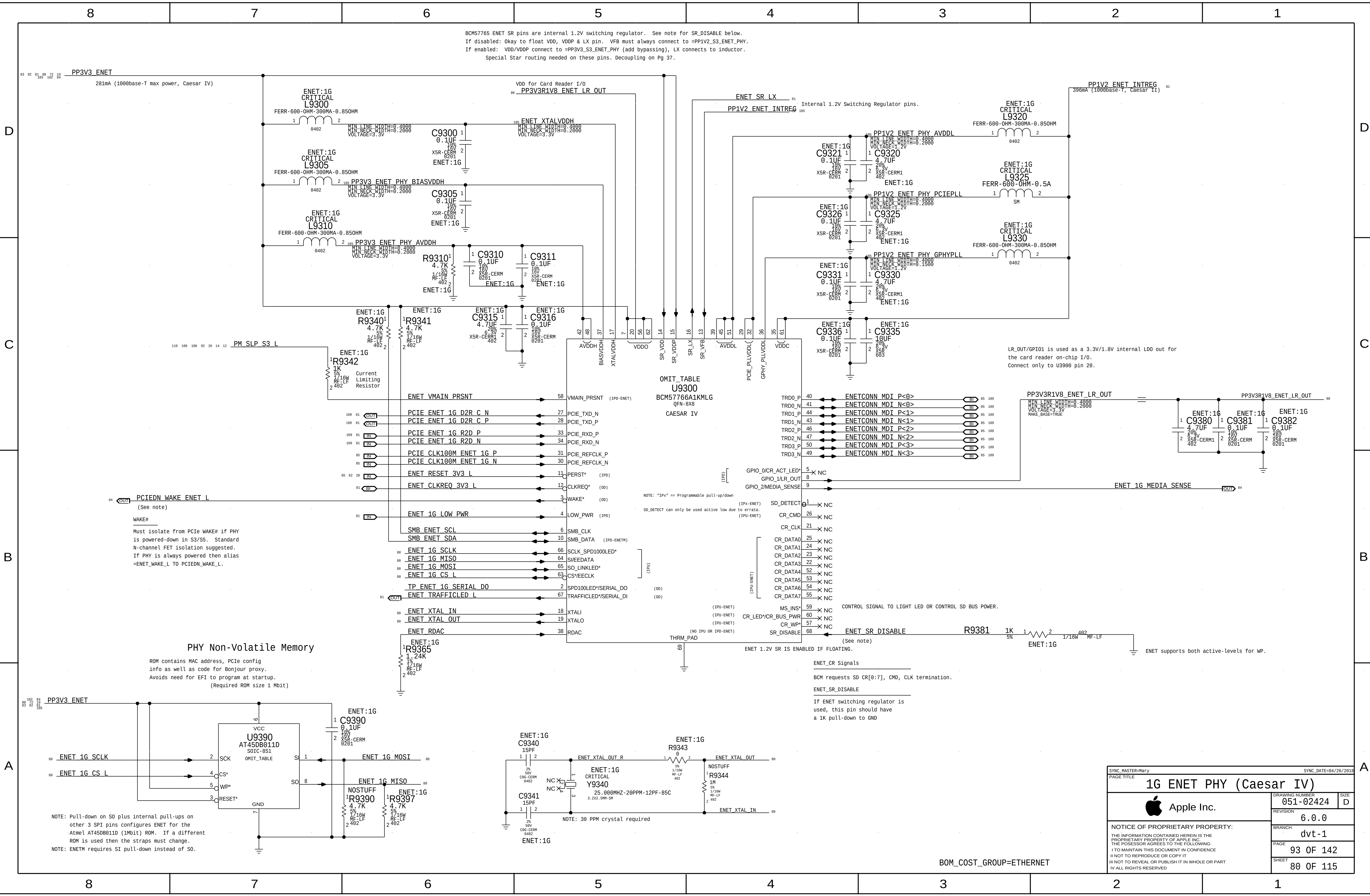
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S0738	1	RES, THICKFILM, 200KOHM, 0.1%, 1/20W, 0603	R9081	CRITICAL	SSD0_NAND_VCC:2.7V
103S00049	1	RES, THICKFILM, 221KOHM, 0.1%, 1/20W, 0603	R9088	CRITICAL	SSD0_NAND_VCC:2.5V

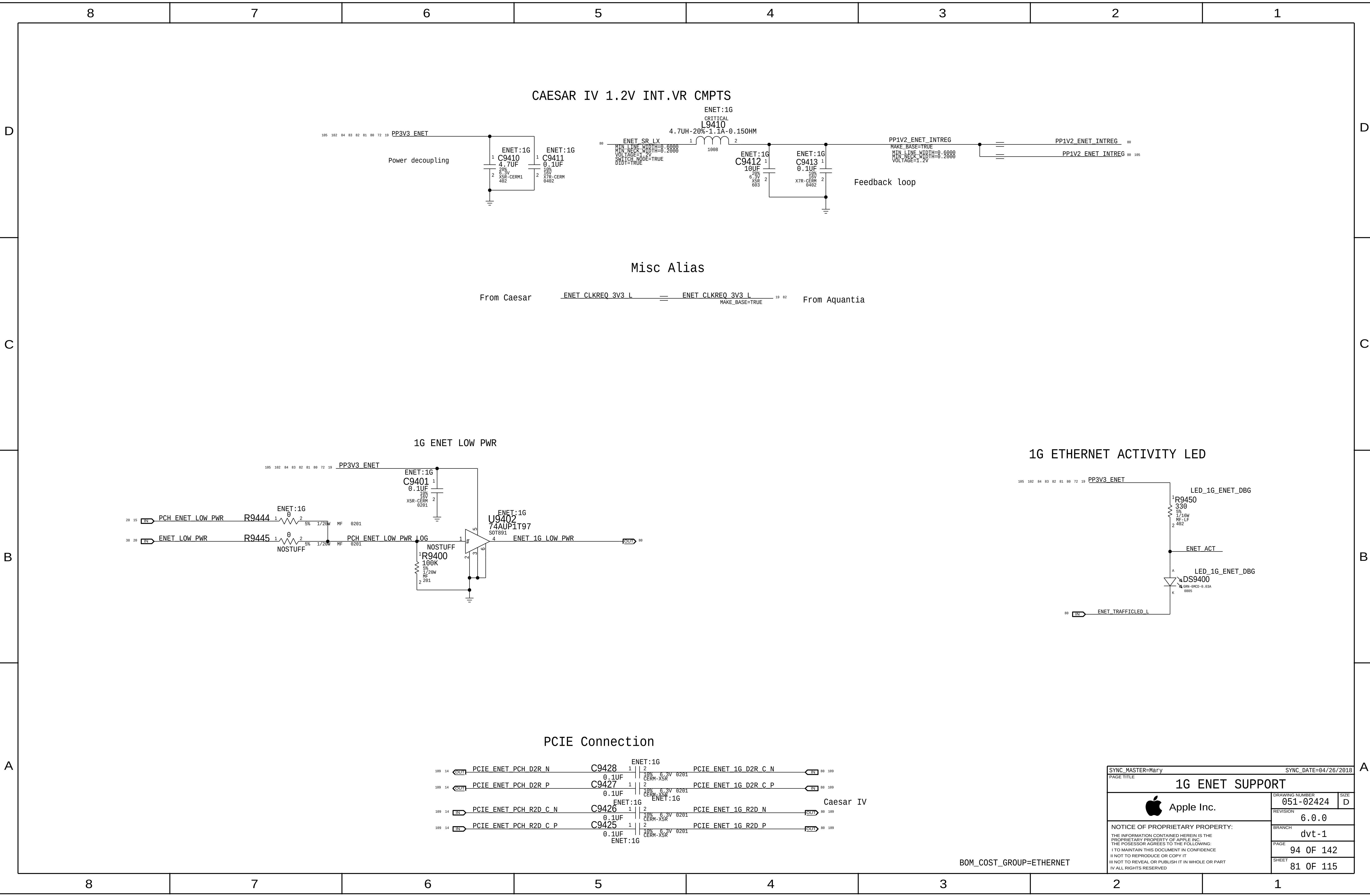
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SSD0 PMIC & VR			
 Apple Inc.	DRAWING NUMBER	051-02424	SIZE D
	REVISION	6.0.0	
	BRANCH	dvt-1	
	PAGE	90 OF 142	
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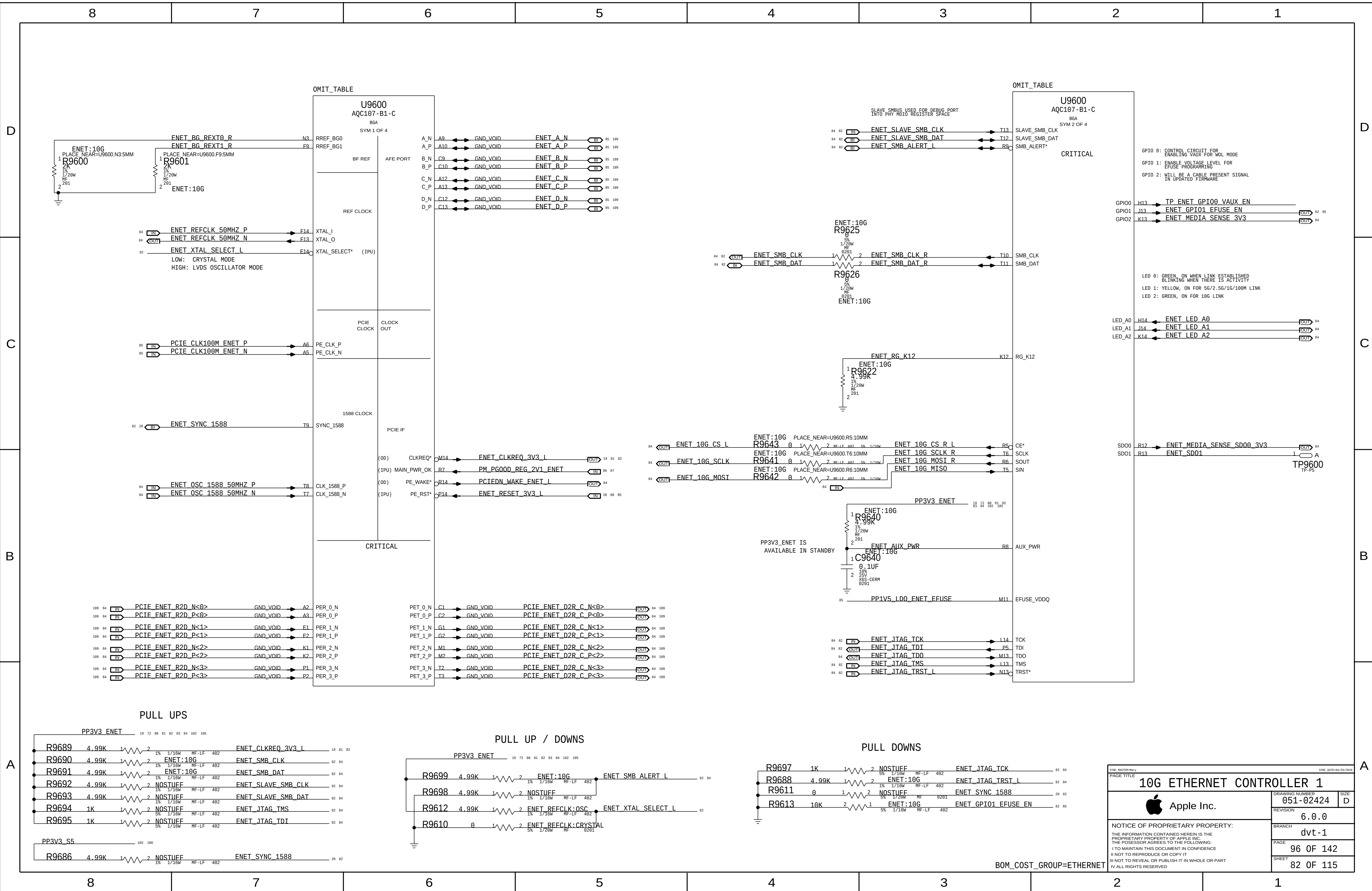
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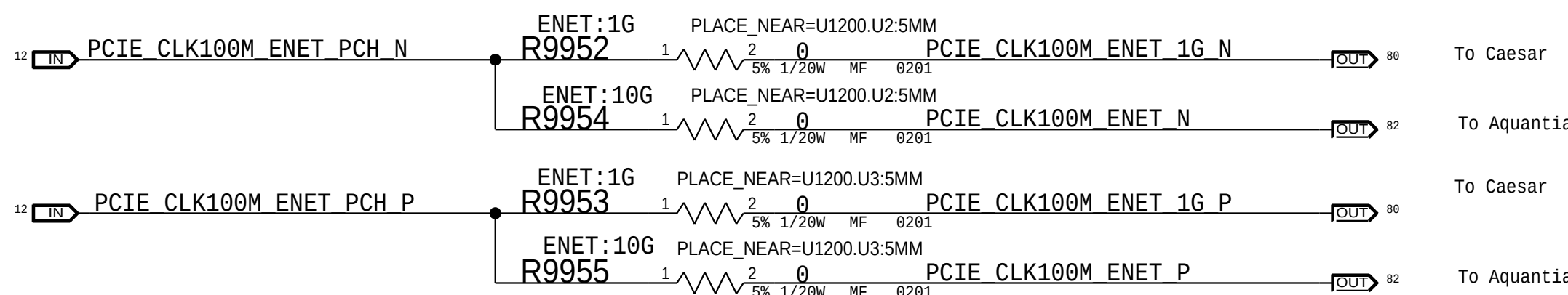
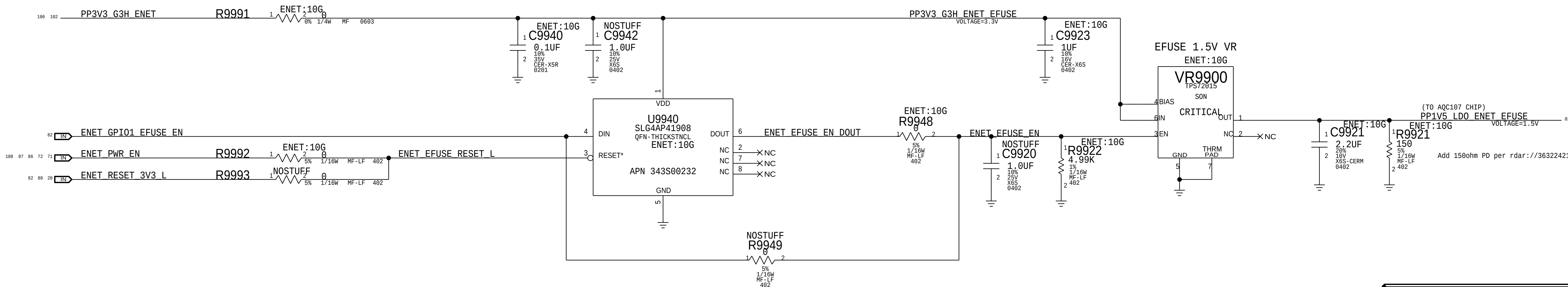
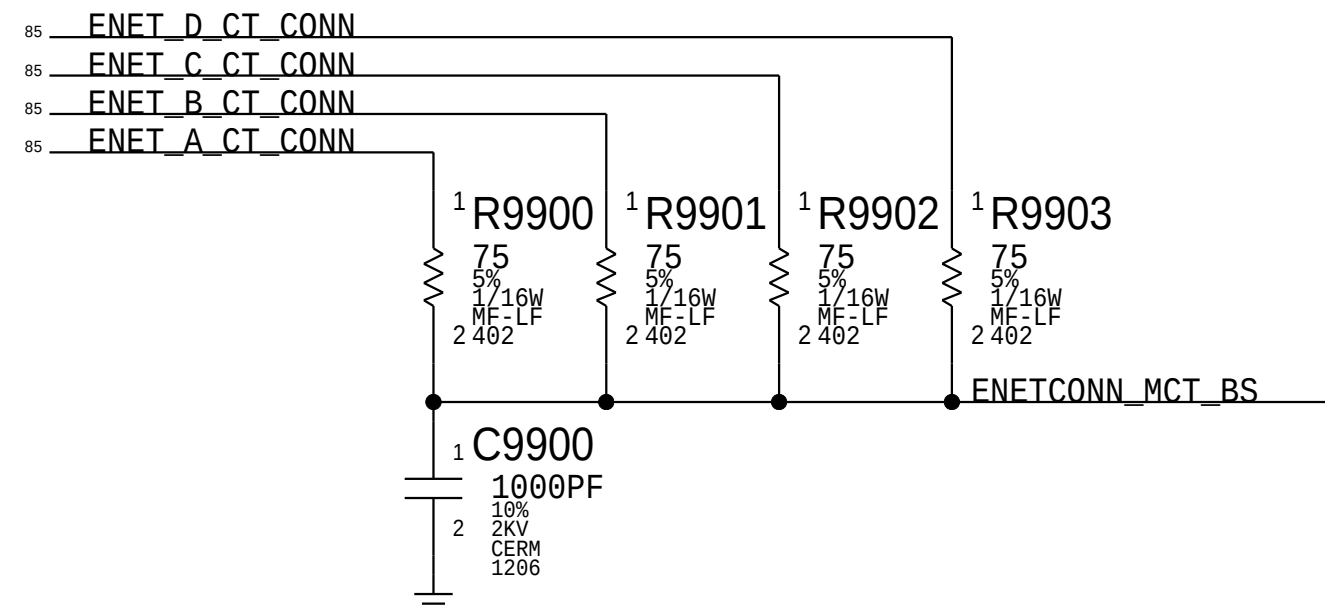
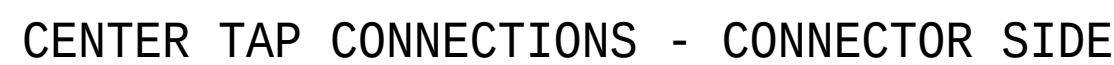
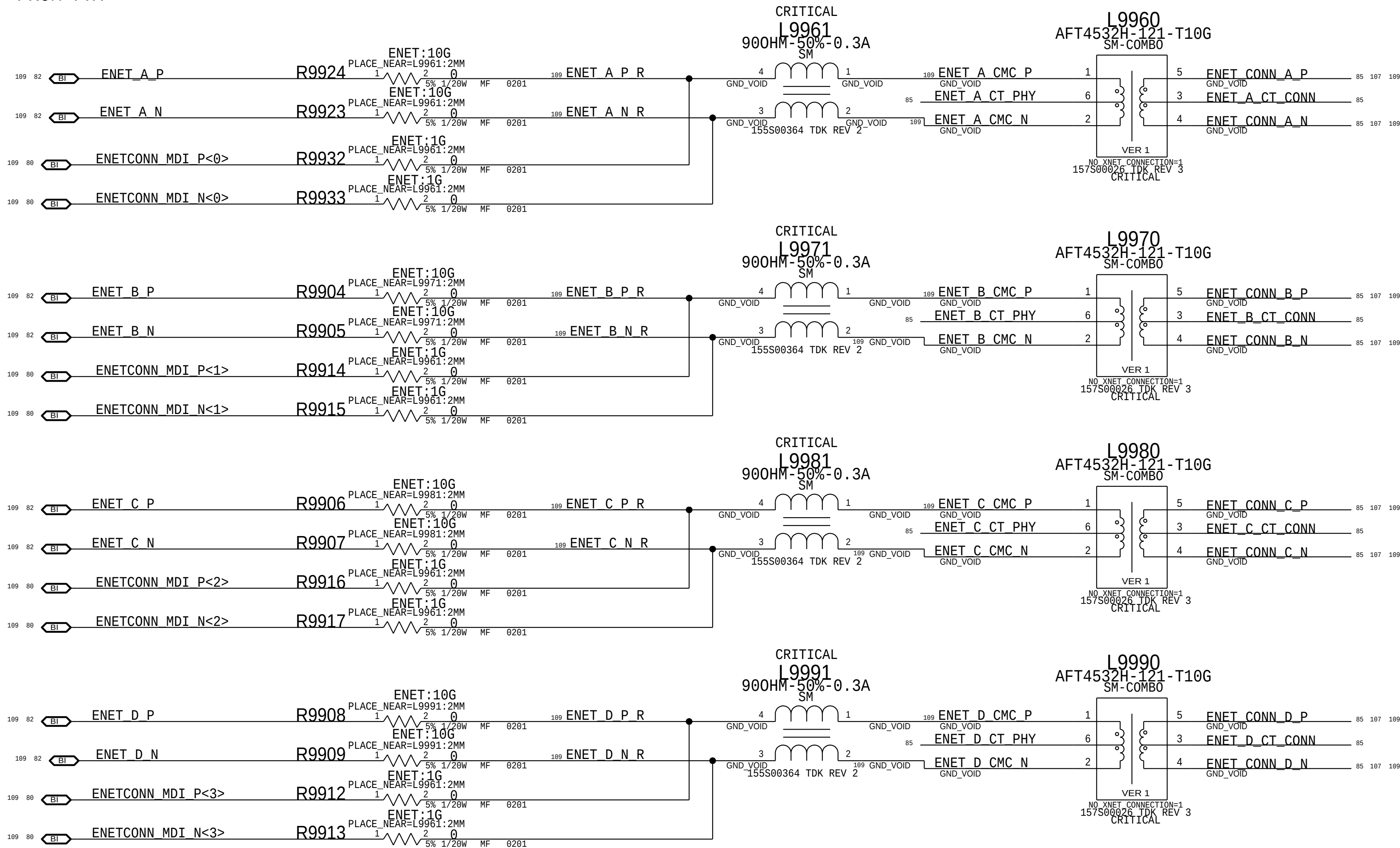
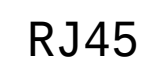


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			BRANCH	dvt-1	
			PAGE	93 OF 142	
			SHEET	80 OF 115	

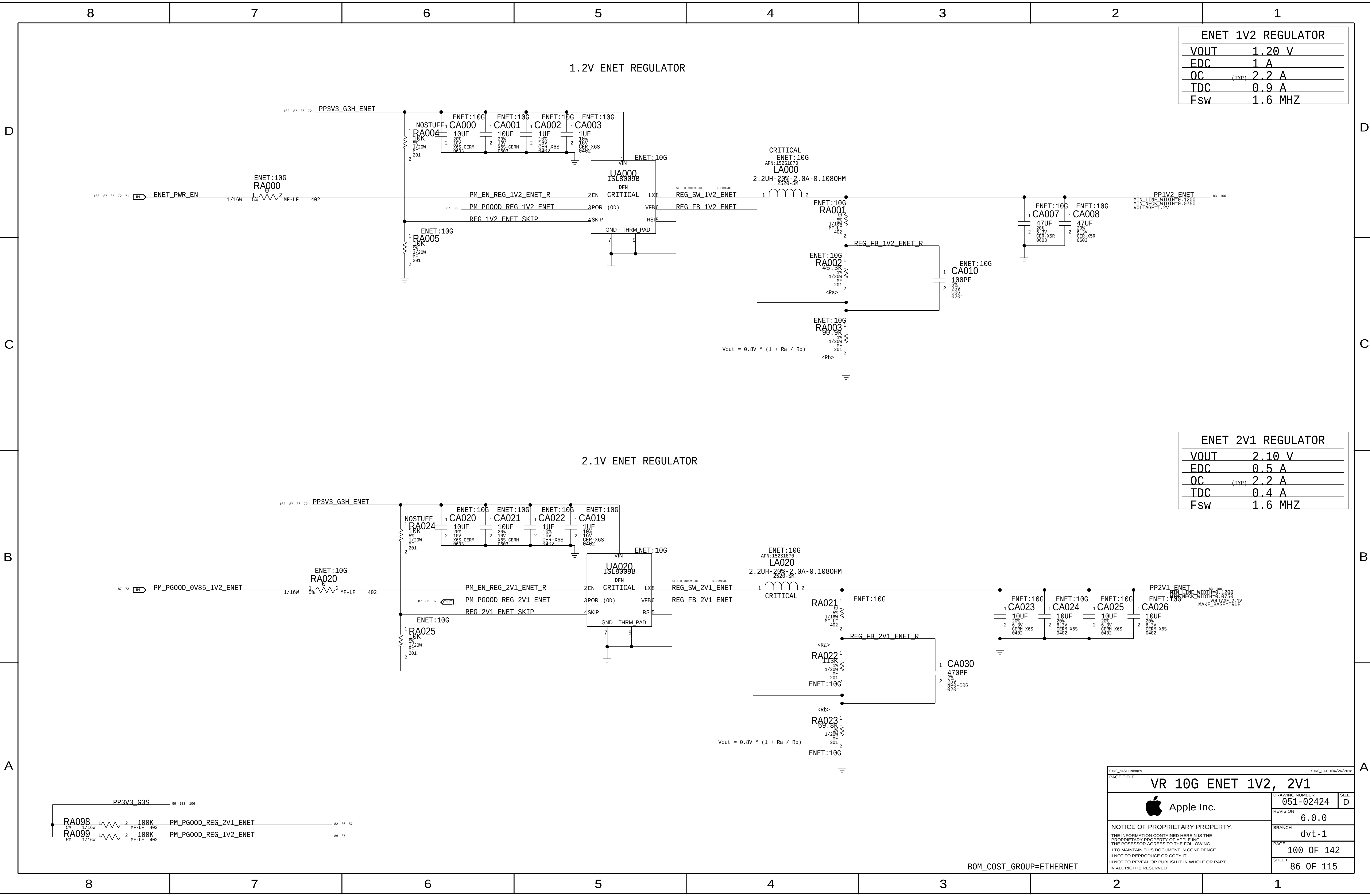
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DOC: MASTER-0007		DOC: 0471-04/06/2018	
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 Apple Inc.	DRAWING NUMBER		SIZE
	051-02424		D
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	SHEET	85 OF 115	

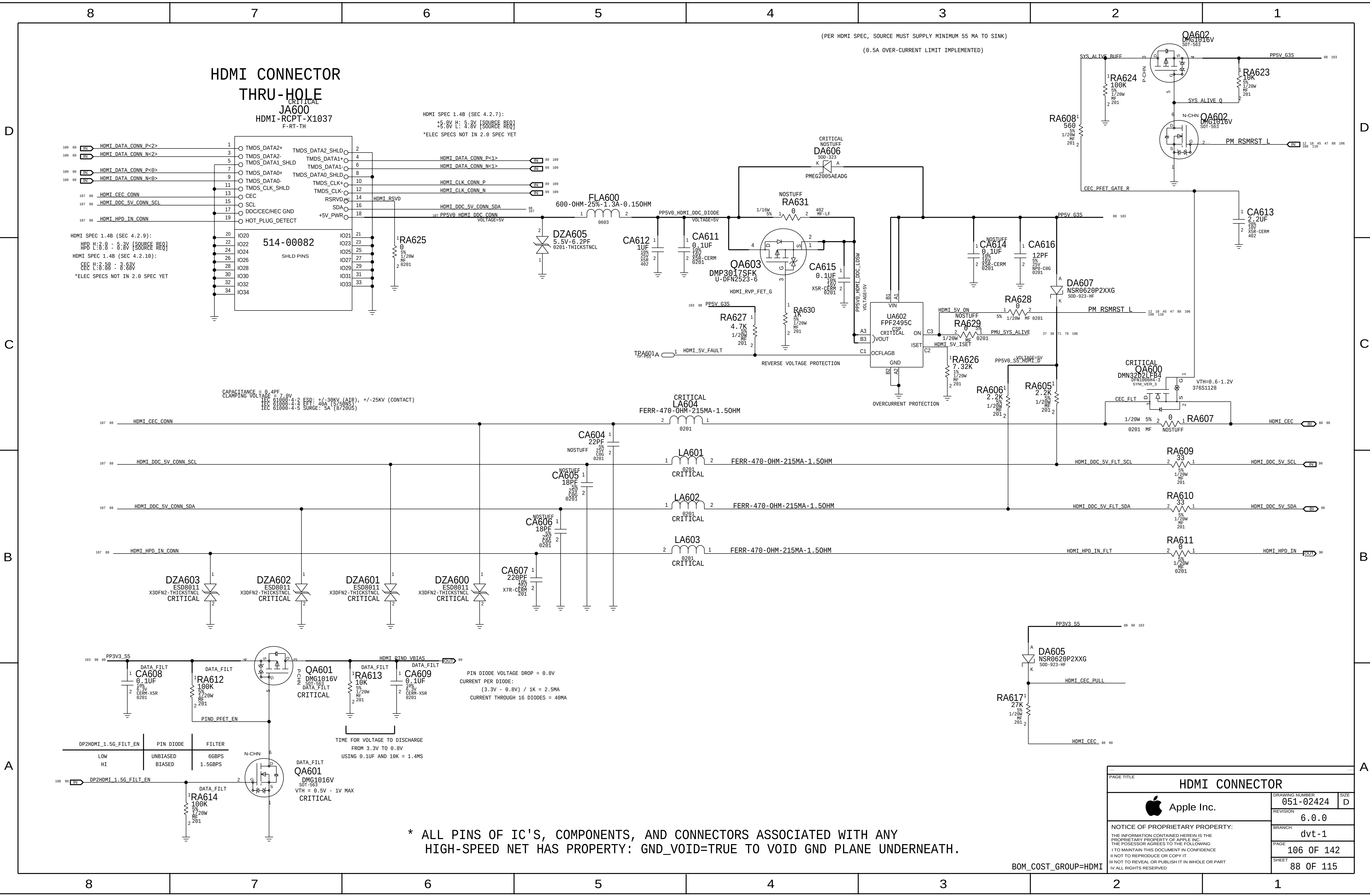


ENET 1V2 REGULATOR	
VOUT	1.20 V
EDC	1 A
OC	2.2 A (TYP)
TDC	0.9 A
Fsw	1.6 MHZ

ENET 2V1 REGULATOR	
VOUT	2.10 V
EDC	0.5 A
OC	2.2 A (TYP)
TDC	0.4 A
Fsw	1.6 MHZ

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	REVISION		6.0.0
	BRANCH		dvt-1
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PAGE		SHEET	
100 OF 142		86 OF 115	

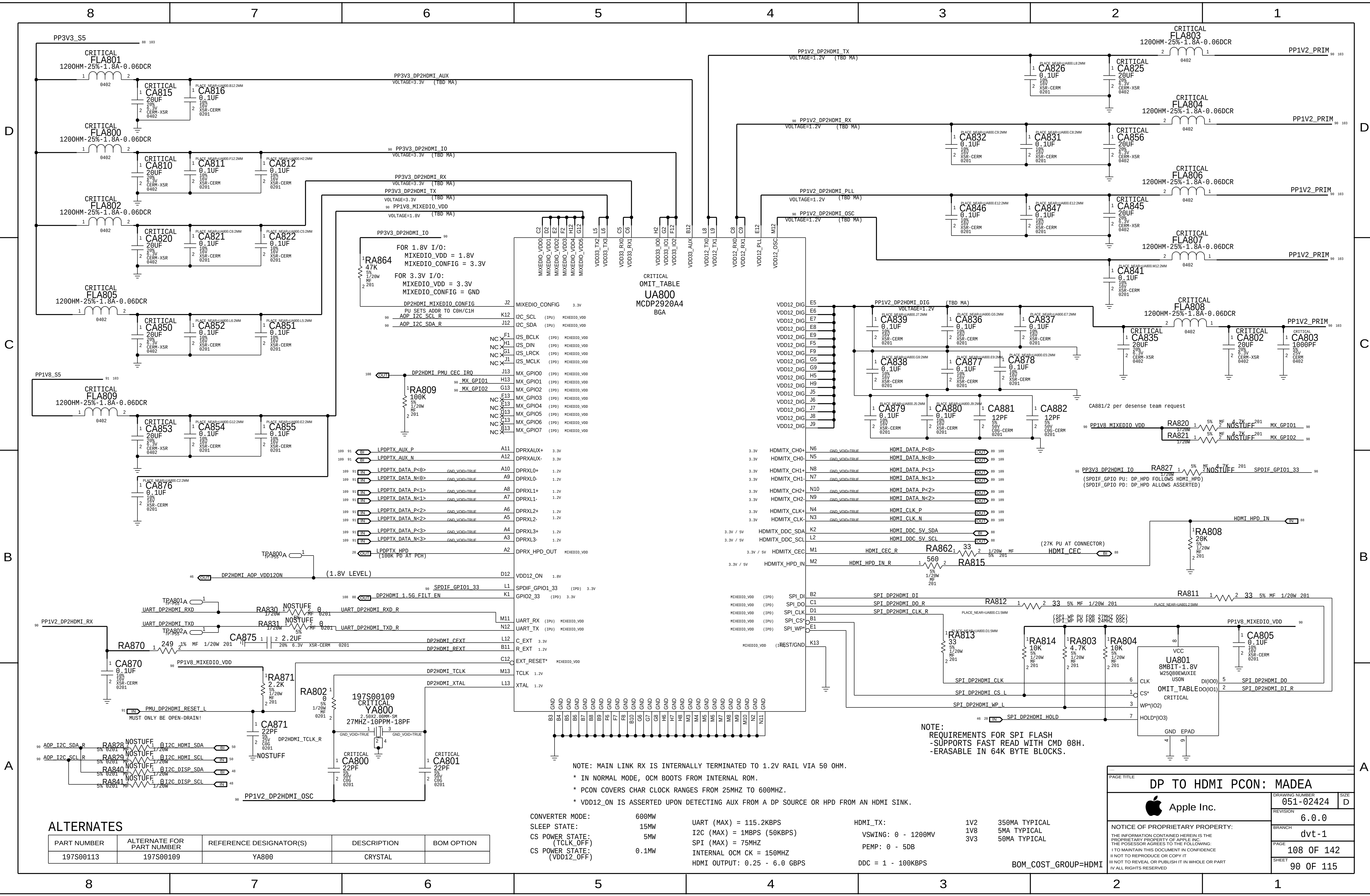
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* ALL PINS OF IC'S, COMPONENTS, AND CONNECTORS ASSOCIATED WITH ANY HIGH-SPEED NET HAS PROPERTY: GND_VOID=TRUE TO VOID GND PLANE UNDERNEATH.

BOM_COST_GROUP=HDMI

PAGE TITLE		
HDMI CONNECTOR		
 Apple Inc.	DRAWING NUMBER	051-02424
	REVISION	6.0.0
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	PAGE	106 OF 142
	SHEET	88 OF 115



ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	REFERENCE DESIGNATOR(S)	DESCRIPTION	BOM OPTION
197S00113	197S00109	YA800	CRYSTAL	

CONVERTER MODE: 600MW

SLEEP STATE: 15MW

CS POWER STATE: 5MW

CS POWER STATE: 0.1MW

UART (MAX) = 115.2KBPS

I2C (MAX) = 1MBPS (50KBPS)

SPI (MAX) = 75MHZ

INTERNAL OCM CK = 150MHZ

HDMI OUTPUT: 0.25 - 6.0 GBPS

HDMI_TX:

VSWING: 0 - 1200MV

PEMP: 0 - 5DB

DDC = 1 - 100KBPS

1V2

1V8

3V3

350MA TYPICAL

5MA TYPICAL

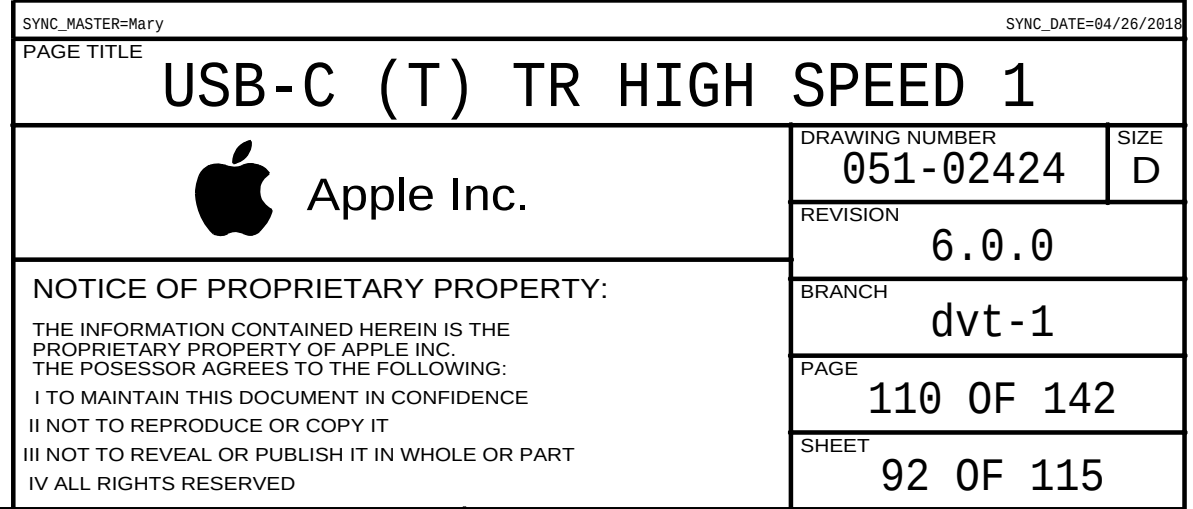
50MA TYPICAL

BOM_COST_GROUP=HDMI

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DRAWING NUMBER		051-02424	
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BRANCH		dvt-1	
PAGE		108 OF 142	
SHEET		90 OF 115	

D

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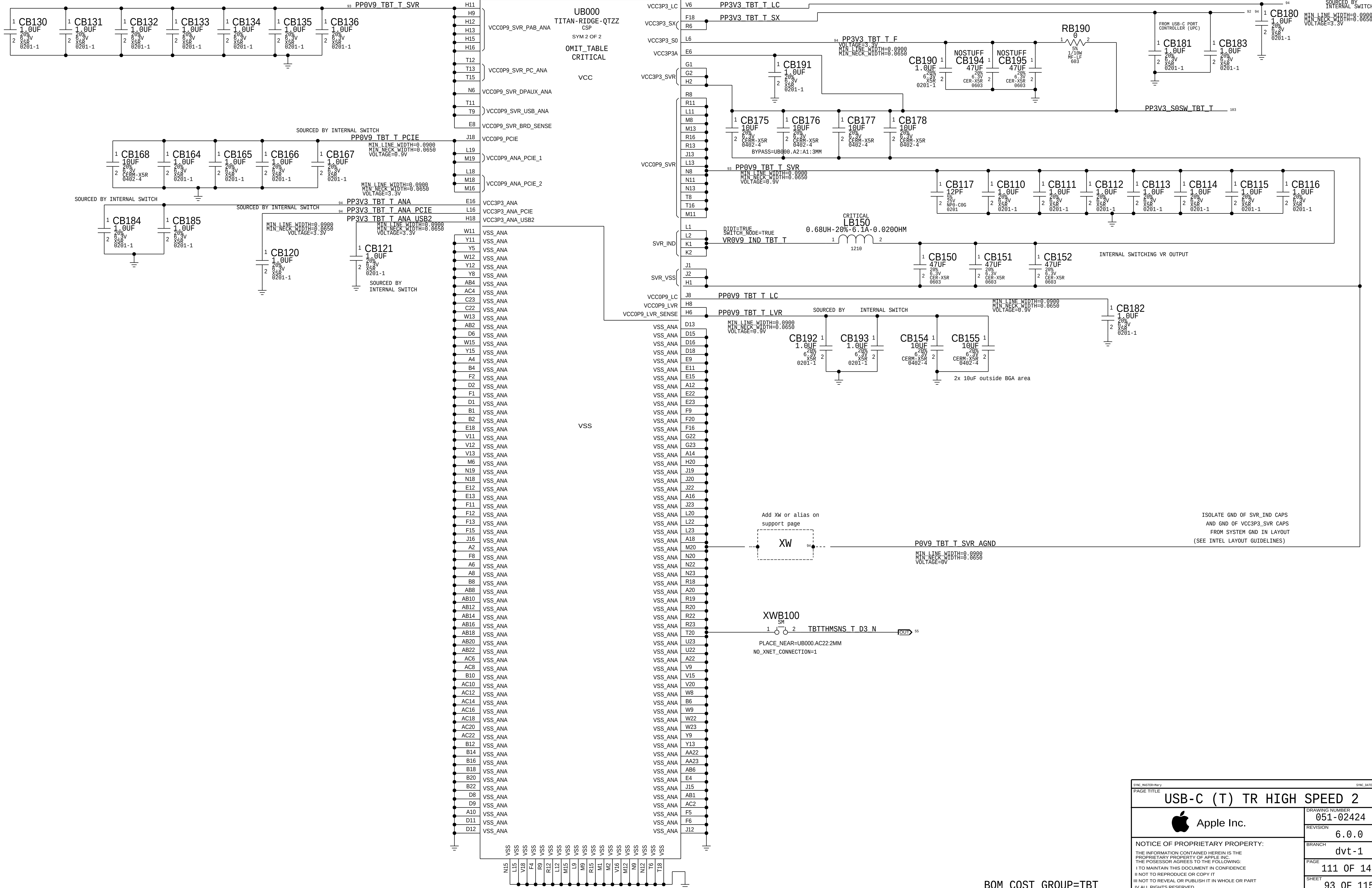
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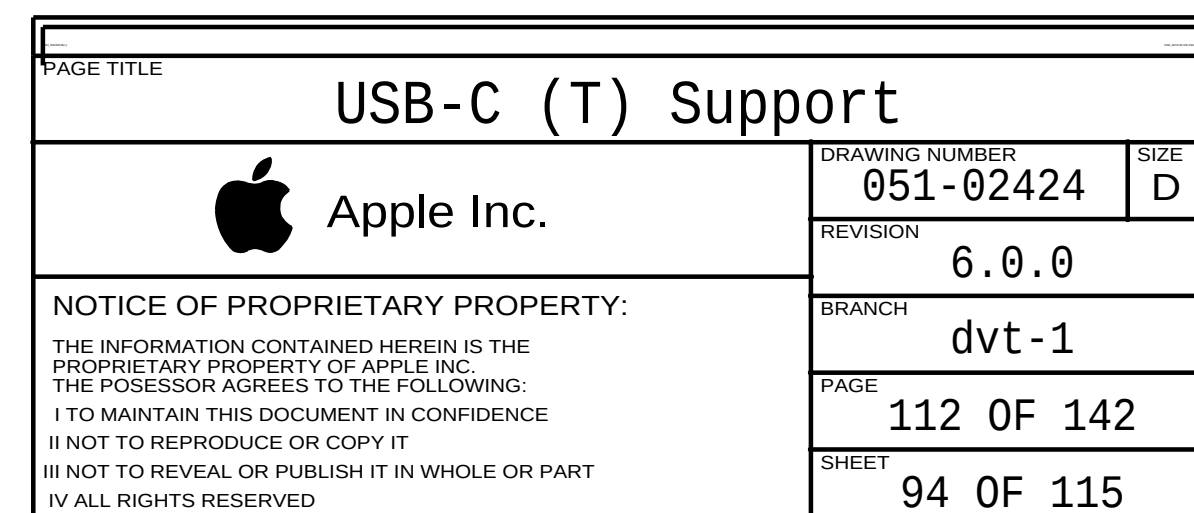
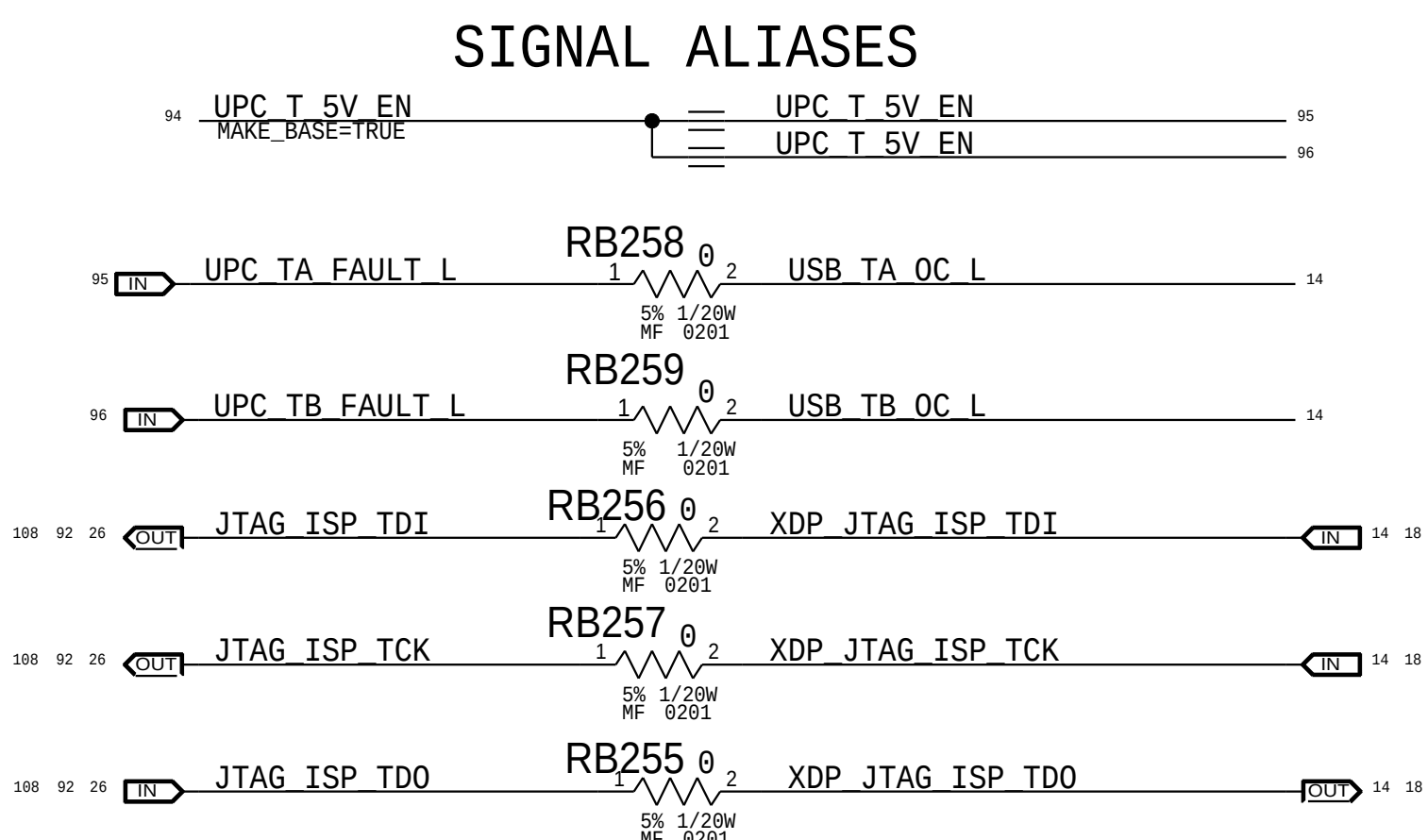
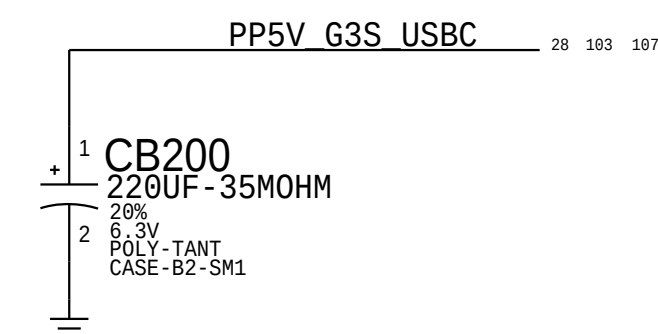
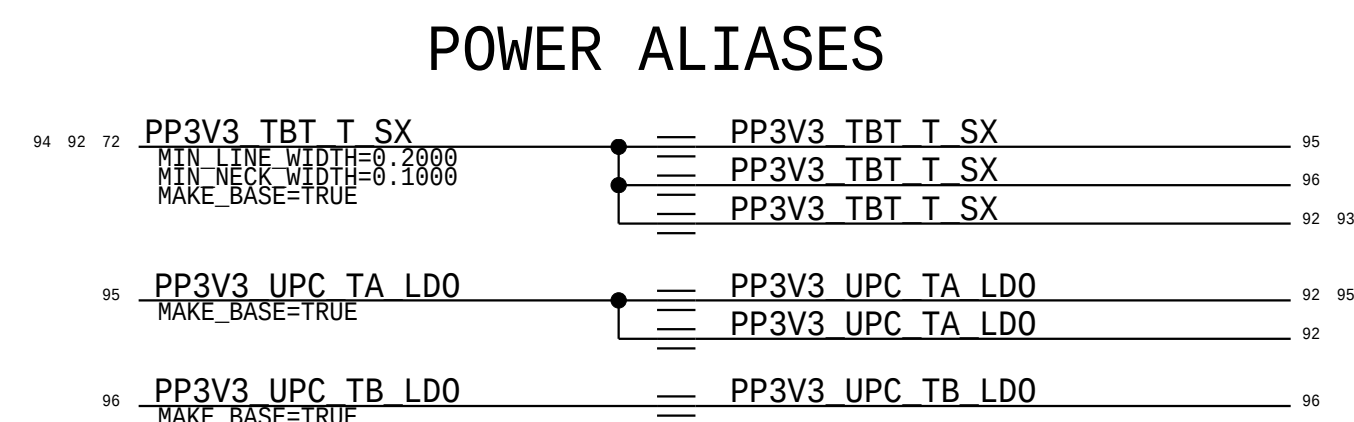
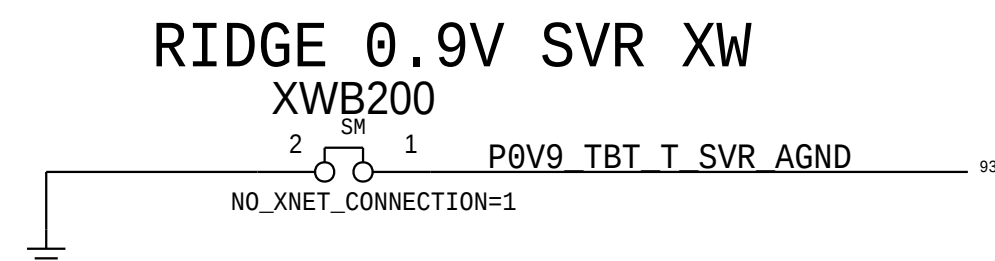
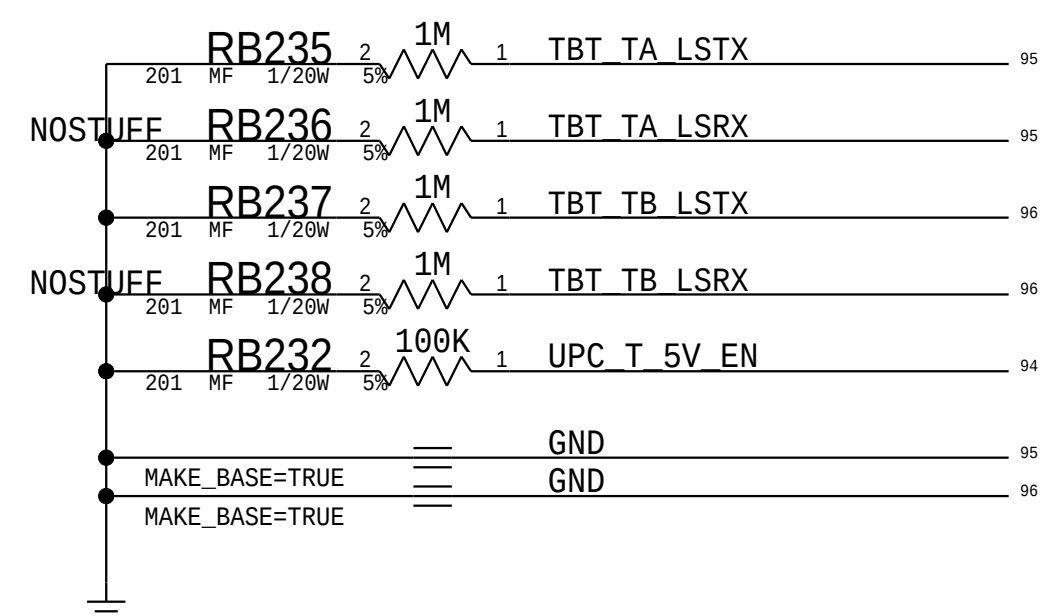
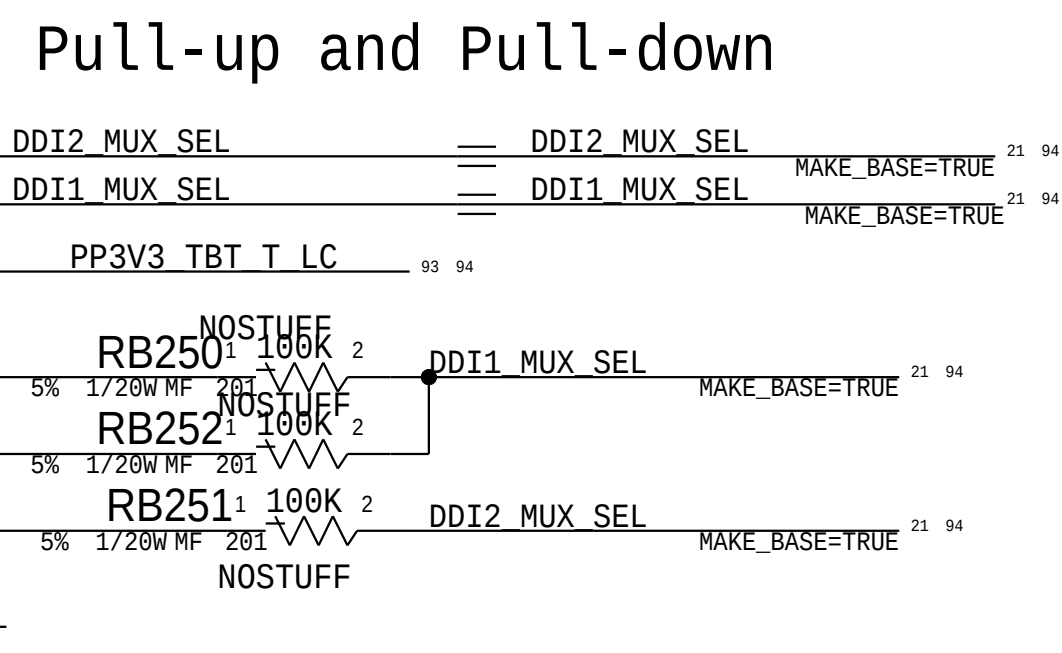
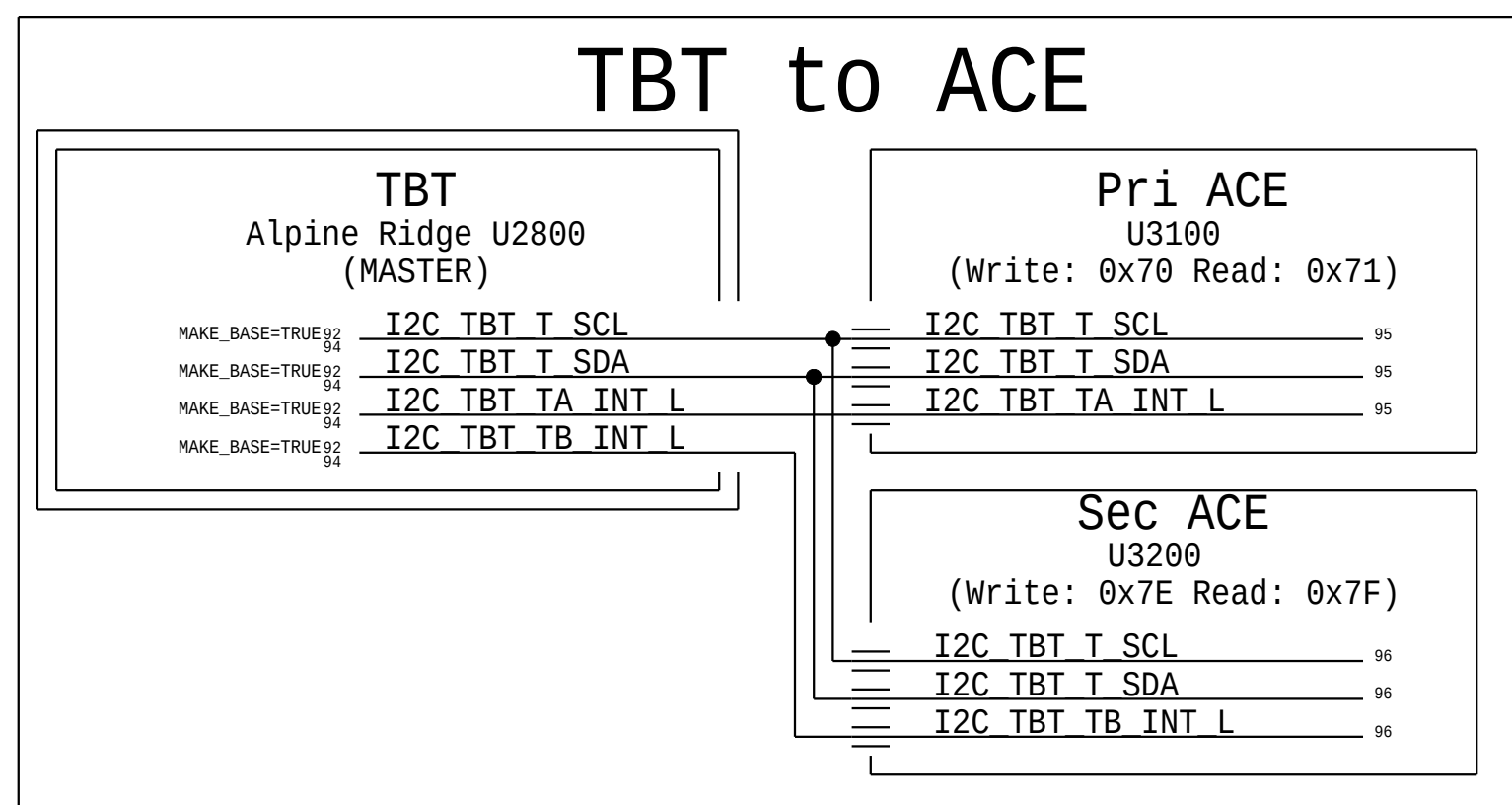
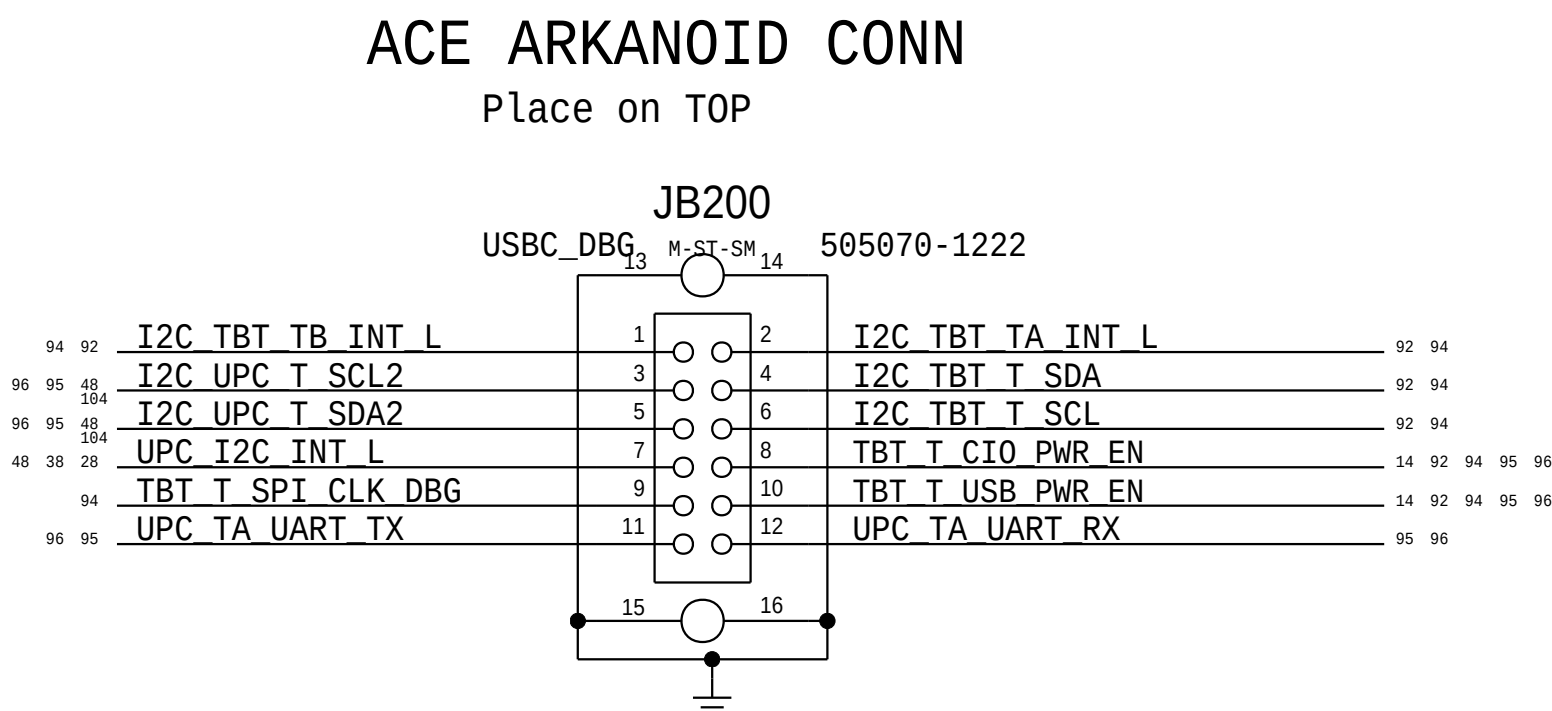
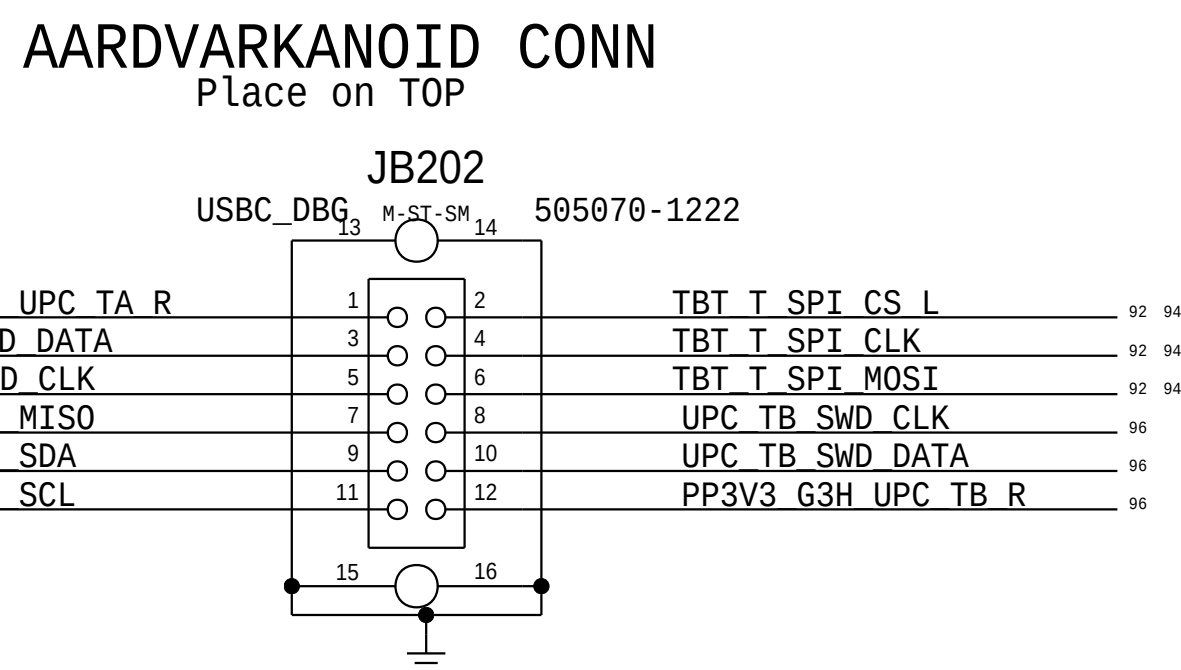
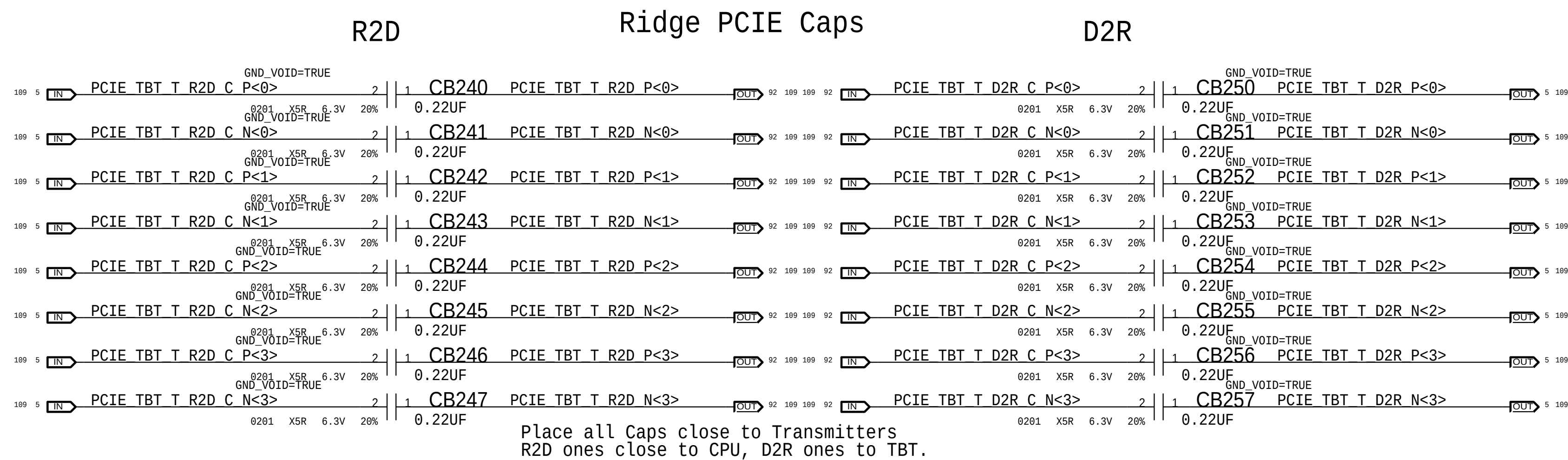
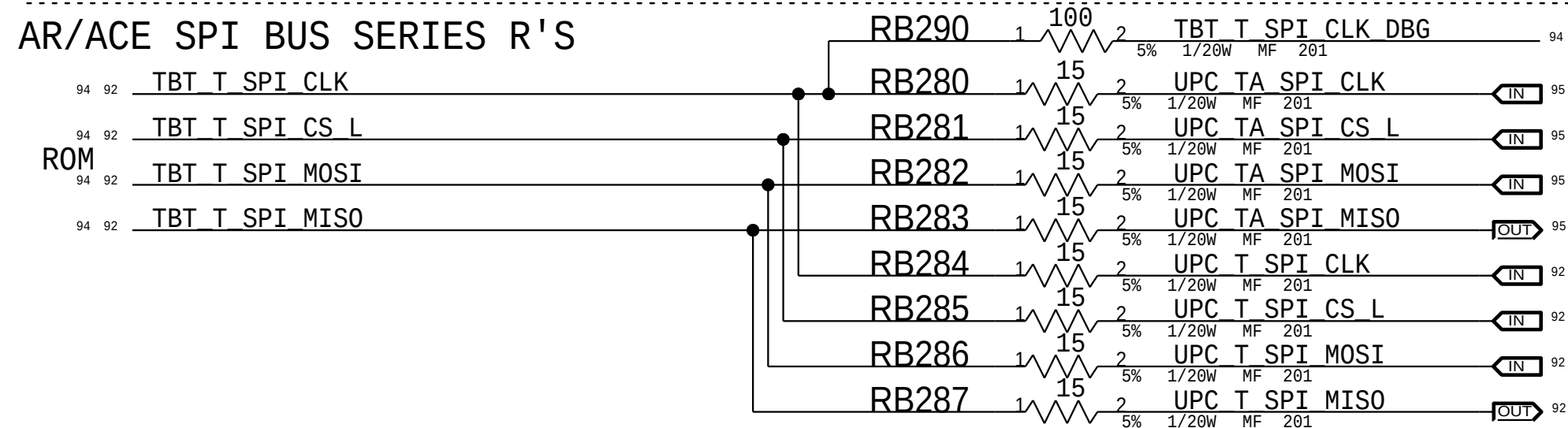
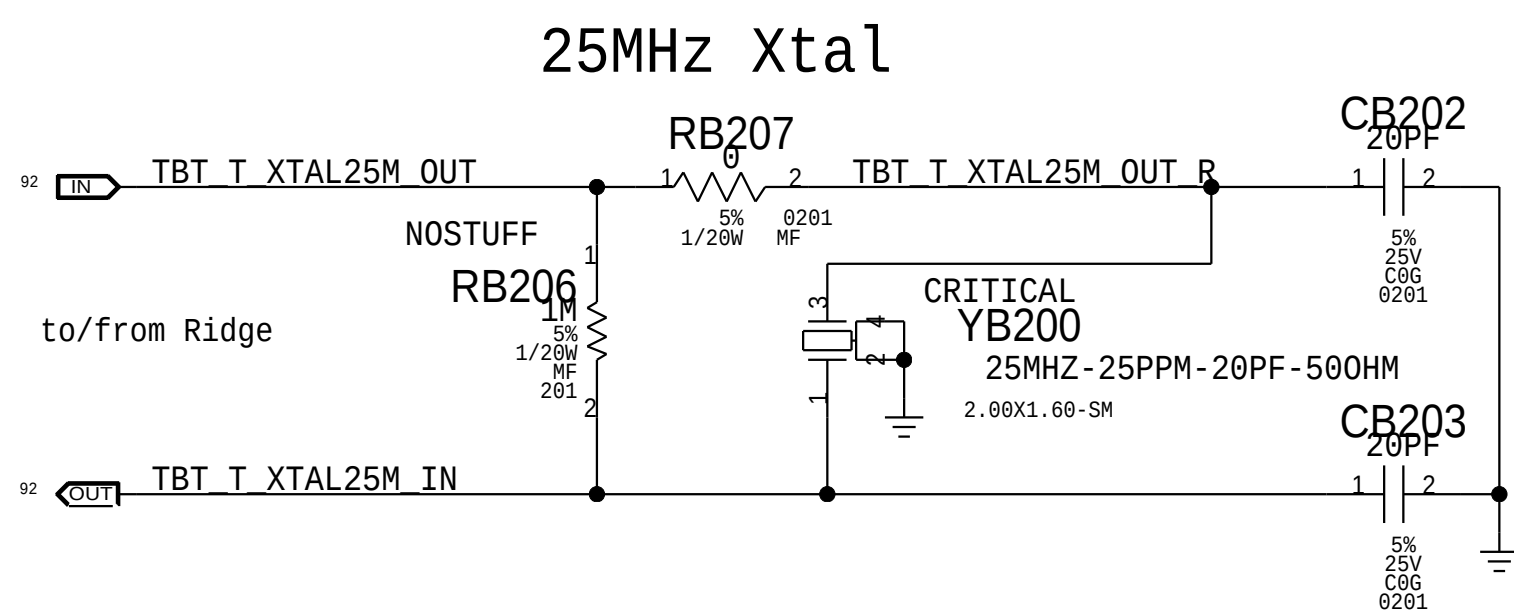
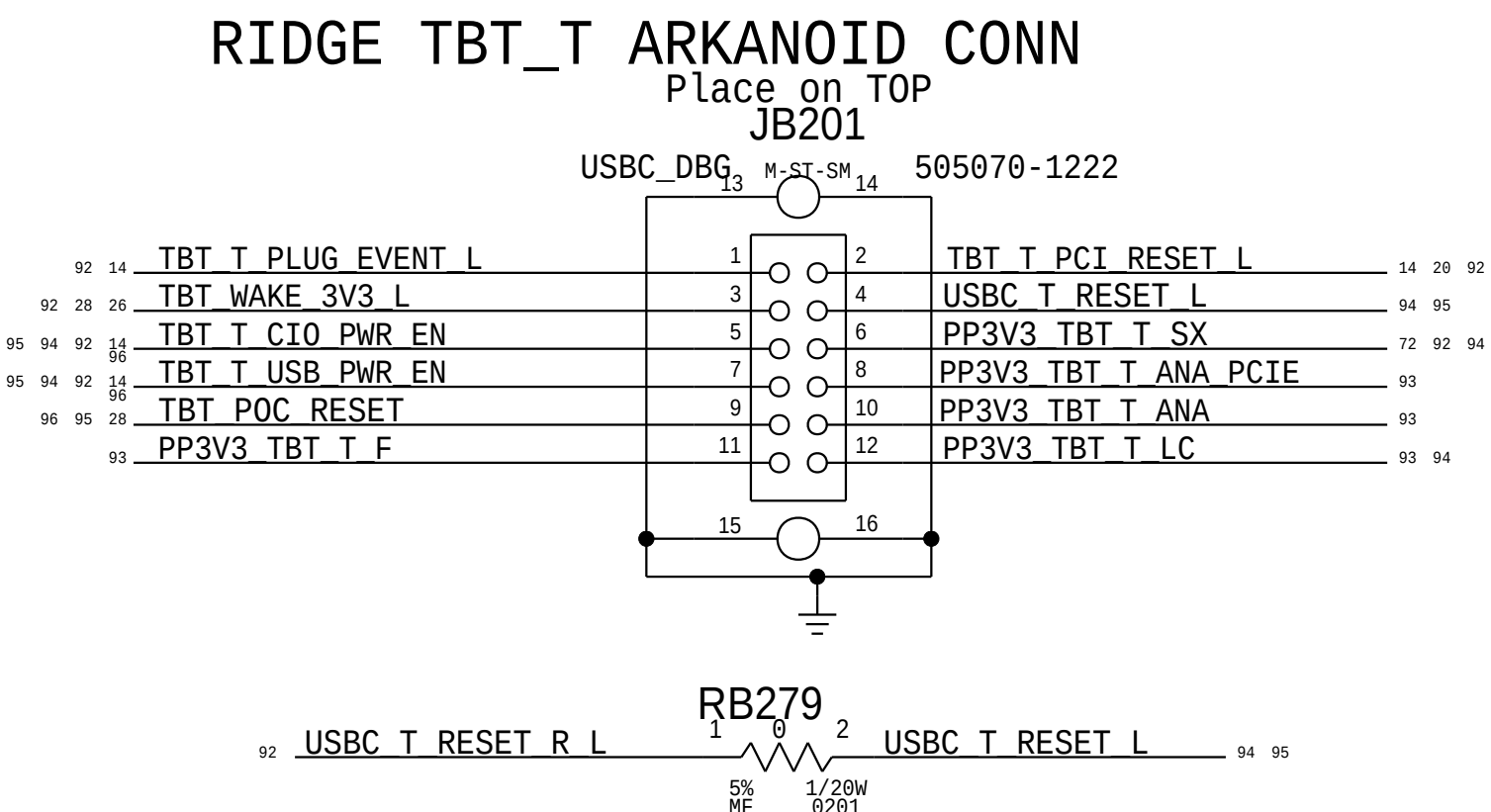
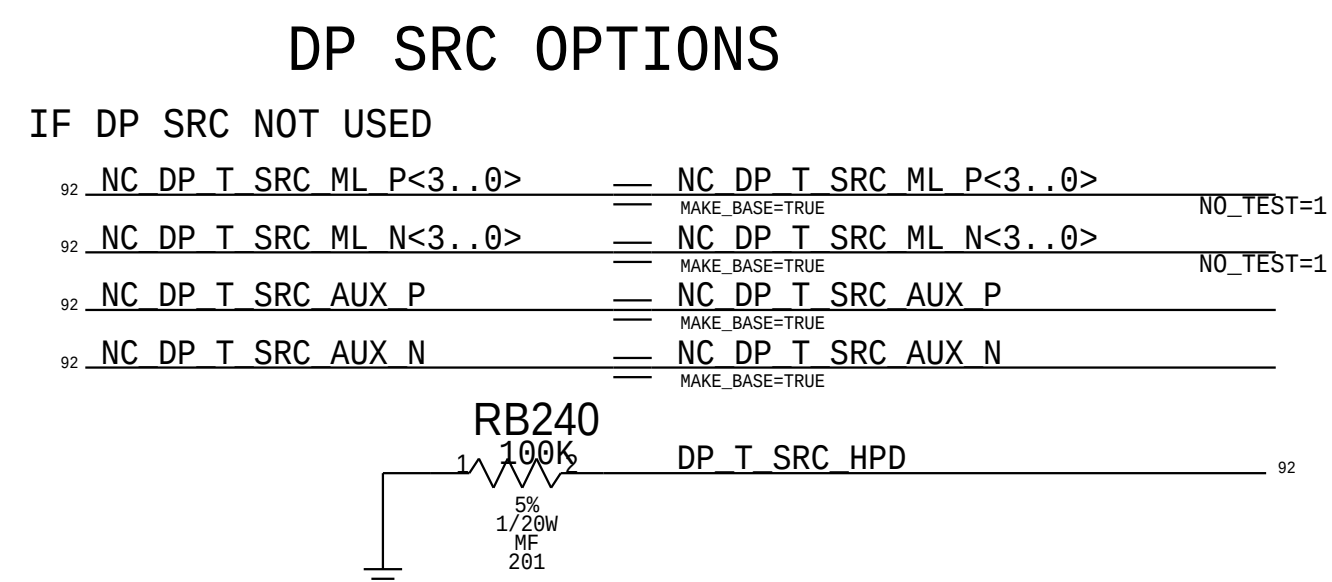
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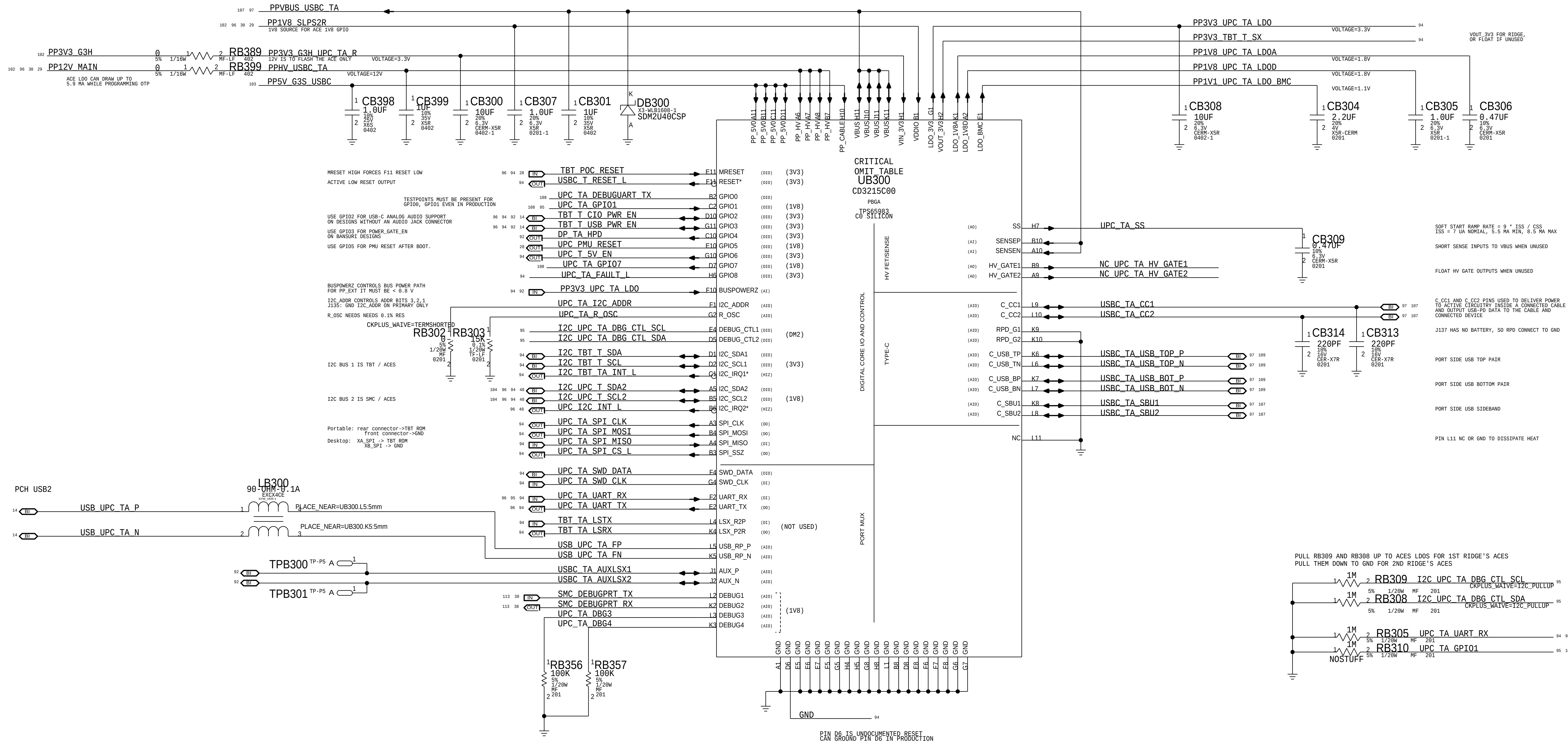
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Apple Inc.			BRANCH			dvt-1		
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T PRIMARY ACE USB-C PORT CONTROLLER (UPC)

J137 USBC SUPPORTS 5V @ 3A
PP12V IS FOR PROGRAMMING ACE ONLY



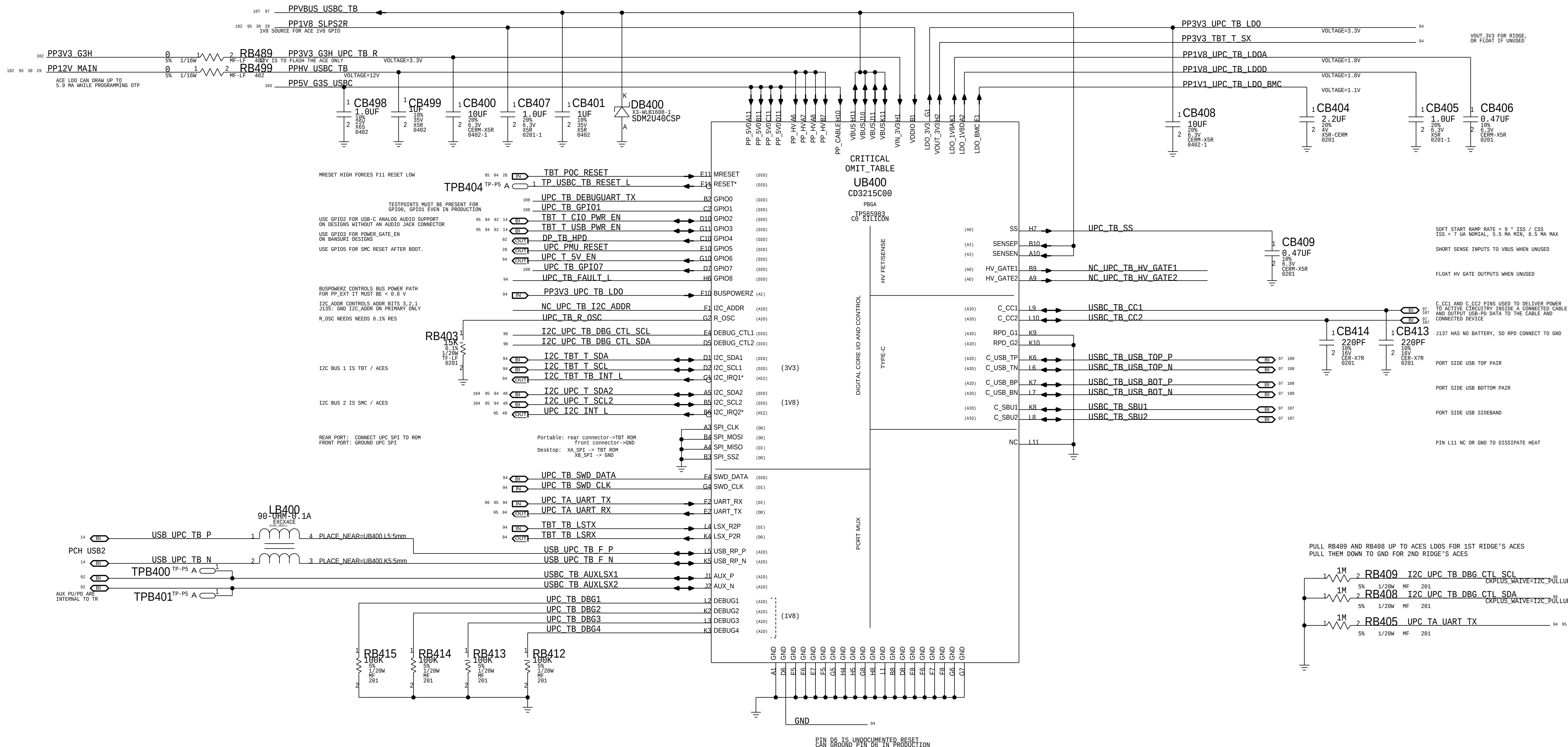
NOTE:
 DEBUG MUX IS CONNECTED TO
 PRIMARY ACE OF 1ST RIDGE (XA ACE)
 R3752-R3755, R3790-R3792 ARE OPTIONAL

BOM_COST_GROUP=USB-C

04/26/2018		Mary	
NOT SYNCED TO OTHER PLATFORMS			
PAGE TITLE			
SB-C (T) PORT CONTROLLER A			
 Apple Inc.		DRAWING NUMBER	
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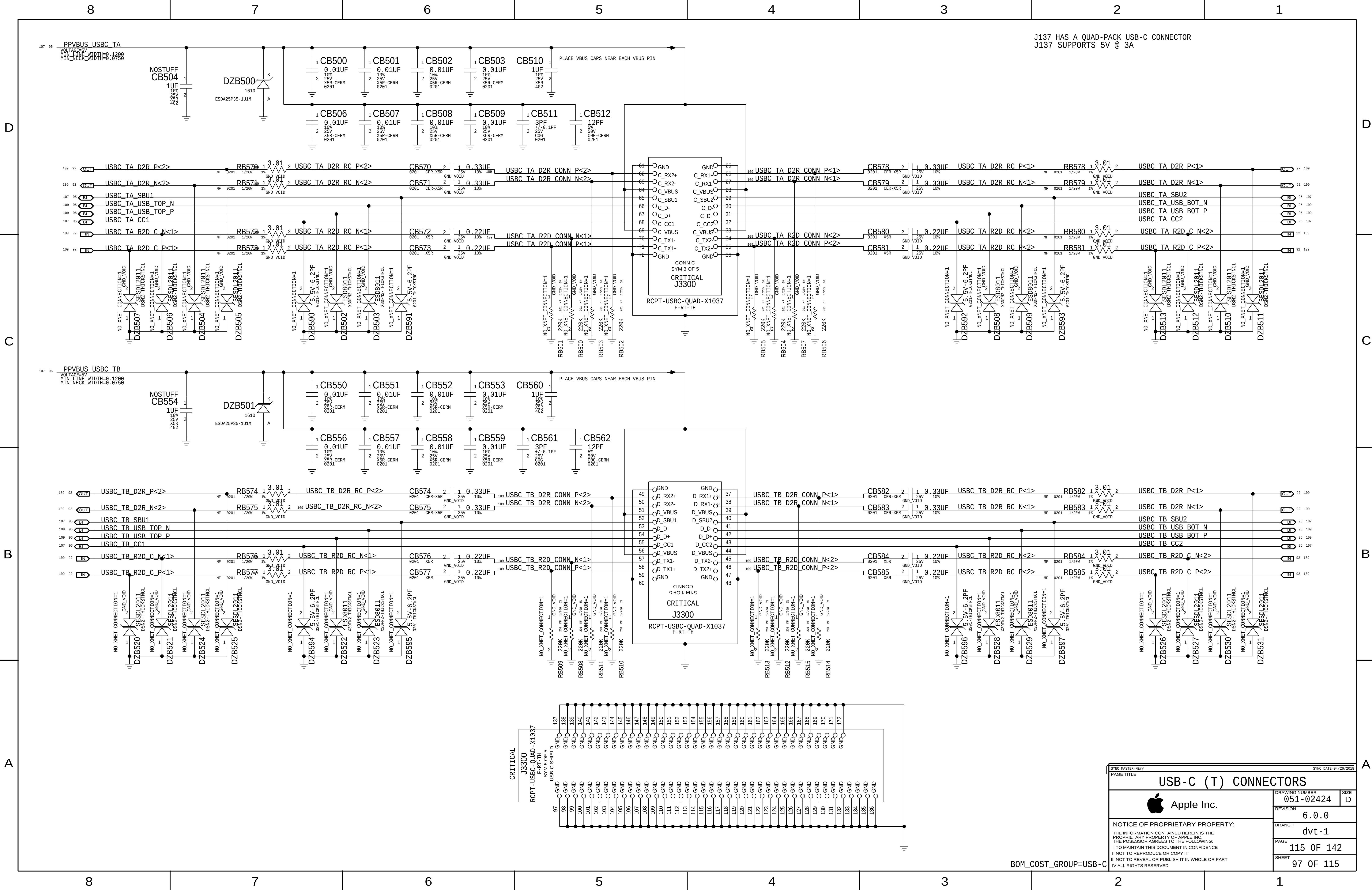
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T SECONDARY ACE USB-C PORT CONTROLLER (UPC)



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 Apple Inc.		BRANCH dvt-1		
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USB-C (T) CONNECTORS		
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	BRANCH	dvt-1
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8		7		6		5		4		3		2		1			
D																D	
C		Place Holder														C	
B																B	
A																A	
8		7		6		5		4		3		2		1			

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dvt-1

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SIZE
D

EMPTY

SYNC_MASTER:PRIV

SYNC_DATE:06/07/2018

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			SHEET			98 OF 115		

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051-02424

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BRANCH
dvt-1

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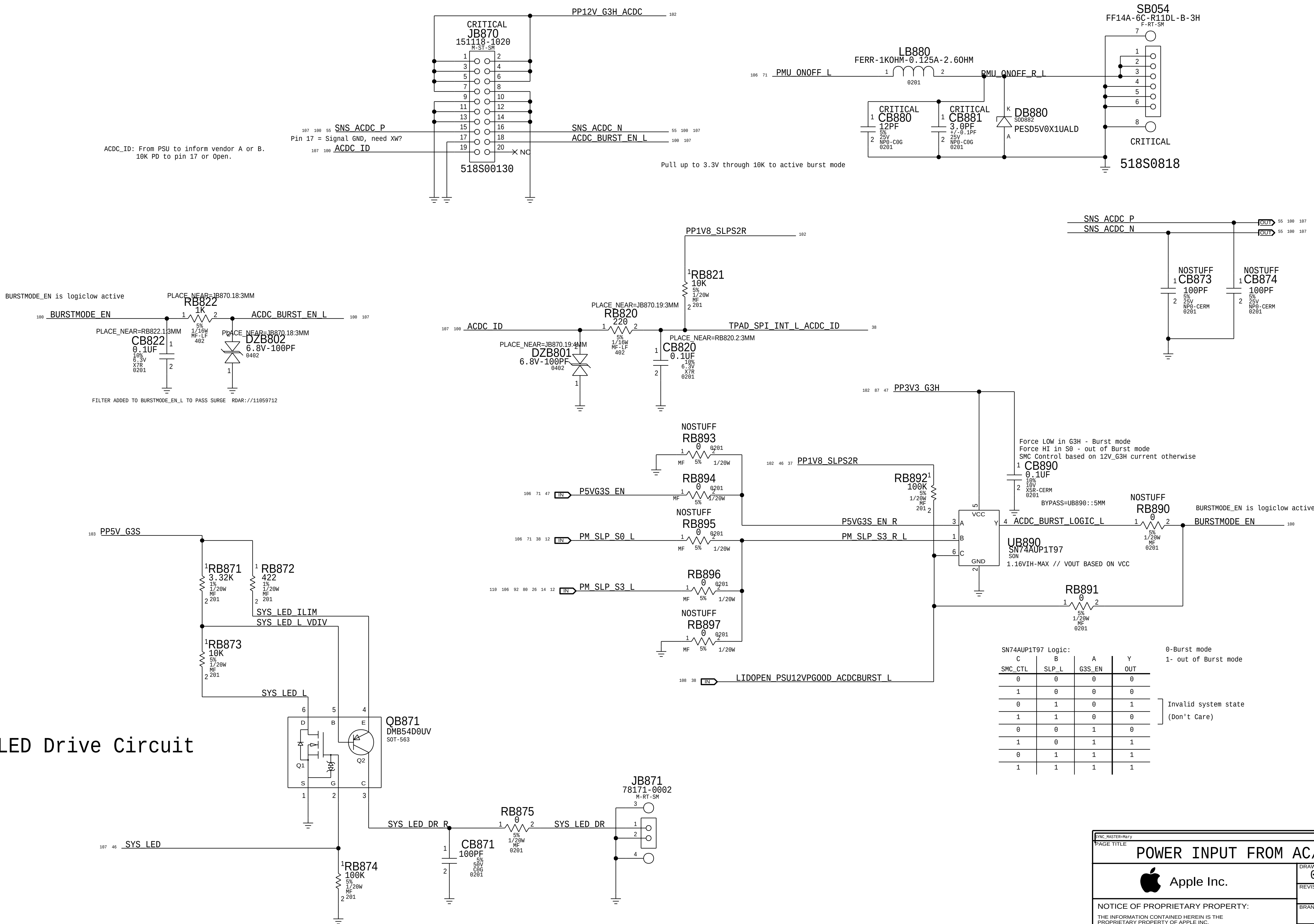
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BOM_COST_GROUP=DEBUG

MLB Power button connector

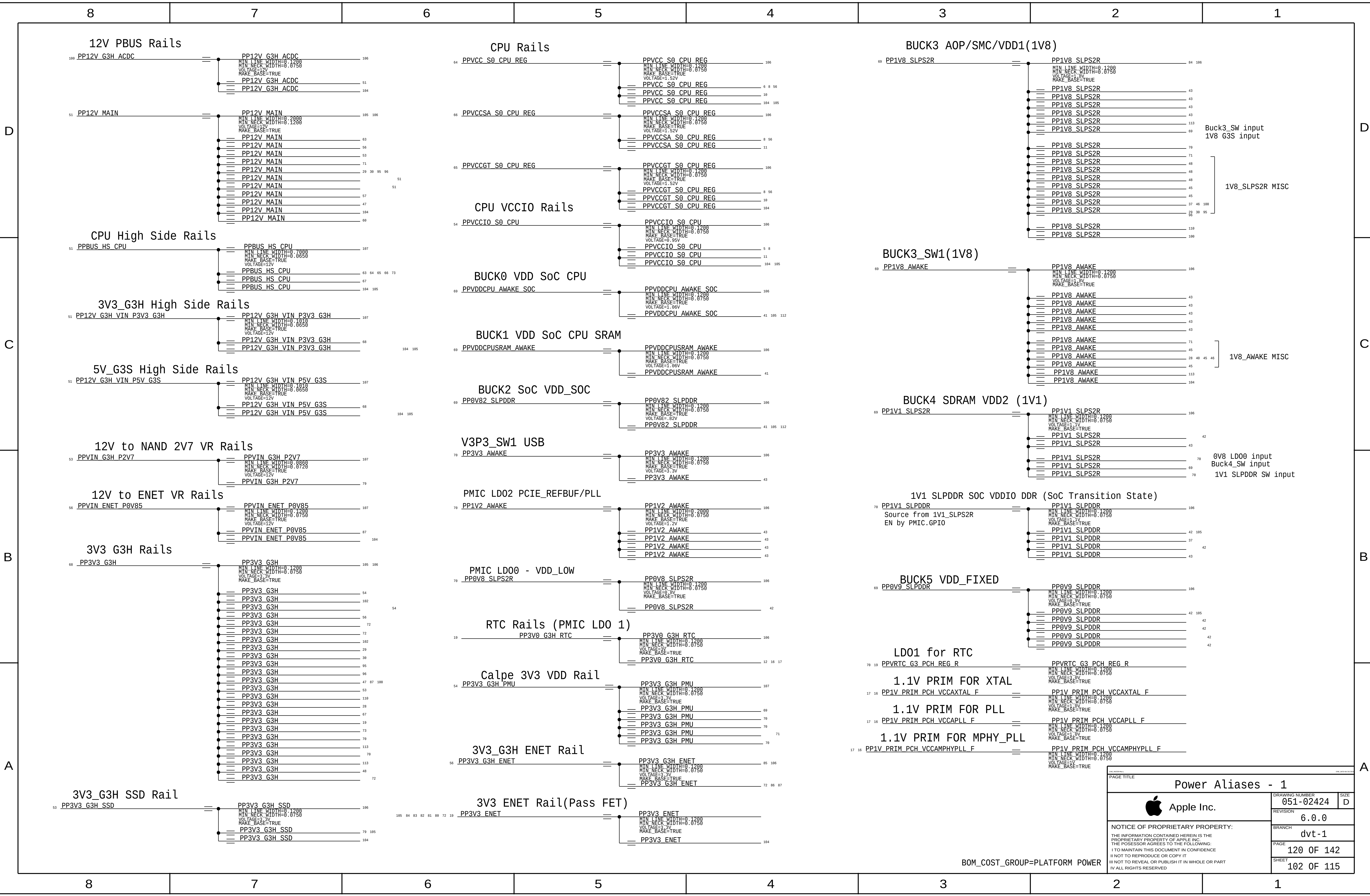
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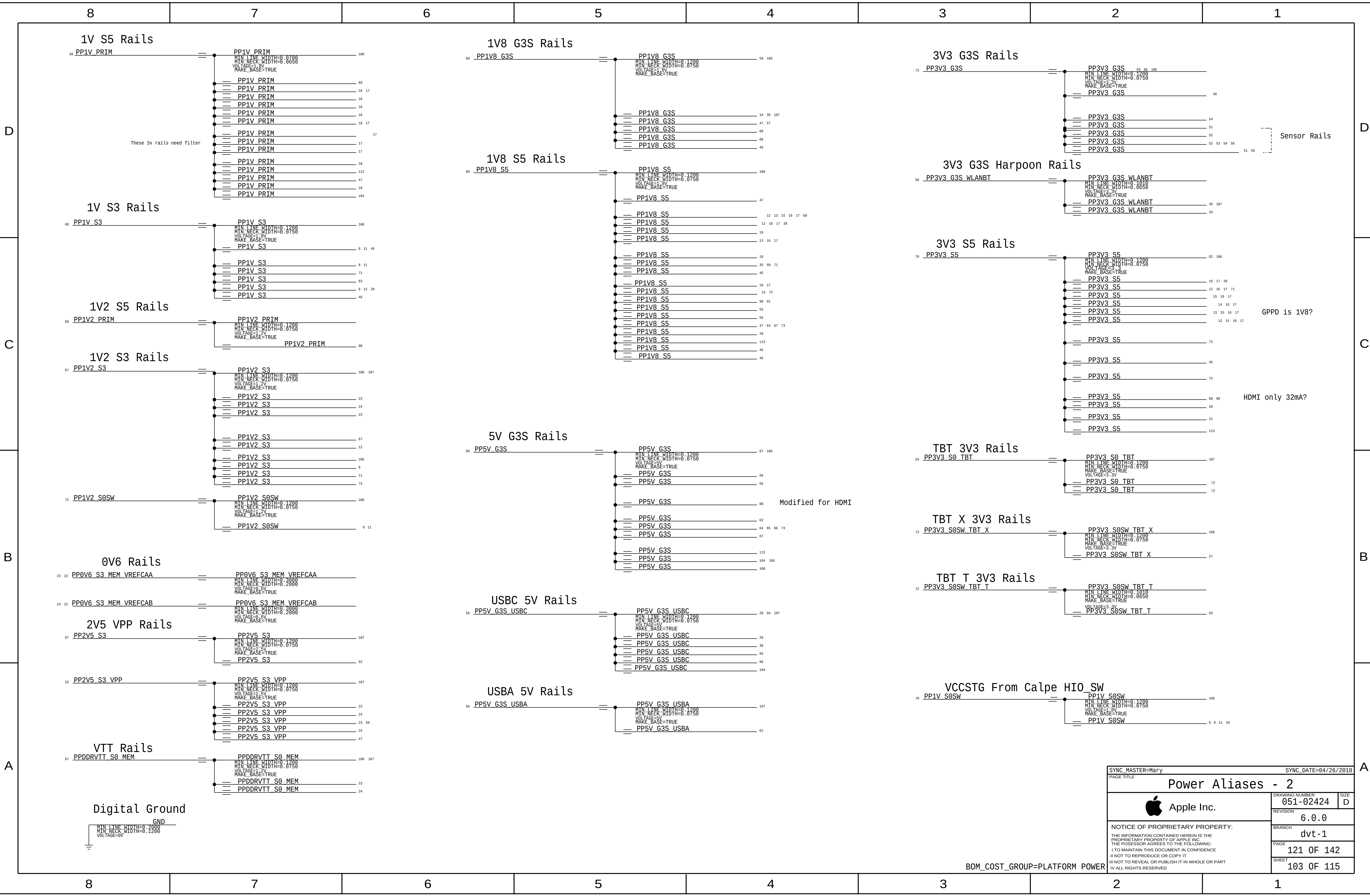


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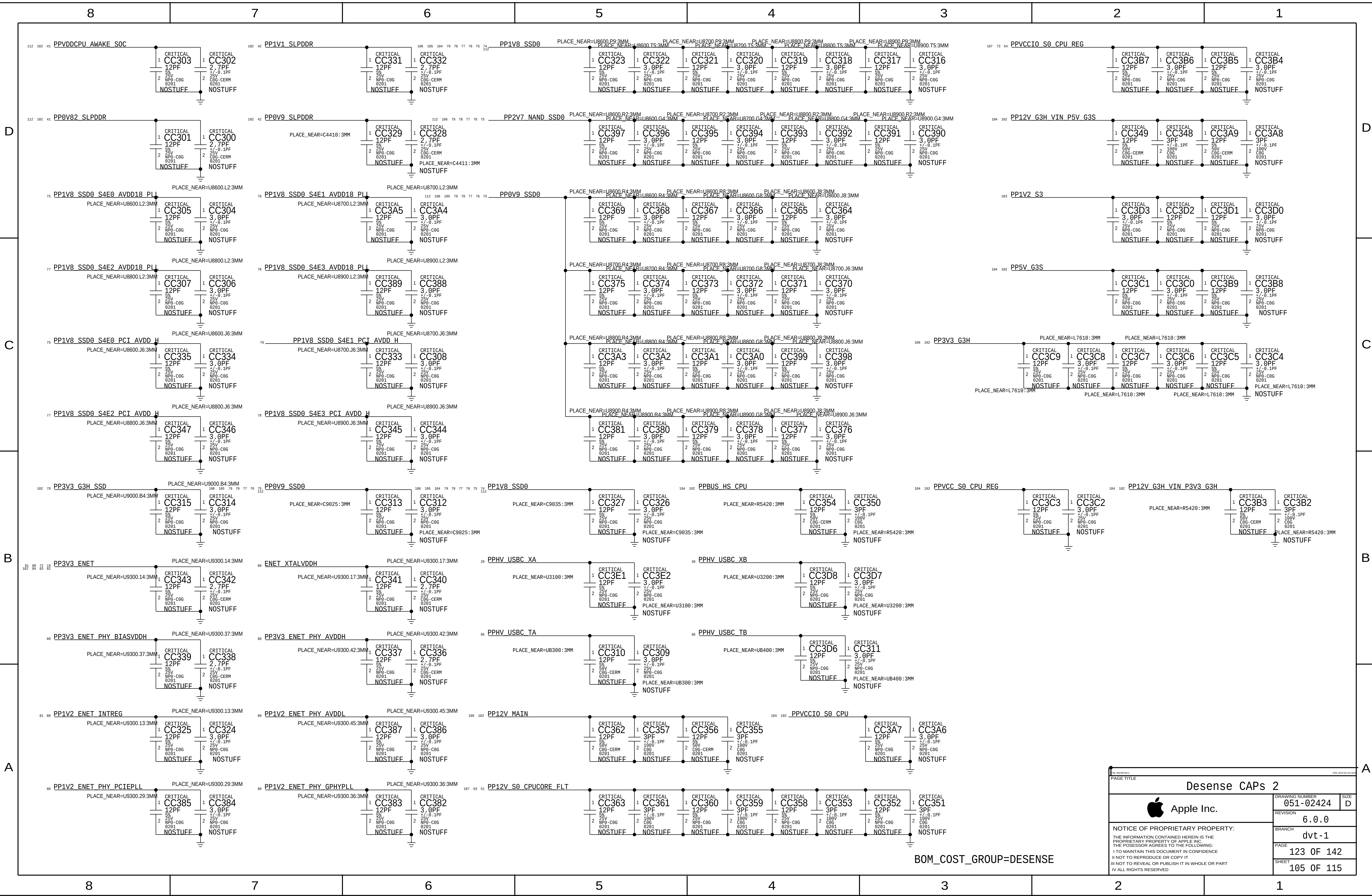
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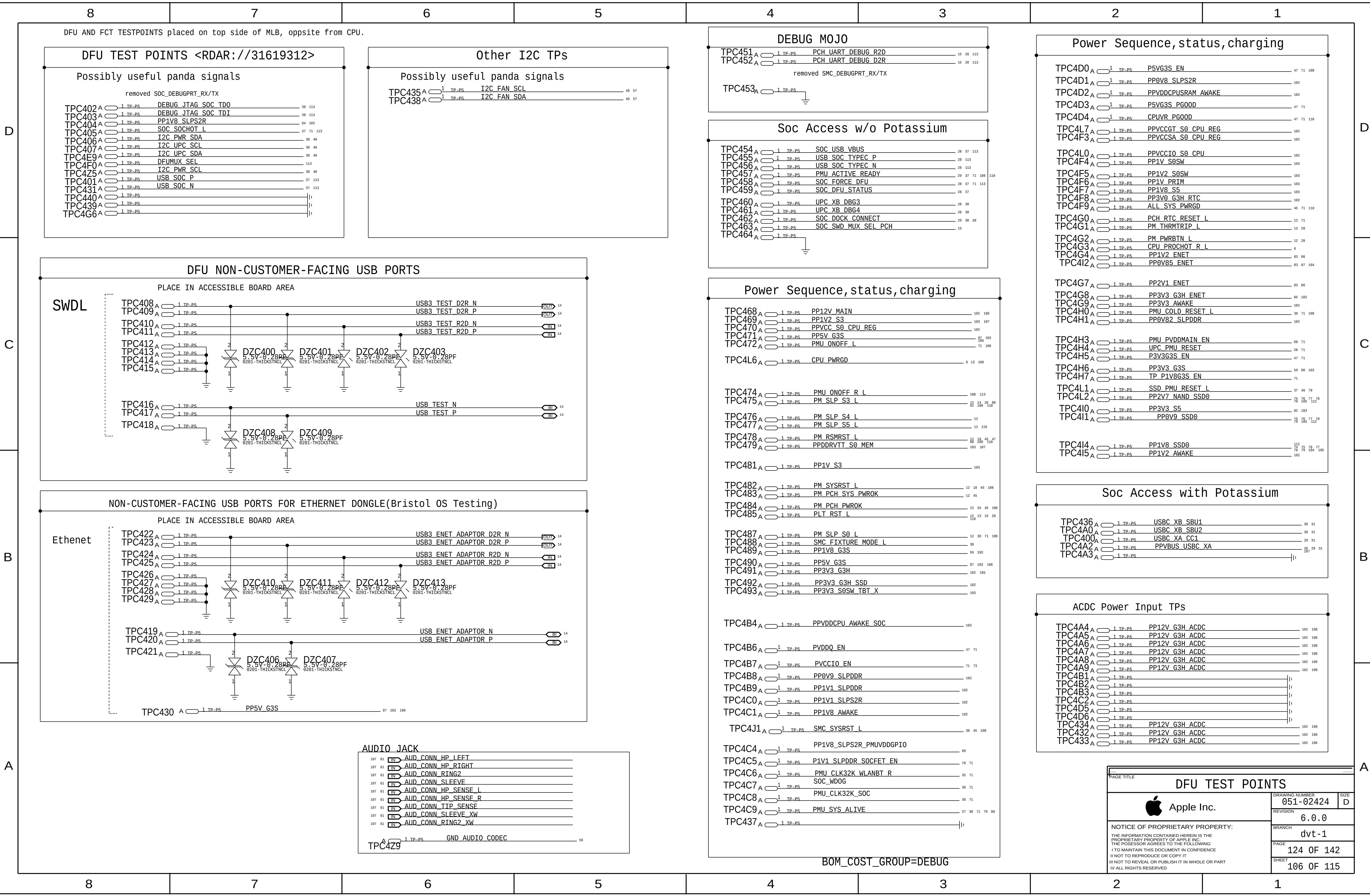


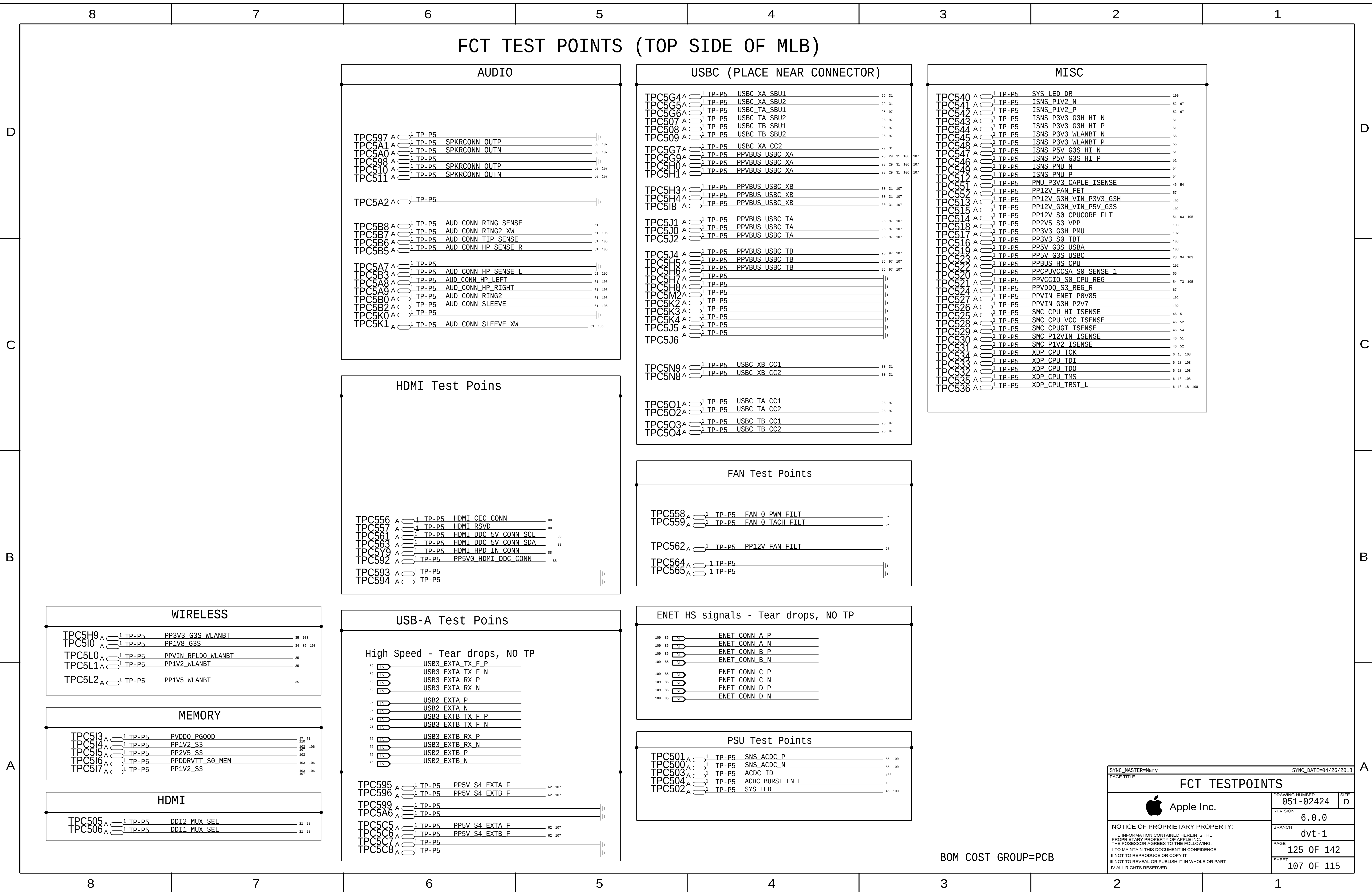


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BOM_COST_GROUP=PLATFORM POWER







ICT TEST POINTS, ICT BOUNDARY SCAN TESTPOINTS

CPU XDP and PCH Test-Points

187	18	6	IN	XDP_CPU_TCK	
18	13	IN	XDP_PCH_TCK		
187	18	6	IN	XDP_CPU_TDI	
187	18	6	IN	XDP_CPU_TDO	
187	18	6	IN	XDP_CPU_TRST_L	
187	18	6	IN	XDP_CPU_TMS	
18	13	IN	XDP_PCH_TMS		
18	13	IN	XDP_PCH_TDI		
18	13	IN	XDP_PCH_TDO		
18	13	6	IN	XDP_CPU_PREQ_L	
18	13	6	IN	XDP_CPU_PRDY_L	
47	45	18	12	PM_RSMRST_L	
110	106	88	IN	PM_PCH_PWROK	
106	45	34	12	IN	PM_SYSRST_L
106	45	18	12	IN	CPU_CFG<3>
18	6	IN			
18	13	IN	PCH_JTAGX		
18	13	IN	PCH_ITP_PMODE		

H9M BOUNDARY SCAN TESTPOINTS ON FCT TESTPOINT PAGE

OTHER ICT TESTPONTs

ACE

26	15	BI	JTAG_TBT_X_TMS	
94	92	26	BI	JTAG_ISP_TDI
94	92	26	BI	JTAG_ISP_TCK
94	92	26	BI	JTAG_ISP_TDO
188	26	BI	TBT_X_TEST_EN	

WLAN

38	36	BI	WLAN_JTAG_TMS	
38	36	BI	WLAN_JTAG_TCK	
38	36	BI	WLAN_JTAG_TDI	
38	36	BI	WLAN_JTAG_TDO	
38	36	BI	WLAN_JTAG_TRST_L	

92	15	BI	JTAG_TBT_T_TMS	
----	----	----	----------------	--

38	BI	TP_JTAG_SOC_TRST_L	
----	----	--------------------	--

45	38	BI	SMC_PCH_SYS_PWROK	
45	38	BI	SMC_PCH_PWROK	
106	45	38	BI	SMC_SYSRST_L

106	13	6	BI	CPU_PWRGD
106	71	38	BI	PMU_COLD_RESET_I
	38		BI	SOC_JTAG_SEL
	37		BI	SOC_TESTMODE

36	35	BI	WLAN_JTAG_SEL	
95	BI	TBT_T_TEST_EN		
188	26	BI	TBT_X_TEST_EN	

TP of HDMI (EE tests)

TPC646	TP	1	DP2HDMI PMU CEC IRQ	98
TPC641	TP	1	DP2HDMI 1.5G FILT_EN	88 98

TP of Calpe PMIC (EE tests)

TPC642	TP	1	TP PP1V8 PMU BUCK3SW5	69
TPC643	TP	1	TP PP1V1 PMU BUCK4SW1	69
TPC644	TP	1	TP PP1V2 PMU BUCK6SW1	69 108
TPC645	TP	1	TP PP1V PMU BUCK8SW1	69 108
TPC645	TP	1	SOC_VDDCPU_SENSE	41 71
TPC676	TP	1	TP P3V3G3W_EN	71
TPC664	TP	1	TP P3V3G3W_PG00D	71
TPC665	TP	1	TP SE_PWR_EN	71
TPC667	TP	1	TP P3V3_S5_EN	71
TPC668	TP	1	TP PVCCOPTIOEDRAM_P2V7NAND_PG00D	71
TPC670	TP	1	TP PEARL_P2V7NAND_EN	71
TPC671	TP	1	TP NAND_DISCHARGE_HDD_PWR_EN	71
TPC672	TP	1	TP NAND_RESET_L_SD_PWR_EN	71
TPC673	TP	1	ENET_PWR_EN	71 72 85 86 87
TPC674	TP	1	TBT_PWR_EN	71 72
TPC675	TP	1	UPC_TA_GPI07	95
TPC679	TP	1	BT_PWR_EN	35 36 71
TPC635	TP	1	WLAN_PWR_EN	35 36 71
TPC669	TP	1	UPC_TB_GPI07	96
TPC680	TP	1	TP PP1V2 PMU BUCK6SW1	69 108
TPC681	TP	1	TP PP1V PMU BUCK8SW1	69 108
TPC682	TP	1	FAN_PWR_EN	57 71
TPC677	TP	1	PVCCIO_PG00D	47 71
TPC683	TP	1	P3V3MAIN_PG00D	47 71 110
TPC684	TP	1	PVCCPLL0C_EN	71 72
TPC636	TP	1		

Test Points for Calpe breakout

TPC647	TP	1	PMU_VCCIO_ISENSE	46 71 108
TPC648	TP	1	PMU_VCCIO_ISENSE	46 71 108
TPC649	TP	1	PMU_VCCIO_ISENSE	46 71 108
TPC650	TP	1	PMU_P5V_G3S_HI_ISENSE	46 71
TPC651	TP	1	PMU_CPUSA_VSENSE	46 71
TPC651	TP	1	PMU_P5V_USBC_ISENSE	46 71
TPC652	TP	1	PMU_P5V_USBA_ISENSE	46 71
TPC653	TP	1	TP_PMU_AMUX_AY	71
TPC654	TP	1		
TPC655	TP	1	PMU_P3V3_TBT_ISENSE	46 71
TPC656	TP	1	PMU_P3V3_SSD0_ISENSE	46 71
TPC657	TP	1	PMU_PBUS_SSD0_ISENSE	46 71
TPC658	TP	1	PMU_P3V3_G3H_HI_ISENSE	46 71
TPC659	TP	1	PMU_P3V3_WLANBT_ISENSE	46 71
TPC660	TP	1	PMU_P3V3_ENET_ISENSE	46 71
TPC661	TP	1	PMU_P12V_ENET_ISENSE	46 71
TPC662	TP	1	TP_PMU_AMUX_B7	71
TPC663	TP	1	TP_PMU_AMUX_BY	71

TP of H9ML (EE tests)

TPC678	TP	1	PCC_EVENT	38
TPC686	TP	1	TP_SMC_DEBUG	46
TPC687	TP	1	WLAN_CONTEXT_A	35 36 38
TPC688	TP	1	WLAN_CONTEXT_B	35 36 38
TP of CONTEXT_A/B can be used for AOP_UART_TX/RX for debug				
TPC600	TP	1	LIDOPEN_PSU12VPG00D_ACDGBURST_L	38 108
TPC6G8	TP	1	SWD_SOC_SWDIO	38 46 113
TPC6G9	TP	1	SWD_SOC_SWCLK	38 46 113

EE TESTS PCH

TPC631	TP	1	PCIE_CLK100M_WLAN_PCH_N	12 28
TPC632	TP	1	PCIE_CLK100M_WLAN_PCH_P	12 28
TPC633	TP	1	PCIE_CLK100M_SOC_N	12 48
TPC634	TP	1	PCIE_CLK100M_SOC_P	12 48
TPC637	TP	1	PCIE_CLK100M_TBT_X_N	12 26
TPC638	TP	1	PCIE_CLK100M_TBT_X_P	12 26
TPC639	TP	1	PCIE_CLK100M_TBT_T_N	12 92
TPC640	TP	1	PCIE_CLK100M_TBT_T_P	12 92

TPC690	TP	1	SPI_PCHROM_MOST	13 18
TPC691	TP	1	TP_PCH_CLK32K_SUS	12
TPC692	TP	1	SPI_I0<2>	13 18
TPC693	TP	1	SPI_I0<3>	13

TP Debug ACE Nets (EE tests)

TPC622	TP	1	UPC_XA_DEBUGUART_TX	29
TPC623	TP	1	PMU_ACTIVE_READY	29 37 71 106 110
TPC624	TP	1	UPC_XB_DEBUGUART_TX	30
TPC625	TP	1	UPC_XB_GPI01	30
Test Points for Ace XA/XB GPI00/1, must have in production.				
TPC626	TP	1	UPC_TA_DEBUGUART_TX	95
TPC627	TP	1	UPC_TA_GPI01	95
TPC628	TP	1	UPC_TB_DEBUGUART_TX	96
TPC629	TP	1	UPC_TB_GPI01	96
Test Points for Ace XA/XB GPI00/1, must have in production.				

SENSOR TPs

TPC694	TP	1	SNS_P12VG3H_P	51
TPC695	TP	1	SNS_P12VG3H_N	51
TPC6C9	TP	1	ISNS_HS_COMPUTING_P	51
TPC6C8	TP	1	ISNS_HS_COMPUTING_N	51
TPC6D2	TP	1	CPUVR_ISNS_R_P	52
TPC6D1	TP	1	CPUVR_ISNS_R_N	52
TPC6D0	TP	1	ISNS_P3V3_G3W_SSD0_P	52
TPC6D3	TP	1	ISNS_P3V3_G3W_SSD0_N	53
TPC6D6	TP	1	ISNS_PPBUS_MAIN_SSD0_P	53
TPC6D5	TP	1	ISNS_PPBUS_MAIN_SSD0_N	53
TPC6D4	TP	1	ISNS_TBT_P	54
TPC6D8	TP	1	ISNS_TBT_N	54
TPC6D7	TP	1	ISNS_CPUVCCIO_P	54
TPC6E0	TP	1	ISNS_CPUVCCIO_N	54
TPC6E1	TP	1	CPUGT_ISNS_P	54
TPC6D9	TP	1	CPUGT_ISNS_N	54
TPC6E4	TP	1	ISNS_P5V_USBC_P	56
TPC6E3	TP	1	ISNS_P5V_USBC_N	56
TPC6E2	TP	1	ISNS_P5V_USBA_P	56
TPC6E6	TP	1	ISNS_P5V_USBA_N	56
TPC6E5	TP	1	ISNS_P3V3_ENET_P	56
TPC6E8	TP	1	ISNS_P3V3_ENET_N	56
TPC6E9	TP	1	ISNS_P12V_ENET_P	56
TPC6E7	TP	1	ISNS_P12V_ENET_N	56
TPC6F0	TP	1	PMU_CPUSA_VSENSE	46 56
TPC6F1	TP	1	SMC_CPU_VCC_VSENSE	46 56
TPC6F4	TP	1	SMC_CPUGT_VSENSE	46 56
TPC6F2	TP	1	TSNS_T1_DX1_P	55
TPC6F3	TP	1	TSNS_T1_DX2_P	55
TPC6F6	TP	1	TSNS_T1_DX3_P	55
TPC6F7	TP	1	TSNS_T1_DX4_P	55
TPC6F5	TP	1	TSNS_T1_DX5_P	55
TPC6F8	TP	1	TSNS_T1_DX6_P	55
TPC6G0	TP	1	TSNS_T1_DX7_P	55
TPC6G1	TP	1	TSNS_T1_DX8_P	55
TPC6G2	TP	1	TSNS_T1_DN	55
TPC6G4	TP	1	TSNS_T2_DX1_P	55
TPC6G3	TP	1	TSNS_T2_DX4_P	55
TPC6G5	TP	1	TSNS_T2_DN	55
TPC6G6	TP	1	TBTHMSNS_X_D2_P	55
TPC6G7	TP	1	TBTHMSNS_X_D2_N	55
TPC6H1	TP	1	REG_VCCSA_ISNS_1_P	54 66
TPC6H0	TP	1	REG_VCCSA_ISNS_1_N	54 66
TPC6H3	TP	1	ISNS_2V5_S3_P	52
TPC6H2	TP	1	ISNS_2V5_S3_N	52
TPC6H5	TP	1	TBTHMSNS_T_D3_P	55
TPC6H4	TP	1	TBTHMSNS_T_D3_N	55

S4E

TPC619	TP	1	SSD0_CLKREQ00_L	48 46 75
TPC617	TP	1	SSD0_CLKREQ01_L	48 46 76
TPC618	TP	1	SSD0_CLKREQ02_L	48 46 77
TPC630	TP	1	SSD0_CLKREQ03_L	48 46 78
TPC614	TP	1	SSD0_S4E0_DROOP_L	75
TPC618	TP	1	SSD0_S4E1_DROOP_L	76
TPC611	TP	1	SSD0_S4E2_DROOP_L	77
TPC612	TP	1	SSD0_S4E3_DROOP_L	78
TPC685	TP	1	SSD0_OCARINA_VDD_LD0	79
TPC689	TP	1	SSD0_OCARINA_WP_L	75 76 77 78 79
TPC601	TP	1	SSD0_VR_P2V7_EN	79
TPC602	TP	1	SSD0_OCARINA_RESET_L	75 76 77 78 79
TPC603	TP	1	SSD0_VR_P2V7_PG00D	79
TPC604	TP	1	SSD0_OCARINA_TCAL	79
TPC606	TP	1	SSD0_OCARINA_PFN	75 76 77 78 79
TPC605	TP	1	SSD0_OCARINA_FORCE_EN	79
TPC607	TP	1	SSD0_OCARINA_VREF	79
TPC608	TP	1	SSD0_S4E_BOOT2	74 75 76 77 78
TPC609	TP	1	SSD0_S4E0_ZQ_C	75
TPC615	TP	1	SSD0_S4E0_ZQ_L	75
TPC610	TP	1	SSD0_S4E1_ZQ_C	76
TPC613	TP	1	SSD0_S4E1_ZQ_L	76
TPC620	TP	1	SSD0_S4E2_ZQ_C	77
TPC666	TP	1	SSD0_S4E2_ZQ_L	77
TPC621	TP	1	SSD0_S4E3_ZQ_C	78
TPC696	TP	1	SSD0_S4E3_ZQ_L	78
TPC697	TP	1	SSD0_S4E0_PCIE_RESREF	75
TPC698	TP	1	SSD0_S4E1_PCIE_RESREF	76
TPC6A1	TP	1	SSD0_S4E2_PCIE_RESREF	77
TPC6A0	TP	1	SSD0_S4E3_PCIE_RESREF	78
TPC699	TP	1	I2C_SSD_SCL	38 48
TPC6A2	TP	1	I2C_SSD_SDA	38 48
TPC6A3	TP	1	SSD_BFH	37 74 75 76 77 78
TPC6A4	TP	1	SSD0_PCIE_RESET_L	48 46 75 76 77 78
TPC6A5	TP	1	SSD0_STG01_ADDR	79
TPC6B0	TP	1	TP_SSD0_OCARINA_VR2_DIS	79
TPC6B1	TP	1	SSD0_OCARINA_LPB_L	74 75 76 77 78 79

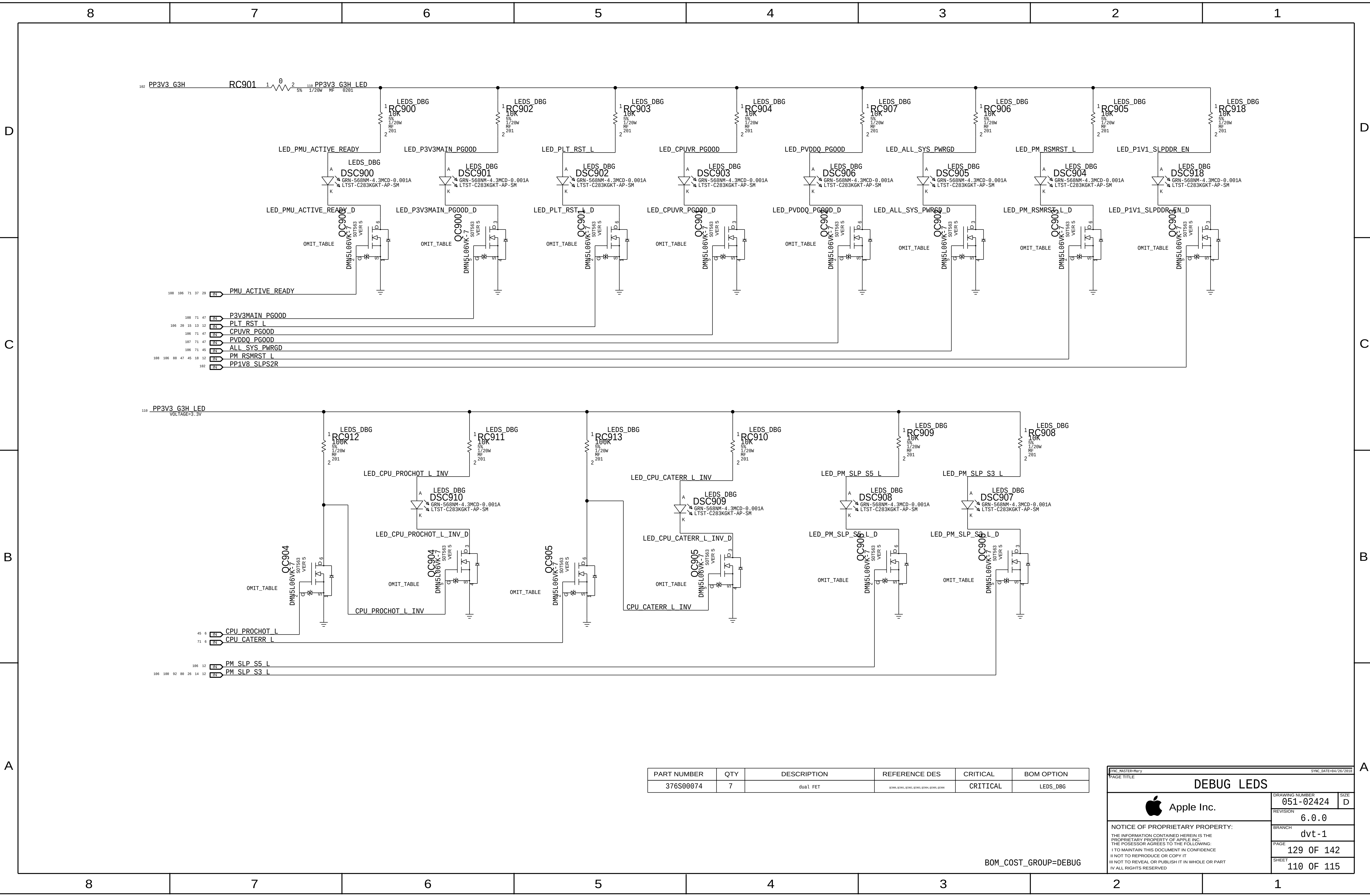
SSD BOUNDARY SCAN Test-Points

TPC6A6	TP	1	SSD0_S4E_JTAG_TRST_L	74 75 76 77 78
TPC6A9	TP	1	SSD0_S4E_JTAG_SEL	74 75 76 77 78
TPC6A8	TP	1	SSD0_SWDIO_UART_D2R	38 74 75 76 77 78
TPC6A7	TP	1	SSD0_SWCLK_UART_R2D	38 74 75 76 77 78
TPC6B2	TP	1	TP_SSD0_S4E0_ANI1_VREF	75
TPC6B3	TP	1	TP_SSD0_S4E0_ANI0_VREF	75
TPC6B6	TP	1	TP_SSD0_S4E1_ANI1_VREF	76
TPC6B5	TP	1	TP_SSD0_S4E1_ANI0_VREF	76
TPC6B4	TP	1	TP_SSD0_S4E2_ANI1_VREF	77
TPC6B7	TP	1	TP_SSD0_S4E2_ANI0_VREF	77
TPC6B8	TP	1	TP_SSD0_S4E3_ANI1_VREF	78
TPC6B9	TP	1	TP_SSD0_S4E3_ANI0_VREF	78
TPC6C1	TP	1	SSD0_S4E0_SWD_UID0	75
TPC6C0	TP	1	SSD0_S4E0_SWD_UID1	75
TPC6C2	TP	1	SSD0_S4E1_SWD_UID0	76
TPC6C3	TP	1	SSD0_S4E1_SWD_UID1	76
TPC6C4	TP	1	SSD0_S4E2_SWD_UID0	77
TPC6C6	TP	1	SSD0_S4E2_SWD_UID1	77
TPC6C7	TP	1	SSD0_S4E3_SWD_UID0	78
TPC6C5	TP	1	SSD0_S4E3_SWD_UID1	78

BOM_COST_GROUP=PCB

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	8	7	6	5	4	3	2	1				
	ICT TESTPOINTS , High Speed NO_TEST											
D	CPU		PCH	Titan Ridges		USB-C CONN		H9M	D			
	PEG		CLK	PCIE		USB-C		PCIE				
	84 6	IN	PCIE ENET D2R N<3..0>	NO_TEST=1	28 26	IN	PCIE TBT X R2D N<3..0>	NO_TEST=1	46 40	IN	PCIE SOC R2D N<3..0>	NO_TEST=1
	84 5	IN	PCIE ENET D2R P<3..0>	NO_TEST=1	28 26	IN	PCIE TBT X R2D P<3..0>	NO_TEST=1	46 40	IN	PCIE SOC R2D P<3..0>	NO_TEST=1
C	84 5	IN	PCIE ENET R2D C N<3..0>	NO_TEST=1	28 26	IN	PCIE TBT X D2R C N<3..0>	NO_TEST=1	46 40	IN	PCIE SOC D2R C N<3..0>	NO_TEST=1
	84 5	IN	PCIE ENET R2D C P<3..0>	NO_TEST=1	28 26	IN	PCIE TBT X D2R C P<3..0>	NO_TEST=1	46 40	IN	PCIE SOC D2R C P<3..0>	NO_TEST=1
	94 5	IN	PCIE TBT T D2R N<3..0>	NO_TEST=1	94 92	IN	PCIE TBT T R2D N<3..0>	NO_TEST=1	78 77 76 75 40	IN	PCIE SSD0 D2R N<3..0>	NO_TEST=1
	94 5	IN	PCIE TBT T D2R P<3..0>	NO_TEST=1	94 92	IN	PCIE TBT T R2D P<3..0>	NO_TEST=1	78 77 76 75 40	IN	PCIE SSD0 D2R P<3..0>	NO_TEST=1
B	94 5	IN	PCIE TBT T R2D C N<3..0>	NO_TEST=1	94 92	IN	PCIE TBT T D2R C N<3..0>	NO_TEST=1	78 77 76 75 40	IN	PCIE SSD0 R2D C N<3..0>	NO_TEST=1
	94 5	IN	PCIE TBT T R2D C P<3..0>	NO_TEST=1	94 92	IN	PCIE TBT T D2R C P<3..0>	NO_TEST=1	78 77 76 75 40	IN	PCIE SSD0 R2D C P<3..0>	NO_TEST=1
	28 5	IN	PCIE TBT X D2R N<3..0>	NO_TEST=1	94 92	IN	PCIE TBT T D2R C N<3..0>	NO_TEST=1	78 77 76 75 40	IN	PCIE SSD0 R2D C N<3..0>	NO_TEST=1
	28 5	IN	PCIE TBT X D2R P<3..0>	NO_TEST=1	94 92	IN	PCIE TBT T D2R C P<3..0>	NO_TEST=1	78 77 76 75 40	IN	PCIE SSD0 R2D C P<3..0>	NO_TEST=1
A	28 5	IN	PCIE TBT X R2D C N<3..0>	NO_TEST=1	26	IN	DP X SNK0 ML N<3..0>	NO_TEST=1	31 28	IN	USBC XA D2R RC N<2..1>	NO_TEST=1
	28 5	IN	PCIE TBT X R2D C P<3..0>	NO_TEST=1	26	IN	DP X SNK0 ML P<3..0>	NO_TEST=1	31 28	IN	USBC XA D2R RC P<2..1>	NO_TEST=1
	28 5	IN	PCIE TBT X R2D C N<3..0>	NO_TEST=1	26	IN	DP X SNK0 AUXCH N	NO_TEST=1	31 28	IN	USBC XA R2D RC N<2..1>	NO_TEST=1
	28 5	IN	PCIE TBT X R2D C P<3..0>	NO_TEST=1	26	IN	DP X SNK0 AUXCH P	NO_TEST=1	31 28	IN	USBC XA R2D RC P<2..1>	NO_TEST=1
	13 5	IN	DMI S2N N<3..0>	NO_TEST=1	26	IN	DP X SNK1 ML N<3..0>	NO_TEST=1	31 28	IN	USBC XA R2D CONN N<2..1>	NO_TEST=1
	13 5	IN	DMI S2N P<3..0>	NO_TEST=1	26	IN	DP X SNK1 ML P<3..0>	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	13 5	IN	DMI N2S N<3..0>	NO_TEST=1	26	IN	DP X SNK1 AUXCH N	NO_TEST=1	31 28	IN	USBC XA R2D CONN N<2..1>	NO_TEST=1
	13 5	IN	DMI N2S P<3..0>	NO_TEST=1	26	IN	DP X SNK1 AUXCH P	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	21 5	IN	DP DDI1 ML C N<3..0>	NO_TEST=1	26	IN	DP X SNK1 ML N<3..0>	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	21 5	IN	DP DDI1 ML C P<3..0>	NO_TEST=1	26	IN	DP X SNK1 ML P<3..0>	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	21 5	IN	DP DDI1 ML C N<3..0>	NO_TEST=1	26	IN	DP X SNK1 AUXCH N	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	21 5	IN	DP DDI1 ML C P<3..0>	NO_TEST=1	26	IN	DP X SNK1 AUXCH P	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	91 5	IN	DP DDI3 ML C N<3..0>	NO_TEST=1	26	IN	DP X SNK1 AUXCH N	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	91 5	IN	DP DDI3 ML C P<3..0>	NO_TEST=1	26	IN	DP X SNK1 AUXCH P	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	21 5	IN	DP DDI1 AUXCH C N	NO_TEST=1	26	IN	DP X SNK1 AUXCH N	NO_TEST=1	31 28	IN	USBC XA R2D CONN P<2..1>	NO_TEST=1
	21 5											



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S00074	7	dual FET	QC900, QC901, QC902, QC903, QC904, QC905, QC906	CRITICAL	LEDS_DBG

SYNC_MASTER=Mary SYNC_DATE=04/26/2018		
PAGE TITLE		
DEBUG LEDS		
 Apple Inc.	DRAWING NUMBER	051-02424
	REVISION	6.0.0
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	PAGE	129 OF 142
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BOM_COST_GROUP=DEBUG

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C																C	
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PAGE TITLE		
Constraints		
 Apple Inc.	DRAWING NUMBER	051-02424
	REVISION	6.0.0
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BOM_COST_GROUP=PCB

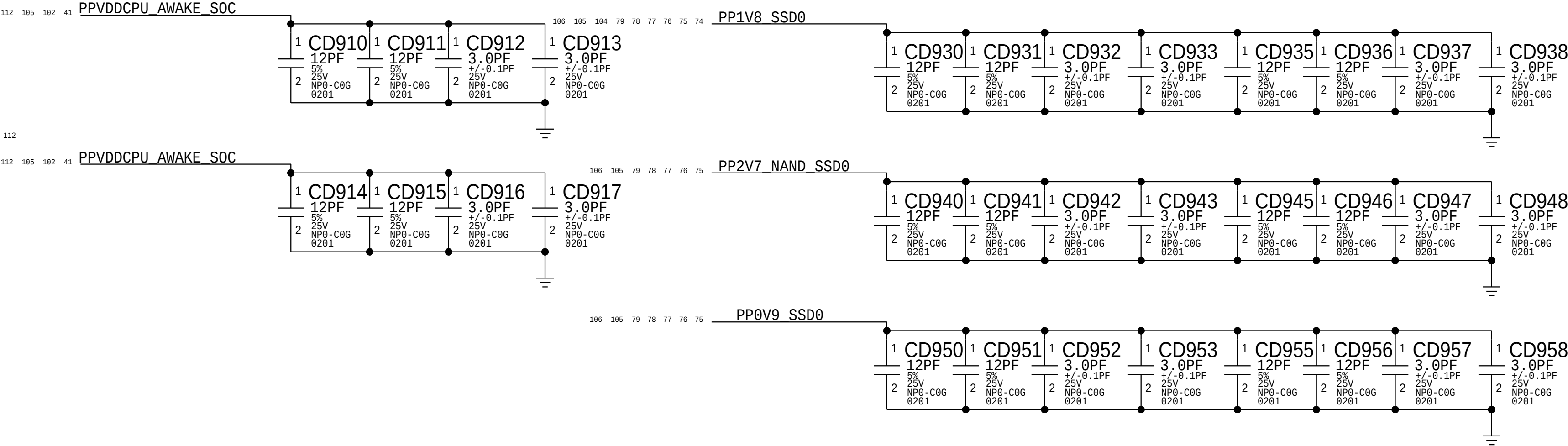
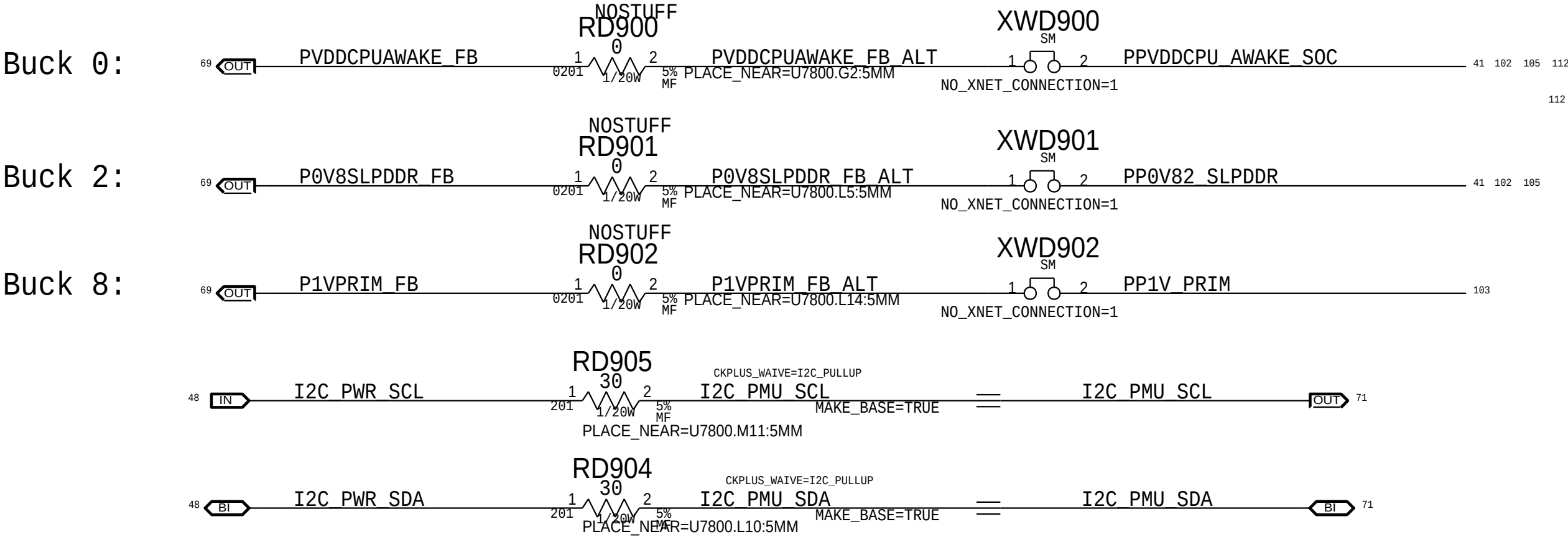
Calpe Dev Support:

H9ML Desense Caps:

SSD/NAND Desense Caps:

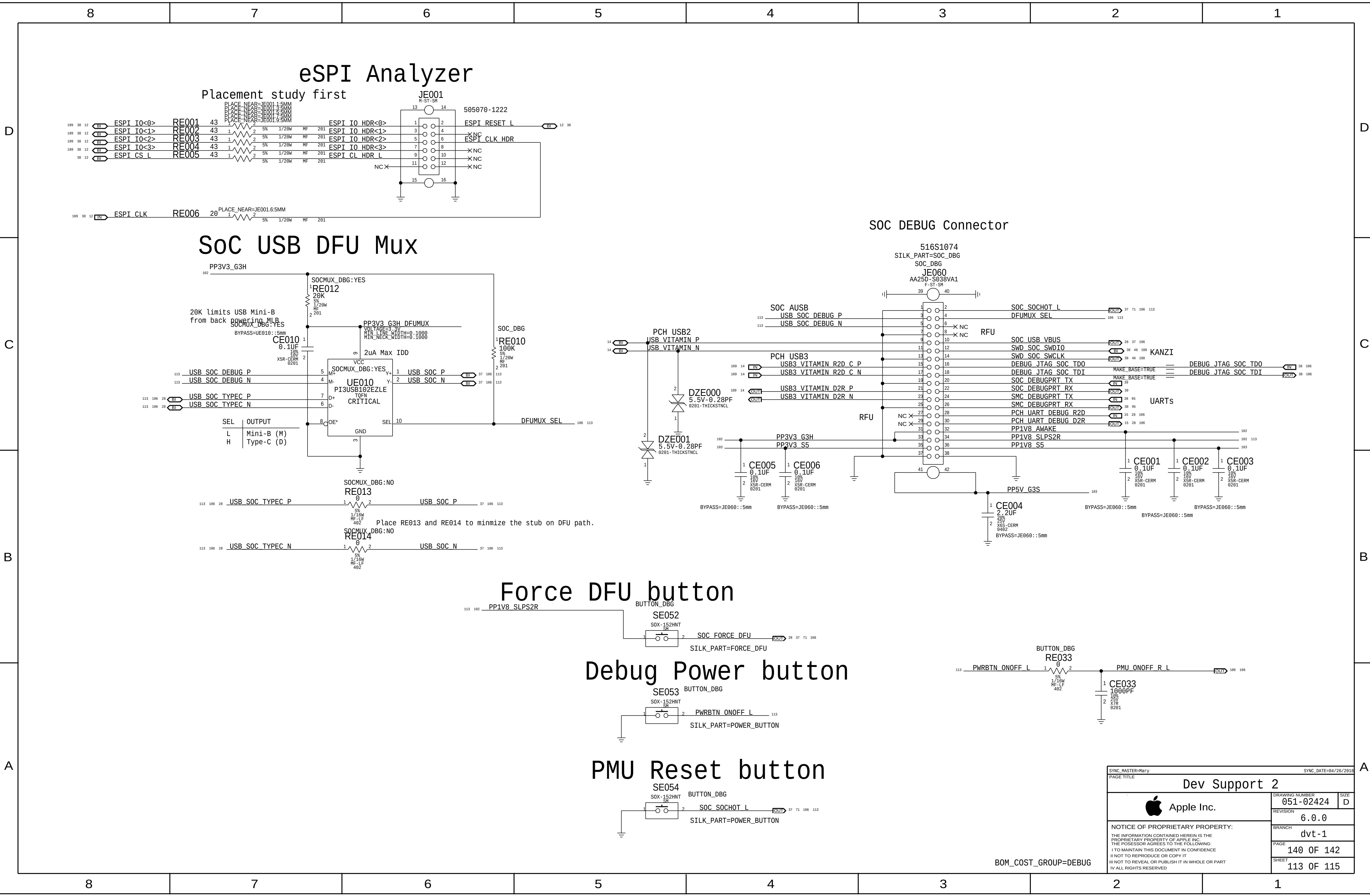
Alternate Feedback:remote sense

Place RD900,RD901 and RD902 to close FB pins to minimize stubs.



SYNC_MASTER=Mary		SYNC_DATE=04/26/2018	
PAGE TITLE			
Dev Support 1			
 Apple Inc.		DRAWING NUMBER	051-02424
		REVISION	6.0.0
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		SHEET	112 OF 115

BOM_COST_GROUP=DEBUG



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D	-->	BOM NUMBER		BOM NAME			BOM OPTIONS			BOM Variants													
		639-06831		PCBA, MLB, 3. 6G_4C, X1036, 128G_WD, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 128_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-													
		639-06830		PCBA, MLB, 3. 6G_4C, X1036, 128G_TB, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 128_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
		639-06849		PCBA, MLB, 3. 6G_4C, X1036, 256G_WD, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 256_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
		639-06832		PCBA, MLB, 3. 6G_4C, X1036, 256G_TB, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 256_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
		639-06833		PCBA, MLB, 3. 6G_4C, X1036, 512G_WD, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 512_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			3.6GHz 4C ENET_1G													
		639-06834		PCBA, MLB, 3. 6G_4C, X1036, 512G_TB, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 512_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-													
		639-06835		PCBA, MLB, 3. 6G_4C, X1036, 1TB_WD, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 1T_SD, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
		639-06836		PCBA, MLB, 3. 6G_4C, X1036, 1TB_TB, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 1T_TB, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
		639-06837		PCBA, MLB, 3. 6G_4C, X1036, 2TB_SM, ENET_1G			COMM_BOM, CPU: 3. 6G, SSD: 2T_SM, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-													
3.6GHz, ENET_10G->	639-06838		PCBA, MLB, 3. 6G_4C, X1036, 128G_WD, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 128_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
	639-06839		PCBA, MLB, 3. 6G_4C, X1036, 128G_TB, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 128_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-06840		PCBA, MLB, 3. 6G_4C, X1036, 256G_WD, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 256_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-06841		PCBA, MLB, 3. 6G_4C, X1036, 256G_TB, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 256_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-06842		PCBA, MLB, 3. 6G_4C, X1036, 512G_WD, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 512_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			3.6GHz 4C ENET_10G														
	639-06843		PCBA, MLB, 3. 6G_4C, X1036, 512G_TB, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 512_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
	639-06844		PCBA, MLB, 3. 6G_4C, X1036, 1TB_WD, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 1T_SD, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-06845		PCBA, MLB, 3. 6G_4C, X1036, 1TB_TB, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 1T_TB, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-06846		PCBA, MLB, 3. 6G_4C, X1036, 2TB_SM, ENET_10G			COMM_BOM, CPU: 3. 6G, SSD: 2T_SM, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
	C	2.8GHz, ENET_1G->	639-05191		PCBA, MLB, 2. 8G, X1036, 128G_WD, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 128_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-												
639-05383			PCBA, MLB, 2. 8G, X1036, 128G_TB, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 128_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
639-05177			PCBA, MLB, 2. 8G, X1036, 256G_WD, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 256_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
639-05385			PCBA, MLB, 2. 8G, X1036, 256G_TB, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 256_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
639-05387			PCBA, MLB, 2. 8G, X1036, 512G_WD, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 512_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			2.8GHz 6C ENET_1G														
639-05391			PCBA, MLB, 2. 8G, X1036, 512G_TB, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 512_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
639-05393			PCBA, MLB, 2. 8G, X1036, 1TB_WD, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 1T_SD, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
639-05397			PCBA, MLB, 2. 8G, X1036, 1TB_TB, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 1T_TB, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
639-05959			PCBA, MLB, 2. 8GHz, X1036, 2TB_SM, ENET_1G			COMM_BOM, CPU: 2. 8G, SSD: 2T_SM, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
2.8GHz, ENET_10G->			639-05189		PCBA, MLB, 2. 8G, X1036, 128G_WD, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 128_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-												
	639-05384		PCBA, MLB, 2. 8G, X1036, 128G_TB, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 128_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-05175		PCBA, MLB, 2. 8G, X1036, 256G_WD, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 256_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-05386		PCBA, MLB, 2. 8G, X1036, 256G_TB, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 256_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-05388		PCBA, MLB, 2. 8G, X1036, 512G_WD, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 512_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			2.8GHz 6C ENET_10G														
	639-05392		PCBA, MLB, 2. 8G, X1036, 512G_TB, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 512_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
	639-05394		PCBA, MLB, 2. 8G, X1036, 1TB_WD, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 1T_SD, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-05398		PCBA, MLB, 2. 8G, X1036, 1TB_TB, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 1T_TB, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-05958		PCBA, MLB, 2. 8GHz, X1036, 2TB_SM, ENET_10G			COMM_BOM, CPU: 2. 8G, SSD: 2T_SM, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
	3.0GHz, ENET_1G->	639-05971		PCBA, MLB, 3. 0GHz, X1036, 256G_WD, ENET_1G			COMM_BOM, CPU: 3. 0G, SSD: 256_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-													
639-05969		PCBA, MLB, 3. 0GHz, X1036, 256G_TB, ENET_1G			COMM_BOM, CPU: 3. 0G, SSD: 256_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																		
639-05977		PCBA, MLB, 3. 0GHz, X1036, 512G_WD, ENET_1G			COMM_BOM, CPU: 3. 0G, SSD: 512_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			3.0GHz 6C ENET_1G															
639-05975		PCBA, MLB, 3. 0GHz, X1036, 512G_TB, ENET_1G			COMM_BOM, CPU: 3. 0G, SSD: 512_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-															
639-05983		PCBA, MLB, 3. 0GHz, X1036, 1T_WD, ENET_1G			COMM_BOM, CPU: 3. 0G, SSD: 1T_SD, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																		
639-05981		PCBA, MLB, 3. 0GHz, X1036, 1T_TB, ENET_1G			COMM_BOM, CPU: 3. 0G, SSD: 1T_TB, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																		
639-05985		PCBA, MLB, 3. 0GHz, X1036, 2T_SM, ENET_1G			COMM_BOM, CPU: 3. 0G, SSD: 2T_SM, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-															
3.0GHz, ENET_10G->		639-05968		PCBA, MLB, 3. 0GHz, X1036, 256G_WD, ENET_10G			COMM_BOM, CPU: 3. 0G, SSD: 256_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-													
		639-05966		PCBA, MLB, 3. 0GHz, X1036, 256G_TB, ENET_10G			COMM_BOM, CPU: 3. 0G, SSD: 256_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
		639-05974		PCBA, MLB, 3. 0GHz, X1036, 512G_WD, ENET_10G			COMM_BOM, CPU: 3. 0G, SSD: 512_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
	639-05972		PCBA, MLB, 3. 0GHz, X1036, 512G_TB, ENET_10G			COMM_BOM, CPU: 3. 0G, SSD: 512_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			3.0GHz 6C ENET_10G														
	639-05980		PCBA, MLB, 3. 0GHz, X1036, 1T_WD, ENET_10G			COMM_BOM, CPU: 3. 0G, SSD: 1T_SD, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
639-05978	PCBA, MLB, 3. 0GHz, X1036, 1T_TB, ENET_10G			COMM_BOM, CPU: 3. 0G, SSD: 1T_TB, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																			
	639-05984			PCBA, MLB, 3. 0GHz, X1036, 2T_SM, ENET_10G			COMM_BOM, CPU: 3. 0G, SSD: 2T_SM, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-													
	3.2GHz, ENET_1G->	639-05997		PCBA, MLB, 3. 2GHz, X1036, 256G_WD, ENET_1G			COMM_BOM, CPU: 3. 2G, SSD: 256_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-													
		639-05995		PCBA, MLB, 3. 2GHz, X1036, 256G_TB, ENET_1G			COMM_BOM, CPU: 3. 2G, SSD: 256_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
		639-06003		PCBA, MLB, 3. 2GHz, X1036, 512G_WD, ENET_1G			COMM_BOM, CPU: 3. 2G, SSD: 512_SD, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																
639-06001		PCBA, MLB, 3. 2GHz, X1036, 512G_TB, ENET_1G			COMM_BOM, CPU: 3. 2G, SSD: 512_TB, X1036_ENET_1G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			3.2GHz 6C ENET_1G															
639-06009		PCBA, MLB, 3. 2GHz, X1036, 1T_WD, ENET_1G			COMM_BOM, CPU: 3. 2G, SSD: 1T_SD, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-															
639-06007		PCBA, MLB, 3. 2GHz, X1036, 1T_TB, ENET_1G			COMM_BOM, CPU: 3. 2G, SSD: 1T_TB, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																		
639-06011		PCBA, MLB, 3. 2GHz, X1036, 2T_SM, ENET_1G			COMM_BOM, CPU: 3. 2G, SSD: 2T_SM, X1036_ENET_1G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-															
3.2GHz, ENET_10G->		639-05994		PCBA, MLB, 3. 2GHz, X1036, 256G_WD, ENET_10G			COMM_BOM, CPU: 3. 2G, SSD: 256_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-													
	639-05992		PCBA, MLB, 3. 2GHz, X1036, 256G_TB, ENET_10G			COMM_BOM, CPU: 3. 2G, SSD: 256_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-06000		PCBA, MLB, 3. 2GHz, X1036, 512G_WD, ENET_10G			COMM_BOM, CPU: 3. 2G, SSD: 512_SD, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			3.3GHz 6C ENET_10G														
	639-05998		PCBA, MLB, 3. 2GHz, X1036, 512G_TB, ENET_10G			COMM_BOM, CPU: 3. 2G, SSD: 512_TB, X1036_ENET_10G, SOC: 1G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
	639-06006		PCBA, MLB, 3. 2GHz, X1036, 1T_WD, ENET_10G			COMM_BOM, CPU: 3. 2G, SSD: 1T_SD, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-06004		PCBA, MLB, 3. 2GHz, X1036, 1T_TB, ENET_10G			COMM_BOM, CPU: 3. 2G, SSD: 1T_TB, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM																	
	639-06010		PCBA, MLB, 3. 2GHz, X1036, 2T_SM, ENET_10G			COMM_BOM, CPU: 3. 2G, SSD: 2T_SM, X1036_ENET_10G, SOC: 2G_DEV, ALTERNATE, DEV_BOM, 1GDBG_BOM			<-														
	Note:SSD SD and WD are the same																						
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639 BOM Configuration 1

Apple Inc.

051-02424

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BOM_COST_GROUP=PCB

